

MOROK TOWER

promin

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Додатки

Ці таблиці є повною згенерованою проєкцією канонічних власників. Кожний запис показує шлях власника та ідентифікатор виконуваного валідатора, посилання на схему, предикат або ідентифікатор мутаційної перевірки.

Власник: `core/promin.manifest.json` | Валідатор: `bundle_digest + core_components`

Трасування канонічних власників

Власник	Єдина функція	Валідатор	SHA256
<code>core/promin.manifest.json</code>	Єдиний власник ідентичності Core bundle і точного digest binding п'яти інших Core artifacts.	<code>urn:promin:contracts:1#/\$defs/StandardManifest</code>	<code>f4441e1ddbee11f9c548a5aeb2a9a15b52db3ebd06ae885b80080bc4f403100d</code>
<code>core/semantic-model.json</code>	Єдиний власник мінімального persistent vocabulary, relation grammar, межі derived projection і provider capability contracts.	<code>urn:promin:contracts:1#/\$defs/SemanticModel</code>	<code>78116797783cf2db1401fe0bbc07df2ed6624679ede4d89f00d73de53a364ef3</code>
<code>core/authority-model.json</code>	Own action capabilities, runtime Grant semantics, trust modes, configured signature-adapter verification, root ceilings, and separation-of-duty constraints.	<code>urn:promin:contracts:1#/\$defs/AuthorityModel</code>	<code>6a0c6f87d8bf5f7257d3aba915bbcb4183ec97f101549a3a933abeb02c32016</code>
<code>core/policy-set.json</code>	Єдиний власник executable cross-record invariants через стабільні validator identifiers без дублювання budget values.	<code>urn:promin:contracts:1#/\$defs/PolicySet</code>	<code>39da9ca0d02c5284fc1a3a4514f211d175dfe92a7533be6b1bff2c5aaca471f1</code>
<code>core/conformance.json</code>	Єдиний власник acceptance predicates, mutation families, structural hard ceilings, scale contracts і production-claim rules.	<code>urn:promin:contracts:1#/\$defs/Conformance</code>	<code>5417f45995b4937ce4a29600cc667d451889cb6e0dd7ef54559cff53e7ee7500</code>
<code>core/contracts.schema.json</code>	Єдиний власник усіх structural, format, kind-specific payload і strict critical-field contracts в одному bundle.	<code>Draft 2020-12 oneOf</code>	<code>3718c1485cc92a2a19ac7b780f8b8050888079b0db5d17319e0b3dfab3d57712</code>

Digest пакета Core: `add3c0dd8891f154ad9f772dcc190d7c596c6b25f13ddc126f38ec3b15137a50`

Вибраний preset поза ідентичністю Core: `presets/semantic-morok-tower.json` | SHA256: `306746fcecf30ca96ad88f11b3e53e855c956fdd55af33e942e65f9edaf6a924a`

Власник: `core/promin.manifest.json` | Валідатор: `canonical_name_only + core_components + bundle_digest`

Каталог схеми - 127 визначень

ID схеми: `urn:promin:contracts:1` | Діалект: `https://json-schema.org/draft/2020-12/schema` |
Визначень: 127

1. ActionCapabilityDef

Опис типу: `type="object"; additionalProperties=false`

Обов'язкові поля: `id`, `purpose`

Властивість	Обов'язков е	Структурне правило
<code>id</code>	так	<code>enum=[16 items]</code>
<code>purpose</code>	так	<code>type="string"; minLength=1</code>
<pre>{ "additionalProperties": false, "properties": { "id": { "enum": ["standard.activate", "authority.manage", "task.plan", "task.execute", "lease.manage",</pre>		


```

        "evidence.publish",
        "validation.evaluate",
        "finding.record",
        "finding.resolve",
        "finding.waive",
        "candidate.promote",
        "release.decide",
        "export.create",
        "migration.manage",
        "projection.rebuild",
        "projection.read"
    ]
},
"purpose": {
    "minLength": 1,
    "type": "string"
}
},
"required": [
    "id",
    "purpose"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/ActionCapabilityDef](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/ActionCapabilityDef](#)

2. Activation

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, canonical_name, core_bundle_digest, preset_digest, init_file_digests, operating_profile, trust_mode, activation_digest, proofs, implementation_closure_digest

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
canonical_name	так	const="promin"
core_bundle_digest	так	\$ref="#/\$defs/Digest"
implementation_closure_digest	так	\$ref="#/\$defs/Digest"
init_file_digests	так	\$ref="#/\$defs/InitFileDigests"
operating_profile	так	\$ref="#/\$defs/Id"
preset_digest	так	\$ref="#/\$defs/Digest"
proofs	так	type="array"; minItems=1; maxItems=32
record_type	так	const="Activation"
trust_mode	так	enum=["local-owner", "team-signed"]

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "canonical_name": {
      "const": "promin"
    },
    "core_bundle_digest": {
      "$ref": "#/$defs/Digest"
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "init_file_digests": {
      "$ref": "#/$defs/InitFileDigests"
    },
    "operating_profile": {
      "$ref": "#/$defs/Id"
    },
    "preset_digest": {
      "$ref": "#/$defs/Digest"
    },
    "proofs": {
      "items": {

```

```

    "$ref": "#/$defs/ActivationProof"
  },
  {
    "maxItems": 32,
    "minItems": 1,
    "type": "array"
  },
  {
    "record_type": {
      "const": "Activation"
    }
  },
  {
    "trust_mode": {
      "enum": [
        "local-owner",
        "team-signed"
      ]
    }
  },
  {
    "required": [
      "record_type",
      "canonical_name",
      "core_bundle_digest",
      "preset_digest",
      "init_file_digests",
      "operating_profile",
      "trust_mode",
      "activation_digest",
      "proofs",
      "implementation_closure_digest"
    ],
    "type": "object"
  }
}

```

Власник: [core/contracts.schema.json#/\\$defs/Activation](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/Activation](#)

3. ActivationProof

Опис типу: `oneOf[2]=#/$defs/LocalActivationProof | #/$defs/SignatureActivationProof`

Обов'язкові поля: -

```

{
  "oneOf": [
    {
      "$ref": "#/$defs/LocalActivationProof"
    },
    {
      "$ref": "#/$defs/SignatureActivationProof"
    }
  ]
}

```

Власник: [core/contracts.schema.json#/\\$defs/ActivationProof](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/ActivationProof](#)

4. Artifact

Опис типу: `type="object"; additionalProperties=false; allOf[3]=keys:else,if,then,title | keys:if,then,title | keys:else,if,then,title`

Обов'язкові поля: `record_type`, `artifact_id`, `artifact_kind`, `digest`, `media_type`, `size_bytes`, `retention_class`, `created_at`

Властивість	Обов'язков е	Структурне правило
<code>artifact_id</code>	так	<code>\$ref="#/\$defs/Id"</code>
<code>artifact_kind</code>	так	<code>enum=["product", "evidence", "log", "report", "diff"]</code>
<code>candidate_delta</code>	ні	<code>\$ref="#/\$defs/CandidateDelta"</code>
<code>created_at</code>	так	<code>\$ref="#/\$defs/Timestamp"</code>
<code>digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>evidence_binding</code>	ні	<code>\$ref="#/\$defs/EvidenceBinding"</code>
<code>evidence_class</code>	ні	<code>enum=["product-execution", "validator", "harness-generated", "migration"]</code>
<code>evidence_purpose</code>	ні	<code>enum=["diagnostic", "gate", "product"]</code>
<code>media_type</code>	так	<code>type="string"; minLength=1</code>
<code>outcome</code>	ні	<code>enum=["pass", "fail", "blocked", "skipped", "error"]</code>
<code>product_credit_eligible</code>	ні	<code>type="boolean"</code>

Властивість	Обов'язков е	Структурне правило
record_type	так	const="Artifact"
retention_class	так	enum=["ephemeral", "project", "release", "audit"]
size_bytes	так	type="integer"; minimum=0
stale	ні	type="boolean"
unresolved	ні	type="boolean"
<pre> { "additionalProperties": false, "allOf": [{ "else": { "not": { "anyOf": [{ "required": ["evidence_binding"] }, { "required": ["evidence_purpose"] }, { "required": ["outcome"] }, { "required": ["stale"] }], "required": ["unresolved"] }, "required": ["evidence_class"] }, "required": ["product_credit_eligible"] }], "if": { "properties": { "artifact_kind": { "const": "evidence" } }, "required": ["artifact_kind"] }, "then": { "required": ["evidence_binding", "evidence_purpose", "outcome", "stale", "unresolved", "evidence_class", "product_credit_eligible"] }, "title": "вимагати незмінні metadata evidence, достатні для replay" } </pre>		

```

    },
    {
      "if": {
        "properties": {
          "product_credit_eligible": {
            "const": true
          }
        },
        "required": [
          "product_credit_eligible"
        ]
      },
      "then": {
        "properties": {
          "artifact_kind": {
            "const": "evidence"
          },
          "evidence_binding": {
            "required": [
              "provider_invocations"
            ]
          },
          "evidence_class": {
            "const": "product-execution"
          },
          "evidence_purpose": {
            "const": "product"
          },
          "outcome": {
            "const": "pass"
          },
          "stale": {
            "const": false
          },
          "unresolved": {
            "const": false
          }
        },
        "required": [
          "evidence_binding",
          "evidence_purpose",
          "outcome",
          "stale",
          "unresolved",
          "evidence_class",
          "product_credit_eligible"
        ]
      },
      "title": "product credit потребує свіжого, вирішеного, успішного evidence виконання продукту"
    },
    {
      "else": {
        "not": {
          "required": [
            "candidate_delta"
          ]
        }
      },
      "if": {
        "properties": {
          "artifact_kind": {
            "const": "diff"
          }
        },
        "required": [
          "artifact_kind"
        ]
      },
      "then": {
        "required": [
          "candidate_delta"
        ]
      },
      "title": "diff Artifact is the canonical Candidate delta change manifest"
    }
  ],
  "properties": {

```

```

"artifact_id": {
  "$ref": "#/$defs/Id"
},
"artifact_kind": {
  "enum": [
    "product",
    "evidence",
    "log",
    "report",
    "diff"
  ]
},
"candidate_delta": {
  "$ref": "#/$defs/CandidateDelta"
},
"created_at": {
  "$ref": "#/$defs/Timestamp"
},
"digest": {
  "$ref": "#/$defs/Digest"
},
"evidence_binding": {
  "$ref": "#/$defs/EvidenceBinding"
},

```

```

"evidence_class": {
  "enum": [
    "product-execution",
    "validator",
    "harness-generated",
    "migration"
  ]
},
"evidence_purpose": {
  "enum": [
    "diagnostic",
    "gate",
    "product"
  ]
},
"media_type": {
  "minLength": 1,
  "type": "string"
},
"outcome": {
  "enum": [
    "pass",
    "fail",
    "blocked",
    "skipped",
    "error"
  ]
},

```

```

"product_credit_eligible": {
  "type": "boolean"
},
"record_type": {
  "const": "Artifact"
},
"retention_class": {
  "enum": [
    "ephemeral",
    "project",
    "release",
    "audit"
  ]
},
"size_bytes": {
  "minimum": 0,
  "type": "integer"
},
"stale": {
  "type": "boolean"
},
"unresolved": {
  "type": "boolean"
}
},
"required": [

```

```

    "record_type",
    "artifact_id",

    "artifact_kind",
    "digest",
    "media_type",
    "size_bytes",
    "retention_class",
    "created_at"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/Artifact | Валідатор: urn:promin:contracts:1#/\$defs/Artifact

5. AuthorityInit

Опис типу: type="object"; additionalProperties=false; allOf[1]=keys;else;if,then

Обов'язкові поля: record_type, trust_mode, subjects, roots

Властивість	Обов'язков е	Структурне правило
keys	ні	type="array"; minItems=1; maxItems=64
record_type	так	const="AuthorityInit"
roots	так	type="array"; minItems=1; maxItems=16
subjects	так	type="array"; minItems=1; maxItems=64
team_policy	ні	\$ref="#/\$defs/TeamPolicy"
trust_mode	так	enum=["local-owner", "team-signed"]

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "allOf": [
          {
            "not": {
              "required": [
                "team_policy"
              ]
            }
          },
          {
            "not": {
              "required": [
                "keys"
              ]
            }
          }
        ]
      },
      "if": {
        "properties": {
          "trust_mode": {
            "const": "team-signed"
          }
        }
      }
    },
    {
      "then": {
        "required": [
          "team_policy",
          "keys"
        ]
      }
    }
  ],
  "properties": {
    "keys": {
      "items": {
        "$ref": "#/$defs/TrustKey"
      },
      "maxItems": 64,
      "minItems": 1,
      "type": "array"
    },

```

```

"record_type": {
  "const": "AuthorityInit"
},
"roots": {
  "items": {
    "$ref": "#/$defs/RootAuthority"
  },
  "maxItems": 16,
  "minItems": 1,
  "type": "array"
},
"subjects": {
  "items": {
    "$ref": "#/$defs/SubjectInit"
  },
  "maxItems": 64,
  "minItems": 1,
  "type": "array"
},
"team_policy": {
  "$ref": "#/$defs/TeamPolicy"
},
"trust_mode": {
  "enum": [
    "local-owner",
    "team-signed"
  ]
},
"required": [
  "record_type",
  "trust_mode",
  "subjects",
  "roots"
],
"type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/AuthorityInit | Валідатор: urn:promin:contracts:1#/\$defs/AuthorityInit

6. AuthorityModel

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, schema_ref, document, version, function, default, capabilities, command_capability_rules, command_effect_scope_rules, bootstrap_rule, trust_modes, role_presets_non_authoritative, separation_of_duties, model_tier_rule, command_mutation_claim_rule, continuation_access_rule, standard_release_decision_rule, release_evidence_authority_rule, canonical_timestamp_contract, delegation_depth_max, derived_event_store_policy_contract, derived_runtime_policy_contract, event_contract, grant_contract, scope_contract, separation_of_duties_contract

Властивість	Обов'язков	Структурне правило
bootstrap_rule	так	const="authority.json configures root ceilings and Activation only. Root command proofs bind one command intent and may issue only the first authority.manage Grant. Every Grant proof binds claim_digest; later Grants cite an active authority.manage issuer Grant or team signatures verified by the exact signature adapter selected in technologies.json."
canonical_timestamp_contract	так	const=schema(keys:calendar,format,formatter_api,fractional_seconds,normalization,offset,parser_api,pattern,real_calendar_validation,resolution_seconds,year_max,year_min)
capabilities	так	const=[16 items]
command_capability_rules	так	const=[11 items]
command_effect_scope_rules	так	const=[11 items]
command_mutation_claim_rule	так	type="object"; additionalProperties=false
continuation_access_rule	так	const=schema(keys:authority_effect,authorization_rule,capability_id,default_budget_source,default_depth_source,default_ttl_seconds,depth_max_source,depth_min,hard_budget_source,max_continuation_state_bytes_source,max_query_bytes_source,max_resume_binding_bytes,max_resume_binding_fields,max_resume_binding_key_chars,max_resume_binding_value_bytes,min_result_bytes,persistent_entity_types_source,ranking_algorithm_id,required_claims,required_resume_binding_fields,revocation_rule,secret_lifecycle,selected_profile_digest_rule,selected_profile_id_source,state_digest_rule,state_persistence,token_envelope,token_max_bytes_source,token_version,traversal_algorithm_id,ttl_min_seconds,ttl_seconds_max)

Властивість	Обов'язков е	Структурне правило
default	так	const="deny"
delegation_depth_max	так	const=8
derived_event_store_policy_contract	так	const=schema(keys:allowed_state_binding_leaf_types_source,command_conditional_fields_source,command_mutation_fields_source,event_contract_source,lease_bound_command_kinds_source,lease_bound_task_transition_states_source,persistence,policy_digest_rule,required_bindings,scope_contract_source,timestamp_contract_source)
derived_runtime_policy_contract	так	const=schema(keys:capability_ids_source,delegation_depth_source,grant_contract_source,persistence,policy_digest_rule,required_bindings,scope_contract_source,separation_of_duties_capability_ids_source,separation_of_duties_contract_source,separation_of_duties_source,timestamp_contract_source)
document	так	const="promin.authority-model"
event_contract	так	const=schema(keys:auxiliary_event_kinds,command_bytes_max,command_conditional_fields,command_required_fields,command_to_primary_event,derived_tail_batch_threshold,derived_tail_byte_threshold,envelope_bytes_max,event_batch_commitment_rule,event_batch_required_fields,event_required_fields,events_per_batch_max,genesis_event_semantic_digest,genesis_previous_authority_commitment,lease_bound_command_kinds_source,lease_bound_task_transition_states_source,primary_event_count_per_command,requested_scope_items_max_source,runtime_overlay_compaction_depth,state_binding_algorithm,state_binding_algorithm_contract,state_binding_bytes_max,state_binding_identity_rule,state_binding_identity_rules,state_binding_leaf_type_source,state_binding_updates_per_batch_max,state_binding_value_rules,timestamp_contract_source)
function	так	const="Own action capabilities, runtime Grant semantics, trust modes, configured signature-adaptor verification, root ceilings, and separation-of-duty constraints."
grant_contract	так	const=schema(keys:claim_digest_rule,nonce,required_fields,scope_items_max_source,trust_proof_kinds,trust_proofs_max,trust_proofs_min,trust_proofs_unique,unknown_proof_kind)
model_tier_rule	так	const="Потужність моделі змінює декомпозицію, контекст, паралельність і глибину аналізу, але ніколи не змінює повноваження."
record_type	так	const="AuthorityModel"
release_evidence_authority_rule	так	const=schema(keys:attestation_algorithm,attestation_domain,decision_capability,evidence_capability,evidence_key_scope,required_claim_bindings,required_evidence_role_count,role_subject_key_uniqueness,separation_rule,unsigned_or_replayed)
role_presets_non_authoritative	так	const=schema(keys:auditor,root-owner,validator,worker)
schema_ref	так	const="urn:promin:contracts:1#/\$defs/AuthorityModel"
scope_contract	так	const=schema(keys:all_selector,conjunctive_effect_scope,duplicate_selector,grant_scope_items_max,kinds,path_selector,project_selector,requested_scope_items_max,selector_fields,unique_items,unknown_kind,value_max_length,value_min_length,wildcard_outside_all)
separation_of_duties	так	const=[5 items]
separation_of_duties_contract	так	const=schema(keys:base_fields,capability_values_max,external_capability_ids,rules_max,supported_rule_fields,unknown_rule_field)
standard_release_decision_rule	так	const=schema(keys:chain,chronology,decision_outcomes,derived_truths,evidence_binding,history_and_current_state,independent_trust_binding,location,product_acceptance_pass,product_public_approval,publication_rule,rule,signature_algorithm,signature_provider_id,standard_name,version_source)
trust_modes	так	const=schema(keys:local-owner,team-signed)
version	так	const="1.0.0-alpha.1"
<pre>{ "additionalProperties": false, "properties": { "bootstrap_rule": { "const": "authority.json configures root ceilings and Activation only. Root command proofs bind one command intent and may issue only the first authority.manage Grant. Every Grant proof binds claim_digest; later Grants cite an active authority.manage issuer Grant or team signatures verified by the exact signature adaptor selected in technologies.json." }, "canonical_timestamp_contract": { "const": { "calendar": "proleptic-gregorian", "format": "utc-second-z", "formatter_api": "promin.canonical.format_utc_second(value: datetime) -> str", "fractional_seconds": "reject", </pre>		


```

    "normalization": "identity-round-trip",
    "offset": "Z-only",
    "parser_api": "promin.canonical.parse_utc_second(value: str) -> timezone-aware UTC datetime",
    "pattern": "^[0-9]{4}-[0-9]{2}-[0-9]{2}T[0-9]{2}:[0-9]{2}:[0-9]{2}Z$",
    "real_calendar_validation": true,
    "resolution_seconds": 1,
    "year_max": 9999,
    "year_min": 1
  }
},
"capabilities": {
  "const": [
    {
      "id": "standard.activate",
      "purpose": "Активувати одну конфігурацію репозиторію, прив'язану до digest."
    },

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    {
      "id": "authority.manage",
      "purpose": "Видавати або відкликати runtime Grants у межах налаштованого root ceiling."
    },
    {
      "id": "task.plan",
      "purpose": "Створювати, ділити, впорядковувати або блокувати Tasks."
    },
    {
      "id": "task.execute",
      "purpose": "Виконувати один Task у межах його WorkCard і чинної Lease."
    },
    {
      "id": "lease.manage",
      "purpose": "Отримувати, підтверджувати heartbeat, закривати, завершувати строк або відкликати Lease."
    },
    {
      "id": "evidence.publish",
      "purpose": "Публікувати evidence Artifacts, прив'язані до candidate."
    },
    {
      "id": "validation.evaluate",
      "purpose": "Оцінювати gates без повноважень на promotion або release."
    },
    {
      "id": "finding.record",
      "purpose": "Записувати Findings."
    },
  ],

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    {
      "id": "finding.resolve",
      "purpose": "Resolve one Finding only with evidence bound to that Finding's canonical digest."
    },
    {
      "id": "finding.waive",
      "purpose": "Waive one Finding only through an expiring exception-policy Decision, Finding-bound evidence, and separation of duties."
    },
    {
      "id": "candidate.promote",
      "purpose": "Просувати перевірений Candidate."
    },
    {
      "id": "release.decide",
      "purpose": "Record an immutable historical release judgement bound to an exact acceptance-closure and provider-binding digest; current eligibility remains derived."
    },
    {
      "id": "export.create",
      "purpose": "Створювати очищений зовнішній пакет."
    },
    {
      "id": "migration.manage",
      "purpose": "Виконувати cutover, rollback і deterministic recutover."
    },
    {
      "id": "projection.rebuild",
      "purpose": "Перебудовувати або перевіряти ненормативні проєкції."
    },
  ],

```

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    {
      "id": "projection.read",

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    "purpose": "Read one bounded current projection or resume its continuation under the exact
current Grant claim; it grants no mutation, validation, promotion, product acceptance, or standard
distribution authority."
  }
]
},
"command_capability_rules": {
  "const": [
    {
      "capability_id": "task.plan",
      "command_kind": "task.record"
    },
    {
      "capability_id": "task.plan",
      "command_kind": "task.transition"
    },
    {
      "capability_id": "task.execute",
      "command_kind": "candidate.record"
    },
    {
      "capability_by_value": {
        "diff": "task.execute",
        "evidence": "evidence.publish",
        "log": "task.execute",
        "product": "task.execute",
        "report": "projection.rebuild"
      }
    },
    {
      "command_kind": "artifact.record",
      "payload_field": "artifact_kind"
    },
    {
      "capability_by_value": {
        "execution": "task.execute",
        "validation": "validation.evaluate"
      },
      "command_kind": "run.record",
      "payload_field": "run_kind"
    },
    {
      "capability_id": "validation.evaluate",
      "command_kind": "gate.record"
    },
    {
      "capability_id": "finding.record",
      "command_kind": "finding.record"
    },
    {
      "capability_by_value": {
        "accept": "validation.evaluate",
        "promote": "candidate.promote",
        "reject": "validation.evaluate",
        "release": "release.decide",
        "resolve": "finding.resolve",
        "revoke": "authority.manage",
        "waive": "finding.waive"
      }
    },
    {
      "command_kind": "decision.record",
      "payload_field": "decision_kind"
    },
    {
      "capability_id": "authority.manage",
      "command_kind": "grant.issue"
    },
    {
      "capability_id": "authority.manage",
      "command_kind": "grant.revoke"
    },
    {
      "capability_id": "lease.manage",
      "command_kind": "lease.record"
    }
  ]
},
"command_effect_scope_rules": {
  "const": [
    {
      "command_kind": "task.record",

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    "id_field": "task_id",
    "kind": "task",
    "mode": "payload-id"
  },
  {
    "command_kind": "task.transition",

    "id_field": "task_id",
    "kind": "task",
    "mode": "payload-id"
  },
  {
    "command_kind": "candidate.record",
    "id_field": "candidate_id",
    "kind": "candidate",
    "mode": "payload-id"
  },
  {
    "command_kind": "artifact.record",
    "id_field": "artifact_id",
    "kind": "artifact",
    "mode": "payload-id"
  },
  {
    "command_kind": "run.record",
    "mode": "payload-multi-id",
    "targets": [
      {
        "id_field": "task_id",
        "kind": "task"
      },
      {
        "id_field": "candidate_digest",
        "kind": "candidate"
      }
    ]
  },
  {
    "command_kind": "gate.record",
    "id_field": "candidate_digest",
    "kind": "candidate",
    "mode": "payload-id"
  },
  {
    "command_kind": "finding.record",
    "id_field": "finding_id",
    "kind": "finding",
    "mode": "payload-id"
  },
  {
    "command_kind": "decision.record",
    "grant_target_type": "Grant",
    "id_field": "target_id",
    "kind_field": "target_type",
    "kind_map": {
      "Candidate": "candidate",
      "Finding": "finding",
      "Task": "task"
    },
    "mode": "payload-dynamic-id-or-referenced-grant-scope"
  },
  {
    "command_kind": "grant.issue",

    "mode": "payload-scope",
    "scope_field": "scope"
  },
  {
    "command_kind": "grant.revoke",
    "grant_id_field": "grant_id",
    "mode": "referenced-grant-scope"
  },
  {
    "command_kind": "lease.record",
    "id_field": "task_id",
    "kind": "task",
    "mode": "payload-id"
  }
]

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    },
    "command_mutation_claim_rule": {
      "additionalProperties": false,
      "properties": {
        "binding_equalities": {
          "items": {
            "minLength": 1,
            "type": "string"
          },
          "maxItems": 16,
          "minItems": 8,
          "type": "array",
          "uniqueItems": true
        }
      }
    },
    "claim_rule": {
      "minLength": 1,
      "type": "string"
    },
    "command_authorization_source": {
      "const": "CommandRequest.authorization.grant_id + grant_claim_digest"
    },
    "grant_separation_rule": {
      "minLength": 1,
      "type": "string"
    },
    "holder_grant_capability": {
      "const": "task.execute"
    },
    "holder_grant_field": {
      "const": "holder_authorization.grant_id"
    },
    "lease_bound_command_kinds": {
      "items": {
        "enum": [
          "task.record",
          "task.transition",
          "candidate.record",
          "artifact.record",
          "run.record",
          "gate.record",
          "finding.record",
          "decision.record",
          "grant.issue",
          "grant.revoke",
          "lease.record"
        ]
      },
      "maxItems": 11,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "lease_bound_task_transition_states": {
      "items": {
        "enum": [
          "BLOCKED",
          "CANCELLED",
          "COMPLETED",
          "LEASED",
          "PLANNED",
          "READY",
          "RUNNING"
        ]
      },
      "maxItems": 7,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "lease_required_state": {
      "const": "ACTIVE"
    },
    "required_command_fields": {
      "const": [
        "workcard_task_id",
        "holder_authorization",
        "lease_id",

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```

        "lease_generation",
        "fencing_token",
        "workcard_digest",
        "context_digest"
    ],
    "type": "array"
},
"required_workcard_fields": {
    "const": [
        "holder_grant_id",
        "lease_id",
        "lease_generation",
        "fencing_token",
        "context_digest"
    ],
    "type": "array"
},
"stale_or_unresolved": {
    "const": "reject"
}
},
"required": [
    "binding_equalities",
    "claim_rule",
    "command_authorization_source",
    "grant_separation_rule",
    "holder_grant_capability",
    "holder_grant_field",
    "lease_bound_command_kinds",
    "lease_bound_task_transition_states",
    "lease_required_state",
    "required_command_fields",
    "required_workcard_fields",
    "stale_or_unresolved"
],
"type": "object"
},
"continuation_access_rule": {
    "const": {
        "authority_effect": false,
        "authorization_rule": "Initial retrieval and every continuation resume require a current
projection.read Grant whose subject, grant_id, claim_digest, capability, Activation, conjunctive requested
scope, interval, and revocation state match the token claims exactly.",
        "capability_id": "projection.read",
        "default_budget_source": "selected Preset profile max_context_bytes, max_entities, max_relations,
max_fanout_per_entity, and top_k",
        "default_depth_source": "selected Preset profile default_dependency_depth",
        "default_ttl_seconds": 900,
        "depth_max_source": "Conformance.dependency_depth_hard_max",
        "depth_min": 1,
        "hard_budget_source": "Conformance.workcard_hard_ceiling",
        "max_continuation_state_bytes_source": "Conformance.scale_contracts.workcard.continuation_state_byt
es_max",
        "max_query_bytes_source": "PolicySet.derived_result_contracts.RetrievalPage.query_bytes_max",
        "max_resume_binding_bytes": 4096,
        "max_resume_binding_fields": 8,
        "max_resume_binding_key_chars": 64,
        "max_resume_binding_value_bytes": 512,
        "min_result_bytes": 1024,
        "persistent_entity_types_source": "SemanticModel.persistent_entities.kind-in-owner-order",
        "ranking_algorithm_id": "bm25-v1",
        "required_claims": [
            "activation_digest",
            "subject_id",
            "grant_id",
            "grant_claim_digest",
            "capability_id",
            "requested_scope",
            "normalized_query_digest",
            "cursor",
            "head_event_id",
            "head_sequence",
            "head_digest",
            "projection_digest",
            "ranking_id",
            "traversal_id",
            "dependency_depth",
            "budgets",

```

<pre> "issued_at", "expires_at", </pre>	
<pre> "ttl_seconds", "revocation_epoch", "implementation_closure_digest", "state_id"], "required_resume_binding_fields": ["activation_digest", "capability_id", "grant_claim_digest", "grant_id", "implementation_closure_digest", "requested_scope_digest", "revocation_epoch", "subject_id"], "revocation_rule": "Expired, revoked, substituted, stale, mismatched, or unresolved Grant claims reject before projection continuation.", "secret_lifecycle": "Generate one random 32-byte secret per Activation, store it only under . promin/state/secrets with restrictive permissions, never derive it from public package or control digests, never journal or export it, and rotate it with Activation.", "selected_profile_digest_rule": "sha256(canonical exact selected Preset profile object)", "selected_profile_id_source": "Activation.operating_profile", "state_digest_rule": "sha256(canonical full ContinuationTokenClaims); the compact token MAC binds token version, opaque state_id, state digest, Activation, subject, Grant claim, and expiry", "state_persistence": "derived-live-only", "token_envelope": "promin-v2.<base64url opaque 128-bit state handle>.<lowercase state digest>.<base64url HMAC-SHA256>", "token_max_bytes_source": "Conformance.scale_contracts.workcard.continuation_token_bytes_max", "token_version": 2, "traversal_algorithm_id": "typed-bfs-v2", "ttl_min_seconds": 1, "ttl_seconds_max": 3600 } </pre>	
<pre> }, "default": { "const": "deny" }, "delegation_depth_max": { "const": 8 }, "derived_event_store_policy_contract": { "const": { "allowed_state_binding_leaf_types_source": "Activation plus SemanticModel.persistent_entities.kind plus Relation", "command_conditional_fields_source": "event_contract.command_conditional_fields", "command_mutation_fields_source": "command_mutation_claim_rule.required_command_fields", "event_contract_source": "event_contract", "lease_bound_command_kinds_source": "command_mutation_claim_rule.lease_bound_command_kinds", "lease_bound_task_transition_states_source": "command_mutation_claim_rule.lease_bound_task_transiti on_states", "persistence": "derived-live-only", "policy_digest_rule": "sha256(canonical EventStorePolicy without policy_digest)", "required_bindings": ["authority_model_digest", "max_command_bytes", "max_envelope_bytes", "max_state_binding_bytes", "max_state_binding_updates_per_batch", "max_events_per_batch", "max_requested_scope_items", "derived_tail_batch_threshold", "derived_tail_byte_threshold", "runtime_overlay_compaction_depth", </pre>	
<pre> "command_required_fields", "command_conditional_fields", "command_mutation_fields", "lease_bound_command_kinds", "lease_bound_task_transition_states", "command_to_primary_event", "allowed_state_binding_leaf_types", "state_binding_identity_rules", "state_binding_algorithm_contract", "state_binding_value_rules", "canonical_timestamp_contract", "genesis_previous_authority_commitment", </pre>	

```

        "genesis_event_semantic_digest",
        "policy_digest"
    ],
    "scope_contract_source": "scope_contract",
    "timestamp_contract_source": "canonical_timestamp_contract"
}
},
"derived_runtime_policy_contract": {
    "const": {
        "capability_ids_source": "capabilities[].id-in-owner-order",
        "delegation_depth_source": "delegation_depth_max",
        "grant_contract_source": "grant_contract",
        "persistence": "derived-live-only",
        "policy_digest_rule": "sha256(canonical AuthorityRuntimePolicy without policy_digest)",
        "required_bindings": [
            "authority_model_digest",
            "capability_ids",
            "separation_of_duties",
            "separation_of_duties_capability_ids",
            "separation_of_duties_contract",
            "delegation_depth_max",
            "scope_contract",
            "grant_contract",
            "canonical_timestamp_contract",
            "policy_digest"
        ],
        "scope_contract_source": "scope_contract",
        "separation_of_duties_capability_ids_source": "capabilities[].id plus
separation_of_duties_contract.external_capability_ids",
        "separation_of_duties_contract_source": "separation_of_duties_contract",
        "separation_of_duties_source": "separation_of_duties-in-owner-order",
        "timestamp_contract_source": "canonical_timestamp_contract"
    }
},
"document": {
    "const": "promin.authority-model"
},
"event_contract": {
    "const": {
        "auxiliary_event_kinds": [
            "relation.recorded"
        ],
        "command_bytes_max": 1048576,
        "command_conditional_fields": [
            "definition_digest"
        ],
        "command_required_fields": [
            "record_type",
            "command_id",
            "command_kind",
            "subject_id",
            "activation_digest",
            "idempotency_key",
            "requested_scope",
            "expected_head_digest",
            "issued_at",
            "payload",
            "intent_digest",
            "authorization"
        ],
        "command_to_primary_event": {
            "artifact.record": "artifact.recorded",
            "candidate.record": "candidate.recorded",
            "decision.record": "decision.recorded",
            "finding.record": "finding.recorded",
            "gate.record": "gate.recorded",
            "grant.issue": "grant.issued",
            "grant.revoke": "grant.revoked",
            "lease.record": "lease.recorded",
            "run.record": "run.recorded",
            "task.record": "task.recorded",
            "task.transition": "task.transitioned"
        }
    },
    "derived_tail_batch_threshold": 128,
    "derived_tail_byte_threshold": 8388608,
    "envelope_bytes_max": 2097152,
    "event_batch_commitment_rule": {

```

```

    "authority_commitment": "sha256(canonical object with exactly previous_authority_commitment,
sequence, previous_batch_digest, event_count, event_semantic_digest, state_binding_update_count,
state_binding_digest)",
    "cumulative_event_count": "previous cumulative_event_count plus current events length",
    "cumulative_state_binding_update_count": "previous cumulative state-binding update count plus
current state_binding_delta length",
    "event_semantic_digest": "ordered append-only digest over every canonical Event",
    "previous_authority_commitment": "exact previous EventBatch authority_commitment; sequence one
uses genesis_previous_authority_commitment",
    "state_binding_delta": "nonempty canonical ordered unique set/delete updates over changed typed
authority leaves only",
    "state_binding_digest": "exact typed-sparse-merkle-v1 post-batch authority root using runtime
domain separators for leaf and internal nodes; never a full serialized state hash"
  },
  "event_batch_required_fields": [
    "record_type",
    "batch_id",
    "sequence",
    "previous_digest",
    "previous_authority_commitment",
    "authority_commitment",
    "cumulative_event_count",
    "event_semantic_digest",
    "state_binding_delta",
    "cumulative_state_binding_update_count",
    "state_binding_digest",
    "created_at",
    "command_id",
    "idempotency_key",
    "activation_record_digest",

    "events",
    "subject_id",
    "command_intent_digest",
    "command_digest",
    "authorization_digest"
  ],
  "event_required_fields": [
    "record_type",
    "event_id",
    "event_kind",
    "activation_digest",
    "payload"
  ],
  "events_per_batch_max": 20,
  "genesis_event_semantic_digest": "e3b0c44298fclcl49afb4c8996fb92427ae41e4649b934ca495991b7852b855",

  "genesis_previous_authority_commitment": "e3b0c44298fclcl49afb4c8996fb92427ae41e4649b934ca495991b7
852b855",
  "lease_bound_command_kinds_source": "command_mutation_claim_rule.lease_bound_command_kinds",
  "lease_bound_task_transition_states_source": "command_mutation_claim_rule.lease_bound_task_transiti
on_states",
  "primary_event_count_per_command": 1,
  "requested_scope_items_max_source": "scope_contract.requested_scope_items_max",
  "runtime_overlay_compaction_depth": 64,
  "state_binding_algorithm": "typed-sparse-merkle-v1",
  "state_binding_algorithm_contract": {
    "delete_rule": "delete maps the keyed leaf to empty_leaf",
    "delta_order": [
      "leaf_type-ascending",
      "leaf_id-ascending"
    ]
  },

  "depth": 256,
  "depth_encoding": "unsigned-16-bit-big-endian",
  "empty_leaf_domain_utf8": "promin:typed-sparse-merkle-v1:empty-leaf",
  "internal_node_domain_utf8": "promin:typed-sparse-merkle-v1:node\u0000",
  "internal_node_rule": "sha256(domain + depth_uint16_be + left_32_bytes + right_32_bytes)",
  "leaf_domain_utf8": "promin:typed-sparse-merkle-v1:leaf\u0000",
  "leaf_key_rule": "sha256(canonical bytes of exact object leaf_type and leaf_id)",
  "set_leaf_rule": "sha256(domain + leaf_key_32_bytes + value_digest_32_bytes)"
},
"state_binding_bytes_max": 1048576,
"state_binding_identity_rule": "direct uses the exact id field; typed_digest_id uses prefix plus
first 48 lowercase hex characters of sha256(canonical ordered id_fields array)",
"state_binding_identity_rules": {
  "Activation": {
    "id_fields": [
      "activation_digest"
    ]
  }
}

```



```

    "identity_mode": "direct"
  },
  "Artifact": {
    "id_fields": [
      "artifact_id"
    ],
    "identity_mode": "direct"
  },
  "Candidate": {
    "id_fields": [
      "candidate_id"
    ],

```

```

    "identity_mode": "direct"
  },
  "Decision": {
    "id_fields": [
      "decision_id"
    ],
    "identity_mode": "direct"
  },
  "Finding": {
    "id_fields": [
      "finding_id"
    ],
    "identity_mode": "direct"
  },
  "GateResult": {
    "digest_algorithm": "sha256",
    "digest_hex_chars": 48,
    "digest_input": "canonical-ordered-id-field-values",
    "id_fields": [
      "gate_id",
      "run_id"
    ],
    "identity_mode": "typed_digest_id",
    "prefix": "gate-result:"
  },
  "Grant": {
    "id_fields": [
      "grant_id"

```

```

    ],
    "identity_mode": "direct"
  },
  "Lease": {
    "id_fields": [
      "lease_id"
    ],
    "identity_mode": "direct"
  },
  "Relation": {
    "id_fields": [
      "relation_id"
    ],
    "identity_mode": "direct"
  },
  "Run": {
    "id_fields": [
      "run_id"
    ],
    "identity_mode": "direct"
  },
  "Task": {
    "id_fields": [
      "task_id"
    ],
    "identity_mode": "direct"
  }
},

```

```

"state_binding_leaf_type_source": "Activation plus SemanticModel.persistent_entities.kind plus
Relation",
"state_binding_updates_per_batch_max": 20,
"state_binding_value_rules": {
  "Activation": {
    "event_kinds": [],
    "value_definition": "Activation",
    "value_digest_rule": "sha256(canonical installed Activation record); installed before sequence
one and immutable for the journal"

```

```

    },
    "Artifact": {
      "event_kinds": [
        "artifact.recorded"
      ],
      "value_definition": "Artifact",
      "value_digest_rule": "sha256(canonical exact Event.payload Artifact)"
    },
    "Candidate": {
      "event_kinds": [
        "candidate.recorded"
      ],
      "value_definition": "Candidate",
      "value_digest_rule": "sha256(canonical exact Event.payload Candidate)"
    },
    "Decision": {
      "event_kinds": [
        "decision.recorded"
      ],
      "value_definition": "Decision",
      "value_digest_rule": "sha256(canonical exact Event.payload Decision)"
    },
    "Finding": {
      "event_kinds": [
        "finding.recorded"
      ],
      "value_definition": "Finding",
      "value_digest_rule": "sha256(canonical exact Event.payload Finding)"
    },
    "GateResult": {
      "event_kinds": [
        "gate.recorded"
      ],
      "value_definition": "GateResult",
      "value_digest_rule": "sha256(canonical exact Event.payload GateResult)"
    },
    "Grant": {
      "event_kinds": [
        "grant.issued",
        "grant.revoked"
      ],
      "value_definition": "GrantAuthorityState",
      "value_digest_rule": "sha256(canonical exact GrantAuthorityState containing the current Grant
and null or exact current GrantRevocation)"
    },
    "Lease": {
      "event_kinds": [
        "lease.recorded"
      ],
      "value_definition": "Оренда",
      "value_digest_rule": "sha256(canonical exact Event.payload post-state Lease)"
    },
    "Relation": {
      "event_kinds": [
        "relation.recorded"
      ],
      "value_definition": "Відношення",
      "value_digest_rule": "sha256(canonical exact Event.payload Relation)"
    },
    "Run": {
      "event_kinds": [
        "run.recorded"
      ],
      "value_definition": "Run",
      "value_digest_rule": "sha256(canonical exact Event.payload Run)"
    },
    "Task": {
      "event_kinds": [
        "task.recorded",
        "task.transitioned"
      ],
      "value_definition": "Завдання",
      "value_digest_rule": "sha256(canonical exact Event.payload post-state Task)"
    }
  },
  "timestamp_contract_source": "canonical_timestamp_contract"
}

```

```

"function": {
  "const": "Own action capabilities, runtime Grant semantics, trust modes, configured
signature-adaptor verification, root ceilings, and separation-of-duty constraints."
},
"grant_contract": {
  "const": {
    "claim_digest_rule": "sha256(canonical Grant without claim_digest and trust_proofs)",
    "nonce": {
      "max_length": 128,
      "min_length": 16,
      "pattern": "^[A-Za-z0-9._:-]{16,128}$"
    },
    "required_fields": [
      "record_type",
      "grant_id",
      "subject_id",
      "capability_id",
      "scope",
      "activation_digest",
      "issued_at",
      "expires_at",
      "nonce",
      "claim_digest",
      "trust_proofs"
    ],
    "scope_items_max_source": "scope_contract.grant_scope_items_max",
    "trust_proof_kinds": {
      "issuer-grant": {
        "required_fields": [
          "kind",
          "issuer_grant_id",
          "issuer_signed_claim_digest",
          "signed_claim_digest"
        ]
      },
      "local-root": {
        "required_fields": [
          "kind",
          "root_subject_id",
          "authority_init_digest",
          "signed_claim_digest"
        ]
      },
      "signature": {
        "required_fields": [
          "kind",
          "key_id",
          "algorithm",
          "signed_digest",
          "signature"
        ]
      },
      "signature_max_length": 8192,
      "signature_min_length": 1
    }
  },
  "trust_proofs_max": 32,
  "trust_proofs_min": 1,
  "trust_proofs_unique": true,
  "unknown_proof_kind": "reject"
},
"model_tier_rule": {
  "const": "Потужність моделі змінює декомпозицію, контекст, паралельність і глибину аналізу, але
ніколи не змінює повноваження."
},
"record_type": {
  "const": "AuthorityModel"
},
"release_evidence_authority_rule": {
  "const": {
    "attestation_algorithm": "Ed25519",
    "attestation_domain": "promin.release-evidence.producer-attestation.v1",
    "decision_capability": "standard.distribute",
    "evidence_capability": "evidence.produce",
    "evidence_key_scope": "Every evidence producer key is independently pinned to an exact subject,
evidence role, platform, public key, validity interval, and revocation state.",
    "required_claim_bindings": [
      "candidate_binding_digest",

```

```

        "record_type",
        "evidence_role",
        "platform",
        "invocation_id",
        "payload_digest",
        "raw_artifact_manifest_digest"
    ],
    "required_evidence_role_count": 7,
    "role_subject_key_uniqueness": "one-role-one-subject-one-key",
    "separation_rule": "For one candidate, an evidence-producing subject or key cannot hold standard.distribute, and a distribution decider subject or key cannot hold evidence.produce.",
    "unsigned_or_replayed": "reject"
}
},
"role_presets_non_authoritative": {
    "const": {
        "auditor": [
            "projection.read",
            "finding.record",
            "finding.resolve",
            "export.create"
        ],
        "root-owner": [
            "standard.activate",
            "authority.manage",
            "task.plan",
            "lease.manage",
            "finding.resolve",
            "finding.waive",
            "candidate.promote",
            "release.decide",
            "export.create",
            "migration.manage",
            "projection.rebuild",
            "projection.read"
        ],
        "validator": [
            "projection.read",
            "validation.evaluate",
            "evidence.publish",
            "finding.record"
        ],
        "worker": [
            "projection.read",
            "task.execute",
            "evidence.publish"
        ]
    }
},
"schema_ref": {
    "const": "urn:promin:contracts:1#/$defs/AuthorityModel"
},
"scope_contract": {
    "const": {
        "all_selector": {
            "exclusive": true,
            "kind": "all",
            "value": "*"
        },
        "conjunctive_effect_scope": true,
        "duplicate_selector": "reject",
        "grant_scope_items_max": 32,
        "kinds": [
            "project",
            "path",
            "task",
            "candidate",
            "artifact",
            "finding",
            "all"
        ],
        "path_selector": {
            "absolute": "reject",
            "dot_segments": "reject",
            "kind": "path",
            "normalization": "NFC POSIX relative path"
        }
    }
},

```

```

    "project_selector": {
      "items_max": 1,
      "kind": "project"
    },
    "requested_scope_items_max": 32,
    "selector_fields": [
      "kind",
      "value"
    ],
    "unique_items": true,
    "unknown_kind": "reject",
    "value_max_length": 4096,
    "value_min_length": 1,
    "wildcard_outside_all": "reject"
  }
},
"separation_of_duties": {

```

```

  "const": [
    {
      "forbid_lease_derived_capabilities": [
        "validation.evaluate",
        "finding.resolve",
        "finding.waive",
        "candidate.promote",
        "release.decide"
      ],
      "id": "SOD-001",
      "rule": "lease-is-not-acceptance",
      "scope_kind": "lease"
    },
    {
      "forbid_same_subject_for_same_candidate": [
        "task.execute",
        "release.decide"
      ],
      "id": "SOD-002",
      "rule": "executor-not-release-decider",
      "scope_kind": "candidate"
    },
    {
      "forbid_same_subject_for_same_candidate": [
        "validation.evaluate",
        "candidate.promote"
      ],
      "id": "SOD-003",

```

```

      "rule": "validator-not-self-promoter",
      "scope_kind": "candidate"
    },
    {
      "forbid_same_subject_for_same_finding": [
        "finding.record",
        "finding.waive"
      ],
      "id": "SOD-004",
      "rule": "finder-not-sole-waiver",
      "scope_kind": "finding"
    },
    {
      "forbid_same_key_capabilities": [
        "evidence.produce",
        "standard.distribute"
      ],
      "forbid_same_subject_for_same_candidate": [
        "evidence.produce",
        "standard.distribute"
      ],
      "id": "SOD-005",
      "rule": "evidence-producer-not-distribution-decider",
      "scope_kind": "candidate-and-key"
    }
  ]
},
"separation_of_duties_contract": {

```

```

  "const": {
    "base_fields": [
      "id",
      "rule",

```

```

    "scope_kind"
  ],
  "capability_values_max": 32,
  "external_capability_ids": [
    "evidence.produce",
    "standard.distribute"
  ],
  "rules_max": 64,
  "supported_rule_fields": {
    "forbid_lease_derived_capabilities": {
      "cardinality": "unique-list",
      "items_min": 1
    },
    "forbid_same_key_capabilities": {
      "cardinality": "unique-pair",
      "items_exact": 2
    },
    "forbid_same_subject_for_same_candidate": {
      "cardinality": "unique-pair",
      "items_exact": 2
    },
    "forbid_same_subject_for_same_finding": {
      "cardinality": "unique-pair",
      "items_exact": 2
    }
  },
  "unknown_rule_field": "reject"
},
"standard_release_decision_rule": {
  "const": {
    "chain": "exact candidate bytes -> StandardReleaseCandidateBinding -> seven unique stable-read authority-role records plus one non-authoritative exact eight-lane matrix and four CP314 supplemental records -> physically resolved StandardReleaseEvidenceManifest -> signed approve or reject StandardReleaseDecision -> independently pinned trust root -> derived distribution status",
    "chronology": "Every evidence invocation starts no later than it completes; its attestation time equals completion and stays within bounded verifier skew. Nested physical runs complete no later than their outer saturation audit, human review occurs inside its signed invocation, and max_evidence_completed_at is derived from all seven authority records, all nested physical records, and all four CP314 supplemental records. decided_at follows that maximum, is no more than 300 seconds beyond verifier time, and every producer/decision key is valid at its event time.",
    "decision_outcomes": [
      "approve",
      "reject"
    ],
    "derived_truths": "ExactPackageVerification is recomputed by the verifier and multi-platform implementation closure is derived only from fully authenticated Windows and Linux records; neither is an independently supplied required role or public command.",
    "evidence_binding": "Every authority, supplemental, matrix, and nested physical record binds exact path, SHA-256, and byte count and is read through a bounded handle-relative no-follow component walk. The manifest rejects symlink/junction/reparse/root substitution and NFC/casefold collisions, validates exact Draft 2020-12 schemas, verifies domain-separated producer attestations, binds producer/tool, role, platform, invocation, raw artifacts, and exact candidate, and runs role-specific semantic recomputation. The aggregate matrix remains non-authoritative with pass_credit=false and all acceptance/public flags false.",
    "history_and_current_state": "The signed StandardReleaseDecision and its digest remain immutable history. Current distribution eligibility is a derived view and becomes false when evidence, trust pin, candidate identity, chronology, key validity, or any required gate no longer validates.",
    "independent_trust_binding": "Manifest construction and closure verification require an independently operator-pinned SHA-256 for the exact trust configuration bytes. Exactly seven evidence roles have one distinct key, subject, and public key each; CP313 and CP314 observations for one role reuse only that role's configured producer. A mismatched, missing, duplicate, over-broad, or invalid scope cannot yield evidence credit or approval.",
    "location": "external-to-canonical-standard-folder",
    "product_acceptance_pass": false,
    "product_public_approval": "not_approved",
    "publication_rule": "Only outcome approve signed by an active configured Ed25519 key with standard.distribute capability and verified against an independently pinned trust-root SHA-256 yields approved distribution status. Missing, rejected, stale, unsigned, mismatched, unpinned, fabricated, or unresolved records fail closed.",
    "rule": "StandardReleaseDecision is external to the canonical standard bytes, binds candidate and evidence manifest digests, includes explicit outcome and anti-replay nonce, proves decider authority through a configured trust root and signature provider, and cannot elevate a trust root delivered by the same untrusted source without an independent digest pin.",
    "signature_algorithm": "Ed25519",
    "signature_provider_id": "cryptography-ed25519-v1",
    "standard_name": "promin",
    "version_source": "StandardReleaseCandidateBinding.version"
  }
}

```

```

    }
  },
  "trust_modes": {
    "const": {
      "local-owner": {
        "boundary": "root authority однієї машини під контролем власника",
        "signer_required": false
      },
      "team-signed": {
        "boundary": "кілька учасників або віддалений транспорт",
        "minimum_threshold": 1,
        "provider_capability": "signature",
        "provider_resolution": "Resolve the exact technologies.json signature binding for init and every
ActivationGuard, service, CLI, replay, and mutation path. A process-local injected adapter is explicitly
non-reconstructable and valid only while bound to the same project, Activation, provider, and identity;
cross-process CLI workflows require a reconstructable configured adapter.",
        "signer_required": true,
        "unsupported_binding": "reject-before-workflow"
      }
    }
  },
  "version": {
    "const": "1.0.0-alpha.1"
  }
},
"required": [
  "record_type",
  "schema_ref",
  "document",
  "version",
  "function",
  "default",
  "capabilities",
  "command_capability_rules",
  "command_effect_scope_rules",
  "bootstrap_rule",
  "trust_modes",
  "role_presets_non_authoritative",
  "separation_of_duties",
  "model_tier_rule",
  "command_mutation_claim_rule",
  "continuation_access_rule",
  "standard_release_decision_rule",
  "release_evidence_authority_rule",
  "canonical_timestamp_contract",
  "delegation_depth_max",
  "derived_event_store_policy_contract",
  "derived_runtime_policy_contract",
  "event_contract",
  "grant_contract",
  "scope_contract",
  "separation_of_duties_contract"
],
"type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/AuthorityModel | Валідатор: urn:promin:contracts:1#/\$defs/AuthorityModel

7. AuthorityRuntimePolicy

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, authority_model_digest, capability_ids, separation_of_duties, separation_of_duties_capability_ids, separation_of_duties_contract, delegation_depth_max, scope_contract, grant_contract, canonical_timestamp_contract, policy_digest

Властивість	Обов'язков е	Структурне правило
authority_model_digest	так	\$ref="#/\$defs/Digest"
canonical_timestamp_contract	так	const=schema(keys:calendar,format,formatter_api,fractional_seconds,normali zation,offset,parser_api,pattern,real_calendar_validation,resolution_seconds,year _max,year_min)
capability_ids	так	const=[16 items]
delegation_depth_max	так	type="integer"; const=8

Властивість	Обов'язков е	Структурне правило
grant_contract	так	const=schema(keys:claim_digest_rule,nonce,required_fields,scope_items_max_source,trust_proof_kinds,trust_proofs_max,trust_proofs_min,trust_proofs_unique,unknown_proof_kind)
policy_digest	так	\$ref="#/\$defs/Digest"
record_type	так	const="AuthorityRuntimePolicy"
scope_contract	так	const=schema(keys:all_selector,conjunctive_effect_scope,duplicate_selector,grant_scope_items_max,kinds,path_selector,project_selector,requested_scope_items_max,selector_fields,unique_items,unknown_kind,value_max_length,value_min_length,wildcard_outside_all)
separation_of_duties	так	const=[5 items]
separation_of_duties_capability_ids	так	const=[18 items]
separation_of_duties_contract	так	const=schema(keys:base_fields,capability_values_max,external_capability_ids,rules_max,supported_rule_fields,unknown_rule_field)
<pre> { "additionalProperties": false, "properties": { "authority_model_digest": { "\$ref": "#/\$defs/Digest" }, "canonical_timestamp_contract": { "const": { "calendar": "proleptic-gregorian", "format": "utc-second-z", "formatter_api": "promin.canonical.format_utc_second(value: datetime) -> str", "fractional_seconds": "reject", "normalization": "identity-round-trip", "offset": "Z-only", "parser_api": "promin.canonical.parse_utc_second(value: str) -> timezone-aware UTC datetime", "pattern": "^[0-9]{4}-[0-9]{2}-[0-9]{2}T[0-9]{2}::[0-9]{2}::[0-9]{2}Z\$", "real_calendar_validation": true, "resolution_seconds": 1, "year_max": 9999, "year_min": 1 } }, "capability_ids": { "const": ["standard.activate", "authority.manage", "task.plan", "task.execute", "lease.manage", "evidence.publish", "validation.evaluate", "finding.record", "finding.resolve", "finding.waive", "candidate.promote", "release.decide", "export.create", "migration.manage", "projection.rebuild", "projection.read"] }, "delegation_depth_max": { "const": 8, "type": "integer" }, "grant_contract": { "const": { "claim_digest_rule": "sha256(canonical Grant without claim_digest and trust_proofs)", "nonce": { "max_length": 128, "min_length": 16, "pattern": "^[A-Za-z0-9._:-]{16,128}\$" }, "required_fields": ["record_type", </pre>		


```

    "grant_id",
    "subject_id",
    "capability_id",
    "scope",
    "activation_digest",
    "issued_at",
    "expires_at",
    "nonce",
    "claim_digest",
    "trust_proofs"
  ],
  "scope_items_max_source": "scope_contract.grant_scope_items_max",
  "trust_proof_kinds": {
    "issuer-grant": {
      "required_fields": [
        "kind",
        "issuer_grant_id",
        "issuer_signed_claim_digest",
        "signed_claim_digest"
      ]
    },
    "local-root": {
      "required_fields": [
        "kind",
        "root_subject_id",
        "authority_init_digest",
        "signed_claim_digest"
      ]
    }
  ]
}

```

```

},
"signature": {
  "required_fields": [
    "kind",
    "key_id",
    "algorithm",
    "signed_digest",
    "signature"
  ],
  "signature_max_length": 8192,
  "signature_min_length": 1
},
"trust_proofs_max": 32,
"trust_proofs_min": 1,
"trust_proofs_unique": true,
"unknown_proof_kind": "reject"
},
"policy_digest": {
  "$ref": "#/$defs/Digest"
},
"record_type": {
  "const": "AuthorityRuntimePolicy"
},
"scope_contract": {
  "const": {
    "all_selector": {

```

```

      "exclusive": true,
      "kind": "all",
      "value": "*"
    },
    "conjunctive_effect_scope": true,
    "duplicate_selector": "reject",
    "grant_scope_items_max": 32,
    "kinds": [
      "project",
      "path",
      "task",
      "candidate",
      "artifact",
      "finding",
      "all"
    ],
    "path_selector": {
      "absolute": "reject",
      "dot_segments": "reject",
      "kind": "path",
      "normalization": "NFC POSIX relative path"
    }
  }
},

```

```

"project_selector": {
  "items_max": 1,
  "kind": "project"
},
"requested_scope_items_max": 32,
"selector_fields": [
  "kind",
  "value"
],
"unique_items": true,
"unknown_kind": "reject",
"value_max_length": 4096,
"value_min_length": 1,
"wildcard_outside_all": "reject"
}
},
"separation_of_duties": {
  "const": [
    {
      "forbid_lease_derived_capabilities": [
        "validation.evaluate",
        "finding.resolve",
        "finding.waive",
        "candidate.promote",
        "release.decide"
      ],
      "id": "SOD-001",
      "rule": "lease-is-not-acceptance",
      "scope_kind": "lease"
    },
    {
      "forbid_same_subject_for_same_candidate": [
        "task.execute",
        "release.decide"
      ],
      "id": "SOD-002",
      "rule": "executor-not-release-decider",
      "scope_kind": "candidate"
    },
    {
      "forbid_same_subject_for_same_candidate": [
        "validation.evaluate",
        "candidate.promote"
      ],
      "id": "SOD-003",
      "rule": "validator-not-self-promoter",
      "scope_kind": "candidate"
    },
    {
      "forbid_same_subject_for_same_finding": [
        "finding.record",
        "finding.waive"
      ],
      "id": "SOD-004",
      "rule": "finder-not-sole-waiver",
      "scope_kind": "finding"
    },
    {
      "forbid_same_key_capabilities": [
        "evidence.produce",
        "standard.distribute"
      ],
      "id": "SOD-005",
      "rule": "evidence-producer-not-distribution-decider",
      "scope_kind": "candidate-and-key"
    }
  ]
},
"separation_of_duties_capability_ids": {
  "const": [
    "standard.activate",
    "authority.manage",
    "task.plan",

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```

        "task.execute",
        "lease.manage",
        "evidence.publish",
        "validation.evaluate",
        "finding.record",
        "finding.resolve",
        "finding.waive",
        "candidate.promote",
        "release.decide",
        "export.create",
        "migration.manage",
        "projection.rebuild",
        "projection.read",

        "evidence.produce",
        "standard.distribute"
    ]
},
"separation_of_duties_contract": {
    "const": {
        "base_fields": [
            "id",
            "rule",
            "scope_kind"
        ],
        "capability_values_max": 32,
        "external_capability_ids": [
            "evidence.produce",
            "standard.distribute"
        ],
        "rules_max": 64,
        "supported_rule_fields": {
            "forbid_lease_derived_capabilities": {
                "cardinality": "unique-list",
                "items_min": 1
            },
            "forbid_same_key_capabilities": {
                "cardinality": "unique-pair",
                "items_exact": 2
            },
            "forbid_same_subject_for_same_candidate": {
                "cardinality": "unique-pair",
                "items_exact": 2
            },
            "forbid_same_subject_for_same_finding": {
                "cardinality": "unique-pair",
                "items_exact": 2
            }
        },
        "unknown_rule_field": "reject"
    }
},
"required": [
    "record_type",
    "authority_model_digest",
    "capability_ids",
    "separation_of_duties",
    "separation_of_duties_capability_ids",
    "separation_of_duties_contract",
    "delegation_depth_max",
    "scope_contract",
    "grant_contract",
    "canonical_timestamp_contract",
    "policy_digest"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Власник: [core/contracts.schema.json#/\\$defs/AuthorityRuntimePolicy](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/AuthorityRuntimePolicy](#)

8. AuthorizationRef

Опис типу: [oneOf\[2\]=#/\\$defs/GrantAuthorization](#) | [#/\\$defs/RootAuthorization](#)

Обов'язкові поля: -

```
{
  "oneOf": [
    {
      "$ref": "#/$defs/GrantAuthorization"
    },
    {
      "$ref": "#/$defs/RootAuthorization"
    }
  ]
}
```

Власник: core/contracts.schema.json#/\$defs/AuthorizationRef | Валідатор: urn:promin:contracts:1#/\$defs/AuthorizationRef

9. Budget

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: max_bytes, max_entities, max_relations, max_fanout_per_entity, top_k

Властивість	Обов'язков е	Структурне правило
max_bytes	так	type="integer"; minimum=1024
max_entities	так	type="integer"; minimum=1
max_fanout_per_entity	так	type="integer"; minimum=1
max_relations	так	type="integer"; minimum=0
top_k	так	type="integer"; minimum=1

```
{
  "additionalProperties": false,
  "properties": {
    "max_bytes": {
      "minimum": 1024,
      "type": "integer"
    },
    "max_entities": {
      "minimum": 1,
      "type": "integer"
    },
    "max_fanout_per_entity": {
      "minimum": 1,
      "type": "integer"
    },
    "max_relations": {
      "minimum": 0,
      "type": "integer"
    },
    "top_k": {
      "minimum": 1,
      "type": "integer"
    }
  },
  "required": [
    "max_bytes",
    "max_entities",
    "max_relations",
    "max_fanout_per_entity",
    "top_k"
  ],
  "type": "object"
}
```

Власник: core/contracts.schema.json#/\$defs/Budget | Валідатор: urn:promin:contracts:1#/\$defs/Budget

10. Candidate

Опис типу: type="object"; additionalProperties=false; allOf[2]=keys:else,if,then,title | keys:else,if,then,title

Обов'язкові поля: record_type, candidate_id, candidate_digest, inventory_digest, product_root_digest, control_excluded, candidate_recipe_digest, consistency_mode, creditable

Властивість	Обов'язков е	Структурне правило
baseline_kind	ні	enum=["specification", "product", "hybrid"]
candidate_digest	так	\$ref="#/\$defs/Digest"
candidate_id	так	\$ref="#/\$defs/Id"

Властивість	Обов'язков е	Структурне правило
candidate_recipe_digest	так	\$ref="#/\$defs/Digest"
consistency_mode	так	enum=["observational-best-effort", "immutable-vcs-tree"]
control_excluded	так	const=true
creditable	так	type="boolean"
inventory_digest	так	\$ref="#/\$defs/Digest"
product_root_digest	так	\$ref="#/\$defs/Digest"
record_type	так	const="Candidate"
snapshot_digest	ні	\$ref="#/\$defs/Digest"
snapshot_provider_id	ні	\$ref="#/\$defs/Id"
specification_digest	ні	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [
            "specification_digest"
          ]
        }
      },
      "if": {
        "properties": {
          "baseline_kind": {
            "enum": [
              "specification",
              "hybrid"
            ]
          }
        },
        "required": [
          "baseline_kind"
        ]
      },
      "then": {
        "required": [
          "specification_digest"
        ]
      }
    },
    {
      "title": "Candidate specification baseline binding is exact"
    },
    {
      "else": {
        "not": {
          "anyOf": [
            {
              "required": [
                "snapshot_provider_id"
              ]
            },
            {
              "required": [
                "snapshot_digest"
              ]
            }
          ]
        },
        "properties": {
          "creditable": {
            "const": false
          }
        }
      },
      "if": {
        "properties": {
          "consistency_mode": {
            "enum": [
              "immutable-vcs-tree"
            ]
          }
        }
      }
    }
  ]
}

```

```

    }
  },
  "required": [
    "consistency_mode"
  ]
},
"then": {
  "properties": {
    "creditable": {
      "const": true
    }
  },
  "required": [
    "snapshot_provider_id",
    "snapshot_digest"
  ]
},
"title": "creditable Candidate requires provider-proven immutable snapshot"
}
],
"properties": {
  "baseline_kind": {
    "enum": [
      "specification",
      "product",

```

```

      "hybrid"
    ]
  },
  "candidate_digest": {
    "$ref": "#/$defs/Digest"
  },
  "candidate_id": {
    "$ref": "#/$defs/Id"
  },
  "candidate_recipe_digest": {
    "$ref": "#/$defs/Digest"
  },
  "consistency_mode": {
    "enum": [
      "observational-best-effort",
      "immutable-vcs-tree"
    ]
  },
  "control_excluded": {
    "const": true
  },
  "creditable": {
    "type": "boolean"
  },
  "inventory_digest": {
    "$ref": "#/$defs/Digest"
  },
  "product_root_digest": {

```

```

    "$ref": "#/$defs/Digest"
  },
  "record_type": {
    "const": "Candidate"
  },
  "snapshot_digest": {
    "$ref": "#/$defs/Digest"
  },
  "snapshot_provider_id": {
    "$ref": "#/$defs/Id"
  },
  "specification_digest": {
    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "record_type",
  "candidate_id",
  "candidate_digest",
  "inventory_digest",
  "product_root_digest",
  "control_excluded",
  "candidate_recipe_digest",
  "consistency_mode",
  "creditable"

```

```

    },
    "type": "object"
  }

```

Власник: core/contracts.schema.json#/\$defs/Candidate | Валідатор: urn:promin:contracts:1#/\$defs/Candidate

11. CandidateDelta

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: base_candidate_digest, new_candidate_digest, workcard_digest, changed_paths

Властивість	Обов'язков е	Структурне правило
base_candidate_digest	так	\$ref="#/\$defs/Digest"
changed_paths	так	type="array"; minItems=1; maxItems=100000; uniqueItems=true
new_candidate_digest	так	\$ref="#/\$defs/Digest"
workcard_digest	так	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "properties": {
    "base_candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "changed_paths": {
      "items": {
        "$ref": "#/$defs/RelativePath"
      },
      "maxItems": 100000,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "new_candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "workcard_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "base_candidate_digest",
    "new_candidate_digest",
    "workcard_digest",
    "changed_paths"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/CandidateDelta | Валідатор: urn:promin:contracts:1#/\$defs/CandidateDelta

12. CandidateRecipe

Опис типу: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then,title

Обов'язкові поля: inventory_mode, include, exclude, symlink_policy, path_identity, collision_policy, product_identity_excludes_control_state

Властивість	Обов'язков е	Структурне правило
collision_policy	так	const="reject-nfc-and-casefold-collisions"
exclude	так	type="array"; minItems=1; maxItems=128; uniqueItems=true
include	так	type="array"; minItems=1; maxItems=64; uniqueItems=true
inventory_mode	так	const="explicit"
path_identity	так	const="nfc-posix-relative"
product_identity_excludes_control_state	так	const=true
snapshot_consistency	ні	enum=["observational-best-effort", "immutable-vcs-tree"]
snapshot_provider_id	ні	\$ref="#/\$defs/Id"
symlink_policy	так	enum=["reject", "hash-link-metadata"]

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [
            "snapshot_provider_id"
          ]
        }
      },
      "if": {
        "properties": {
          "snapshot_consistency": {
            "enum": [
              "immutable-vcs-tree"
            ]
          }
        },
        "required": [
          "snapshot_consistency"
        ]
      },
      "then": {
        "required": [
          "snapshot_provider_id"
        ]
      }
    },
    {
      "title": "immutable snapshot recipe requires exact provider"
    }
  ],
  "properties": {
    "collision_policy": {
      "const": "reject-nfc-and-casefold-collisions"
    },
    "exclude": {
      "items": {
        "$ref": "#/$defs/RelativePath"
      },
      "maxItems": 128,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "include": {
      "items": {
        "$ref": "#/$defs/RelativePath"
      },
      "maxItems": 64,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "inventory_mode": {
      "const": "explicit"
    },
    "path_identity": {
      "const": "nfc-posix-relative"
    },
    "product_identity_excludes_control_state": {
      "const": true
    },
    "snapshot_consistency": {
      "enum": [
        "observational-best-effort",
        "immutable-vcs-tree"
      ]
    },
    "snapshot_provider_id": {
      "$ref": "#/$defs/Id"
    },
    "symlink_policy": {
      "enum": [
        "reject",
        "hash-link-metadata"
      ]
    }
  },

```



```

"required": [
  "inventory_mode",
  "include",
  "exclude",
  "symlink_policy",
  "path_identity",

  "collision_policy",
  "product_identity_excludes_control_state"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/CandidateRecipe](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/CandidateRecipe](#)

13. CapacityReconciliation

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: reconciled_by, grant_id, grant_claim_digest, reconciled_at, generation, fencing_token

Властивість	Обов'язков е	Структурне правило
fencing_token	так	type="integer"; minimum=1
generation	так	type="integer"; minimum=1
grant_claim_digest	так	\$ref="#/\$defs/Digest"
grant_id	так	\$ref="#/\$defs/Id"
reconciled_at	так	\$ref="#/\$defs/Timestamp"
reconciled_by	так	\$ref="#/\$defs/Id"

```

{
  "additionalProperties": false,
  "properties": {
    "fencing_token": {
      "minimum": 1,
      "type": "integer"
    },
    "generation": {
      "minimum": 1,
      "type": "integer"
    },
    "grant_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "grant_id": {
      "$ref": "#/$defs/Id"
    },
    "reconciled_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "reconciled_by": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "reconciled_by",
    "grant_id",
    "grant_claim_digest",

    "reconciled_at",
    "generation",
    "fencing_token"
  ],
  "type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/CapacityReconciliation](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/CapacityReconciliation](#)

14. CloseAck

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: acknowledged_by, grant_id, grant_claim_digest, acknowledged_at

Властивість	Обов'язков е	Структурне правило
acknowledged_at	так	\$ref="#/\$defs/Timestamp"
acknowledged_by	так	\$ref="#/\$defs/Id"
grant_claim_digest	так	\$ref="#/\$defs/Digest"
grant_id	так	\$ref="#/\$defs/Id"
<pre>{ "additionalProperties": false, "properties": { "acknowledged_at": { "\$ref": "#/\$defs/Timestamp" }, "acknowledged_by": { "\$ref": "#/\$defs/Id" }, "grant_claim_digest": { "\$ref": "#/\$defs/Digest" }, "grant_id": { "\$ref": "#/\$defs/Id" } }, "required": ["acknowledged_by", "grant_id", "grant_claim_digest", "acknowledged_at"], "type": "object" }</pre>		
Власник: core/contracts.schema.json#/\$defs/CloseAck Валідатор: urn:promin:contracts:1#/\$defs/CloseAck		

15. CommandCapabilityRule

Опис типу: type="object"; additionalProperties=false; oneOf[2]=keys:not,required | keys:not,required

Обов'язкові поля: command_kind

Властивість	Обов'язков е	Структурне правило
capability_by_value	ні	type="object"; additionalProperties=schema(keys:\$ref)
capability_id	ні	\$ref="#/\$defs/Id"
command_kind	так	\$ref="#/\$defs/Id"
payload_field	ні	\$ref="#/\$defs/Id"
<pre>{ "additionalProperties": false, "oneOf": [{ "not": { "anyOf": [{ "required": ["payload_field"] }, { "required": ["capability_by_value"] }] }, "required": ["capability_id"] }, { "not": { "required": ["capability_id"] } }], "type": "object" }</pre>		

```

    "required": [
      "payload_field",
      "capability_by_value"
    ]
  },
  "properties": {
    "capability_by_value": {
      "additionalProperties": {
        "$ref": "#/$defs/Id"
      },
      "minProperties": 1,
      "type": "object"
    },
    "capability_id": {
      "$ref": "#/$defs/Id"
    },
    "command_kind": {
      "$ref": "#/$defs/Id"
    },
    "payload_field": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "command_kind"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/CommandCapabilityRule | Валідатор: urn:promin:contracts:1#/\$defs/CommandCapabilityRule

16. CommandEffectScopeRule

Опис типу: type="object"; additionalProperties=false; allOf[6]=keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then

Обов'язкові поля: command_kind, mode

Властивість	Обов'язков е	Структурне правило
command_kind	так	\$ref="#/\$defs/Id"
grant_id_field	ні	\$ref="#/\$defs/Id"
grant_target_type	ні	const="Grant"
id_field	ні	\$ref="#/\$defs/Id"
kind	ні	\$ref="#/\$defs/Id"
kind_field	ні	\$ref="#/\$defs/Id"
kind_map	ні	type="object"; additionalProperties=schema(keys:\$ref)
mode	так	enum=["payload-id", "payload-multi-id", "payload-dynamic-id", "payload-scope", "referenced-grant-scope", "payload-dynamic-id-or-referenced-grant-scope"]
scope_field	ні	\$ref="#/\$defs/Id"
targets	ні	type="array"; minItems=1; maxItems=8; uniqueItems=true

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "if": {
        "properties": {
          "mode": {
            "const": "payload-id"
          }
        }
      },
      "then": {
        "required": [
          "kind",
          "id_field"
        ]
      }
    }
  ],
}

```

```

{
  "if": {
    "properties": {
      "mode": {
        "const": "payload-multi-id"
      }
    }
  },
  "then": {
    "required": [
      "targets"
    ]
  }
},
{
  "if": {
    "properties": {
      "mode": {
        "const": "payload-dynamic-id"
      }
    }
  },
  "then": {
    "required": [
      "kind_field",
      "kind_map",
      "id_field"
    ]
  }
},
{
  "if": {
    "properties": {
      "mode": {
        "const": "payload-scope"
      }
    }
  },
  "then": {
    "required": [
      "scope_field"
    ]
  }
},
{
  "if": {
    "properties": {
      "mode": {
        "const": "referenced-grant-scope"
      }
    }
  },
  "then": {
    "required": [
      "grant_id_field"
    ]
  }
},
{
  "if": {
    "properties": {
      "mode": {
        "const": "payload-dynamic-id-or-referenced-grant-scope"
      }
    }
  },
  "then": {
    "required": [
      "kind_field",
      "kind_map",
      "id_field",
      "grant_target_type"
    ]
  }
},
],
"properties": {

```

Власник: `core/contracts.schema.json#/$defs/CommandEffectScopeRule` | Валідатор: `urn:promin:contracts:1#/$defs/CommandEffectScopeRule`

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
authorization	так	\$ref="#/\$defs/AuthorizationRef"
command_id	так	\$ref="#/\$defs/Id"
command_kind	так	enum=["task.record", "task.transition", "candidate.record", "artifact.record", "run.record", "gate.record", "finding.record", "decision.record", "grant.issue", "grant.revoke", "lease.record"]

Властивість	Обов'язков е	Структурне правило
context_digest	ні	\$ref="#/\$defs/Digest"
definition_digest	ні	\$ref="#/\$defs/Digest"
expected_head_digest	так	oneOf[2]=#/\$defs/Digest type:null
fencing_token	ні	type="integer"; minimum=1
holder_authorization	ні	\$ref="#/\$defs/GrantAuthorization"
idempotency_key	так	type="string"; pattern="^[A-Za-z0-9._-]{16,128}\$"
intent_digest	так	\$ref="#/\$defs/Digest"
issued_at	так	\$ref="#/\$defs/Timestamp"
lease_generation	ні	type="integer"; minimum=1
lease_id	ні	\$ref="#/\$defs/Id"
payload	так	складене структурне визначення
record_type	так	const="CommandRequest"
requested_scope	так	type="array"; minItems=1; maxItems=32
subject_id	так	\$ref="#/\$defs/Id"
workcard_digest	ні	\$ref="#/\$defs/Digest"
workcard_task_id	ні	\$ref="#/\$defs/Id"

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "if": {
        "properties": {
          "command_kind": {
            "const": "task.record"
          }
        }
      },
      "then": {
        "properties": {
          "payload": {
            "$ref": "#/$defs/Task"
          }
        }
      }
    },
    {
      "if": {
        "properties": {
          "command_kind": {
            "const": "task.transition"
          }
        }
      },
      "then": {
        "properties": {
          "payload": {
            "$ref": "#/$defs/TaskTransition"
          }
        }
      }
    },
    {
      "if": {
        "properties": {
          "command_kind": {
            "const": "candidate.record"
          }
        }
      },
      "then": {
        "properties": {
          "payload": {
            "$ref": "#/$defs/Candidate"
          }
        }
      }
    }
  ]
}

```

```

},
{
  "if": {
    "properties": {
      "command_kind": {
        "const": "artifact.record"

```

```

      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Artifact"
      }
    }
  }
},
{
  "if": {
    "properties": {
      "command_kind": {
        "const": "run.record"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Run"
      }
    }
  }
},
{
  "if": {
    "properties": {
      "command_kind": {
        "const": "gate.record"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/GateResult"
      }
    }
  }
},
{
  "if": {
    "properties": {
      "command_kind": {
        "const": "finding.record"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Finding"
      }
    }
  }
},
{
  "if": {
    "properties": {
      "command_kind": {
        "const": "decision.record"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Decision"
      }
    }
  }
}

```

```

    "properties": {
      "command_kind": {
        "const": "gate.record"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/GateResult"
      }
    }
  }
},
{
  "if": {
    "properties": {
      "command_kind": {
        "const": "finding.record"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Finding"
      }
    }
  }
}

```

```

    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Decision"
      }
    }
  }
}

```

```

    }
  }
},
{
  "if": {
    "properties": {
      "command_kind": {
        "const": "grant.issue"
      }
    }
  },
  "then": {

```

```

    "properties": {
      "payload": {
        "$ref": "#/$defs/Grant"
      }
    }
  },
},
{
  "if": {
    "properties": {
      "command_kind": {
        "const": "grant.revoke"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/GrantRevocation"
      }
    }
  }
},
{
  "if": {
    "properties": {
      "command_kind": {
        "const": "lease.record"

```

```

    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Lease"
      }
    }
  },
},
{
  "if": {
    "properties": {
      "command_kind": {
        "enum": [
          "artifact.record",
          "run.record",
          "gate.record",
          "finding.record"
        ]
      }
    },
    "required": [
      "command_kind"
    ]
  },
  "then": {

```

```

    "required": [
      "workcard_task_id",
      "holder_authorization",
      "lease_id",
      "lease_generation",
      "fencing_token",
      "workcard_digest",
      "context_digest"

```



```

    ],
    "title": "вимагати поточну Lease й обмежений WorkCard для прямої мутації"
  },
  {
    "if": {
      "properties": {
        "command_kind": {
          "const": "task.transition"
        },
        "payload": {
          "properties": {
            "to_state": {
              "enum": [
                "LEASED",
                "RUNNING",
                "COMPLETED"
              ]
            }
          }
        }
      },
      "required": [
        "to_state"
      ]
    },
    "required": [
      "command_kind",
      "payload"
    ]
  },
  "then": {
    "required": [
      "workcard_task_id",
      "holder_authorization",
      "lease_id",
      "lease_generation",
      "fencing_token",
      "workcard_digest",
      "context_digest"
    ]
  },
  "title": "вимагати поточну Lease й обмежений WorkCard для переходу execution"
},
{
  "if": {
    "not": {
      "anyOf": [
        {
          "properties": {
            "command_kind": {
              "enum": [
                "artifact.record",
                "run.record",
                "gate.record",
                "finding.record"
              ]
            }
          },
          "required": [
            "command_kind"
          ]
        },
        {
          "properties": {
            "command_kind": {
              "const": "task.transition"
            },
            "payload": {
              "properties": {
                "to_state": {
                  "enum": [
                    "LEASED",
                    "RUNNING",
                    "COMPLETED"
                  ]
                }
              }
            }
          }
        }
      ]
    }
  }
}

```

```

        },
        "required": [
            "to_state"
        ]
    }
},
"required": [
    "command_kind",
    "payload"
]
}
]
}
},
"then": {
    "not": {
        "anyOf": [
            {
                "required": [
                    "workcard_task_id"
                ]
            },
            {
                "required": [
                    "holder_authorization"
                ]
            }
        ],
        {

```

```

            "required": [
                "lease_id"
            ]
        },
        {
            "required": [
                "lease_generation"
            ]
        },
        {
            "required": [
                "fencing_token"
            ]
        },
        {
            "required": [
                "workcard_digest"
            ]
        },
        {
            "required": [
                "context_digest"
            ]
        }
    ]
}
},
"title": "заборонити mutation binding для команди без Lease"

```

```

},
{
    "if": {
        "properties": {
            "command_kind": {
                "enum": [
                    "run.record",
                    "gate.record"
                ]
            }
        }
    },
    "then": {
        "required": [
            "definition_digest"
        ]
    },
    "title": "run and gate commands bind an immutable definition digest"
},
{
    "if": {
        "properties": {

```

```

        "command_kind": {
          "enum": [
            "task.record",
            "task.transition",
            "candidate.record",
            "artifact.record",
            "finding.record",
            "decision.record",
            "grant.issue",
            "grant.revoke",
            "lease.record"
          ]
        },
        "then": {
          "not": {
            "required": [
              "definition_digest"
            ]
          }
        },
        "title": "other commands cannot claim a gate run definition"
      }
    ],
    "properties": {
      "activation_digest": {
        "$ref": "#/$defs/Digest"
      },
      "authorization": {
        "$ref": "#/$defs/AuthorizationRef"
      },
      "command_id": {
        "$ref": "#/$defs/Id"
      }
    }
  },
  "command_kind": {
    "enum": [
      "task.record",
      "task.transition",
      "candidate.record",
      "artifact.record",
      "run.record",
      "gate.record",
      "finding.record",
      "decision.record",
      "grant.issue",
      "grant.revoke",
      "lease.record"
    ]
  },
  "context_digest": {
    "$ref": "#/$defs/Digest"
  },
  "definition_digest": {
    "$ref": "#/$defs/Digest"
  },
  "expected_head_digest": {
    "oneOf": [
      {
        "$ref": "#/$defs/Digest"
      },
      {

```

```

        "type": "null"
      }
    ]
  },
  "fencing_token": {
    "minimum": 1,
    "type": "integer"
  },
  "holder_authorization": {
    "$ref": "#/$defs/GrantAuthorization"
  },
  "idempotency_key": {
    "pattern": "^[A-Za-z0-9._-]{16,128}$",
    "type": "string"
  },

```

```

    "intent_digest": {
      "$ref": "#/$defs/Digest"
    },
    "issued_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "lease_generation": {
      "minimum": 1,
      "type": "integer"
    },
    "lease_id": {
      "$ref": "#/$defs/Id"
    },
    "payload": {},
    "record_type": {
      "const": "CommandRequest"
    },
    "requested_scope": {
      "items": {
        "$ref": "#/$defs/ScopeSelector"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array"
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    },
    "workcard_digest": {
      "$ref": "#/$defs/Digest"
    },
    "workcard_task_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "record_type",
    "command_id",
    "command_kind",
    "subject_id",
    "activation_digest",
    "idempotency_key",
    "requested_scope",
    "expected_head_digest",
    "issued_at",
    "payload",
    "intent_digest",
    "authorization"
  ],
  "type": "object",
  "x-promin-max-canonical-bytes": 1048576
}

```

Власник: core/contracts.schema.json#/\$defs/CommandRequest | Валідатор: urn:promin:contracts:1#/\$defs/CommandRequest

18. CommandResult

Опис типу: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then,title

Обов'язкові поля: record_type, command_id, outcome, batch_digest, reason_codes, facts, created_at

Властивість	Обов'язков е	Структурне правило
batch_digest	так	oneOf[2]=#/\$defs/Digest type:null
command_id	так	\$ref="#/\$defs/Id"
created_at	так	\$ref="#/\$defs/Timestamp"
facts	так	type="object"
operation_metrics	ні	\$ref="#/\$defs/OperationMetrics"
outcome	так	enum=["committed", "rejected", "idempotent-replay", "conflict"]
reason_codes	так	type="array"; minItems=0; maxItems=32; uniqueItems=true
record_type	так	const="CommandResult"
<pre> { "additionalProperties": false, </pre>		

```

"allOf": [
  {
    "else": {
      "not": {
        "required": [
          "operation_metrics"
        ]
      }
    },
    "if": {
      "properties": {
        "outcome": {
          "enum": [
            "committed",
            "idempotent-replay"
          ]
        }
      },
      "required": [
        "outcome"
      ]
    },
    "then": {
      "required": [
        "operation_metrics"
      ]
    },
    "title": "committed command result carries exact operation metrics"
  }
],
"properties": {
  "batch_digest": {
    "oneOf": [
      {
        "$ref": "#/$defs/Digest"
      },
      {
        "type": "null"
      }
    ]
  },
  "command_id": {
    "$ref": "#/$defs/Id"
  },
  "created_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "facts": {
    "type": "object"
  },
  "operation_metrics": {
    "$ref": "#/$defs/OperationMetrics"
  },
  "outcome": {
    "enum": [
      "committed",
      "rejected",
      "idempotent-replay",
      "conflict"
    ]
  },
  "reason_codes": {
    "items": {
      "$ref": "#/$defs/Id"
    },
    "maxItems": 32,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "record_type": {
    "const": "CommandResult"
  }
},
"required": [
  "record_type",
  "command_id",
  "outcome",

```

```

    "batch_digest",
    "reason_codes",
    "facts",
    "created_at"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/CommandResult | Валідатор: urn:promin:contracts:1#/\$defs/CommandResult

19. Conformance

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, schema_ref, document, version, function, structural_budgets, workcard_hard_ceiling, scale_contracts, reference_benchmarks, required_acceptance, mutation_families, production_claim_rule, dependency_depth_hard_max, acceptance_proof_class_counts, acceptance_proof_classes

Властивість	Обов'язков	Структурне правило
acceptance_proof_class_counts	так	const=schema(keys:direct-semantic,human-decision,local-runtime-reexecute,migration-external-no-credit,package-static,saturation-signed,signed-role-platfor m-evidence,standard-release-signed-manifest)
acceptance_proof_classes	так	const=schema(keys:activation-digest-binding,artifact-kind-resolves-correct-cap ability,authenticated-role-platform-evidence-attestations,authorization-binds-ex act-grant-claim,bounded-incremental-commit-and-compaction,bounded-workcar d-through-depth-twelve,broad-query-bounded-seed-refinement,candidate-delta- within-task-and-grant-scope,candidate-evidence-decision-acyclic-chain,candidat e-evidence-provider-binding,candidate-recipe-applied-once,canonical-exact-zip- distribution,canonical-name-only,command-capability-and-scope-resolution,com mand-effect-target-in-requested-scope,configuration-exact-idempotency,conjun ctive-scope-containment,continuation-secret-subject-grant-binding,continuation -token-bytes-at-most-256,continuation-v2-complete-page-union,creditable-cand idate-immutable-vcs-tree,current-interpretter-and-clean-install-modes,current-re lease-closure-revalidation,decision-after-evidence-with-bounded-skew,decision- grant-matches-command-authorization,decision-kind-target-scope-binding,depe ndency-graph-acyclic-and-current-ready-frontier,depth-aware-seed-budget,dete rministic-activation-identity,deterministic-replay,evidence-manifest-exact-eight- lane-matrix-resolution,evidence-manifest-four-cp314-supplemental-records,evid ence-manifest-nested-and-supplemental-chronology,evidence-manifest-path-di gest-size-closure,evidence-manifest-physical-resolution,exact-command-primar y-event-correspondence,exact-five-init-records,exact-one-selected-preset,exact -six-base-command-surface,exact-six-core-artifacts,explicit-scale-marker-selecti on,external-standard-release-decision-exact-binding,finding-bound-resolution-a nd-waiver-evidence,fresh-semantic-control-state-per-saturation-iteration,grant-i ssuance-chain-closure,grant-scope-expiry-nonce-proof,handle-relative-evidence -root-safety,historical-decision-current-eligibility-separation,idempotent-state-co mmands,implementation-closure-bound-and-current,installed-platform-observat ion-closure,inventory-stream-memory-amplification-at-most-32,lease-generatio n-fence-and-close-ack,lease-holder-has-task-execute-grant,lossless-continuation -in-reference-100k-scenario,mutation-workcard-requires-current-lease,no-false- success,non-skipped-credit-result-has-evidence,normalization-collision-rejection, offline-installed-no-degradation,offline-release-online-compatibility,one-comm and-one-atomic-batch,physical-100k-one-artifact-proxy-per-file,policy-owned-tas k-and-lease-state-machines,preset-outside-core-digest,product-public-approval- separation,profile-bound-physical-100k-performance,profile-default-with-core-d epth-twelve,projection-disposable-and-current,provider-substitution-invalidates- derived-state,raw-scale-summary-recomputation,read-workcard-without-lease,r equired-no-degradation-tests-fail-closed,required-provider-preflight-before-com mit,rollback-and-recutover-equivalence,root-bootstrap-command-limited,root-ce iling-without-bootstrap-grant-cycle,safe-recursive-export,selected-provider-adap ter-evidence,semver-from-candidate-binding,seven-distinct-role-producer-scope s,signed-standard-decision-authority-proof,single-derived-platform-closure-own er,single-pass-inventory,standard-distribution-separate-from-product-acceptanc e,strict-critical-fields,team-signatures-cryptographically-verified,team-signed-cli- end-to-end,three-zero-new-saturation-iterations,truthful-operation-reporting,tru thful-process-invocation-status,typed-human-document-verification,typed-refere nce-domain-range-closure,verified-head-checkpoint-delta-replay,verified-invent ory-result-provenance,windows-linux-exact-zip-matrix,zero-compiled-cache-files, zero-core-provider-lock-in,zero-dead-evidence-budget-fields,zero-forbidden-nam e-occurrences,zero-product-scan-during-init,zero-unowned-normative-facts)
dependency_depth_hard_max	так	type="integer"; const=12
document	так	const="promin.conformance"
function	так	const="Єдиний власник acceptance predicates, mutation families, structural hard ceilings, scale contracts i production-claim rules."
mutation_families	так	const=[77 items]; minItems=77; maxItems=77

Властивість	Обов'язков е	Структурне правило
production_claim_rule	так	const="Standard distribution success never grants product acceptance. Live product credit separately requires a creditable immutable-vcs-tree Candidate, all P0/P1 findings closed with exact target-bound evidence, current project release closure eligibility, rollback and recutover equivalence, and an authorized human project Decision. product_acceptance_pass remains false for promin v1 distribution evidence."
record_type	так	const="Conformance"
reference_benchmarks	так	const=schema(keys:core_valid_relation_count,depths_tested,file_count,forced_c ontinuation,inventory_passes,performance_profiles,physical_relation_corpus,qu ery_count,raw_file_proxy_ratio,rebuild_product_passes,reference_targets_are_e nvironment_specific,semantic_corpus,silent_truncations_max,synthetic_task_rat io)
required_acceptance	так	const=[102 items]; minItems=102; maxItems=102
scale_contracts	так	const=schema(keys:event_commit,init,inventory,normal_one_shot_replay,proje ction,replay,saturation,workcard)
schema_ref	так	const="urn:promin:contracts:1#/\$defs/Conformance"
structural_budgets	так	const=schema(keys:authority_bearing_init_records,base_user_commands_max, core_normative_artifacts,persistent_entity_kinds_max,relation_kinds_max,select ed_presets)
version	так	const="1.0.0-alpha.1"
workcard_hard_ceiling	так	const=schema(keys:max_bytes,max_entities,max_fanout_per_entity,max_relati ons,top_k)
<pre> { "additionalProperties": false, "properties": { "acceptance_proof_class_counts": { "const": { "direct-semantic": 35, "human-decision": 9, "local-runtime-reexecute": 30, "migration-external-no-credit": 1, "package-static": 4, "saturation-signed": 1, "signed-role-platform-evidence": 17, "standard-release-signed-manifest": 5 } }, "acceptance_proof_classes": { "const": { "activation-digest-binding": "direct-semantic", "artifact-kind-resolves-correct-capability": "local-runtime-reexecute", "authenticated-role-platform-evidence-attestations": "signed-role-platform-evidence", "authorization-binds-exact-grant-claim": "local-runtime-reexecute", "bounded-incremental-commit-and-compaction": "signed-role-platform-evidence", "bounded-workcard-through-depth-twelve": "direct-semantic", "broad-query-bounded-seed-refinement": "direct-semantic", "candidate-delta-within-task-and-grant-scope": "direct-semantic", "candidate-evidence-decision-acyclic-chain": "human-decision", "candidate-evidence-provider-binding": "local-runtime-reexecute", "candidate-recipe-applied-once": "direct-semantic", "canonical-exact-zip-distribution": "direct-semantic", "canonical-name-only": "direct-semantic", "command-capability-and-scope-resolution": "local-runtime-reexecute", "command-effect-target-in-requested-scope": "local-runtime-reexecute", "configuration-exact-idempotency": "local-runtime-reexecute", "conjunctive-scope-containment": "local-runtime-reexecute", "continuation-secret-subject-grant-binding": "direct-semantic", "continuation-token-bytes-at-most-256": "direct-semantic", "continuation-v2-complete-page-union": "direct-semantic", "creditable-candidate-immutable-vcs-tree": "direct-semantic", "current-interpreter-and-clean-install-modes": "signed-role-platform-evidence", "current-release-closure-revalidation": "human-decision", "decision-after-evidence-with-bounded-skew": "human-decision", "decision-grant-matches-command-authorization": "local-runtime-reexecute", "decision-kind-target-scope-binding": "local-runtime-reexecute", "dependency-graph-acyclic-and-current-ready-frontier": "direct-semantic", "depth-aware-seed-budget": "local-runtime-reexecute", "deterministic-activation-identity": "local-runtime-reexecute", "deterministic-replay": "local-runtime-reexecute", "evidence-manifest-exact-eight-lane-matrix-resolution": "standard-release-signed-manifest", "evidence-manifest-four-cp314-supplemental-records": "standard-release-signed-manifest", </pre>		

```
"evidence-manifest-nested-and-supplemental-chronology": "standard-release-signed-manifest",
"evidence-manifest-path-digest-size-closure": "standard-release-signed-manifest",
"evidence-manifest-physical-resolution": "signed-role-platform-evidence",
"exact-command-primary-event-correspondence": "local-runtime-reexecute",
"exact-five-init-records": "direct-semantic",
"exact-one-selected-preset": "direct-semantic",
"exact-six-base-command-surface": "direct-semantic",
```

```
"exact-six-core-artifacts": "direct-semantic",
"explicit-scale-marker-selection": "direct-semantic",
"external-standard-release-decision-exact-binding": "human-decision",
"finding-bound-resolution-and-waiver-evidence": "direct-semantic",
"fresh-semantic-control-state-per-saturation-iteration": "saturation-signed",
"grant-issuance-chain-closure": "local-runtime-reexecute",
"grant-scope-expiry-nonce-proof": "local-runtime-reexecute",
"handle-relative-evidence-root-safety": "signed-role-platform-evidence",
"historical-decision-current-eligibility-separation": "human-decision",
"idempotent-state-commands": "local-runtime-reexecute",
"implementation-closure-bound-and-current": "direct-semantic",
"installed-platform-observation-closure": "signed-role-platform-evidence",
"inventory-stream-memory-amplification-at-most-32": "direct-semantic",
"lease-generation-fence-and-close-ack": "local-runtime-reexecute",
"lease-holder-has-task-execute-grant": "local-runtime-reexecute",
"lossless-continuation-in-reference-100k-scenario": "direct-semantic",
"mutation-workcard-requires-current-lease": "direct-semantic",
"no-false-success": "direct-semantic",
"non-skipped-credit-result-has-evidence": "local-runtime-reexecute",
"normalization-collision-rejection": "direct-semantic",
"offline-installed-no-degradation": "signed-role-platform-evidence",
"offline-release-online-compatibility": "signed-role-platform-evidence",
"one-command-one-atomic-batch": "local-runtime-reexecute",
"physical-100k-one-artifact-proxy-per-file": "signed-role-platform-evidence",
"policy-owned-task-and-lease-state-machines": "local-runtime-reexecute",
"preset-outside-core-digest": "direct-semantic",
"product-public-approval-separation": "human-decision",
"profile-bound-physical-100k-performance": "signed-role-platform-evidence",
```

```
"profile-default-with-core-depth-twelve": "direct-semantic",
"projection-disposable-and-current": "local-runtime-reexecute",
"provider-substitution-invalidates-derived-state": "local-runtime-reexecute",
"raw-scale-summary-recomputation": "signed-role-platform-evidence",
"read-workcard-without-lease": "direct-semantic",
"required-no-degradation-tests-fail-closed": "signed-role-platform-evidence",
"required-provider-preflight-before-commit": "local-runtime-reexecute",
"rollback-and-recutover-equivalence": "migration-external-no-credit",
"root-bootstrap-command-limited": "local-runtime-reexecute",
"root-ceiling-without-bootstrap-grant-cycle": "local-runtime-reexecute",
"safe-recursive-export": "direct-semantic",
"selected-provider-adapter-evidence": "direct-semantic",
"semver-from-candidate-binding": "human-decision",
"seven-distinct-role-producer-scopes": "standard-release-signed-manifest",
"signed-standard-decision-authority-proof": "human-decision",
"single-derived-platform-closure-owner": "signed-role-platform-evidence",
"single-pass-inventory": "local-runtime-reexecute",
"standard-distribution-separate-from-product-acceptance": "human-decision",
"strict-critical-fields": "direct-semantic",
"team-signatures-cryptographically-verified": "local-runtime-reexecute",
"team-signed-cli-end-to-end": "direct-semantic",
"three-zero-new-saturation-iterations": "signed-role-platform-evidence",
"truthful-operation-reporting": "local-runtime-reexecute",
"truthful-process-invocation-status": "signed-role-platform-evidence",
"typed-human-document-verification": "signed-role-platform-evidence",
"typed-reference-domain-range-closure": "direct-semantic",
"verified-head-checkpoint-delta-replay": "direct-semantic",
"verified-inventory-result-provenance": "direct-semantic",
```

```
"windows-linux-exact-zip-matrix": "signed-role-platform-evidence",
"zero-compiled-cache-files": "package-static",
"zero-core-provider-lock-in": "package-static",
"zero-dead-evidence-budget-fields": "direct-semantic",
"zero-forbidden-name-occurrences": "package-static",
"zero-product-scan-during-init": "local-runtime-reexecute",
"zero-unowned-normative-facts": "package-static"
```

```
}
},
"dependency_depth_hard_max": {
  "const": 12,
  "type": "integer"
},
"document": {
```



```

    "const": "promin.conformance"
  },
  "function": {
    "const": "Єдиний власник acceptance predicates, mutation families, structural hard ceilings, scale
contracts i production-claim rules."
  },
  "mutation_families": {
    "const": [
      "status-credit-contradiction",
      "missing-critical-field",
      "unknown-critical-field",
      "activation-drift",
      "grant-replay",
      "scope-escalation",
      "expired-grant",
      "revoked-grant",
      "stale-fence",
      "candidate-drift",
      "provider-substitution",
      "event-fork",
      "batch-command-mismatch",
      "duplicate-idempotency-key",
      "projection-corruption",
      "projection-staleness",
      "reference-break",
      "relation-domain-break",
      "context-flood",
      "symlink-core-input",
      "shadow-core-file",
      "symlink-product-entry",
      "secret-export",
      "archive-expansion",
      "preset-authority-escalation",
      "normalization-key-collision",
      "path-casefold-collision",
      "forbidden-name-reintroduction",
      "reinit-configuration-conflict",
      "clock-order-violation",
      "capability-confusion",
      "command-scope-escalation",
      "grant-issuer-chain-break",
      "unverified-signature-acceptance",
      "provider-declared-but-unavailable",
      "illegal-task-transition",
      "illegal-lease-transition",
      "decision-target-confusion",
      "credit-without-evidence",
      "command-primary-event-mismatch",
      "duplicate-event-id",
      "scope-selector-drop-broadening",
      "authorization-grant-claim-substitution",
      "command-effect-outside-requested-scope",
      "lease-holder-grant-mismatch",
      "decision-grant-provenance-mismatch",
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      "continuation-duplicate-emission",
      "candidate-recipe-bypass",
      "observational-candidate-pass-credit",
      "snapshot-provider-substitution",
      "unsupported-provider-adapter",
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      "inventory-graph-inflation",
      "workcard-budget-bypass",
      "fabricated-evidence-digest",
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      "unconfigured-standard-decision-key",

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"standard-decision-version-replay",
"candidate-decision-cycle",
"broad-query-corpus-pagination",
"continuation-token-context-inflation",
"inventory-stream-materialization",
"pdf-header-only-credit",
"scale-environment-skip-credit",
"content-search-omission"
],
"items": {
  "enum": [
    "status-credit-contradiction",
    "missing-critical-field",
    "unknown-critical-field",
    "activation-drift",
    "grant-replay",
    "scope-escalation",
    "expired-grant",
    "revoked-grant",
    "stale-fence",
    "candidate-drift",
    "provider-substitution",

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"event-fork",
"batch-command-mismatch",
"duplicate-idempotency-key",
"projection-corruption",
"projection-staleness",
"reference-break",
"relation-domain-break",
"context-flood",
"symlink-core-input",
"shadow-core-file",
"symlink-product-entry",
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"archive-expansion",
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"path-casefold-collision",
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"reinit-configuration-conflict",
"clock-order-violation",
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"illegal-task-transition",
"illegal-lease-transition",
"decision-target-confusion",
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"command-primary-event-mismatch",
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"candidate-delta-scope-escape",
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"inventory-provenance-override",
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"inventory-graph-inflation",
"workcard-budget-bypass",
"fabricated-evidence-digest",

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        "unsigned-standard-decision",
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        "standard-decision-version-replay",
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        "continuation-token-context-inflation",
        "inventory-stream-materialization",
        "pdf-header-only-credit",
        "scale-environment-skip-credit",
        "content-search-omission"
    ]
},
"maxItems": 77,
"minItems": 77
},
"production_claim_rule": {
    "const": "Standard distribution success never grants product acceptance. Live product credit
separately requires a creditable immutable-vcs-tree Candidate, all P0/P1 findings closed with exact
target-bound evidence, current project release closure eligibility, rollback and recutover equivalence,
and an authorized human project Decision. product_acceptance_pass remains false for promin v1 distribution
evidence."
},
"record_type": {
    "const": "Conformance"
},
"reference_benchmarks": {
    "const": {
        "core_valid_relation_count": 198999,
        "depths_tested": [
            1,
            2,
            3,
            4,
            5,
            6,
            7,
            8,
            9,
            10,
            11,
            12
        ],
        "file_count": 100000,
        "forced_continuation": true,
        "inventory_passes": 1,
        "performance_profiles": {
            "portable-local-v1": {
                "commit_bytes_per_changed_record_max": 32768,
                "commit_p95_ms_max": 500,
                "commit_p99_ms_max": 1000,
                "compatible_platforms": [
                    "linux",
                    "windows"
                ],
                "database_bytes_max": 402653184,
                "p50_ms_max": 100,
                "p95_ms_max": 150,
                "p99_ms_max": 500,
                "peak_rss_bytes_max": 805306368,
                "projection_amplification_max": 20,
                "requires_same_runner_no_degradation": true,
                "runtime_checkpoint_count_max": 24,
                "semantic_inflation_max": 3.25,
                "semantic_ingestion_seconds_max": 600
            }
        },
        "physical_relation_corpus": {
            "artifact_target_coverage": 1,
            "expected_counts_source": "fixture-manifest",
            "generation": "explicit-authorized-command-events",
            "relation_count": 198971,
            "relation_kind": "READS",
            "separate_from_raw_inventory": true,
            "task_count": 1567,
            "total_core_valid_relations": 198999
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        "query_count": 600,
        "raw_file_proxy_ratio": 1,

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"rebuild_product_passes": 0,
"reference_targets_are_environment_specific": true,
"semantic_corpus": {
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    1,
    2,
    3,
    4,
    5,
    6,
    7,
    8,
    9,
    10,
    11,
    12
  ],
  "expected_counts_source": "fixture-manifest",
  "relation_count": 28,
  "relation_kind": "DEPENDS_ON",
  "separate_from_raw_inventory": true,
  "task_count": 32
},
"silent_truncations_max": 0,
"synthetic_task_ratio": 0
}
},
"required_acceptance": {
  "const": [
    "canonical-name-only",
    "exact-six-core-artifacts",
    "preset-outside-core-digest",
    "exact-one-selected-preset",
    "exact-five-init-records",
    "zero-product-scan-during-init",
    "single-pass-inventory",
    "no-false-success",
    "strict-critical-fields",
    "activation-digest-binding",
    "root-ceiling-without-bootstrap-grant-cycle",
    "grant-scope-expiry-nonce-proof",
    "lease-generation-fence-and-close-ack",
    "candidate-evidence-provider-binding",
    "one-command-one-atomic-batch",
    "deterministic-replay",
    "projection-disposable-and-current",
    "bounded-workcard-through-depth-twelve",
    "typed-reference-domain-range-closure",
    "dependency-graph-acyclic-and-current-ready-frontier",
    "safe-recursive-export",
    "idempotent-state-commands",
    "normalization-collision-rejection",
    "provider-substitution-invalidates-derived-state",
    "rollback-and-recutover-equivalence",
    "zero-forbidden-name-occurrences",
    "zero-unowned-normative-facts",
    "zero-core-provider-lock-in",
    "zero-compiled-cache-files",
    "physical-100k-one-artifact-proxy-per-file",
    "external-standard-release-decision-exact-binding",
    "root-bootstrap-command-limited",
    "deterministic-activation-identity",
    "configuration-exact-idempotency",
    "truthful-operation-reporting",
    "depth-aware-seed-budget",
    "lossless-continuation-in-reference-100k-scenario",
    "read-workcard-without-lease",
    "mutation-workcard-requires-current-lease",
    "command-capability-and-scope-resolution",
    "grant-issuance-chain-closure",
    "team-signatures-cryptographically-verified",
    "required-provider-preflight-before-commit",
    "policy-owned-task-and-lease-state-machines",
    "decision-kind-target-scope-binding",
    "non-skipped-credit-result-has-evidence",
    "exact-command-primary-event-correspondence",
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"authorization-binds-exact-grant-claim",
"command-effect-target-in-requested-scope",
"lease-holder-has-task-execute-grant",
"decision-grant-matches-command-authorization",
"artifact-kind-resolves-correct-capability",
"continuation-v2-complete-page-union",
"candidate-recipe-applied-once",
"creditable-candidate-immutable-vcs-tree",
"selected-provider-adapter-evidence",
"team-signed-cli-end-to-end",
"current-release-closure-revalidation",
"finding-bound-resolution-and-waiver-evidence",
"candidate-delta-within-task-and-grant-scope",
"verified-head-checkpoint-delta-replay",
"profile-default-with-core-depth-twelve",
"canonical-exact-zip-distribution",
"current-interpretter-and-clean-install-modes",

"required-no-degradation-tests-fail-closed",
"continuation-secret-subject-grant-binding",
"implementation-closure-bound-and-current",
"verified-inventory-result-provenance",
"exact-six-base-command-surface",
"zero-dead-evidence-budget-fields",
"profile-bound-physical-100k-performance",
"windows-linux-exact-zip-matrix",
"fresh-semantic-control-state-per-saturation-iteration",
"three-zero-new-saturation-iterations",
"standard-distribution-separate-from-product-acceptance",
"candidate-evidence-decision-acyclic-chain",
"evidence-manifest-physical-resolution",
"evidence-manifest-exact-eight-lane-matrix-resolution",
"evidence-manifest-four-cp314-supplemental-records",
"evidence-manifest-path-digest-size-closure",
"evidence-manifest-nested-and-supplemental-chronology",
"seven-distinct-role-producer-scopes",
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"semver-from-candidate-binding",
"typed-human-document-verification",
"broad-query-bounded-seed-refinement",
"continuation-token-bytes-at-most-256",
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"explicit-scale-marker-selection",
"authenticated-role-platform-evidence-attestations",
"installed-platform-observation-closure",
"handle-relative-evidence-root-safety",

"raw-scale-summary-recomputation",
"bounded-incremental-commit-and-compaction",
"offline-installed-no-degradation",
"offline-release-online-compatibility",
"decision-after-evidence-with-bounded-skew",
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"historical-decision-current-eligibility-separation"
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    "preset-outside-core-digest",
    "exact-one-selected-preset",
    "exact-five-init-records",
    "zero-product-scan-during-init",
    "single-pass-inventory",
    "no-false-success",
    "strict-critical-fields",
    "activation-digest-binding",
    "root-ceiling-without-bootstrap-grant-cycle",
    "grant-scope-expiry-nonce-proof",
    "lease-generation-fence-and-close-ack",
    "candidate-evidence-provider-binding",
    "one-command-one-atomic-batch",
    "deterministic-replay",

    "projection-disposable-and-current",
    "bounded-workcard-through-depth-twelve",
    "typed-reference-domain-range-closure",
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"safe-recursive-export",
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 "zero-unowned-normative-facts",
 "zero-core-provider-lock-in",
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 "physical-100k-one-artifact-proxy-per-file",
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 "configuration-exact-idempotency",
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 "mutation-workcard-requires-current-lease",
 "command-capability-and-scope-resolution",
 "grant-issuance-chain-closure",
 "team-signatures-cryptographically-verified",
 "required-provider-preflight-before-commit",
 "policy-owned-task-and-lease-state-machines",

"decision-kind-target-scope-binding",
 "non-skipped-credit-result-has-evidence",
 "exact-command-primary-event-correspondence",
 "conjunctive-scope-containment",
 "authorization-binds-exact-grant-claim",
 "command-effect-target-in-requested-scope",
 "lease-holder-has-task-execute-grant",
 "decision-grant-matches-command-authorization",
 "artifact-kind-resolves-correct-capability",
 "continuation-v2-complete-page-union",
 "candidate-recipe-applied-once",
 "creditable-candidate-immutable-vcs-tree",
 "selected-provider-adapter-evidence",
 "team-signed-cli-end-to-end",
 "current-release-closure-revalidation",
 "finding-bound-resolution-and-waiver-evidence",
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 "canonical-exact-zip-distribution",
 "current-interpreter-and-clean-install-modes",
 "required-no-degradation-tests-fail-closed",
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 "implementation-closure-bound-and-current",
 "verified-inventory-result-provenance",
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 "zero-dead-evidence-budget-fields",
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"windows-linux-exact-zip-matrix",
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 "three-zero-new-saturation-iterations",
 "standard-distribution-separate-from-product-acceptance",
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 "evidence-manifest-physical-resolution",
 "evidence-manifest-exact-eight-lane-matrix-resolution",
 "evidence-manifest-four-cp314-supplemental-records",
 "evidence-manifest-path-digest-size-closure",
 "evidence-manifest-nested-and-supplemental-chronology",
 "seven-distinct-role-producer-scopes",
 "signed-standard-decision-authority-proof",
 "semver-from-candidate-binding",
 "typed-human-document-verification",
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 "continuation-token-bytes-at-most-256",
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 "explicit-scale-marker-selection",
 "authenticated-role-platform-evidence-attestations",
 "installed-platform-observation-closure",
 "handle-relative-evidence-root-safety",
 "raw-scale-summary-recomputation",
 "bounded-incremental-commit-and-compaction",
 "offline-installed-no-degradation",
 "offline-release-online-compatibility",
 "decision-after-evidence-with-bounded-skew",

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        "single-derived-platform-closure-owner",
        "truthful-process-invocation-status",
        "product-public-approval-separation",
        "historical-decision-current-eligibility-separation"
    ]
},
"maxItems": 102,
"minItems": 102
},
"scale_contracts": {
    "const": {
        "event_commit": {
            "complexity": "0(batch)",
            "head_lookup": "0(1)"
        },
        "init": {
            "complexity": "0(core+plan)",
            "project_tree_passes": 0
        },
        "inventory": {
            "complexity": "0(files+bytes)",
            "creditable_consistency_modes": [
                "immutable-vcs-tree"
            ],
            "derived_artifact_proxies_per_raw_file": 1,
            "manifest_verified_jsonl": true,
            "memory_amplification_max": 32,
            "observational_mode_pass_credit": false,
            "project_tree_passes_max": 1,
            "search_text_bytes_max": 4096,
            "synthetic_relations_per_raw_file": 0,
            "synthetic_tasks_per_raw_file": 0
        },
        "normal_one_shot_replay": {
            "checkpoint_authoritative": false,
            "complexity": "0(delta-since-verified-checkpoint)",
            "full_replay_allowed": "doctor, validate, recovery, or verified fallback only"
        },
        "projection": {
            "public_rebuild_ingress": "verified InventoryResult or persisted verified manifest only",
            "rebuild_product_tree_passes": 0,
            "reserved_provenance_rejected": true
        },
        "replay": {
            "complexity": "0(events)"
        },
        "saturation": {
            "fresh_control_state_per_iteration": true,
            "init_product_tree_scans": 0,
            "semantic_state_reused": false
        },
        "workcard": {
            "broad_query_behavior": "deterministic bounded top-k seeds, refinement_required=true, no corpus pagination",
            "complexity": "0(top_k+capped_typed_closure)",
            "continuation_state_bytes_max": 16384,
            "continuation_token_bytes_max": 256,
            "corpus_independent_output_bound": true,
            "seed_budget": "effective_top_k=min(profile.top_k,max(2,floor(max_entities/(1+observed_branching*requested_depth))))",
            "selected_closure_union_completeness": 1
        }
    },
    "schema_ref": {
        "const": "urn:promin:contracts:1#/$defs/Conformance"
    },
    "structural_budgets": {
        "const": {
            "authority_bearing_init_records": 5,
            "base_user_commands_max": 10,
            "core_normative_artifacts": 6,
            "persistent_entity_kinds_max": 9,
            "relation_kinds_max": 6,
            "selected_presets": 1
        }
    }
},

```

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    "version": {
      "const": "1.0.0-alpha.1"
    },
    "workcard_hard_ceiling": {
      "const": {
        "max_bytes": 16384,
        "max_entities": 32,
        "max_fanout_per_entity": 8,
        "max_relations": 48,
        "top_k": 12
      }
    }
  },
  "required": [
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    "document",
    "version",
    "function",
    "structural_budgets",
    "workcard_hard_ceiling",
    "scale_contracts",
    "reference_benchmarks",
    "required_acceptance",
    "mutation_families",
    "production_claim_rule",
    "dependency_depth_hard_max",
    "acceptance_proof_class_counts",
    "acceptance_proof_classes"
  ],
  "type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/Conformance](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/Conformance](#)

20. ContinuationTokenClaims

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, token_version, activation_digest, subject_id, grant_id, grant_claim_digest, capability_id, requested_scope, normalized_query_digest, cursor, head_event_id, head_sequence, head_digest, projection_digest, ranking_id, traversal_id, dependency_depth, budgets, issued_at, expires_at, ttl_seconds, revocation_epoch, implementation_closure_digest, state_id

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
budgets	так	\$ref="#/\$defs/Budget"
capability_id	так	const="projection.read"
cursor	так	type="integer"; minimum=0
dependency_depth	так	type="integer"; minimum=1; maximum=12
expires_at	так	\$ref="#/\$defs/Timestamp"
grant_claim_digest	так	\$ref="#/\$defs/Digest"
grant_id	так	\$ref="#/\$defs/Id"
head_digest	так	oneOf[2]=#/\$defs/Digest type:null
head_event_id	так	oneOf[2]=#/\$defs/Id type:null
head_sequence	так	type="integer"; minimum=0
implementation_closure_digest	так	\$ref="#/\$defs/Digest"
issued_at	так	\$ref="#/\$defs/Timestamp"
normalized_query_digest	так	\$ref="#/\$defs/Digest"
projection_digest	так	\$ref="#/\$defs/Digest"
ranking_id	так	const="bm25-v1"
record_type	так	const="ContinuationTokenClaims"
requested_scope	так	type="array"; minItems=1; maxItems=32
revocation_epoch	так	\$ref="#/\$defs/Digest"
state_id	так	\$ref="#/\$defs/Id"

Властивість	Обов'язков е	Структурне правило
subject_id	так	\$ref="#/\$defs/Id"
token_version	так	type="integer"; const=2
traversal_id	так	const="typed-bfs-v2"
ttl_seconds	так	type="integer"; minimum=1; maximum=3600
<pre> { "additionalProperties": false, "properties": { "activation_digest": { "\$ref": "#/\$defs/Digest" }, "budgets": { "\$ref": "#/\$defs/Budget" }, "capability_id": { "const": "projection.read" }, "cursor": { "minimum": 0, "type": "integer" }, "dependency_depth": { "maximum": 12, "minimum": 1, "type": "integer" }, "expires_at": { "\$ref": "#/\$defs/Timestamp" }, "grant_claim_digest": { "\$ref": "#/\$defs/Digest" }, "grant_id": { "\$ref": "#/\$defs/Id" }, "head_digest": { "oneOf": [{ "\$ref": "#/\$defs/Digest" }, { "type": "null" }] }, "head_event_id": { "oneOf": [{ "\$ref": "#/\$defs/Id" }, { "type": "null" }] }, "head_sequence": { "minimum": 0, "type": "integer" }, "implementation_closure_digest": { "\$ref": "#/\$defs/Digest" }, "issued_at": { "\$ref": "#/\$defs/Timestamp" }, "normalized_query_digest": { "\$ref": "#/\$defs/Digest" }, "projection_digest": { "\$ref": "#/\$defs/Digest" }, "ranking_id": { "const": "bm25-v1" } } } </pre>		

```

    },
    "record_type": {
      "const": "ContinuationTokenClaims"
    },
    "requested_scope": {
      "items": {
        "$ref": "#/$defs/ScopeSelector"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array"
    },
    "revocation_epoch": {
      "$ref": "#/$defs/Digest"
    },
    "state_id": {
      "$ref": "#/$defs/Id"
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    },
    "token_version": {
      "const": 2,
      "type": "integer"
    },
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      "const": "typed-bfs-v2"
    },
    "ttl_seconds": {
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      "minimum": 1,
      "type": "integer"
    }
  },
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    "activation_digest",
    "subject_id",
    "grant_id",
    "grant_claim_digest",
    "capability_id",
    "requested_scope",
    "normalized_query_digest",
    "cursor",
    "head_event_id",
    "head_sequence",
    "head_digest",
    "projection_digest",
    "ranking_id",
    "traversal_id",
    "dependency_depth",
    "budgets",
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    "revocation_epoch",
    "implementation_closure_digest",
    "state_id"
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  "type": "object",
  "x-promin-max-state-bytes": 16384,
  "x-promin-max-token-bytes": 256,
  "x-promin-persistence": "derived-live-only"
}

```

Власник: [core/contracts.schema.json#/\\$defs/ContinuationTokenClaims](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/ContinuationTokenClaims](#)

21. ContinueResult

Опис типу: type="object"; additionalProperties=false; allOf[2]=keys:else,if,then | keys:if,then

Обов'язкові поля: record_type, status, subject_id, activation_digest, work_card, continuation

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
continuation	так	oneOf[2]=type:null type:string
record_type	так	const="ContinueResult"
status	так	enum=["ready", "empty"]
subject_id	так	\$ref="#/\$defs/Id"
work_card	так	oneOf[2]=#/\$defs/WorkCard type:null

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "properties": {
          "continuation": {
            "type": "null"
          },
          "work_card": {
            "type": "null"
          }
        }
      },
      "if": {
        "properties": {
          "status": {
            "const": "ready"
          }
        },
        "required": [
          "status"
        ]
      },
      "then": {
        "properties": {
          "work_card": {
            "$ref": "#/$defs/WorkCard"
          }
        }
      }
    },
    {
      "if": {
        "properties": {
          "continuation": {
            "type": "string"
          }
        },
        "required": [
          "continuation"
        ]
      },
      "then": {
        "properties": {
          "work_card": {
            "properties": {
              "truncated": {
                "const": true
              }
            },
            "required": [
              "truncated"
            ]
          }
        }
      }
    }
  ],
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "continuation": {
      "oneOf": [

```

```

    {
      "type": "null"
    },
    {
      "maxLength": 256,
      "minLength": 1,
      "pattern": "^promin-v2\\. [A-Za-z0-9_-]{22}\\. [0-9a-f]{64}\\. [A-Za-z0-9_-]{43}$",
      "type": "string"
    }
  ]
},
"record_type": {
  "const": "ContinueResult"
},
"status": {
  "enum": [
    "ready",
    "empty"
  ]
}
},
"subject_id": {
  "$ref": "#/$defs/Id"
},
"work_card": {
  "oneOf": [
    {
      "$ref": "#/$defs/WorkCard"
    },
    {
      "type": "null"
    }
  ]
}
},
"required": [
  "record_type",
  "status",
  "subject_id",
  "activation_digest",
  "work_card",
  "continuation"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Власник: core/contracts.schema.json#/\$defs/ContinueResult | Валідатор: urn:promin:contracts:1#/\$defs/ContinueResult

22. CoreComponent

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: path, sha256

Властивість	Обов'язков е	Структурне правило
path	так	enum=["semantic-model.json", "authority-model.json", "policy-set.json", "contracts.schema.json", "conformance.json"]
sha256	так	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "properties": {
    "path": {
      "enum": [
        "semantic-model.json",
        "authority-model.json",
        "policy-set.json",
        "contracts.schema.json",
        "conformance.json"
      ]
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "path",

```

```

    "sha256"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/CoreComponent | Валідатор: urn:promin:contracts:1#/\$defs/CoreComponent

23. CurrentInventory

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, candidate_digest, manifest_digest, inventory_digest, stream_digest, stream_bytes, entry_count, projection_row_count, project_tree_passes, inventory_passes, rebuild_product_passes, search_text_rule, projection_authoritative

Властивість	Обов'язков е	Структурне правило
candidate_digest	так	\$ref="#/\$defs/Digest"
entry_count	так	type="integer"; minimum=0
inventory_digest	так	\$ref="#/\$defs/Digest"
inventory_passes	так	type="integer"; const=1
manifest_digest	так	\$ref="#/\$defs/Digest"
project_tree_passes	так	type="integer"; minimum=0; maximum=1
projection_authoritative	так	const=false
projection_row_count	так	type="integer"; minimum=0
rebuild_product_passes	так	type="integer"; const=0
record_type	так	const="CurrentInventory"
search_text_rule	так	const="exact NFC-normalized canonical path, UTF-8 bounded by Conformance scale_contracts.inventory.search_text_bytes_max"
stream_bytes	так	type="integer"; minimum=1
stream_digest	так	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "entry_count": {
      "minimum": 0,
      "type": "integer"
    },
    "inventory_digest": {
      "$ref": "#/$defs/Digest"
    },
    "inventory_passes": {
      "const": 1,
      "type": "integer"
    },
    "manifest_digest": {
      "$ref": "#/$defs/Digest"
    },
    "project_tree_passes": {
      "maximum": 1,
      "minimum": 0,
      "type": "integer"
    },
    "projection_authoritative": {
      "const": false
    },
    "projection_row_count": {
      "minimum": 0,
      "type": "integer"
    },
    "rebuild_product_passes": {
      "const": 0,
      "type": "integer"
    },
    "record_type": {
      "const": "CurrentInventory"
    }
  }
}

```

```

    "search_text_rule": {
      "const": "exact NFC-normalized canonical path, UTF-8 bounded by Conformance
scale_contracts.inventory.search_text_bytes_max"
    },
    "stream_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "stream_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "record_type",
    "candidate_digest",
    "manifest_digest",
    "inventory_digest",
    "stream_digest",
    "stream_bytes",
    "entry_count",
    "projection_row_count",
    "project_tree_passes",
    "inventory_passes",
    "rebuild_product_passes",
    "search_text_rule",
    "projection_authoritative"
  ],
  "type": "object",
  "x-promin-ingress-operations": [
    "rebuild"
  ],
  "x-promin-persistence": "derived-rebuild-only"
}

```

Власник: core/contracts.schema.json#/\$defs/CurrentInventory | Валідатор: urn:promin:contracts:1#/\$defs/CurrentInventory

24. Decision

Опис типу: type="object"; additionalProperties=false; allOf[7]=keys:if,then | keys:if,then | keys:if,then | keys:else,if,then,title | keys:else,if,then,title | keys:else,if,then,title | keys:else,if,then,title

Обов'язкові поля: record_type, decision_id, decision_kind, subject_id, grant_id, grant_claim_digest, activation_digest, target_type, target_id, target_digest, rationale, evidence_artifacts, created_at

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
candidate_digest	ні	\$ref="#/\$defs/Digest"
created_at	так	\$ref="#/\$defs/Timestamp"
decision_id	так	\$ref="#/\$defs/Id"
decision_kind	так	enum=["accept", "reject", "resolve", "waive", "promote", "release", "revoke"]
evidence_artifacts	так	type="array"; minItems=1; maxItems=64; uniqueItems=true
exception_policy_digest	ні	\$ref="#/\$defs/Digest"
expires_at	ні	\$ref="#/\$defs/Timestamp"
finding_digest	ні	\$ref="#/\$defs/Digest"
grant_claim_digest	так	\$ref="#/\$defs/Digest"
grant_id	так	\$ref="#/\$defs/Id"
provider_binding_digest	ні	\$ref="#/\$defs/Digest"
rationale	так	type="string"; minLength=1
record_type	так	const="Decision"
release_closure_digest	ні	\$ref="#/\$defs/Digest"
subject_id	так	\$ref="#/\$defs/Id"
target_digest	так	\$ref="#/\$defs/Digest"
target_id	так	\$ref="#/\$defs/Id"
target_type	так	enum=["Завдання", "Candidate", "Finding", "Grant"]
{ "additionalProperties": false,		

```

"allOf": [
  {
    "if": {
      "properties": {
        "decision_kind": {
          "enum": [
            "resolve",
            "waive"
          ]
        }
      }
    },
    "then": {
      "properties": {
        "target_type": {
          "const": "Finding"
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "decision_kind": {
          "enum": [
            "promote",
            "release"
          ]
        }
      }
    },
    "then": {
      "properties": {
        "target_type": {
          "const": "Candidate"
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "decision_kind": {
          "enum": [
            "accept",
            "reject"
          ]
        }
      }
    },
    "then": {
      "properties": {
        "target_type": {
          "enum": [
            "Завдання",
            "Candidate"
          ]
        }
      }
    }
  },
  {
    "else": {
      "not": {
        "anyOf": [
          {
            "required": [
              "release_closure_digest"
            ]
          },
          {
            "required": [
              "provider_binding_digest"
            ]
          }
        ]
      }
    }
  }
]

```

```

    "if": {
      "properties": {
        "decision_kind": {
          "const": "release"
        }
      },
      "required": [
        "decision_kind"
      ]
    },
    "then": {
      "required": [
        "release_closure_digest",
        "provider_binding_digest"
      ]
    },
    "title": "release Decision binds exact historical closure and providers"
  },
  {
    "else": {
      "not": {
        "required": [
          "finding_digest"
        ]
      }
    },
    "if": {
      "properties": {
        "decision_kind": {
          "enum": [
            "resolve",
            "waive"
          ]
        }
      }
    }
  }
}

```

```

    }
  },
  "required": [
    "decision_kind"
  ]
},
"then": {
  "required": [
    "finding_digest"
  ]
},
"title": "Finding disposition binds exact canonical Finding digest"
},
{
  "else": {
    "not": {
      "anyOf": [
        {
          "required": [
            "exception_policy_digest"
          ]
        },
        {
          "required": [
            "expires_at"
          ]
        }
      ]
    }
  }
}

```

```

    }
  },
  "if": {
    "properties": {
      "decision_kind": {
        "const": "waive"
      }
    },
    "required": [
      "decision_kind"
    ]
  },
  "then": {
    "required": [
      "exception_policy_digest",
      "expires_at"
    ]
  }
}

```



```

    ]
  },
  "title": "waiver binds exception policy and expiry"
},
{
  "else": {
    "required": [
      "candidate_digest"
    ]
  },
  "if": {
    "properties": {
      "decision_kind": {
        "const": "revoke"
      }
    },
    "required": [
      "decision_kind"
    ]
  },
  "then": {
    "not": {
      "required": [
        "candidate_digest"
      ]
    },
    "properties": {
      "target_type": {
        "const": "Grant"
      }
    }
  },
  "title": "Grant revocation has no Candidate binding"
}
],
"properties": {
  "activation_digest": {
    "$ref": "#/$defs/Digest"
  },

```

```

    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "created_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "decision_id": {
      "$ref": "#/$defs/Id"
    },
    "decision_kind": {
      "enum": [
        "accept",
        "reject",
        "resolve",
        "waive",
        "promote",
        "release",
        "revoke"
      ]
    },
    "evidence_artifacts": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "artifact_id": {
            "$ref": "#/$defs/Id"
          },
          "artifact_record_digest": {

```

```

            "$ref": "#/$defs/Digest"
          }
        },
        "required": [
          "artifact_id",
          "artifact_record_digest"
        ],
        "type": "object"
      },
      "maxItems": 64,

```

```

    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "exception_policy_digest": {
    "$ref": "#/$defs/Digest"
  },
  "expires_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "finding_digest": {
    "$ref": "#/$defs/Digest"
  },
  "grant_claim_digest": {
    "$ref": "#/$defs/Digest"
  },
  "grant_id": {
    "$ref": "#/$defs/Id"
  }

```

```

  },
  "provider_binding_digest": {
    "$ref": "#/$defs/Digest"
  },
  "rationale": {
    "minLength": 1,
    "type": "string"
  },
  "record_type": {
    "const": "Decision"
  },
  "release_closure_digest": {
    "$ref": "#/$defs/Digest"
  },
  "subject_id": {
    "$ref": "#/$defs/Id"
  },
  "target_digest": {
    "$ref": "#/$defs/Digest"
  },
  "target_id": {
    "$ref": "#/$defs/Id"
  },
  "target_type": {
    "enum": [
      "Завдання",
      "Candidate",
      "Finding",

```

```

      "Grant"
    ]
  }
},
"required": [
  "record_type",
  "decision_id",
  "decision_kind",
  "subject_id",
  "grant_id",
  "grant_claim_digest",
  "activation_digest",
  "target_type",
  "target_id",
  "target_digest",
  "rationale",
  "evidence_artifacts",
  "created_at"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/Decision](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/Decision](#)

25. Digest

Опис типу: type="string"; pattern="^[0-9a-f]{64}\$"

Обов'язкові поля: -

```

{
  "pattern": "^[0-9a-f]{64}$",

```

```
"type": "string"
}
```

Власник: [core/contracts.schema.json#/\\$defs/Digest](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/Digest](#)

26. DoctorResult

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, status, claim_scope, source_scope, report_authoritative, pass_credit, product_acceptance_pass, product_public_approval, activation_digest, implementation_closure_digest, head, recovery, replay, projection, provider_adapters, metrics, components, archival_diagnostics, operation_metrics, product_tree_scans

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
archival_diagnostics	так	type="object"; additionalProperties=false
claim_scope	так	const="runtime-health-only"
components	так	type="object"; additionalProperties=false
head	так	type="object"; additionalProperties=false
implementation_closure_digest	так	\$ref="#/\$defs/Digest"
metrics	так	type="object"; additionalProperties=false
operation_metrics	так	oneOf[2]=#/\$defs/OperationMetrics type:null
pass_credit	так	const=false
product_acceptance_pass	так	const=false
product_public_approval	так	const="not_approved"
product_tree_scans	так	const=0
projection	так	oneOf[2]=type:object type:object
provider_adapters	так	type="array"; minItems=1; maxItems=32
record_type	так	const="DoctorResult"
recovery	так	type="object"; additionalProperties=false
replay	так	oneOf[2]=type:null type:object
report_authoritative	так	const=false
source_scope	так	const="installed-core-and-live-operational-state-only"
status	так	enum=["healthy", "degraded", "failed", "incomplete"]

```
{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "archival_diagnostics": {
      "additionalProperties": false,
      "properties": {
        "affects_current_status": {
          "const": false
        }
      },
      "entries": {
        "items": {
          "additionalProperties": false,
          "properties": {
            "detail_digest": {
              "$ref": "#/$defs/Digest"
            },
            "diagnostic_id": {
              "$ref": "#/$defs/Id"
            },
            "recorded_at": {
              "$ref": "#/$defs/Timestamp"
            },
            "status": {
              "enum": [
                "healthy",
                "degraded",
                "failed",
```

```

        "incomplete"
      ]
    }
  },
  "required": [
    "diagnostic_id",
    "status",
    "recorded_at",
    "detail_digest"
  ],
  "type": "object"
},
"maxItems": 256,
"minItems": 0,
"type": "array"
},
"pass_credit": {
  "const": false
}
},
"required": [
  "affects_current_status",
  "pass_credit",
  "entries"
],
"type": "object"

```

```

},
"claim_scope": {
  "const": "runtime-health-only"
},
"components": {
  "additionalProperties": false,
  "properties": {
    "core": {
      "additionalProperties": false,
      "properties": {
        "observed_digest": {
          "oneOf": [
            {
              "$ref": "#/$defs/Digest"
            },
            {
              "type": "null"
            }
          ]
        }
      }
    },
    "reason_codes": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 32,
      "minItems": 0,
      "type": "array",
      "uniqueItems": true
    }
  }
}

```

```

},
"status": {
  "enum": [
    "healthy",
    "degraded",
    "failed",
    "incomplete"
  ]
}
},
"required": [
  "status",
  "reason_codes",
  "observed_digest"
],
"type": "object"
},
"init": {
  "additionalProperties": false,
  "properties": {
    "observed_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        }
      ]
    }
  }
}

```

```

    },
    {
      "type": "null"
    }
  ]
},
"reason_codes": {
  "items": {
    "$ref": "#/$defs/Id"
  },
  "maxItems": 32,
  "minItems": 0,
  "type": "array",
  "uniqueItems": true
},
"status": {
  "enum": [
    "healthy",
    "degraded",
    "failed",
    "incomplete"
  ]
}
},
"required": [
  "status",
  "reason_codes",
  "observed_digest"
],
"type": "object"
},
"projection": {

```

```

  "additionalProperties": false,
  "properties": {
    "observed_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    },
    "reason_codes": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 32,
      "minItems": 0,
      "type": "array",
      "uniqueItems": true
    },
    "status": {
      "enum": [
        "healthy",
        "degraded",
        "failed",
        "incomplete"
      ]
    }
  ]
}

```

```

},
"required": [
  "status",
  "reason_codes",
  "observed_digest"
],
"type": "object"
},
"providers": {
  "additionalProperties": false,
  "properties": {
    "observed_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },

```

```

        {
          "type": "null"
        }
      ]
    },
    "reason_codes": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 32,
      "minItems": 0,
      "type": "array",
      "uniqueItems": true
    },
    "status": {
      "enum": [
        "healthy",
        "degraded",
        "failed",
        "incomplete"
      ]
    }
  },
  "required": [
    "status",
    "reason_codes",
    "observed_digest"
  ],
  "type": "object"
},
"recovery": {
  "additionalProperties": false,
  "properties": {
    "observed_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    },
    "reason_codes": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 32,
      "minItems": 0,
      "type": "array",
      "uniqueItems": true
    },
    "status": {
      "enum": [
        "healthy",
        "degraded",
        "failed",
        "incomplete"
      ]
    }
  },
  "required": [
    "status",
    "reason_codes",
    "observed_digest"
  ],
  "type": "object"
},
"replay": {
  "additionalProperties": false,
  "properties": {
    "observed_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {

```

```

        "type": "null"
      }
    ]
  },
  "reason_codes": {
    "items": {
      "$ref": "#/$defs/Id"
    },
    "maxItems": 32,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "status": {
    "enum": [
      "healthy",
      "degraded",
      "failed",

```

```

      "incomplete"
    ]
  }
},
"required": [
  "status",
  "reason_codes",
  "observed_digest"
],
"type": "object"
}
},
"required": [
  "core",
  "init",
  "providers",
  "recovery",
  "replay",
  "projection"
],
"type": "object"
},
"head": {
  "additionalProperties": false,
  "properties": {
    "batch_digest": {
      "oneOf": [

```

```

        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    },
    "batch_id": {
      "oneOf": [
        {
          "$ref": "#/$defs/Id"
        },
        {
          "type": "null"
        }
      ]
    },
    "sequence": {
      "minimum": 0,
      "type": "integer"
    }
  },
  "required": [
    "sequence",
    "batch_id",
    "batch_digest"
  ],
  "type": "object"

```

```

},
"implementation_closure_digest": {
  "$ref": "#/$defs/Digest"
}

```

```

    },
    "metrics": {
      "additionalProperties": false,
      "properties": {
        "core_artifacts_verified": {
          "minimum": 0,
          "type": "integer"
        },
        "event_batches": {
          "minimum": 0,
          "type": "integer"
        },
        "init_records_verified": {
          "minimum": 0,
          "type": "integer"
        },
        "projection_entities": {
          "oneOf": [
            {
              "minimum": 0,
              "type": "integer"
            },
            {
              "type": "null"
            }
          ]
        },
        "projection_relations": {
          "oneOf": [
            {
              "minimum": 0,
              "type": "integer"
            },
            {
              "type": "null"
            }
          ]
        },
        "provider_healthchecks_executed": {
          "minimum": 0,
          "type": "integer"
        }
      },
      "required": [
        "core_artifacts_verified",
        "event_batches",
        "init_records_verified",
        "projection_entities",
        "projection_relations",
        "provider_healthchecks_executed"
      ],
      "type": "object"
    },
    "operation_metrics": {
      "oneOf": [
        {
          "$ref": "#/$defs/OperationMetrics"
        },
        {
          "type": "null"
        }
      ]
    },
    "pass_credit": {
      "const": false
    },
    "product_acceptance_pass": {
      "const": false
    },
    "product_public_approval": {
      "const": "not_approved"
    },
    "product_tree_scans": {
      "const": 0
    },
    "projection": {
      "oneOf": [
        {

```



```

"additionalProperties": false,
"properties": {
  "projection_authoritative": {
    "const": false
  },
  "status": {
    "const": "missing"
  }
},
"required": [
  "status",
  "projection_authoritative"
],
"type": "object"
},
{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "built_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "entity_count": {
      "minimum": 0,
      "type": "integer"
    },
    "event_count": {
      "minimum": 0,
      "type": "integer"
    },
    "head_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    },
    "head_sequence": {
      "minimum": 0,
      "type": "integer"
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "incremental_changed_records": {
      "minimum": 0,
      "type": "integer"
    },
    "incremental_commit_count": {
      "minimum": 0,
      "type": "integer"
    },
    "inventory_entries": {
      "minimum": 0,
      "type": "integer"
    },
    "inventory_passes": {
      "minimum": 0,
      "type": "integer"
    },
    "inventory_proxies": {
      "minimum": 0,
      "type": "integer"
    },
    "inventory_relations": {
      "minimum": 0,
      "type": "integer"
    },
    "product_passes": {
      "minimum": 0,
      "type": "integer"
    }
  },

```

```

    "projection_authoritative": {
      "const": false
    },
    "projection_compaction_count": {
      "minimum": 0,
      "type": "integer"
    },
    "projection_db_bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "raw_file_proxy_ratio": {
      "minimum": 0,
      "type": "number"
    },
    "relation_count": {
      "minimum": 0,
      "type": "integer"
    },
    "semantic_digest": {
      "$ref": "#/$defs/Digest"
    },
    "semantic_digest_algorithm": {
      "maxLength": 128,
      "minLength": 1,
      "type": "string"
    },
    "storage_layout": {
      "maxLength": 128,
      "minLength": 1,
      "type": "string"
    },
    "synthetic_task_count": {
      "const": 0,
      "type": "integer"
    },
    "synthetic_task_ratio": {
      "const": 0.0,
      "type": "number"
    }
  },
  "required": [
    "activation_digest",
    "head_digest",
    "head_sequence",
    "semantic_digest",
    "entity_count",
    "relation_count",
    "inventory_entries",
    "inventory_proxies",
    "inventory_relations",
    "inventory_passes",
    "product_passes",
    "event_count",
    "built_at",
    "projection_authoritative",
    "implementation_closure_digest",
    "semantic_digest_algorithm",
    "storage_layout",
    "incremental_commit_count",
    "incremental_changed_records",
    "projection_compaction_count",
    "raw_file_proxy_ratio",
    "synthetic_task_count",
    "synthetic_task_ratio",
    "projection_db_bytes"
  ],
  "type": "object"
}
],
},
"provider_adapters": {
  "items": {
    "additionalProperties": false,
    "oneOf": [
      {
        "properties": {

```

```

        "adapter_id": {
            "const": "python-runtime-v1"
        },
        "capability_id": {
            "const": "control-runtime"
        },
        "identity_kind": {
            "const": "file-digest"
        },
        "invocation_kind": {
            "const": "python-runtime"
        },
        "operation": {
            "const": "healthcheck"
        },
        "operation_contract_digest": {
            "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
        },
        "protocol_id": {
            "const": "promin.control-runtime.v1"
        }
    },
    {
        "properties": {
            "adapter_id": {
                "const": "python-runtime-v1"
            },
            "capability_id": {
                "const": "shape-validation"
            },
            "identity_kind": {
                "const": "file-digest"
            },
            "invocation_kind": {
                "const": "python-runtime"
            },
            "operation": {
                "const": "healthcheck"
            },
            "operation_contract_digest": {
                "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
            },
            "protocol_id": {
                "const": "promin.shape-validation.v1"
            }
        }
    },
    {
        "properties": {
            "adapter_id": {
                "const": "python-runtime-v1"
            },
            "capability_id": {
                "const": "content-identity"
            },
            "identity_kind": {
                "const": "file-digest"
            },
            "invocation_kind": {
                "const": "python-runtime"
            },
            "operation": {
                "const": "healthcheck"
            },
            "operation_contract_digest": {
                "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
            },
            "protocol_id": {
                "const": "promin.content-identity.v1"
            }
        }
    },
    {
        "properties": {
            "adapter_id": {
                "const": "python-runtime-v1"
            }
        }
    }
}

```

```

    },
    "capability_id": {
      "const": "local-serialization"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
    },
    "protocol_id": {
      "const": "promin.local-serialization.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {

```

```

      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "query-projection"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
    },
    "protocol_id": {
      "const": "promin.query-projection.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "git-executable-v1"
    },
    "capability_id": {

```

```

      "const": "filesystem-inventory"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "053c4181d6d35bd4ca5a13b8a1f18c19571c21b3426644bfa6bda99066dce5b3"
    },
    "protocol_id": {
      "const": "promin.filesystem-inventory.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-export-scan-v1"
    },
    "capability_id": {
      "const": "export-scan"
    }
  }
}

```

```

    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "116e5e8a519592f63585f0fae7b90f86863ba3369f0b006913a0e4f7d51eaf2e"
    },
    "protocol_id": {
      "const": "promin.export-scan.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "0e2de1aec47aec67eacbec7838d7f7f19f457bcff9c3fae57c9e88a72b6bb6be"
    },
    "protocol_id": {
      "const": "promin.signature.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-module-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "module-file-digest"
    },
    "invocation_kind": {
      "const": "python-module"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "0e2de1aec47aec67eacbec7838d7f7f19f457bcff9c3fae57c9e88a72b6bb6be"
    },
    "protocol_id": {
      "const": "promin.signature.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-build-dependency-v1"
    },
    "capability_id": {
      "const": "build-dependency"
    },
    "identity_kind": {

```

```

        "const": "file-digest"
    },
    "invocation_kind": {
        "const": "executable"
    },
    "operation": {
        "const": "healthcheck"
    },
    "operation_contract_digest": {
        "const": "116e5e8a519592f63585f0fae7b90f86863ba3369f0b006913a0e4f7d51eaf2e"
    },
    "protocol_id": {
        "const": "promin.build-dependency.v1"
    }
}
},
"properties": {
    "adapter_id": {
        "$ref": "#/$defs/Id"
    },
    "authoritative": {
        "const": false
    },
    "capability_id": {
        "enum": [
            "control-runtime",
            "shape-validation",
            "content-identity",
            "local-serialization",
            "query-projection",
            "filesystem-inventory",
            "export-scan",
            "signature",
            "build-dependency"
        ]
    }
},

```

```

"completed_at": {
    "$ref": "#/$defs/Timestamp"
},
"dependency_receipt_digest": {
    "$ref": "#/$defs/Digest"
},
"exit_code": {
    "oneOf": [
        {
            "type": "integer"
        },
        {
            "type": "null"
        }
    ]
},
"identity_digest": {
    "$ref": "#/$defs/Digest"
},
"identity_kind": {
    "enum": [
        "file-digest",
        "module-file-digest",
        "distribution-tree-digest",
        "module-closure-digest",
        "platform-attestation"
    ]
},

```

```

"invocation_kind": {
    "enum": [
        "python-runtime",
        "python-module",
        "executable",
        "platform-service"
    ]
},
"invocation_receipt_digest": {
    "$ref": "#/$defs/Digest"
},
"invoked": {

```

```

    "const": true
  },
  "operation": {
    "const": "healthcheck"
  },
  "operation_contract_digest": {
    "$ref": "#/$defs/Digest"
  },
  "outcome": {
    "enum": [
      "healthy",
      "degraded",
      "failed"
    ]
  },
  "pass_credit": {

```

```

    "const": false
  },
  "persistence_scope": {
    "const": "content-addressed-receipt"
  },
  "protocol_id": {
    "$ref": "#/$defs/Id"
  },
  "provider_id": {
    "$ref": "#/$defs/Id"
  },
  "reconstructable": {
    "const": true
  },
  "started_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "stderr_digest": {
    "$ref": "#/$defs/Digest"
  },
  "stderr_size_bytes": {
    "maximum": 1048576,
    "minimum": 0,
    "type": "integer"
  },
  "stdout_digest": {
    "$ref": "#/$defs/Digest"
  },

```

```

    "stdout_size_bytes": {
      "maximum": 1048576,
      "minimum": 0,
      "type": "integer"
    }
  },
  "required": [
    "capability_id",
    "provider_id",
    "invocation_kind",
    "identity_kind",
    "identity_digest",
    "adapter_id",
    "protocol_id",
    "operation",
    "operation_contract_digest",
    "invocation_receipt_digest",
    "dependency_receipt_digest",
    "persistence_scope",
    "reconstructable",
    "invoked",
    "started_at",
    "completed_at",
    "outcome",
    "exit_code",
    "stdout_digest",
    "stdout_size_bytes",
    "stderr_digest",

```

```

    "stderr_size_bytes",
    "authoritative",
    "pass_credit"
  ],
  "type": "object"

```

```

    },
    "maxItems": 32,
    "minItems": 1,
    "type": "array"
  },
  "record_type": {
    "const": "DoctorResult"
  },
  "recovery": {
    "additionalProperties": false,
    "properties": {
      "batch_digest": {
        "oneOf": [
          {
            "$ref": "#/$defs/Digest"
          },
          {
            "type": "null"
          }
        ]
      }
    }
  },
  "batch_id": {
    "oneOf": [

```

```

      {
        "$ref": "#/$defs/Id"
      },
      {
        "type": "null"
      }
    ]
  },
  "sequence": {
    "minimum": 0,
    "type": "integer"
  }
},
"required": [
  "sequence",
  "batch_id",
  "batch_digest"
],
"type": "object"
},
"replay": {
  "oneOf": [
    {
      "type": "null"
    },
    {
      "additionalProperties": false,
      "properties": {

```

```

        "event_batches": {
          "minimum": 0,
          "type": "integer"
        },
        "head": {
          "additionalProperties": false,
          "properties": {
            "batch_digest": {
              "oneOf": [
                {
                  "$ref": "#/$defs/Digest"
                },
                {
                  "type": "null"
                }
              ]
            }
          }
        },
        "batch_id": {
          "oneOf": [
            {
              "$ref": "#/$defs/Id"
            },
            {
              "type": "null"
            }
          ]
        }
      }
    ]
  }
}

```



```

    },
    "sequence": {
        "minimum": 0,
        "type": "integer"
    }
},
"required": [
    "sequence",
    "batch_id",
    "batch_digest"
],
"type": "object"
},
"record_type": {
    "const": "ReplayResult"
},
"runtime_checkpoint": {
    "oneOf": [
        {
            "additionalProperties": false,
            "properties": {
                "authoritative": {
                    "const": false
                },
                "batch_threshold": {
                    "minimum": 1,
                    "type": "integer"
                },
                "byte_threshold": {
                    "minimum": 1,
                    "type": "integer"
                },
                "checkpoint_bytes": {
                    "minimum": 0,
                    "type": "integer"
                },
                "checkpoint_count": {
                    "minimum": 0,
                    "type": "integer"
                },
                "tail_batches": {
                    "minimum": 0,
                    "type": "integer"
                },
                "tail_bytes": {
                    "minimum": 0,
                    "type": "integer"
                },
                "written": {
                    "const": false
                }
            }
        },
        {
            "additionalProperties": false,
            "properties": {
                "authoritative": {
                    "const": false
                },
                "batch_threshold": {
                    "minimum": 1,
                    "type": "integer"
                },
                "byte_threshold": {
                    "minimum": 1,
                    "type": "integer"
                },
                "checkpoint_bytes": {
                    "minimum": 0,
                    "type": "integer"
                },
                "checkpoint_count": {
                    "minimum": 0,
                    "type": "integer"
                },
                "tail_batches": {
                    "minimum": 0,
                    "type": "integer"
                },
                "tail_bytes": {
                    "minimum": 0,
                    "type": "integer"
                },
                "written": {
                    "const": false
                }
            }
        }
    ]
},
"required": [
    "authoritative",
    "batch_threshold",
    "byte_threshold",
    "checkpoint_bytes",
    "checkpoint_count",
    "tail_batches",
    "tail_bytes",
    "written"
],
"type": "object"
},
{
    "additionalProperties": false,
    "properties": {
        "authoritative": {
            "const": false
        },
        "batch_threshold": {
            "minimum": 1,
            "type": "integer"
        },
        "byte_threshold": {
            "minimum": 1,
            "type": "integer"
        },
        "checkpoint_bytes": {
            "minimum": 0,
            "type": "integer"
        },
        "checkpoint_count": {
            "minimum": 0,
            "type": "integer"
        },
        "tail_batches": {
            "minimum": 0,
            "type": "integer"
        },
        "tail_bytes": {
            "minimum": 0,
            "type": "integer"
        },
        "written": {
            "const": false
        }
    }
}

```

```

    },
    "checkpoint_bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "checkpoint_count": {
      "minimum": 0,
      "type": "integer"
    },
  },

```

```

    "compacted_tail_batches": {
      "minimum": 0,
      "type": "integer"
    },
    "compacted_tail_bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "tail_batches": {
      "minimum": 0,
      "type": "integer"
    },
    "tail_bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "written": {
      "const": true
    }
  },
  "required": [
    "authoritative",
    "batch_threshold",
    "byte_threshold",
    "checkpoint_bytes",
    "checkpoint_count",
    "tail_batches",
    "tail_bytes",

```

```

    "written",
    "compacted_tail_batches",
    "compacted_tail_bytes"
  ],
  "type": "object"
}
]
},
"runtime_checkpoint_authoritative": {
  "const": false
},
"runtime_checkpoint_written": {
  "type": "boolean"
},
"status": {
  "const": "pass"
}
},
"required": [
  "record_type",
  "status",
  "event_batches",
  "head",
  "runtime_checkpoint_written",
  "runtime_checkpoint",
  "runtime_checkpoint_authoritative"
],
"type": "object"

```

```

}
]
},
"report_authoritative": {
  "const": false
},
"source_scope": {
  "const": "installed-core-and-live-operational-state-only"
},
"status": {
  "enum": [
    "healthy",

```

```

        "degraded",
        "failed",
        "incomplete"
    ]
},
"required": [
    "record_type",
    "status",
    "claim_scope",
    "source_scope",
    "report_authoritative",
    "pass_credit",
    "product_acceptance_pass",
    "product_public_approval",
    "activation_digest",

    "implementation_closure_digest",
    "head",
    "recovery",
    "replay",
    "projection",
    "provider_adapters",
    "metrics",
    "components",
    "archival_diagnostics",
    "operation_metrics",
    "product_tree_scans"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Власник: `core/contracts.schema.json#/$defs/DoctorResult` | Валідатор: `urn:promin:contracts:1#/$defs/DoctorResult`

27. EffectTarget

Опис типу: `type="object"; additionalProperties=false`

Обов'язкові поля: `kind`, `id_field`

Властивість	Обов'язков е	Структурне правило
<code>id_field</code>	так	<code>\$ref="#/\$defs/Id"</code>
<code>kind</code>	так	<code>\$ref="#/\$defs/Id"</code>

```

{
  "additionalProperties": false,
  "properties": {
    "id_field": {
      "$ref": "#/$defs/Id"
    },
    "kind": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "kind",
    "id_field"
  ],
  "type": "object"
}

```

Власник: `core/contracts.schema.json#/$defs/EffectTarget` | Валідатор: `urn:promin:contracts:1#/$defs/EffectTarget`

28. Event

Опис типу: `type="object"; additionalProperties=false; allOf[12]=keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then | keys:if,then`

Обов'язкові поля: `record_type`, `event_id`, `event_kind`, `activation_digest`, `payload`

Властивість	Обов'язков е	Структурне правило
<code>activation_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>event_id</code>	так	<code>\$ref="#/\$defs/Id"</code>

Властивість	Обов'язков е	Структурне правило
event_kind	так	enum=["artifact.recorded", "candidate.recorded", "decision.recorded", "finding.recorded", "gate.recorded", "grant.issued", "grant.revoked", "lease.recorded", "run.recorded", "task.recorded", "task.transitioned", "relation.recorded"]
payload	так	складене структурне визначення
record_type	так	const="Event"
<pre> { "additionalProperties": false, "allOf": [{ "if": { "properties": { "event_kind": { "const": "task.recorded" } } }, "then": { "properties": { "payload": { "\$ref": "#/\$defs/Task" } } } }, { "if": { "properties": { "event_kind": { "const": "task.transitioned" } } }, "then": { "properties": { "payload": { "\$ref": "#/\$defs/TaskTransition" } } } }, { "if": { "properties": { "event_kind": { "const": "candidate.recorded" } } }, "then": { "properties": { "payload": { "\$ref": "#/\$defs/Candidate" } } } }, { "if": { "properties": { "event_kind": { "const": "artifact.recorded" } } }, "then": { "properties": { "payload": { "\$ref": "#/\$defs/Artifact" } } } }] } </pre>		

```

{
  "if": {
    "properties": {
      "event_kind": {
        "const": "run.recorded"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Run"
      }
    }
  }
},
{

```

```

  "if": {
    "properties": {
      "event_kind": {
        "const": "gate.recorded"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/GateResult"
      }
    }
  }
},
{
  "if": {
    "properties": {
      "event_kind": {
        "const": "finding.recorded"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Finding"
      }
    }
  }
}

```

```

}
},
{
  "if": {
    "properties": {
      "event_kind": {
        "const": "decision.recorded"
      }
    }
  },
  "then": {
    "properties": {
      "payload": {
        "$ref": "#/$defs/Decision"
      }
    }
  }
},
{
  "if": {
    "properties": {
      "event_kind": {
        "const": "grant.issued"
      }
    }
  },
  "then": {

```

```

    "properties": {
      "payload": {
        "$ref": "#/$defs/Grant"
      }
    }
  }
}

```

```

    }
  },
  {
    "if": {
      "properties": {
        "event_kind": {
          "const": "grant.revoked"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/GrantRevocation"
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "event_kind": {
          "const": "lease.recorded"

```

```

        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/Lease"
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "event_kind": {
          "const": "relation.recorded"
        }
      }
    },
    "then": {
      "properties": {
        "payload": {
          "$ref": "#/$defs/Relation"
        }
      }
    }
  }
],

```

```

"properties": {
  "activation_digest": {
    "$ref": "#/$defs/Digest"
  },
  "event_id": {
    "$ref": "#/$defs/Id"
  },
  "event_kind": {
    "enum": [
      "artifact.recorded",
      "candidate.recorded",
      "decision.recorded",
      "finding.recorded",
      "gate.recorded",
      "grant.issued",
      "grant.revoked",
      "lease.recorded",
      "run.recorded",
      "task.recorded",
      "task.transitioned",
      "relation.recorded"
    ]
  },
  "payload": {},
  "record_type": {
    "const": "Event"

```

```

    }
  },
  "required": [
    "record_type",
    "event_id",
    "event_kind",
    "activation_digest",
    "payload"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/Event | Валідатор: urn:promin:contracts:1#/\$defs/Event

29. EventBatch

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, batch_id, sequence, previous_digest, previous_authority_commitment, authority_commitment, cumulative_event_count, event_semantic_digest, state_binding_delta, cumulative_state_binding_update_count, state_binding_digest, created_at, command_id, idempotency_key, activation_record_digest, events, subject_id, command_intent_digest, command_digest, authorization_digest

Властивість	Обов'язков е	Структурне правило
activation_record_digest	так	\$ref="#/\$defs/Digest"
authority_commitment	так	\$ref="#/\$defs/Digest"
authorization_digest	так	\$ref="#/\$defs/Digest"
batch_id	так	\$ref="#/\$defs/Id"
command_digest	так	\$ref="#/\$defs/Digest"
command_id	так	\$ref="#/\$defs/Id"
command_intent_digest	так	\$ref="#/\$defs/Digest"
created_at	так	\$ref="#/\$defs/Timestamp"
cumulative_event_count	так	type="integer"; minimum=1
cumulative_state_binding_update_count	так	type="integer"; minimum=1
event_semantic_digest	так	\$ref="#/\$defs/Digest"
events	так	type="array"; minItems=1; maxItems=20
idempotency_key	так	type="string"; pattern="^[A-Za-z0-9._-]{16,128}\$"
previous_authority_commitment	так	\$ref="#/\$defs/Digest"
previous_digest	так	oneOf[2]=#/\$defs/Digest type:null
record_type	так	const="EventBatch"
sequence	так	type="integer"; minimum=1
state_binding_delta	так	type="array"; minItems=1; maxItems=20; uniqueItems=true
state_binding_digest	так	\$ref="#/\$defs/Digest"
subject_id	так	\$ref="#/\$defs/Id"

```

{
  "additionalProperties": false,
  "properties": {
    "activation_record_digest": {
      "$ref": "#/$defs/Digest"
    },
    "authority_commitment": {
      "$ref": "#/$defs/Digest"
    },
    "authorization_digest": {
      "$ref": "#/$defs/Digest"
    },
    "batch_id": {
      "$ref": "#/$defs/Id"
    },
    "command_digest": {
      "$ref": "#/$defs/Digest"
    },
    "command_id": {
      "$ref": "#/$defs/Id"
    }
  }
}

```

```

    },
    "command_intent_digest": {
      "$ref": "#/$defs/Digest"
    },
    "created_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "cumulative_event_count": {
      "minimum": 1,
      "type": "integer"
    },
    "cumulative_state_binding_update_count": {
      "minimum": 1,
      "type": "integer"
    },
    "event_semantic_digest": {
      "$ref": "#/$defs/Digest"
    },
    "events": {
      "items": {
        "$ref": "#/$defs/Event"
      },
      "maxItems": 20,
      "minItems": 1,
      "type": "array"
    },
    "idempotency_key": {
      "pattern": "^[A-Za-z0-9._-]{16,128}$",
      "type": "string"
    },
    "previous_authority_commitment": {
      "$ref": "#/$defs/Digest"
    },
    "previous_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {
          "type": "null"
        }
      ]
    },
    "record_type": {
      "const": "EventBatch"
    },
    "sequence": {
      "minimum": 1,
      "type": "integer"
    },
    "state_binding_delta": {
      "items": {
        "$ref": "#/$defs/StateBindingUpdate"
      },
      "maxItems": 20,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "state_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "record_type",
    "batch_id",
    "sequence",
    "previous_digest",
    "previous_authority_commitment",
    "authority_commitment",
    "cumulative_event_count",
    "event_semantic_digest",
    "state_binding_delta",
    "cumulative_state_binding_update_count",

```



```

    "state_binding_digest",
    "created_at",
    "command_id",
    "idempotency_key",
    "activation_record_digest",
    "events",
    "subject_id",
    "command_intent_digest",
    "command_digest",
    "authorization_digest"
  ],
  "type": "object",
  "x-promin-max-canonical-bytes": 2097152,
  "x-promin-max-state-binding-bytes": 1048576,
  "x-promin-primary-event-count-per-command": 1
}

```

Власник: `core/contracts.schema.json#/$defs/EventBatch` | Валідатор: `urn:promin:contracts:1#/$defs/EventBatch`

30. EventStorePolicy

Опис типу: `type="object"; additionalProperties=false`

Обов'язкові поля: `record_type`, `authority_model_digest`, `max_command_bytes`, `max_envelope_bytes`, `max_state_binding_bytes`, `max_state_binding_updates_per_batch`, `max_events_per_batch`, `max_requested_scope_items`, `derived_tail_batch_threshold`, `derived_tail_byte_threshold`, `runtime_overlay_compaction_depth`, `command_required_fields`, `command_conditional_fields`, `command_mutation_fields`, `lease_bound_command_kinds`, `lease_bound_task_transition_states`, `command_to_primary_event`, `allowed_state_binding_leaf_types`, `state_binding_identity_rules`, `state_binding_algorithm_contract`, `state_binding_value_rules`, `canonical_timestamp_contract`, `genesis_previous_authority_commitment`, `genesis_event_semantic_digest`, `policy_digest`

Властивість	Обов'язков е	Структурне правило
<code>allowed_state_binding_leaf_types</code>	так	<code>const=["Activation", "Завдання", "Candidate", "Artifact", "Run", "Finding", "Decision", "GateResult", "Grant", "Оренда", "Відношення"]</code>
<code>authority_model_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>canonical_timestamp_contract</code>	так	<code>const=schema(keys:calendar,format,formatter_api,fractional_seconds,normализation,offset,parser_api,pattern,real_calendar_validation,resolution_seconds,year_max,year_min)</code>
<code>command_conditional_fields</code>	так	<code>const=["definition_digest"]</code>
<code>command_mutation_fields</code>	так	<code>const=["workcard_task_id", "holder_authorization", "lease_id", "lease_generation", "fencing_token", "workcard_digest", "context_digest"]</code>
<code>command_required_fields</code>	так	<code>const=["record_type", "command_id", "command_kind", "subject_id", "activation_digest", "idempotency_key", "requested_scope", "expected_head_digest", "issued_at", "payload", "intent_digest", "authorization"]</code>
<code>command_to_primary_event</code>	так	<code>const=[11 items]</code>
<code>derived_tail_batch_threshold</code>	так	<code>type="integer"; const=128</code>
<code>derived_tail_byte_threshold</code>	так	<code>type="integer"; const=8388608</code>
<code>genesis_event_semantic_digest</code>	так	<code>const="e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855"</code>
<code>genesis_previous_authority_com mitment</code>	так	<code>const="e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855"</code>
<code>lease_bound_command_kinds</code>	так	<code>const=["artifact.record", "run.record", "gate.record", "finding.record"]</code>
<code>lease_bound_task_transition_state s</code>	так	<code>const=["LEASED", "RUNNING", "COMPLETED"]</code>
<code>max_command_bytes</code>	так	<code>type="integer"; const=1048576</code>
<code>max_envelope_bytes</code>	так	<code>type="integer"; const=2097152</code>
<code>max_events_per_batch</code>	так	<code>type="integer"; const=20</code>
<code>max_requested_scope_items</code>	так	<code>type="integer"; const=32</code>
<code>max_state_binding_bytes</code>	так	<code>type="integer"; const=1048576</code>
<code>max_state_binding_updates_per_ batch</code>	так	<code>type="integer"; const=20</code>
<code>policy_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>record_type</code>	так	<code>const="EventStorePolicy"</code>

Властивість	Обов'язков е	Структурне правило
runtime_overlay_compaction_dep th	так	type="integer"; const=64
state_binding_algorithm_contract	так	const=schema(keys:delete_rule,delta_order,depth,depth_encoding,empty_leaf_ domain_utf8,internal_node_domain_utf8,internal_node_rule,leaf_domain_utf8,le af_key_rule,set_leaf_rule)
state_binding_identity_rules	так	const=schema(keys:Activation,Artifact,Candidate,Decision,Finding,GateResult, Grant,Lease,Relation,Run,Task)
state_binding_value_rules	так	const=schema(keys:Activation,Artifact,Candidate,Decision,Finding,GateResult, Grant,Lease,Relation,Run,Task)
<pre> { "additionalProperties": false, "properties": { "allowed_state_binding_leaf_types": { "const": ["Activation", "Завдання", "Candidate", "Artifact", "Run", "Finding", "Decision", "GateResult", "Grant", "Оренда", "Відношення"] }, "authority_model_digest": { "\$ref": "#/\$defs/Digest" }, "canonical_timestamp_contract": { "const": { "calendar": "proleptic-gregorian", "format": "utc-second-z", "formatter_api": "promin.canonical.format_utc_second(value: datetime) -> str", "fractional_seconds": "reject", "normalization": "identity-round-trip", "offset": "Z-only", "parser_api": "promin.canonical.parse_utc_second(value: str) -> timezone-aware UTC datetime", "pattern": "^[0-9]{4}-[0-9]{2}-[0-9]{2}T[0-9]{2}::[0-9]{2}::[0-9]{2}Z\$", "real_calendar_validation": true, "resolution_seconds": 1, "year_max": 9999, "year_min": 1 } }, "command_conditional_fields": { "const": ["definition_digest"] }, "command_mutation_fields": { "const": ["workcard_task_id", "holder_authorization", "lease_id", "lease_generation", "fencing_token", "workcard_digest", "context_digest"] }, "command_required_fields": { "const": ["record_type", "command_id", "command_kind", "subject_id", "activation_digest", "idempotency_key", "requested_scope", "expected_head_digest", "issued_at", </pre>		

```

        "payload",
        "intent_digest",
        "authorization"
    ]
},
"command_to_primary_event": {
    "const": [
        [
            "artifact.record",
            "artifact.recorded"
        ],
        [
            "candidate.record",
            "candidate.recorded"
        ],
        [
            "decision.record",
            "decision.recorded"
        ],
        [

```

```

            "finding.record",
            "finding.recorded"
        ],
        [
            "gate.record",
            "gate.recorded"
        ],
        [
            "grant.issue",
            "grant.issued"
        ],
        [
            "grant.revoke",
            "grant.revoked"
        ],
        [
            "lease.record",
            "lease.recorded"
        ],
        [
            "run.record",
            "run.recorded"
        ],
        [
            "task.record",
            "task.recorded"
        ],
        [

```

```

            "task.transition",
            "task.transitioned"
        ]
    ]
},
"derived_tail_batch_threshold": {
    "const": 128,
    "type": "integer"
},
"derived_tail_byte_threshold": {
    "const": 8388608,
    "type": "integer"
},
"genesis_event_semantic_digest": {
    "const": "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855"
},
"genesis_previous_authority_commitment": {
    "const": "e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855"
},
"lease_bound_command_kinds": {
    "const": [
        "artifact.record",
        "run.record",
        "gate.record",
        "finding.record"
    ]
},
"lease_bound_task_transition_states": {

```

```

    "const": [
      "LEASED",
      "RUNNING",
      "COMPLETED"
    ]
  },
  "max_command_bytes": {
    "const": 1048576,
    "type": "integer"
  },
  "max_envelope_bytes": {
    "const": 2097152,
    "type": "integer"
  },
  "max_events_per_batch": {
    "const": 20,
    "type": "integer"
  },
  "max_requested_scope_items": {
    "const": 32,
    "type": "integer"
  },
  "max_state_binding_bytes": {
    "const": 1048576,
    "type": "integer"
  },
  "max_state_binding_updates_per_batch": {
    "const": 20,
    "type": "integer"
  },
  "policy_digest": {
    "$ref": "#/$defs/Digest"
  },
  "record_type": {
    "const": "EventStorePolicy"
  },
  "runtime_overlay_compaction_depth": {
    "const": 64,
    "type": "integer"
  },
  "state_binding_algorithm_contract": {
    "const": {
      "delete_rule": "delete maps the keyed leaf to empty_leaf",
      "delta_order": [
        "leaf_type-ascending",
        "leaf_id-ascending"
      ],
      "depth": 256,
      "depth_encoding": "unsigned-16-bit-big-endian",
      "empty_leaf_domain_utf8": "promin:typed-sparse-merkle-v1:empty-leaf",
      "internal_node_domain_utf8": "promin:typed-sparse-merkle-v1:node\u0000",
      "internal_node_rule": "sha256(domain + depth_uint16_be + left_32_bytes + right_32_bytes)",
      "leaf_domain_utf8": "promin:typed-sparse-merkle-v1:leaf\u0000",
      "leaf_key_rule": "sha256(canonical bytes of exact object leaf_type and leaf_id)",
      "set_leaf_rule": "sha256(domain + leaf_key_32_bytes + value_digest_32_bytes)"
    }
  },
  "state_binding_identity_rules": {
    "const": {
      "Activation": {
        "id_fields": [
          "activation_digest"
        ],
        "identity_mode": "direct"
      },
      "Artifact": {
        "id_fields": [
          "artifact_id"
        ],
        "identity_mode": "direct"
      },
      "Candidate": {
        "id_fields": [
          "candidate_id"
        ],
        "identity_mode": "direct"
      },
      "Decision": {

```

```

    "id_fields": [
        "decision_id"
    ],
    "identity_mode": "direct"
},
"Finding": {
    "id_fields": [
        "finding_id"
    ],
    "identity_mode": "direct"
},
"GateResult": {
    "digest_algorithm": "sha256",
    "digest_hex_chars": 48,
    "digest_input": "canonical-ordered-id-field-values",
    "id_fields": [
        "gate_id",
        "run_id"
    ],
    "identity_mode": "typed_digest_id",
    "prefix": "gate-result:"
},
"Grant": {
    "id_fields": [
        "grant_id"
    ],
    "identity_mode": "direct"
},
"Lease": {
    "id_fields": [
        "lease_id"
    ],
    "identity_mode": "direct"
},
"Relation": {
    "id_fields": [
        "relation_id"
    ],
    "identity_mode": "direct"
},
"Run": {
    "id_fields": [
        "run_id"
    ],
    "identity_mode": "direct"
},
"Task": {
    "id_fields": [
        "task_id"
    ],
    "identity_mode": "direct"
}
},
"state_binding_value_rules": {
    "const": {
        "Activation": {
            "event_kinds": [],
            "value_definition": "Activation",
            "value_digest_rule": "sha256(canonical installed Activation record); installed before sequence
one and immutable for the journal"
        },
        "Artifact": {
            "event_kinds": [
                "artifact.recorded"
            ],
            "value_definition": "Artifact",
            "value_digest_rule": "sha256(canonical exact Event.payload Artifact)"
        },
        "Candidate": {
            "event_kinds": [
                "candidate.recorded"
            ],
            "value_definition": "Candidate",
            "value_digest_rule": "sha256(canonical exact Event.payload Candidate)"
        },
        "Decision": {

```

```

    "event_kinds": [
      "decision.recorded"
    ],
    "value_definition": "Decision",
    "value_digest_rule": "sha256(canonical exact Event.payload Decision)"
  },
  "Finding": {
    "event_kinds": [
      "finding.recorded"
    ],
    "value_definition": "Finding",
    "value_digest_rule": "sha256(canonical exact Event.payload Finding)"
  },
  "GateResult": {
    "event_kinds": [
      "gate.recorded"
    ],
    "value_definition": "GateResult",
    "value_digest_rule": "sha256(canonical exact Event.payload GateResult)"
  },
  "Grant": {
    "event_kinds": [
      "grant.issued",
      "grant.revoked"
    ],
    "value_definition": "GrantAuthorityState",
    "value_digest_rule": "sha256(canonical exact GrantAuthorityState containing the current Grant
and null or exact current GrantRevocation)"
  },
  "Lease": {
    "event_kinds": [
      "lease.recorded"
    ],
    "value_definition": "Оренда",
    "value_digest_rule": "sha256(canonical exact Event.payload post-state Lease)"
  },
  "Relation": {
    "event_kinds": [
      "relation.recorded"
    ],
    "value_definition": "Відношення",
    "value_digest_rule": "sha256(canonical exact Event.payload Relation)"
  },
  "Run": {
    "event_kinds": [
      "run.recorded"
    ],
    "value_definition": "Run",
    "value_digest_rule": "sha256(canonical exact Event.payload Run)"
  },
  "Task": {
    "event_kinds": [
      "task.recorded",
      "task.transitioned"
    ],
    "value_definition": "Завдання",
    "value_digest_rule": "sha256(canonical exact Event.payload post-state Task)"
  }
}
},
"required": [
  "record_type",
  "authority_model_digest",
  "max_command_bytes",
  "max_envelope_bytes",
  "max_state_binding_bytes",
  "max_state_binding_updates_per_batch",
  "max_events_per_batch",
  "max_requested_scope_items",
  "derived_tail_batch_threshold",
  "derived_tail_byte_threshold",
  "runtime_overlay_compaction_depth",
  "command_required_fields",
  "command_conditional_fields",
  "command_mutation_fields",
  "lease_bound_command_kinds",

```

```

    "lease_bound_task_transition_states",
    "command_to_primary_event",
    "allowed_state_binding_leaf_types",
    "state_binding_identity_rules",
    "state_binding_algorithm_contract",
    "state_binding_value_rules",
    "canonical_timestamp_contract",
    "genesis_previous_authority_commitment",
    "genesis_event_semantic_digest",
    "policy_digest"
  ],
  "type": "object",
  "x-promin-persistence": "derived-live-only"
}

```

Власник: [core/contracts.schema.json#/\\$defs/EventStorePolicy](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/EventStorePolicy](#)

31. EvidenceBinding

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: activation_digest, implementation_closure_digest, candidate_digest, policy_digest, tool_digest, input_digests

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
candidate_digest	так	\$ref="#/\$defs/Digest"
finding_digest	ні	\$ref="#/\$defs/Digest"
implementation_closure_digest	так	\$ref="#/\$defs/Digest"
input_digests	так	type="array"; minItems=0; maxItems=128; uniqueItems=true
policy_digest	так	\$ref="#/\$defs/Digest"
provider_invocations	ні	type="array"; minItems=1; maxItems=32; uniqueItems=true
tool_digest	так	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "finding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "input_digests": {
      "items": {
        "$ref": "#/$defs/Digest"
      },
      "maxItems": 128,
      "minItems": 0,
      "type": "array",
      "uniqueItems": true
    },
    "policy_digest": {
      "$ref": "#/$defs/Digest"
    },
    "provider_invocations": {
      "items": {
        "$ref": "#/$defs/ProviderInvocationEvidence"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "tool_digest": {
      "$ref": "#/$defs/Digest"
    }
  }
}

```

```

},
"required": [
  "activation_digest",
  "implementation_closure_digest",
  "candidate_digest",
  "policy_digest",
  "tool_digest",
  "input_digests"
],
"type": "object"
}

```

Власник: `core/contracts.schema.json#/$defs/EvidenceBinding` | Валідатор: `urn:promin:contracts:1#/$defs/EvidenceBinding`

32. EvidenceProducerAttestation

Опис типу: `type="object"; additionalProperties=false`

Обов'язкові поля: `record_type`, `claim_domain`, `candidate_binding_digest`, `evidence_role`, `evidence_record_type`, `platform`, `invocation_id`, `payload_digest`, `raw_artifact_manifest_digest`, `producer_id`, `trust_root_id`, `signature_provider_id`, `key_id`, `public_key`, `nonce`, `signed_at`, `signed_claim_digest`, `signature`

Властивість	Обов'язков е	Структурне правило
<code>candidate_binding_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>claim_domain</code>	так	<code>const="promin.release-evidence.producer-attestation.v1"</code>
<code>evidence_record_type</code>	так	<code>enum=["HumanDocumentVerification", "NoDegradationResult", "PlatformVerificationResult", "SaturationAudit", "SaturationEvidence"]</code>
<code>evidence_role</code>	так	<code>enum=["human-documents", "linux", "linux-no-degradation", "physical-scale", "saturation-audit", "windows", "windows-no-degradation"]</code>
<code>invocation_id</code>	так	<code>\$ref="#/\$defs/Id"</code>
<code>key_id</code>	так	<code>\$ref="#/\$defs/Id"</code>
<code>nonce</code>	так	<code>type="string"; minLength=24; maxLength=128</code>
<code>payload_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>platform</code>	так	<code>enum=["cross-platform", "darwin", "linux", "windows"]</code>
<code>producer_id</code>	так	<code>\$ref="#/\$defs/Id"</code>
<code>public_key</code>	так	<code>type="string"; pattern="^[A-Za-z0-9+/{43}=\$"; minLength=44; maxLength=44</code>
<code>raw_artifact_manifest_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>record_type</code>	так	<code>const="EvidenceProducerAttestation"</code>
<code>signature</code>	так	<code>type="string"; minLength=80; maxLength=128</code>
<code>signature_provider_id</code>	так	<code>const="cryptography-ed25519-v1"</code>
<code>signed_at</code>	так	<code>\$ref="#/\$defs/Timestamp"</code>
<code>signed_claim_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>trust_root_id</code>	так	<code>\$ref="#/\$defs/Id"</code>

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "claim_domain": {
      "const": "promin.release-evidence.producer-attestation.v1"
    },
    "evidence_record_type": {
      "enum": [
        "HumanDocumentVerification",
        "NoDegradationResult",
        "PlatformVerificationResult",
        "SaturationAudit",
        "SaturationEvidence"
      ]
    },
    "evidence_role": {
      "enum": [
        "human-documents",
        "linux",
        "linux-no-degradation",

```



```

    "physical-scale",
    "saturation-audit",
    "windows",
    "windows-no-degradation"
  ]

```

```

  },
  "invocation_id": {
    "$ref": "#/$defs/Id"
  },
  "key_id": {
    "$ref": "#/$defs/Id"
  },
  "nonce": {
    "maxLength": 128,
    "minLength": 24,
    "type": "string"
  },
  "payload_digest": {
    "$ref": "#/$defs/Digest"
  },
  "platform": {
    "enum": [
      "cross-platform",
      "darwin",
      "linux",
      "windows"
    ]
  },
  "producer_id": {
    "$ref": "#/$defs/Id"
  },
  "public_key": {
    "maxLength": 44,

```

```

    "minLength": 44,
    "pattern": "^[A-Za-z0-9+/{43}=$",
    "type": "string"
  },
  "raw_artifact_manifest_digest": {
    "$ref": "#/$defs/Digest"
  },
  "record_type": {
    "const": "EvidenceProducerAttestation"
  },
  "signature": {
    "maxLength": 128,
    "minLength": 80,
    "type": "string"
  },
  "signature_provider_id": {
    "const": "cryptography-ed25519-v1"
  },
  "signed_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "signed_claim_digest": {
    "$ref": "#/$defs/Digest"
  },
  "trust_root_id": {
    "$ref": "#/$defs/Id"
  }
},

```

```

"required": [
  "record_type",
  "claim_domain",
  "candidate_binding_digest",
  "evidence_role",
  "evidence_record_type",
  "platform",
  "invocation_id",
  "payload_digest",
  "raw_artifact_manifest_digest",
  "producer_id",
  "trust_root_id",
  "signature_provider_id",
  "key_id",
  "public_key",
  "nonce",

```

```

    "signed_at",
    "signed_claim_digest",
    "signature"
  ],
  "type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/EvidenceProducerAttestation](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/EvidenceProducerAttestation](#)

33. Finding

Опис типу: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then

Обов'язкові поля: record_type, finding_id, status, severity, blocking, statement, activation_digest, candidate_digest, evidence_artifacts, created_at

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
blocking	так	type="boolean"
candidate_digest	так	\$ref="#/\$defs/Digest"
created_at	так	\$ref="#/\$defs/Timestamp"
disposition_decision_id	ні	\$ref="#/\$defs/Id"
evidence_artifacts	так	type="array"; minItems=1; maxItems=64; uniqueItems=true
finding_id	так	\$ref="#/\$defs/Id"
record_type	так	const="Finding"
severity	так	enum=["P0", "P1", "P2", "P3"]
statement	так	type="string"; minLength=1
status	так	enum=["OPEN", "RESOLVED", "WAIVED"]

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [
            "disposition_decision_id"
          ]
        }
      },
      "if": {
        "properties": {
          "status": {
            "enum": [
              "RESOLVED",
              "WAIVED"
            ]
          }
        }
      },
      "then": {
        "properties": {
          "blocking": {
            "const": false
          }
        }
      },
      "required": [
        "disposition_decision_id"
      ]
    }
  ],
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "blocking": {
      "type": "boolean"
    },
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "created_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "evidence_artifacts": {
      "type": "array",
      "minItems": 1,
      "maxItems": 64,
      "uniqueItems": true
    },
    "finding_id": {
      "$ref": "#/$defs/Id"
    },
    "record_type": {
      "const": "Finding"
    },
    "severity": {
      "enum": ["P0", "P1", "P2", "P3"]
    },
    "statement": {
      "type": "string",
      "minLength": 1
    },
    "status": {
      "enum": ["OPEN", "RESOLVED", "WAIVED"]
    }
  }
}

```

```

    },
    "created_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "disposition_decision_id": {
      "$ref": "#/$defs/Id"
    },
    "evidence_artifacts": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "artifact_id": {
            "$ref": "#/$defs/Id"
          },

```

```

          "artifact_record_digest": {
            "$ref": "#/$defs/Digest"
          }
        },
        "required": [
          "artifact_id",
          "artifact_record_digest"
        ],
        "type": "object"
      },
      "maxItems": 64,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "finding_id": {
      "$ref": "#/$defs/Id"
    },
    "record_type": {
      "const": "Finding"
    },
    "severity": {
      "enum": [
        "P0",
        "P1",
        "P2",
        "P3"
      ]
    }
  ]

```

```

    },
    "statement": {
      "minLength": 1,
      "type": "string"
    },
    "status": {
      "enum": [
        "OPEN",
        "RESOLVED",
        "WAIVED"
      ]
    }
  },
  "required": [
    "record_type",
    "finding_id",
    "status",
    "severity",
    "blocking",
    "statement",
    "activation_digest",
    "candidate_digest",
    "evidence_artifacts",
    "created_at"
  ],
  "type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/Finding](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/Finding](#)

34. GateEvidenceArtifactBinding

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: artifact_id, artifact_record_digest, run_id, run_digest

Властивість	Обов'язков	Структурне правило
artifact_id	так	\$ref="#/\$defs/Id"
artifact_record_digest	так	\$ref="#/\$defs/Digest"
run_digest	так	\$ref="#/\$defs/Digest"
run_id	так	\$ref="#/\$defs/Id"

```
{
  "additionalProperties": false,
  "properties": {
    "artifact_id": {
      "$ref": "#/$defs/Id"
    },
    "artifact_record_digest": {
      "$ref": "#/$defs/Digest"
    },
    "run_digest": {
      "$ref": "#/$defs/Digest"
    },
    "run_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "artifact_id",
    "artifact_record_digest",
    "run_id",
    "run_digest"
  ],
  "type": "object"
}
```

Власник: [core/contracts.schema.json#/\\$defs/GateEvidenceArtifactBinding](#) | Валідатор:
[urn:promin:contracts:1#/\\$defs/GateEvidenceArtifactBinding](#)

35. GateResult

Опис типу: type="object"; additionalProperties=false; oneOf[4]=keys:properties,required | keys:properties,required | keys:properties,required | keys:properties,required

Обов'язкові поля: record_type, task_id, gate_id, run_id, run_digest, definition_digest, status, outcome, pass_credit, activation_digest, candidate_digest, policy_digest, tool_digest, evidence_class, evidence_artifacts

Властивість	Обов'язков	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
candidate_digest	так	\$ref="#/\$defs/Digest"
definition_digest	так	\$ref="#/\$defs/Digest"
evidence_artifacts	так	type="array"; minItems=0; maxItems=64; uniqueItems=true
evidence_class	так	enum=["product-execution", "validator", "harness-generated", "migration"]
gate_id	так	\$ref="#/\$defs/Id"
outcome	так	enum=["pass", "fail", "blocked", "skipped"]
pass_credit	так	type="boolean"
policy_digest	так	\$ref="#/\$defs/Digest"
reason	ні	type="string"; minLength=1; maxLength=2048
record_type	так	const="GateResult"
run_digest	так	\$ref="#/\$defs/Digest"
run_id	так	\$ref="#/\$defs/Id"
status	так	enum=["pass", "fail", "blocked", "skipped"]
task_id	так	\$ref="#/\$defs/Id"
tool_digest	так	\$ref="#/\$defs/Digest"

```
{
  "additionalProperties": false,
  "oneOf": [
    {
      "properties": {
        "evidence_artifacts": {
```

```

        "minItems": 1
      },
      "outcome": {
        "const": "pass"
      },
      "status": {
        "const": "pass"
      }
    },
    "required": [
      "status",
      "outcome",
      "pass_credit"
    ]
  },
  {
    "properties": {
      "evidence_artifacts": {
        "minItems": 1
      },
      "outcome": {
        "const": "fail"
      }
    }
  }

```

```

    },
    "pass_credit": {
      "const": false
    },
    "status": {
      "const": "fail"
    }
  },
  "required": [
    "status",
    "outcome",
    "pass_credit"
  ]
},
{
  "properties": {
    "evidence_artifacts": {
      "minItems": 1
    },
    "outcome": {
      "const": "blocked"
    },
    "pass_credit": {
      "const": false
    },
    "status": {
      "const": "blocked"
    }
  }
}

```

```

},
"required": [
  "status",
  "outcome",
  "pass_credit",
  "reason"
]
},
{
  "properties": {
    "evidence_artifacts": {
      "maxItems": 0
    },
    "outcome": {
      "const": "skipped"
    },
    "pass_credit": {
      "const": false
    },
    "status": {
      "const": "skipped"
    }
  },
  "required": [
    "status",
    "outcome",
    "pass_credit",
    "reason"
  ]
}

```

```

    ]
  },
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "definition_digest": {
      "$ref": "#/$defs/Digest"
    },
    "evidence_artifacts": {
      "items": {
        "$ref": "#/$defs/GateEvidenceArtifactBinding"
      },
      "maxItems": 64,
      "minItems": 0,
      "type": "array",
      "uniqueItems": true
    },
    "evidence_class": {
      "enum": [
        "product-execution",
        "validator",
        "harness-generated",
        "migration"
      ]
    }
  }
}

```

```

    ]
  },
  "gate_id": {
    "$ref": "#/$defs/Id"
  },
  "outcome": {
    "enum": [
      "pass",
      "fail",
      "blocked",
      "skipped"
    ]
  },
  "pass_credit": {
    "type": "boolean"
  },
  "policy_digest": {
    "$ref": "#/$defs/Digest"
  },
  "reason": {
    "maxLength": 2048,
    "minLength": 1,
    "type": "string"
  },
  "record_type": {
    "const": "GateResult"
  },
  "run_digest": {

```

```

    "$ref": "#/$defs/Digest"
  },
  "run_id": {
    "$ref": "#/$defs/Id"
  },
  "status": {
    "enum": [
      "pass",
      "fail",
      "blocked",
      "skipped"
    ]
  },
  "task_id": {
    "$ref": "#/$defs/Id"
  },
  "tool_digest": {
    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "record_type",

```

```

"task_id",
"gate_id",
"run_id",
"run_digest",
"definition_digest",
"status",

"outcome",
"pass_credit",
"activation_digest",
"candidate_digest",
"policy_digest",
"tool_digest",
"evidence_class",
"evidence_artifacts"
],
"type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/GateResult | Валідатор: urn:promin:contracts:1#/\$defs/GateResult

36. GateRunDefinition

Опис типу: type="object"; additionalProperties=false; oneOf[4]=keys:properties,required | keys:properties,required | keys:properties,required | keys:properties,required; allOf[1]=keys:if,then

Обов'язкові поля: definition_kind, definition_id, owner_kind, owner_digest, defined_at_head_digest, gate_id, run_kind, expected_evidence_class, expected_evidence_purpose, product_credit_required, target_kind, target_digest, target_scope, candidate_digest, policy_digest, tool_digest, implementation_closure_digest, provider_binding_digest, input_digests, activation_digest

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
candidate_digest	так	\$ref="#/\$defs/Digest"
defined_at_head_digest	так	oneOf[2]=#/\$defs/Digest type:null
definition_id	так	\$ref="#/\$defs/Id"
definition_kind	так	const="GateRunDefinition"
expected_evidence_class	так	enum=["product-execution", "validator", "harness-generated", "migration"]
expected_evidence_purpose	так	enum=["diagnostic", "gate", "product"]
gate_id	так	\$ref="#/\$defs/Id"
implementation_closure_digest	так	\$ref="#/\$defs/Digest"
input_digests	так	type="array"; minItems=0; maxItems=128; uniqueItems=true
owner_digest	так	\$ref="#/\$defs/Digest"
owner_kind	так	const="Завдання"
policy_digest	так	\$ref="#/\$defs/Digest"
product_credit_required	так	type="boolean"
provider_binding_digest	так	\$ref="#/\$defs/Digest"
run_kind	так	enum=["execution", "validation"]
target_digest	так	\$ref="#/\$defs/Digest"
target_kind	так	enum=["candidate", "finding"]
target_scope	так	type="array"; minItems=1; maxItems=32; uniqueItems=true
tool_digest	так	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "if": {
        "properties": {
          "product_credit_required": {
            "const": true
          }
        }
      },
      "required": [
        "product_credit_required"
      ]
    }
  ],
}

```

```

    "then": {
      "properties": {
        "expected_evidence_class": {
          "const": "product-execution"
        },
        "expected_evidence_purpose": {
          "const": "product"
        }
      }
    }
  },
  "oneOf": [
    {

```

```

      "properties": {
        "expected_evidence_class": {
          "const": "validator"
        },
        "expected_evidence_purpose": {
          "const": "diagnostic"
        }
      },
      "required": [
        "expected_evidence_class",
        "expected_evidence_purpose"
      ]
    },
    {
      "properties": {
        "expected_evidence_class": {
          "const": "validator"
        },
        "expected_evidence_purpose": {
          "const": "gate"
        }
      },
      "required": [
        "expected_evidence_class",
        "expected_evidence_purpose"
      ]
    }
  ],
  {

```

```

    "properties": {
      "expected_evidence_class": {
        "const": "product-execution"
      },
      "expected_evidence_purpose": {
        "const": "gate"
      }
    },
    "required": [
      "expected_evidence_class",
      "expected_evidence_purpose"
    ]
  },
  {
    "properties": {
      "expected_evidence_class": {
        "const": "product-execution"
      },
      "expected_evidence_purpose": {
        "const": "product"
      }
    },
    "required": [
      "expected_evidence_class",
      "expected_evidence_purpose"
    ]
  }
],

```

```

"properties": {
  "activation_digest": {
    "$ref": "#/$defs/Digest"
  },
  "candidate_digest": {
    "$ref": "#/$defs/Digest"
  },

```



```

"defined_at_head_digest": {
  "oneOf": [
    {
      "$ref": "#/$defs/Digest"
    },
    {
      "type": "null"
    }
  ]
},
"definition_id": {
  "$ref": "#/$defs/Id"
},
"definition_kind": {
  "const": "GateRunDefinition"
},
"expected_evidence_class": {
  "enum": [
    "product-execution",
    "validator",
    "harness-generated",

```

```

    "migration"
  ]
},
"expected_evidence_purpose": {
  "enum": [
    "diagnostic",
    "gate",
    "product"
  ]
},
"gate_id": {
  "$ref": "#/$defs/Id"
},
"implementation_closure_digest": {
  "$ref": "#/$defs/Digest"
},
"input_digests": {
  "items": {
    "$ref": "#/$defs/Digest"
  },
  "maxItems": 128,
  "minItems": 0,
  "type": "array",
  "uniqueItems": true
},
"owner_digest": {
  "$ref": "#/$defs/Digest"
},

```

```

"owner_kind": {
  "const": "Завдання"
},
"policy_digest": {
  "$ref": "#/$defs/Digest"
},
"product_credit_required": {
  "type": "boolean"
},
"provider_binding_digest": {
  "$ref": "#/$defs/Digest"
},
"run_kind": {
  "enum": [
    "execution",
    "validation"
  ]
},
"target_digest": {
  "$ref": "#/$defs/Digest"
},
"target_kind": {
  "enum": [
    "candidate",
    "finding"
  ]
},
"target_scope": {

```

```

    "items": {
      "$ref": "#/$defs/ScopeSelector"
    },
    "maxItems": 32,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "tool_digest": {
    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "definition_kind",
  "definition_id",
  "owner_kind",
  "owner_digest",
  "defined_at_head_digest",
  "gate_id",
  "run_kind",
  "expected_evidence_class",
  "expected_evidence_purpose",
  "product_credit_required",
  "target_kind",
  "target_digest",
  "target_scope",
  "candidate_digest",
  "policy_digest",
  "tool_digest",
  "implementation_closure_digest",
  "provider_binding_digest",
  "input_digests",
  "activation_digest"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/GateRunDefinition](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/GateRunDefinition](#)

37. Grant

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, grant_id, subject_id, capability_id, scope, activation_digest, issued_at, expires_at, nonce, claim_digest, trust_proofs

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
capability_id	так	enum=[16 items]
claim_digest	так	\$ref="#/\$defs/Digest"
expires_at	так	\$ref="#/\$defs/Timestamp"
grant_id	так	\$ref="#/\$defs/Id"
issued_at	так	\$ref="#/\$defs/Timestamp"
nonce	так	type="string"; pattern="^[A-Za-z0-9._-]{16,128}\$"; minLength=16; maxLength=128
record_type	так	const="Grant"
scope	так	type="array"; minItems=1; maxItems=32; uniqueItems=true
subject_id	так	\$ref="#/\$defs/Id"
trust_proofs	так	type="array"; minItems=1; maxItems=32; uniqueItems=true

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "capability_id": {
      "enum": [
        "standard.activate",
        "authority.manage",
        "task.plan",
        "task.execute",

```

```

    "lease.manage",
    "evidence.publish",
    "validation.evaluate",
    "finding.record",
    "finding.resolve",
    "finding.waive",
    "candidate.promote",
    "release.decide",
    "export.create",
    "migration.manage",
    "projection.rebuild",
    "projection.read"
  ]
},
"claim_digest": {
  "$ref": "#/$defs/Digest"

```

```

},
"expires_at": {
  "$ref": "#/$defs/Timestamp"
},
"grant_id": {
  "$ref": "#/$defs/Id"
},
"issued_at": {
  "$ref": "#/$defs/Timestamp"
},
"nonce": {
  "maxLength": 128,
  "minLength": 16,
  "pattern": "^[A-Za-z0-9._:-]{16,128}$",
  "type": "string"
},
"record_type": {
  "const": "Grant"
},
"scope": {
  "items": {
    "$ref": "#/$defs/ScopeSelector"
  },
  "maxItems": 32,
  "minItems": 1,
  "type": "array",
  "uniqueItems": true
},

```

```

"subject_id": {
  "$ref": "#/$defs/Id"
},
"trust_proofs": {
  "items": {
    "$ref": "#/$defs/GrantTrustProof"
  },
  "maxItems": 32,
  "minItems": 1,
  "type": "array",
  "uniqueItems": true
}
},
"required": [
  "record_type",
  "grant_id",
  "subject_id",
  "capability_id",
  "scope",
  "activation_digest",
  "issued_at",
  "expires_at",
  "nonce",
  "claim_digest",
  "trust_proofs"
],
"type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/Grant | Валідатор: urn:promin:contracts:1#/\$defs/Grant

38. GrantAuthorityState

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, grant, revocation

Властивість	Обов'язков е	Структурне правило
grant	так	\$ref="#/\$defs/Grant"
record_type	так	const="GrantAuthorityState"
revocation	так	oneOf[2]=#/\$defs/GrantRevocation type:null

```
{
  "additionalProperties": false,
  "properties": {
    "grant": {
      "$ref": "#/$defs/Grant"
    },
    "record_type": {
      "const": "GrantAuthorityState"
    },
    "revocation": {
      "oneOf": [
        {
          "$ref": "#/$defs/GrantRevocation"
        },
        {
          "type": "null"
        }
      ]
    }
  },
  "required": [
    "record_type",
    "grant",
    "revocation"
  ],
  "type": "object"
}
```

Власник: core/contracts.schema.json#/\$defs/GrantAuthorityState | Валідатор: urn:promin:contracts:1#/\$defs/GrantAuthorityState

39. GrantAuthorization

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: kind, grant_id, grant_claim_digest

Властивість	Обов'язков е	Структурне правило
grant_claim_digest	так	\$ref="#/\$defs/Digest"
grant_id	так	\$ref="#/\$defs/Id"
kind	так	const="grant"

```
{
  "additionalProperties": false,
  "properties": {
    "grant_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "grant_id": {
      "$ref": "#/$defs/Id"
    },
    "kind": {
      "const": "grant"
    }
  },
  "required": [
    "kind",
    "grant_id",
    "grant_claim_digest"
  ],
  "type": "object"
}
```

Власник: core/contracts.schema.json#/\$defs/GrantAuthorization | Валідатор: urn:promin:contracts:1#/\$defs/GrantAuthorization

40. GrantRevocation

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: grant_id, decision_id, reason, revoked_at

Властивість	Обов'язков е	Структурне правило
decision_id	так	\$ref="#/\$defs/Id"
grant_id	так	\$ref="#/\$defs/Id"
reason	так	type="string"; minLength=1
revoked_at	так	\$ref="#/\$defs/Timestamp"

```
{
  "additionalProperties": false,
  "properties": {
    "decision_id": {
      "$ref": "#/$defs/Id"
    },
    "grant_id": {
      "$ref": "#/$defs/Id"
    },
    "reason": {
      "minLength": 1,
      "type": "string"
    },
    "revoked_at": {
      "$ref": "#/$defs/Timestamp"
    }
  },
  "required": [
    "grant_id",
    "decision_id",
    "reason",
    "revoked_at"
  ],
  "type": "object"
}
```

Власник: core/contracts.schema.json#/\$defs/GrantRevocation | Валідатор: urn:promin:contracts:1#/\$defs/GrantRevocation

41. GrantTrustProof

Опис типу: oneOf[3]=#/\$defs/IssuerGrantTrustProof | #/\$defs/LocalTrustProof | #/\$defs/SignatureTrustProof

Обов'язкові поля: -

```
{
  "oneOf": [
    {
      "$ref": "#/$defs/IssuerGrantTrustProof"
    },
    {
      "$ref": "#/$defs/LocalTrustProof"
    },
    {
      "$ref": "#/$defs/SignatureTrustProof"
    }
  ]
}
```

Власник: core/contracts.schema.json#/\$defs/GrantTrustProof | Валідатор: urn:promin:contracts:1#/\$defs/GrantTrustProof

42. Healthcheck

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: argv, timeout_ms, expected_exit

Властивість	Обов'язков е	Структурне правило
argv	так	type="array"; minItems=1; maxItems=16
expected_exit	так	type="integer"; minimum=0; maximum=255
timeout_ms	так	type="integer"; minimum=100; maximum=60000

```
{
  "additionalProperties": false,
```

```

"properties": {
  "argv": {
    "items": {
      "minLength": 1,
      "type": "string"
    },
    "maxItems": 16,
    "minItems": 1,
    "type": "array"
  },
  "expected_exit": {
    "maximum": 255,
    "minimum": 0,
    "type": "integer"
  },
  "timeout_ms": {
    "maximum": 60000,
    "minimum": 100,
    "type": "integer"
  }
},
"required": [
  "argv",
  "timeout_ms",
  "expected_exit"
],
"type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/Healthcheck | Валідатор: urn:promin:contracts:1#/\$defs/Healthcheck

43. HumanDocumentVerification

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, status, candidate_binding_digest, documents, deterministic_rebuild, rebuild_digest, page_count, extraction_diagnostics, visual_review_scope, font_bindings, generator_result_digest, render_environment, render_manifest, render_manifest_digest, product_acceptance_pass, producer, invocation, raw_artifact_manifest_digest, producer_attestation, result_digest

Властивість	Обов'язков е	Структурне правило
candidate_binding_digest	так	\$ref="#/\$defs/Digest"
deterministic_rebuild	так	const=true
documents	так	type="array"; minItems=4; maxItems=4; uniqueItems=true
extraction_diagnostics	так	type="object"; additionalProperties=false
font_bindings	так	type="object"; additionalProperties=false
generator_result_digest	так	\$ref="#/\$defs/Digest"
invocation	так	\$ref="#/\$defs/ReleaseEvidenceInvocation"
page_count	так	type="integer"; minimum=1
producer	так	\$ref="#/\$defs/ReleaseEvidenceProducer"
producer_attestation	так	\$ref="#/\$defs/EvidenceProducerAttestation"
product_acceptance_pass	так	const=false
raw_artifact_manifest_digest	так	\$ref="#/\$defs/Digest"
rebuild_digest	так	\$ref="#/\$defs/Digest"
record_type	так	const="HumanDocumentVerification"
render_environment	так	type="object"; additionalProperties=false
render_manifest	так	type="array"; minItems=4; maxItems=4096; uniqueItems=true
render_manifest_digest	так	\$ref="#/\$defs/Digest"
result_digest	так	\$ref="#/\$defs/Digest"
status	так	const="pass"
visual_review_scope	так	type="object"; additionalProperties=false

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_binding_digest": {

```

```

    "$ref": "#/$defs/Digest"
  },
  "deterministic_rebuild": {
    "const": true
  },
  "documents": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "blank_pages": {
          "maxItems": 0,
          "type": "array"
        },
        "extracted_characters": {
          "minimum": 1,
          "type": "integer"
        },
        "extraction_errors": {
          "maxItems": 0,
          "type": "array"
        },
        "page_count": {
          "minimum": 1,
          "type": "integer"
        }
      }
    }
  },

```

```

  },
  "path": {
    "enum": [
      "human/promin_appendices_en.pdf",
      "human/promin_appendices_ua.pdf",
      "human/promin_main_en.pdf",
      "human/promin_main_ua.pdf"
    ]
  },
  "sha256": {
    "$ref": "#/$defs/Digest"
  },
  "size_bytes": {
    "minimum": 1,
    "type": "integer"
  }
},
"required": [
  "path",
  "sha256",
  "size_bytes",
  "page_count",
  "extracted_characters",
  "blank_pages",
  "extraction_errors"
],
"type": "object"
},

```

```

    "maxItems": 4,
    "minItems": 4,
    "type": "array",
    "uniqueItems": true
  },
  "extraction_diagnostics": {
    "additionalProperties": false,
    "properties": {
      "blank_pages": {
        "maxItems": 0,
        "type": "array"
      },
      "documents_parsed": {
        "const": 4
      },
      "errors": {
        "maxItems": 0,
        "type": "array"
      },
      "total_extracted_characters": {
        "minimum": 1,
        "type": "integer"
      }
    }
  },
  "required": [
    "documents_parsed",

```

```

    "total_extracted_characters",
    "blank_pages",

    "errors"
  ],
  "type": "object"
},
"font_bindings": {
  "additionalProperties": false,
  "properties": {
    "bold": {
      "additionalProperties": false,
      "properties": {
        "bytes": {
          "minimum": 1,
          "type": "integer"
        },
        "pdf_font_name": {
          "minLength": 1,
          "type": "string"
        },
        "sha256": {
          "$ref": "#/$defs/Digest"
        }
      },
      "required": [
        "bytes",
        "pdf_font_name",
        "sha256"
      ],
      "type": "object"
    },
    "italic": {
      "additionalProperties": false,
      "properties": {
        "bytes": {
          "minimum": 1,
          "type": "integer"
        },
        "pdf_font_name": {
          "minLength": 1,
          "type": "string"
        },
        "sha256": {
          "$ref": "#/$defs/Digest"
        }
      },
      "required": [
        "bytes",
        "pdf_font_name",
        "sha256"
      ],
      "type": "object"
    },
    "mono": {
      "additionalProperties": false,
      "properties": {
        "bytes": {
          "minimum": 1,
          "type": "integer"
        },
        "pdf_font_name": {
          "minLength": 1,
          "type": "string"
        },
        "sha256": {
          "$ref": "#/$defs/Digest"
        }
      },
      "required": [
        "bytes",
        "pdf_font_name",
        "sha256"
      ],
      "type": "object"
    },
    "regular": {
      "additionalProperties": false,

```



```

    "properties": {
      "bytes": {
        "minimum": 1,
        "type": "integer"
      },
      "pdf_font_name": {
        "minLength": 1,
        "type": "string"
      },
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    },
  },
  "required": [
    "bytes",
    "pdf_font_name",
    "sha256"
  ],
  "type": "object"
}
},
"required": [
  "regular",
  "bold",
  "italic",
  "mono"
],
"type": "object"
},
"generator_result_digest": {
  "$ref": "#/$defs/Digest"
},
"invocation": {
  "$ref": "#/$defs/ReleaseEvidenceInvocation"
},
"page_count": {
  "minimum": 1,

```

```

    "type": "integer"
  },
  "producer": {
    "$ref": "#/$defs/ReleaseEvidenceProducer"
  },
  "producer_attestation": {
    "$ref": "#/$defs/EvidenceProducerAttestation"
  },
  "product_acceptance_pass": {
    "const": false
  },
  "raw_artifact_manifest_digest": {
    "$ref": "#/$defs/Digest"
  },
  "rebuild_digest": {
    "$ref": "#/$defs/Digest"
  },
  "record_type": {
    "const": "HumanDocumentVerification"
  },
  "render_environment": {
    "additionalProperties": false,
    "properties": {
      "font_binding_digest": {
        "$ref": "#/$defs/Digest"
      },
    },
    "generator_sha256": {
      "$ref": "#/$defs/Digest"
    },
  },

```

```

},
"platform": {
  "minLength": 1,
  "type": "string"
},
"python_version": {
  "minLength": 1,
  "type": "string"
},
"reportlab_version": {
  "minLength": 1,
  "type": "string"
}

```

```

    }
  },
  "required": [
    "generator_sha256",
    "reportlab_version",
    "python_version",
    "platform",
    "font_binding_digest"
  ],
  "type": "object"
},
"render_manifest": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "document_path": {

```

```

        "enum": [
          "human/promin_appendices_en.pdf",
          "human/promin_appendices_ua.pdf",
          "human/promin_main_en.pdf",
          "human/promin_main_ua.pdf"
        ]
      },
      "image_bytes": {
        "maximum": 33554432,
        "minimum": 1,
        "type": "integer"
      },
      "image_path": {
        "$ref": "#/$defs/RelativePath"
      },
      "image_sha256": {
        "$ref": "#/$defs/Digest"
      },
      "page": {
        "minimum": 1,
        "type": "integer"
      }
    }
  },
  "required": [
    "document_path",
    "page",
    "image_path",
    "image_sha256",

```

```

    "image_bytes"
  ],
  "type": "object"
},
"maxItems": 4096,
"minItems": 4,
"type": "array",
"uniqueItems": true
},
"render_manifest_digest": {
  "$ref": "#/$defs/Digest"
},
"result_digest": {
  "$ref": "#/$defs/Digest"
},
"status": {
  "const": "pass"
},
"visual_review_scope": {
  "additionalProperties": false,
  "properties": {
    "clipping_detected": {
      "const": false
    },
    "completed": {
      "const": true
    }
  },
  "pages_reviewed": {

```

```

    "minimum": 1,
    "type": "integer"
  },
  "render_manifest_digest": {
    "$ref": "#/$defs/Digest"
  }
}

```

```

    },
    "rendered_page_count": {
      "minimum": 1,
      "type": "integer"
    },
    "reviewed_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "reviewer_id": {
      "$ref": "#/$defs/Id"
    },
    "unreadable_text_detected": {
      "const": false
    }
  },
  "required": [
    "completed",
    "reviewer_id",
    "reviewed_at",
    "rendered_page_count",
    "pages_reviewed",
    "clipping_detected",
    "unreadable_text_detected",
    "render_manifest_digest"
  ],
  "type": "object"
}
},
"required": [
  "record_type",
  "status",
  "candidate_binding_digest",
  "documents",
  "deterministic_rebuild",
  "rebuild_digest",
  "page_count",
  "extraction_diagnostics",
  "visual_review_scope",
  "font_bindings",
  "generator_result_digest",
  "render_environment",
  "render_manifest",
  "render_manifest_digest",
  "product_acceptance_pass",
  "producer",
  "invocation",
  "raw_artifact_manifest_digest",
  "producer_attestation",
  "result_digest"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/HumanDocumentVerification](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/HumanDocumentVerification](#)

44. Id

Опис типу: type="string"; pattern="^[A-Za-z0-9][A-Za-z0-9._-]{0,127}\$"

Обов'язкові поля: -

```

{
  "pattern": "^[A-Za-z0-9][A-Za-z0-9._-]{0,127}$",
  "type": "string"
}

```

Власник: [core/contracts.schema.json#/\\$defs/Id](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/Id](#)

45. ImplementationClosure

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: runtime, interpreter, components, adapters, closure_digest

Властивість	Обов'язков е	Структурне правило
adapters	так	type="array"; minItems=1; maxItems=1; uniqueItems=true

Властивість	Обов'язков е	Структурне правило
closure_digest	так	\$ref="#/\$defs/Digest"
components	так	type="array"; minItems=1; maxItems=4096; uniqueItems=true
interpreter	так	type="object"; additionalProperties=false
runtime	так	type="object"; additionalProperties=false
<pre> { "additionalProperties": false, "properties": { "adapters": { "items": { "additionalProperties": false, "properties": { "adapter_id": { "\$ref": "#/\$defs/Id" }, "capability_id": { "enum": ["control-runtime", "shape-validation", "content-identity", "local-serialization", "query-projection", "filesystem-inventory", "export-scan", "signature", "build-dependency"] }, "dependency_receipt_digest": { "\$ref": "#/\$defs/Digest" }, "digest": { "\$ref": "#/\$defs/Digest" }, "operation_contract_digest": { "\$ref": "#/\$defs/Digest" }, "protocol_id": { "\$ref": "#/\$defs/Id" }, "version": { "minLength": 1, "type": "string" } }, "required": ["adapter_id", "capability_id", "version", "digest", "protocol_id", "operation_contract_digest", "dependency_receipt_digest"], "type": "object" }, "maxItems": 1, "minItems": 1, "type": "array", "uniqueItems": true }, "closure_digest": { "\$ref": "#/\$defs/Digest" }, "components": { "items": { "additionalProperties": false, "properties": { "component_id": { "\$ref": "#/\$defs/Id" }, "component_kind": { "enum": [</pre>		

```

        "native-runtime",
        "provider-file",
        "provider-tree",
        "python-distribution"
    ]
},
"digest": {
    "$ref": "#/$defs/Digest"
},
"size_bytes": {
    "minimum": 1,
    "type": "integer"
},
"source": {
    "maxLength": 4096,
    "minLength": 1,

```

```

        "type": "string"
    },
    "version": {
        "maxLength": 256,
        "minLength": 1,
        "type": "string"
    }
},
"required": [
    "component_id",
    "component_kind",
    "source",
    "version",
    "digest",
    "size_bytes"
],
"type": "object"
},
"maxItems": 4096,
"minItems": 1,
"type": "array",
"uniqueItems": true
},
"interpreter": {
    "additionalProperties": false,
    "properties": {
        "executable_digest": {
            "$ref": "#/$defs/Digest"

```

```

    },
    "name": {
        "minLength": 1,
        "type": "string"
    },
    "version": {
        "minLength": 1,
        "type": "string"
    }
},
"required": [
    "name",
    "version",
    "executable_digest"
],
"type": "object"
},
"runtime": {
    "additionalProperties": false,
    "properties": {
        "digest": {
            "$ref": "#/$defs/Digest"

```

```

    },
    "name": {
        "const": "promin"
    },
    "version": {
        "minLength": 1,
        "type": "string"
    }
},
"required": [
    "name",

```

```

    "version",
    "digest"
  ],
  "type": "object"
}
},
"required": [
  "runtime",
  "interpreter",
  "components",
  "adapters",
  "closure_digest"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/ImplementationClosure](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/ImplementationClosure](#)

46. InitFileDigests

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: project.json, standards.json, technologies.json, authority.json

Властивість	Обов'язков е	Структурне правило
authority.json	так	\$ref="#/\$defs/Digest"
project.json	так	\$ref="#/\$defs/Digest"
standards.json	так	\$ref="#/\$defs/Digest"
technologies.json	так	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "properties": {
    "authority.json": {
      "$ref": "#/$defs/Digest"
    },
    "project.json": {
      "$ref": "#/$defs/Digest"
    },
    "standards.json": {
      "$ref": "#/$defs/Digest"
    },
    "technologies.json": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "project.json",
    "standards.json",
    "technologies.json",
    "authority.json"
  ],
  "type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/InitFileDigests](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/InitFileDigests](#)

47. InventoryInputManifest

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, candidate_digest, inventory_digest, stream_artifact_id, stream_artifact_record_digest, stream_digest, stream_bytes, entry_count, row_schema_digest, consistency_mode, snapshot_provider_id, snapshot_digest

Властивість	Обов'язков е	Структурне правило
candidate_digest	так	\$ref="#/\$defs/Digest"
consistency_mode	так	enum=["immutable-vcs-tree", "observational-best-effort"]
entry_count	так	type="integer"; minimum=0
inventory_digest	так	\$ref="#/\$defs/Digest"
record_type	так	const="InventoryInputManifest"

Властивість	Обов'язков е	Структурне правило
row_schema_digest	так	\$ref="#/\$defs/Digest"
snapshot_digest	так	\$ref="#/\$defs/Digest"
snapshot_provider_id	так	\$ref="#/\$defs/Id"
stream_artifact_id	так	\$ref="#/\$defs/Id"
stream_artifact_record_digest	так	\$ref="#/\$defs/Digest"
stream_bytes	так	type="integer"; minimum=1
stream_digest	так	\$ref="#/\$defs/Digest"
<pre> { "additionalProperties": false, "properties": { "candidate_digest": { "\$ref": "#/\$defs/Digest" }, "consistency_mode": { "enum": ["immutable-vcs-tree", "observational-best-effort"] }, "entry_count": { "minimum": 0, "type": "integer" }, "inventory_digest": { "\$ref": "#/\$defs/Digest" }, "record_type": { "const": "InventoryInputManifest" }, "row_schema_digest": { "\$ref": "#/\$defs/Digest" }, "snapshot_digest": { "\$ref": "#/\$defs/Digest" }, "snapshot_provider_id": { "\$ref": "#/\$defs/Id" }, "stream_artifact_id": { "\$ref": "#/\$defs/Id" }, "stream_artifact_record_digest": { "\$ref": "#/\$defs/Digest" }, "stream_bytes": { "minimum": 1, "type": "integer" }, "stream_digest": { "\$ref": "#/\$defs/Digest" } }, "required": ["record_type", "candidate_digest", "inventory_digest", "stream_artifact_id", "stream_artifact_record_digest", "stream_digest", "stream_bytes", "entry_count", "row_schema_digest", "consistency_mode", "snapshot_provider_id", "snapshot_digest"], "type": "object", "x-promin-ingress-operations": ["rebuild"], </pre>		

```
"x-promin-persistence": "transient-ingress-only"
}
```

Власник: [core/contracts.schema.json#/\\$defs/InventoryInputManifest](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/InventoryInputManifest](#)

48. InventoryJsonlRow

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: path, digest, size, search_text

Властивість	Обов'язков е	Структурне правило
digest	так	\$ref="#/\$defs/Digest"
path	так	\$ref="#/\$defs/RelativePath"
search_text	так	type="string"; minLength=1; maxLength=4096
size	так	type="integer"; minimum=0

```
{
  "additionalProperties": false,
  "properties": {
    "digest": {
      "$ref": "#/$defs/Digest"
    },
    "path": {
      "$ref": "#/$defs/RelativePath"
    },
    "search_text": {
      "maxLength": 4096,
      "minLength": 1,
      "type": "string"
    },
    "size": {
      "minimum": 0,
      "type": "integer"
    }
  },
  "required": [
    "path",
    "digest",
    "size",
    "search_text"
  ],
  "type": "object"
}
```

Власник: [core/contracts.schema.json#/\\$defs/InventoryJsonlRow](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/InventoryJsonlRow](#)

49. InventoryProjectionRow

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, path, digest, size, search_text, semantic_proxy

Властивість	Обов'язков е	Структурне правило
digest	так	\$ref="#/\$defs/Digest"
path	так	\$ref="#/\$defs/RelativePath"
record_type	так	const="InventoryProjectionRow"
search_text	так	type="string"; minLength=1; maxLength=4096
semantic_proxy	так	type="object"; additionalProperties=false
size	так	type="integer"; minimum=0

```
{
  "additionalProperties": false,
  "properties": {
    "digest": {
      "$ref": "#/$defs/Digest"
    },
    "path": {
      "$ref": "#/$defs/RelativePath"
    },
    "record_type": {
      "const": "InventoryProjectionRow"
    }
  }
}
```



```

    },
    "search_text": {
      "maxLength": 4096,
      "minLength": 1,
      "type": "string"
    },
    "semantic_proxy": {
      "additionalProperties": false,
      "properties": {
        "entity_type": {
          "const": "Artifact"
        },
        "id": {
          "$ref": "#/$defs/Id"
        }
      },
      "payload": {
        "$ref": "#/$defs/Artifact"
      }
    },
    "required": [
      "id",
      "entity_type",
      "payload"
    ],
    "type": "object"
  },
  "size": {
    "minimum": 0,
    "type": "integer"
  }
},
"required": [
  "record_type",
  "path",
  "digest",
  "size",
  "search_text",
  "semantic_proxy"
],
"type": "object",
"x-promin-ingress-operations": [
  "rebuild"
],
"x-promin-persistence": "derived-rebuild-only"
}

```

Власник: [core/contracts.schema.json#/\\$defs/InventoryProjectionRow](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/InventoryProjectionRow](#)

50. Invocation

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: kind, value

Властивість	Обов'язков е	Структурне правило
kind	так	enum=["python-runtime", "python-module", "executable", "platform-service"]
value	так	type="string"; minLength=1

```

{
  "additionalProperties": false,
  "properties": {
    "kind": {
      "enum": [
        "python-runtime",
        "python-module",
        "executable",
        "platform-service"
      ]
    },
    "value": {
      "minLength": 1,
      "type": "string"
    }
  },
  "required": [
    "kind",

```

```

    "value"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/Invocation | Валідатор: urn:promin:contracts:1#/\$defs/Invocation

51. IssuerGrantTrustProof

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: kind, issuer_grant_id, issuer_signed_claim_digest, signed_claim_digest

Властивість	Обов'язков е	Структурне правило
issuer_grant_id	так	\$ref="#/\$defs/Id"
issuer_signed_claim_digest	так	\$ref="#/\$defs/Digest"
kind	так	const="issuer-grant"
signed_claim_digest	так	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "properties": {
    "issuer_grant_id": {
      "$ref": "#/$defs/Id"
    },
    "issuer_signed_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "kind": {
      "const": "issuer-grant"
    },
    "signed_claim_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "kind",
    "issuer_grant_id",
    "issuer_signed_claim_digest",
    "signed_claim_digest"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/IssuerGrantTrustProof | Валідатор: urn:promin:contracts:1#/\$defs/IssuerGrantTrustProof

52. Lease

Опис типу: type="object"; additionalProperties=false; allOf[3]=keys:else,if,then | keys:if,then,title | keys:else,if,then,title

Обов'язкові поля: record_type, lease_id, task_id, holder_grant_claim_digest, manager_grant_claim_digest, manager_grant_id, manager_subject_id, holder_subject_id, holder_grant_id, generation, fencing_token, state, acquired_at, heartbeat_at, expires_at, activation_digest

Властивість	Обов'язков е	Структурне правило
acquired_at	так	\$ref="#/\$defs/Timestamp"
activation_digest	так	\$ref="#/\$defs/Digest"
capacity_reconciliation	ні	\$ref="#/\$defs/CapacityReconciliation"
close_ack	ні	\$ref="#/\$defs/CloseAck"
expires_at	так	\$ref="#/\$defs/Timestamp"
fencing_token	так	type="integer"; minimum=1
generation	так	type="integer"; minimum=1
heartbeat_at	так	\$ref="#/\$defs/Timestamp"
holder_grant_claim_digest	так	\$ref="#/\$defs/Digest"
holder_grant_id	так	\$ref="#/\$defs/Id"
holder_subject_id	так	\$ref="#/\$defs/Id"

Властивість	Обов'язков е	Структурне правило
lease_id	так	\$ref="#/\$defs/Id"
manager_grant_claim_digest	так	\$ref="#/\$defs/Digest"
manager_grant_id	так	\$ref="#/\$defs/Id"
manager_subject_id	так	\$ref="#/\$defs/Id"
record_type	так	const="Оренда"
state	так	\$ref="#/\$defs/LeaseState"
task_id	так	\$ref="#/\$defs/Id"
termination	ні	type="object"; additionalProperties=false

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [
            "close_ack"
          ]
        }
      },
      "if": {
        "properties": {
          "state": {
            "const": "CLOSED"
          }
        }
      },
      "then": {
        "required": [
          "close_ack"
        ]
      }
    },
    {
      "if": {
        "properties": {
          "state": {
            "not": {
              "enum": [
                "EXPIRED",
                "REVOKED"
              ]
            }
          }
        }
      },
      "then": {
        "not": {
          "required": [
            "capacity_reconciliation"
          ]
        }
      },
      "title": "capacity reconciliation exists only on reconciled expired or revoked Lease"
    }
  ],
  {
    "else": {
      "not": {
        "required": [
          "termination"
        ]
      }
    },
    "if": {
      "properties": {
        "state": {
          "enum": [
            "EXPIRED",
            "REVOKED"
          ]
        }
      }
    }
  }
}

```

```

    },
    "required": [
      "state"
    ]
  },
  "then": {
    "required": [
      "termination"
    ]
  },
  "title": "terminal Lease carries exact termination provenance"
}
],
"properties": {
  "acquired_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "activation_digest": {
    "$ref": "#/$defs/Digest"
  },
  "capacity_reconciliation": {
    "$ref": "#/$defs/CapacityReconciliation"
  }
}

```

```

  },
  "close_ack": {
    "$ref": "#/$defs/CloseAck"
  },
  "expires_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "fencing_token": {
    "minimum": 1,
    "type": "integer"
  },
  "generation": {
    "minimum": 1,
    "type": "integer"
  },
  "heartbeat_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "holder_grant_claim_digest": {
    "$ref": "#/$defs/Digest"
  },
  "holder_grant_id": {
    "$ref": "#/$defs/Id"
  },
  "holder_subject_id": {
    "$ref": "#/$defs/Id"
  },
  "lease_id": {

```

```

    "$ref": "#/$defs/Id"
  },
  "manager_grant_claim_digest": {
    "$ref": "#/$defs/Digest"
  },
  "manager_grant_id": {
    "$ref": "#/$defs/Id"
  },
  "manager_subject_id": {
    "$ref": "#/$defs/Id"
  },
  "record_type": {
    "const": "Оренда"
  },
  "state": {
    "$ref": "#/$defs/LeaseState"
  },
  "task_id": {
    "$ref": "#/$defs/Id"
  },
  "termination": {
    "additionalProperties": false,
    "properties": {
      "fencing_token": {
        "minimum": 1,
        "type": "integer"
      },
      "generation": {

```

```

        "minimum": 1,
        "type": "integer"
    },
    "grant_claim_digest": {
        "$ref": "#/$defs/Digest"
    },
    "grant_id": {
        "$ref": "#/$defs/Id"
    },
    "state": {
        "enum": [
            "EXPIRED",
            "REVOKED"
        ]
    },
    "terminated_at": {
        "$ref": "#/$defs/Timestamp"
    },
    "terminated_by": {
        "$ref": "#/$defs/Id"
    }
},
"required": [
    "state",
    "terminated_by",
    "grant_id",
    "grant_claim_digest",
    "terminated_at",
    "generation",
    "fencing_token"
],
"type": "object"
}
},
"required": [
    "record_type",
    "lease_id",
    "task_id",
    "holder_grant_claim_digest",
    "manager_grant_claim_digest",
    "manager_grant_id",
    "manager_subject_id",
    "holder_subject_id",
    "holder_grant_id",
    "generation",
    "fencing_token",
    "state",
    "acquired_at",
    "heartbeat_at",
    "expires_at",
    "activation_digest"
],
"type": "object"
}

```

Власник: `core/contracts.schema.json#/$defs/Lease` | Валідатор: `urn:promin:contracts:1#/$defs/Lease`

53. LeaseState

Опис типу: `enum=["ACTIVE", "CLOSED", "CLOSING", "EXPIRED", "REVOKED"]`

Обов'язкові поля: -

```

{
  "enum": [
    "ACTIVE",
    "CLOSED",
    "CLOSING",
    "EXPIRED",
    "REVOKED"
  ]
}

```

Власник: `core/contracts.schema.json#/$defs/LeaseState` | Валідатор: `urn:promin:contracts:1#/$defs/LeaseState`

54. LeaseStateMachine

Опис типу: `const=schema(keys:ACTIVE,CLOSED,CLOSING,EXPIRED,REVOKED)`

Обов'язкові поля: -

```
{
  "const": {
    "ACTIVE": [
      "CLOSING",
      "EXPIRED",
      "REVOKED"
    ],
    "CLOSED": [],
    "CLOSING": [
      "CLOSED",
      "EXPIRED",
      "REVOKED"
    ],
    "EXPIRED": [],
    "REVOKED": []
  }
}
```

Власник: [core/contracts.schema.json#/\\$defs/LeaseStateMachine](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/LeaseStateMachine](#)

55. LicenseBinding

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: expression, source_uris, review_state

Властивість	Обов'язков е	Структурне правило
expression	так	type="string"; minLength=1
review_state	так	enum=["reviewed", "owner-review-required", "source-verified"]
source_uris	так	type="array"; minItems=1; maxItems=8; uniqueItems=true

```
{
  "additionalProperties": false,
  "properties": {
    "expression": {
      "minLength": 1,
      "type": "string"
    },
    "review_state": {
      "enum": [
        "reviewed",
        "owner-review-required",
        "source-verified"
      ]
    },
    "source_uris": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 8,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    }
  },
  "required": [
    "expression",
    "source_uris",
    "review_state"
  ],
  "type": "object"
}
```

Власник: [core/contracts.schema.json#/\\$defs/LicenseBinding](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/LicenseBinding](#)

56. LocalActivationProof

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: kind, subject_id, authority_init_digest

Властивість	Обов'язков е	Структурне правило
authority_init_digest	так	\$ref="#/\$defs/Digest"

Властивість	Обов'язков е	Структурне правило
kind	так	const="local-root"
subject_id	так	\$ref="#/\$defs/Id"
<pre>{ "additionalProperties": false, "properties": { "authority_init_digest": { "\$ref": "#/\$defs/Digest" }, "kind": { "const": "local-root" }, "subject_id": { "\$ref": "#/\$defs/Id" } }, "required": ["kind", "subject_id", "authority_init_digest"], "type": "object" }</pre>		
Власник: core/contracts.schema.json#/\$defs/LocalActivationProof Валідатор: urn:promin:contracts:1#/\$defs/LocalActivationProof		

57. LocalRootCommandProof

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: kind, subject_id, authority_init_digest, signed_intent_digest

Властивість	Обов'язков е	Структурне правило
authority_init_digest	так	\$ref="#/\$defs/Digest"
kind	так	const="local-root-command"
signed_intent_digest	так	\$ref="#/\$defs/Digest"
subject_id	так	\$ref="#/\$defs/Id"
<pre>{ "additionalProperties": false, "properties": { "authority_init_digest": { "\$ref": "#/\$defs/Digest" }, "kind": { "const": "local-root-command" }, "signed_intent_digest": { "\$ref": "#/\$defs/Digest" }, "subject_id": { "\$ref": "#/\$defs/Id" } }, "required": ["kind", "subject_id", "authority_init_digest", "signed_intent_digest"], "type": "object" }</pre>		
Власник: core/contracts.schema.json#/\$defs/LocalRootCommandProof Валідатор: urn:promin:contracts:1#/\$defs/LocalRootCommandProof		

58. LocalTrustProof

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: kind, root_subject_id, authority_init_digest, signed_claim_digest

Властивість	Обов'язков е	Структурне правило
authority_init_digest	так	\$ref="#/\$defs/Digest"
kind	так	const="local-root"
root_subject_id	так	\$ref="#/\$defs/Id"
signed_claim_digest	так	\$ref="#/\$defs/Digest"
<pre>{ "additionalProperties": false, "properties": { "authority_init_digest": { "\$ref": "#/\$defs/Digest" }, "kind": { "const": "local-root" }, "root_subject_id": { "\$ref": "#/\$defs/Id" }, "signed_claim_digest": { "\$ref": "#/\$defs/Digest" } }, "required": ["kind", "root_subject_id", "authority_init_digest", "signed_claim_digest"], "type": "object" }</pre>		
Власник: core/contracts.schema.json#/\$defs/LocalTrustProof Валідатор: urn:promin:contracts:1#/\$defs/LocalTrustProof		

59. NextResult

Опис типу: type="object"; additionalProperties=false; allOf[2]=keys:else,if,then | keys:if,then

Обов'язкові поля: record_type, status, subject_id, activation_digest, work_card, continuation

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
continuation	так	oneOf[2]=type:null type:string
record_type	так	const="NextResult"
status	так	enum=["ready", "empty"]
subject_id	так	\$ref="#/\$defs/Id"
work_card	так	oneOf[2]=#/\$defs/WorkCard type:null
<pre>{ "additionalProperties": false, "allOf": [{ "else": { "properties": { "continuation": { "type": "null" }, "work_card": { "type": "null" } } }, "if": { "properties": { "status": { "const": "ready" } }, "required": ["status"] } }, { "then": { </pre>		


```

        "properties": {
            "work_card": {
                "$ref": "#/$defs/WorkCard"
            }
        }
    },
    {
        "if": {
            "properties": {
                "continuation": {
                    "type": "string"
                }
            },
            "required": [
                "continuation"
            ]
        },
        "then": {
            "properties": {
                "work_card": {
                    "properties": {
                        "truncated": {
                            "const": true
                        }
                    },
                    "required": [
                        "truncated"
                    ]
                }
            }
        }
    }
},
],
"properties": {
    "activation_digest": {
        "$ref": "#/$defs/Digest"
    },
    "continuation": {
        "oneOf": [
            {
                "type": "null"
            },
            {
                "maxLength": 256,
                "minLength": 1,
                "pattern": "^[promin-v2\\.[A-Za-z0-9_-]{22}\\.[0-9a-f]{64}\\.[A-Za-z0-9_-]{43}$",
                "type": "string"
            }
        ]
    },
    "record_type": {
        "const": "NextResult"
    },
    "status": {
        "enum": [
            "ready",
            "empty"
        ]
    }
},
"subject_id": {
    "$ref": "#/$defs/Id"
},
"work_card": {
    "oneOf": [
        {
            "$ref": "#/$defs/WorkCard"
        },
        {
            "type": "null"
        }
    ]
}
},
"required": [
    "record_type",
    "status",

```

```

    "subject_id",
    "activation_digest",
    "work_card",
    "continuation"
  ],
  "type": "object",
  "x-promin-persistence": "derived-live-only"
}

```

Власник: `core/contracts.schema.json#/$defs/NextResult` | Валідатор: `urn:promin:contracts:1#/$defs/NextResult`

60. NoDegradationResult

Опис типу: `type="object"; additionalProperties=false`

Обов'язкові поля: `record_type`, `status`, `platform`, `no_degradation`, `source`, `root`, `artifact_binding`, `candidate_binding_digest`, `test_manifest`, `install_mode`, `required_predicates`, `validation`, `tests`, `passed`, `pass_credit`, `acceptance_pass`, `product_acceptance_pass`, `product_public_approval`, `scale_selection`, `artifact_binding_unchanged`, `standard_distribution_gate_pass`, `phases`, `execution_budget`, `producer`, `invocation`, `raw_artifact_manifest_digest`, `producer_attestation`, `result_digest`

Властивість	Обов'язков е	Структурне правило
<code>acceptance_pass</code>	так	<code>const=false</code>
<code>artifact_binding</code>	так	<code>type="object"</code>
<code>artifact_binding_unchanged</code>	так	<code>const=true</code>
<code>candidate_binding_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>execution_budget</code>	так	<code>type="object"; additionalProperties=false</code>
<code>install_mode</code>	так	<code>const="offline-wheelhouse"</code>
<code>invocation</code>	так	<code>\$ref="#/\$defs/ReleaseEvidenceInvocation"</code>
<code>no_degradation</code>	так	<code>const=true</code>
<code>pass_credit</code>	так	<code>const=false</code>
<code>passed</code>	так	<code>const=true</code>
<code>phases</code>	так	<code>type="array"; minItems=2; maxItems=2</code>
<code>platform</code>	так	<code>enum=["linux", "windows"]</code>
<code>producer</code>	так	<code>\$ref="#/\$defs/ReleaseEvidenceProducer"</code>
<code>producer_attestation</code>	так	<code>\$ref="#/\$defs/EvidenceProducerAttestation"</code>
<code>product_acceptance_pass</code>	так	<code>const=false</code>
<code>product_public_approval</code>	так	<code>const="not_approved"</code>
<code>raw_artifact_manifest_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>record_type</code>	так	<code>const="NoDegradationResult"</code>
<code>required_predicates</code>	так	<code>type="object"; additionalProperties=false</code>
<code>result_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>root</code>	так	<code>type="string"; minLength=1</code>
<code>scale_selection</code>	так	<code>type="object"; additionalProperties=false</code>
<code>source</code>	так	<code>const="live_files_and_executable_tests"</code>
<code>standard_distribution_gate_pass</code>	так	<code>const=false</code>
<code>status</code>	так	<code>const="pass"</code>
<code>test_manifest</code>	так	<code>type="object"; additionalProperties=false</code>
<code>tests</code>	так	<code>type="object"; additionalProperties=false</code>
<code>validation</code>	так	<code>type="object"; additionalProperties=false</code>

```

{
  "additionalProperties": false,
  "properties": {
    "acceptance_pass": {
      "const": false
    },
    "artifact_binding": {
      "type": "object"
    },
    "artifact_binding_unchanged": {
      "const": true
    }
  }
}

```

```

},
"candidate_binding_digest": {
  "$ref": "#/$defs/Digest"
},
"execution_budget": {
  "additionalProperties": false,
  "properties": {
    "completed_at": {},
    "elapsed_ms": {},
    "started_at": {},
    "test_timeout_seconds": {},
    "total_timeout_seconds": {},
    "validation_timeout_seconds": {}
  },
  "required": [
    "total_timeout_seconds",
    "validation_timeout_seconds",
    "test_timeout_seconds",
    "elapsed_ms",
    "started_at",
    "completed_at"
  ],
  "type": "object"
},
"install_mode": {
  "const": "offline-wheelhouse"
},
"invocation": {
  "$ref": "#/$defs/ReleaseEvidenceInvocation"
},
"no_degradation": {
  "const": true
},
"pass_credit": {
  "const": false
},
"passed": {
  "const": true
},
"phases": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "elapsed_ms": {},
      "phase": {},
      "process_group_cleanup": {},
      "returncode": {},
      "status": {},
      "timeout_seconds": {}
    },
    "required": [
      "phase",
      "status",
      "returncode",
      "elapsed_ms",
      "timeout_seconds",
      "process_group_cleanup"
    ],
    "type": "object"
  },
  "maxItems": 2,
  "minItems": 2,
  "type": "array"
},
"platform": {
  "enum": [
    "linux",
    "windows"
  ]
},
"producer": {
  "$ref": "#/$defs/ReleaseEvidenceProducer"
},
"producer_attestation": {
  "$ref": "#/$defs/EvidenceProducerAttestation"
},
"product_acceptance_pass": {

```

```

    "const": false
  },
  "product_public_approval": {
    "const": "not_approved"
  },
  "raw_artifact_manifest_digest": {
    "$ref": "#/$defs/Digest"
  },
  "record_type": {
    "const": "NoDegradationResult"
  },
  "required_predicates": {
    "additionalProperties": false,
    "properties": {
      "acceptance_predicates": {},
      "complete": {},
      "missing_acceptance_predicates": {},
      "missing_mutation_families": {},
      "missing_policies": {},
      "mutation_families": {},
      "policies": {}
    },
    "required": [
      "policies",

```

```

      "acceptance_predicates",
      "mutation_families",
      "missing_policies",
      "missing_acceptance_predicates",
      "missing_mutation_families",
      "complete"
    ],
    "type": "object"
  },
  "result_digest": {
    "$ref": "#/$defs/Digest"
  },
  "root": {
    "minLength": 1,
    "type": "string"
  },
  "scale_selection": {
    "additionalProperties": false,
    "properties": {
      "command": {},
      "missing_workspace_is_failure": {},
      "pass_credit": {},
      "runs_separately": {}
    },
    "required": [
      "command",
      "runs_separately",
      "missing_workspace_is_failure",

```

```

      "pass_credit"
    ],
    "type": "object"
  },
  "source": {
    "const": "live_files_and_executable_tests"
  },
  "standard_distribution_gate_pass": {
    "const": false
  },
  "status": {
    "const": "pass"
  },
  "test_manifest": {
    "additionalProperties": false,
    "properties": {
      "digest": {},
      "file_count": {},
      "files": {}
    },
    "required": [
      "files",
      "file_count",
      "digest"
    ],
    "type": "object"
  },

```

```

},
"tests": {
  "additionalProperties": false,
  "properties": {
    "command": {},
    "elapsed_ms": {},
    "installed_environment": {
      "additionalProperties": false,
      "properties": {
        "build_dependency_source": {
          "const": "offline-wheelhouse"
        },
        "console_script": {
          "additionalProperties": false,
          "properties": {
            "declared": {
              "const": true
            },
            "installed_wrapper_checked": {
              "const": true
            },
            "path": {
              "minLength": 1,
              "type": "string"
            },
            "posix_execute_bits": {
              "type": [
                "boolean",
                "null"
              ]
            }
          }
        }
      }
    }
  }
}

```

```

},
"present": {
  "const": true
},
"sha256": {
  "$ref": "#/$defs/Digest"
},
"size_bytes": {
  "minimum": 1,
  "type": "integer"
},
"source_module_invoked": {
  "const": false
}
},
"required": [
  "present",
  "declared",
  "installed_wrapper_checked",
  "source_module_invoked",
  "posix_execute_bits",
  "path",
  "sha256",
  "size_bytes"
],
"type": "object"
},
"declared_build_dependencies": {

```

```

  "items": {
    "type": "string"
  },
  "minItems": 1,
  "type": "array"
},
"declared_python_requirement": {
  "minLength": 1,
  "type": "string"
},
"declared_runtime_dependencies": {
  "items": {
    "type": "string"
  },
  "minItems": 1,
  "type": "array"
},
"dependency_closure_check": {
  "const": "pass"
}
}

```

```

},
"environment": {
  "const": "clean-venv"
},
"installation_performed": {
  "const": true
},
"installed_environment_observation": {
  "additionalProperties": false,

```

```

  "properties": {
    "build_backend": {
      "additionalProperties": false,
      "properties": {
        "distribution": {
          "const": "setuptools"
        },
        "name": {
          "const": "setuptools.build_meta"
        },
        "record_sha256": {
          "$ref": "#/$defs/Digest"
        },
        "version": {
          "minLength": 1,
          "type": "string"
        }
      }
    },
    "required": [
      "name",
      "distribution",
      "version",
      "record_sha256"
    ],
    "type": "object"
  },
  "console_wrapper": {
    "additionalProperties": false,

```

```

    "properties": {
      "bytes": {
        "minimum": 1,
        "type": "integer"
      },
      "path": {
        "minLength": 1,
        "type": "string"
      },
      "sha256": {
        "$ref": "#/$defs/Digest"
      }
    },
    "required": [
      "path",
      "sha256",
      "bytes"
    ],
    "type": "object"
  },
  "license_closure": {
    "additionalProperties": false,
    "properties": {
      "complete": {
        "const": true
      },
      "digest": {
        "$ref": "#/$defs/Digest"

```

```

    },
    "entries": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "license": {
            "maxLength": 2048,
            "minLength": 1,
            "not": {
              "const": "UNKNOWN"
            },
            "type": "string"

```

```

    },
    "name": {
      "minLength": 1,
      "type": "string"
    },
    "version": {
      "minLength": 1,
      "type": "string"
    }
  },
  "required": [
    "name",
    "version",
    "license"
  ],
  "type": "object"

```

```

    },
    "maxItems": 256,
    "minItems": 1,
    "type": "array"
  }
},
"required": [
  "entries",
  "complete",
  "digest"
],
"type": "object"
},
"observation_digest": {
  "$ref": "#/$defs/Digest"
},
"platform": {
  "additionalProperties": false,
  "properties": {
    "machine": {
      "minLength": 1,
      "type": "string"
    },
    "release": {
      "minLength": 1,
      "type": "string"
    }
  },
  "sys_platform": {

```

```

    "minLength": 1,
    "type": "string"
  },
  "system": {
    "enum": [
      "linux",
      "windows"
    ]
  },
  "tags": {
    "items": {
      "minLength": 1,
      "type": "string"
    },
    "maxItems": 64,
    "minItems": 1,
    "type": "array"
  }
},
"required": [
  "system",
  "machine",
  "release",
  "sys_platform",
  "tags"
],
"type": "object"
},

```

```

"promin": {
  "additionalProperties": false,
  "properties": {
    "installed_file_count": {
      "maximum": 4096,

```

```

        "minimum": 1,
        "type": "integer"
    },
    "installed_files": {
        "items": {
            "additionalProperties": false,
            "properties": {
                "bytes": {
                    "minimum": 0,
                    "type": "integer"
                },
                "path": {
                    "minLength": 1,
                    "type": "string"
                },
                "sha256": {
                    "$ref": "#/$defs/Digest"
                }
            }
        },
        "required": [
            "path",
            "sha256",
            "bytes"
        ]
    }
}

```

```

    ],
    "type": "object"
},
"maxItems": 4096,
"minItems": 1,
"type": "array"
},
"installed_files_digest": {
    "$ref": "#/$defs/Digest"
},
"module_file": {
    "minLength": 1,
    "type": "string"
},
"module_sha256": {
    "$ref": "#/$defs/Digest"
},
"record": {
    "additionalProperties": false,
    "properties": {
        "bytes": {
            "minimum": 1,
            "type": "integer"
        },
        "path": {
            "minLength": 1,
            "pattern": "\\\\.dist-info/RECORD$",
            "type": "string"
        }
    }
}

```

```

    },
    "sha256": {
        "$ref": "#/$defs/Digest"
    }
},
"required": [
    "path",
    "sha256",
    "bytes"
],
"type": "object"
},
"version": {
    "$ref": "#/$defs/SemVer"
}
},
"required": [
    "version",
    "module_file",
    "module_sha256",
    "record",
    "installed_files",
    "installed_files_digest",
    "installed_file_count"
],
"type": "object"
}

```



```

},
"python": {
  "additionalProperties": false,
  "properties": {
    "abi_tag": {
      "minLength": 1,
      "type": "string"
    },
    "base_executable": {
      "minLength": 1,
      "type": "string"
    },
    "base_executable_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "base_prefix": {
      "minLength": 1,
      "type": "string"
    },
    "executable": {
      "minLength": 1,
      "type": "string"
    },
    "executable_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "implementation": {
      "minLength": 1,
      "type": "string"
    }
  },
  "prefix": {
    "minLength": 1,
    "type": "string"
  },
  "version": {
    "minLength": 1,
    "type": "string"
  }
},
"required": [
  "executable",
  "executable_sha256",
  "base_executable",
  "base_executable_sha256",
  "implementation",
  "version",
  "abi_tag",
  "prefix",
  "base_prefix"
],
"type": "object"
},
"sbom": {
  "additionalProperties": false,
  "properties": {
    "components": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "name": {
            "minLength": 1,
            "type": "string"
          },
          "record_sha256": {
            "$ref": "#/$defs/Digest"
          },
          "version": {
            "minLength": 1,
            "type": "string"
          }
        },
        "required": [
          "name",
          "version",
          "record_sha256"
        ],
        "type": "object"
      }
    }
  }
}

```

```

    },
    "maxItems": 256,
    "minItems": 1,
    "type": "array"
  },
  "digest": {
    "$ref": "#/$defs/Digest"
  },
  "format": {

```

```

    "const": "promin-spdx-lite-v1"
  }
},
"required": [
  "format",
  "components",
  "digest"
],
"type": "object"
},
"site_packages": {
  "minLength": 1,
  "type": "string"
},
"sqlite_version": {
  "minLength": 1,
  "type": "string"
},
"transitive_distribution_digest": {
  "$ref": "#/$defs/Digest"
},
"transitive_distributions": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "license": {
        "maxLength": 2048,
        "minLength": 1,

```

```

        "type": "string"
      },
      "name": {
        "minLength": 1,
        "type": "string"
      },
      "record": {
        "additionalProperties": false,
        "properties": {
          "bytes": {
            "minimum": 1,
            "type": "integer"
          },
          "path": {
            "minLength": 1,
            "pattern": "\\..dist-info/RECORD$",
            "type": "string"
          },
          "sha256": {
            "$ref": "#/$defs/Digest"
          }
        },
        "required": [
          "path",
          "sha256",
          "bytes"
        ],
        "type": "object"
      },

```

```

    },
    "requires": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "type": "array"
    },
    "version": {
      "minLength": 1,
      "type": "string"
    }
  }
}

```

```

    },
    "required": [
        "name",
        "version",
        "record",
        "requires",
        "license"
    ],
    "type": "object"
},
"maxItems": 256,
"minItems": 1,
"type": "array"
}
},
"required": [

```

```

    "python",
    "platform",
    "sqlite_version",
    "site_packages",
    "promin",
    "console_wrapper",
    "transitive_distributions",
    "transitive_distribution_digest",
    "sbom",
    "license_closure",
    "build_backend",
    "observation_digest"
],
"type": "object"
},
"interpreter": {
    "additionalProperties": false,
    "properties": {
        "executable": {
            "minLength": 1,
            "type": "string"
        },
        "nested_venv_created": {
            "const": true
        }
    }
},
"required": [
    "executable",

```

```

    "nested_venv_created"
],
"type": "object"
},
"invocations": {
    "items": {
        "additionalProperties": false,
        "properties": {
            "argv_digest": {
                "$ref": "#/$defs/Digest"
            },
            "command": {
                "maxLength": 128,
                "minLength": 1,
                "type": "string"
            },
            "returncode": {
                "const": 0
            },
            "stderr_bytes": {
                "maximum": 1048576,
                "minimum": 0,
                "type": "integer"
            },
            "stderr_sha256": {
                "$ref": "#/$defs/Digest"
            },
            "stdout_bytes": {

```

```

                "maximum": 1048576,
                "minimum": 0,
                "type": "integer"
            },
            "stdout_sha256": {

```

```

        "$ref": "#/$defs/Digest"
    }
},
"required": [
    "command",
    "argv_digest",
    "returncode",
    "stdout_sha256",
    "stdout_bytes",
    "stderr_sha256",
    "stderr_bytes"
],
"type": "object"
},
"maxItems": 7,
"minItems": 7,
"type": "array"
},
"mode": {
    "const": "offline-wheelhouse"
},
"nested_venv_created": {
    "const": true
}

```

```

},
"network_disabled": {
    "const": true
},
"origin_assertion": {
    "additionalProperties": false,
    "properties": {
        "inside_site_packages": {
            "const": true
        },
        "isolated_mode": {
            "const": true
        },
        "module_file": {
            "minLength": 1,
            "type": "string"
        },
        "module_sha256": {
            "$ref": "#/$defs/Digest"
        },
        "outside_install_source": {
            "const": true
        },
        "safe_path": {
            "const": true
        },
        "site_packages": {
            "minLength": 1,

```

```

            "type": "string"
        },
        "source_absent_from_sys_path": {
            "const": true
        },
        "sys_path_digest": {
            "$ref": "#/$defs/Digest"
        }
    }
},
"required": [
    "isolated_mode",
    "safe_path",
    "module_file",
    "module_sha256",
    "site_packages",
    "inside_site_packages",
    "outside_install_source",
    "source_absent_from_sys_path",
    "sys_path_digest"
],
"type": "object"
},
"performed": {
    "const": true
},
"pip_report": {

```

```

"additionalProperties": false,
"properties": {
  "artifact_digest": {
    "$ref": "#/$defs/Digest"
  },
  "report_count": {
    "const": 2
  },
  "reports": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "artifact_digest": {
          "$ref": "#/$defs/Digest"
        },
        "artifacts": {
          "items": {
            "additionalProperties": false,
            "properties": {
              "hashes": {
                "items": {
                  "minLength": 3,
                  "type": "string"
                },
                "type": "array"
              },
              "is_direct": {
                "type": "boolean"
              },
              "name": {
                "minLength": 1,
                "type": "string"
              },
              "version": {
                "minLength": 1,
                "type": "string"
              }
            },
            "required": [
              "name",
              "version",
              "is_direct",
              "hashes"
            ],
            "type": "object"
          },
          "maxItems": 256,
          "minItems": 1,
          "type": "array"
        },
        "bytes": {
          "maximum": 16777216,
          "minimum": 1,
          "type": "integer"
        },
        "install_count": {
          "maximum": 256,
          "minimum": 1,
          "type": "integer"
        },
        "pip_version": {
          "minLength": 1,
          "type": "string"
        },
        "role": {
          "enum": [
            "build-requirements",
            "candidate-install"
          ]
        },
        "sha256": {
          "$ref": "#/$defs/Digest"
        }
      },
      "required": [
        "role",
        "sha256",

```

```

        "bytes",
        "pip_version",
        "install_count",
        "artifacts",
        "artifact_digest"
    ],
    "type": "object"
},
"maxItems": 2,

    "minItems": 2,
    "type": "array"
}
},
"required": [
    "reports",
    "report_count",
    "artifact_digest"
],
"type": "object"
},
"platform": {
    "additionalProperties": false,
    "properties": {
        "os_name": {
            "enum": [
                "nt",
                "posix"
            ]
        },
        "python": {
            "minLength": 1,
            "type": "string"
        },
        "sys_platform": {
            "minLength": 1,
            "type": "string"
        }
    }
}

```

```

},
"required": [
    "os_name",
    "sys_platform",
    "python"
],
"type": "object"
},
"product_acceptance_pass": {
    "const": false
},
"python_distribution_version": {
    "$ref": "#/$defs/SemVer"
},
"resolved_runtime_dependencies": {
    "additionalProperties": false,
    "properties": {
        "cryptography": {
            "additionalProperties": false,
            "properties": {
                "requirement": {
                    "minLength": 1,
                    "type": "string"
                },
                "version": {
                    "minLength": 1,
                    "type": "string"
                }
            }
        }
    }
}

```

```

},
"required": [
    "requirement",
    "version"
],
"type": "object"
},
"jsonschema": {
    "additionalProperties": false,
    "properties": {
        "requirement": {
            "minLength": 1,

```

```

        "type": "string"
      },
      "version": {
        "minLength": 1,
        "type": "string"
      }
    },
    "required": [
      "requirement",
      "version"
    ],
    "type": "object"
  },
  "pypdf": {
    "additionalProperties": false,
    "properties": {
      "requirement": {
        "minLength": 1,
        "type": "string"
      },
      "version": {
        "minLength": 1,
        "type": "string"
      }
    },
    "required": [
      "requirement",
      "version"
    ],
    "type": "object"
  },
  "required": [
    "cryptography",
    "jsonschema",
    "pypdf"
  ],
  "type": "object"
},
"runtime_dependency_source": {
  "const": "offline-wheelhouse"
},
"test_dependencies_installed": {
  "const": true
},
"verified": {
  "const": true
},
"version": {
  "$ref": "#/$defs/SemVer"
},
"wheelhouse_binding": {
  "additionalProperties": false,
  "properties": {
    "file_count": {
      "maximum": 256,
      "minimum": 1,
      "type": "integer"
    },
    "files": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "bytes": {
            "minimum": 1,
            "type": "integer"
          },
          "path": {
            "minLength": 1,
            "type": "string"
          },
          "sha256": {
            "$ref": "#/$defs/Digest"
          }
        },
        "required": [
          "path",

```

```

        "bytes",
        "sha256"
    ],
    "type": "object"
},
"maxItems": 256,
"minItems": 1,
"type": "array"
},
"manifest_sha256": {
    "$ref": "#/$defs/Digest"
}
},
"required": [
    "files",
    "file_count",
    "manifest_sha256"
],
"type": "object"
},
"wheelhouse_lock": {
    "additionalProperties": false,
    "properties": {

```

```

        "copy_verified": {
            "const": true
        },
        "locked_copy_binding": {
            "additionalProperties": false,
            "properties": {
                "file_count": {
                    "maximum": 256,
                    "minimum": 1,
                    "type": "integer"
                },
                "files": {
                    "items": {
                        "additionalProperties": false,
                        "properties": {
                            "bytes": {
                                "minimum": 1,
                                "type": "integer"
                            },
                            "path": {
                                "minLength": 1,
                                "type": "string"
                            },
                            "sha256": {
                                "$ref": "#/$defs/Digest"
                            }
                        }
                    },
                    "required": [

```

```

                        "path",
                        "bytes",
                        "sha256"
                    ],
                    "type": "object"
                },
                "maxItems": 256,
                "minItems": 1,
                "type": "array"
            },
            "manifest_sha256": {
                "$ref": "#/$defs/Digest"
            }
        },
        "required": [
            "files",
            "file_count",
            "manifest_sha256"
        ],
        "type": "object"
    },
    "source_binding": {
        "additionalProperties": false,
        "properties": {
            "file_count": {
                "maximum": 256,

```



```

        "minimum": 1,
        "type": "integer"
    },
    "files": {
        "items": {
            "additionalProperties": false,
            "properties": {
                "bytes": {
                    "minimum": 1,
                    "type": "integer"
                },
                "path": {
                    "minLength": 1,
                    "type": "string"
                },
                "sha256": {
                    "$ref": "#/$defs/Digest"
                }
            },
            "required": [
                "path",
                "bytes",
                "sha256"
            ],
            "type": "object"
        },
        "maxItems": 256,
        "minItems": 1,
        "type": "array"
    },

```

```

        "manifest_sha256": {
            "$ref": "#/$defs/Digest"
        },
        "required": [
            "files",
            "file_count",
            "manifest_sha256"
        ],
        "type": "object"
    },
    "source_unchanged": {
        "const": true
    },
    "required": [
        "source_binding",
        "locked_copy_binding",
        "source_unchanged",
        "copy_verified"
    ],
    "type": "object"
},
"required": [
    "performed",
    "verified",
    "mode",

```

```

    "environment",
    "runtime_dependency_source",
    "build_dependency_source",
    "network_disabled",
    "installation_performed",
    "nested_venv_created",
    "wheelhouse_binding",
    "declared_runtime_dependencies",
    "declared_build_dependencies",
    "declared_python_requirement",
    "resolved_runtime_dependencies",
    "dependency_closure_check",
    "installed_environment_observation",
    "pip_report",
    "wheelhouse_lock",
    "origin_assertion",
    "test_dependencies_installed",
    "version",
    "python_distribution_version",

```

```

        "console_script",
        "invocations",
        "interpreter",
        "platform",
        "product_acceptance_pass"
    ],
    "type": "object"
},
"junit_sha256": {},

```

```

"marker_expression": {},
"optional_platform_tests_excluded": {},
"passed": {},
"process_group_cleanup": {},
"ran": {},
"raw_artifact_manifest_digest": {
    "$ref": "#/$defs/Digest"
},
"raw_artifacts": {
    "items": {
        "additionalProperties": false,
        "properties": {
            "bytes": {
                "maximum": 16777216,
                "minimum": 0,
                "type": "integer"
            },
            "data": {
                "type": "string"
            },
            "encoding": {
                "const": "base64"
            },
            "media_type": {
                "enum": [
                    "application/xml",
                    "text/plain; charset=utf-8"
                ]
            }
        }
    }
}

```

```

},
"role": {
    "enum": [
        "required-junit",
        "required-stdout",
        "required-stderr"
    ]
},
"sha256": {
    "$ref": "#/$defs/Digest"
}
},
"required": [
    "role",
    "media_type",
    "bytes",
    "sha256",
    "encoding",
    "data"
],
"type": "object"
},
"maxItems": 3,
"minItems": 3,
"type": "array"
},
"required_counts": {},
"required_skips_fail_closed": {},

```

```

"returncode": {},
"scale_environment_configured": {},
"selection_expression": {},
"source_tree_shadowing_disabled": {
    "const": true
},
"stderr": {
    "additionalProperties": false,
    "properties": {
        "bytes": {
            "minimum": 0,
            "type": "integer"
        }
    }
}

```

```

    },
    "selection": {
      "enum": [
        "complete",
        "tail"
      ]
    },
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    },
    },
    "text": {
      "type": "string"
    },
    },
    "truncated": {
      "type": "boolean"
    }
  }

```

```

    },
    "required": [
      "text",
      "bytes",
      "sha256",
      "truncated",
      "selection"
    ],
    "type": "object"
  },
  "stdout": {
    "additionalProperties": false,
    "properties": {
      "bytes": {
        "minimum": 0,
        "type": "integer"
      },
      "selection": {
        "enum": [
          "complete",
          "tail"
        ]
      },
      },
      "sha256": {
        "$ref": "#/$defs/Digest"
      },
      },
      "text": {
        "type": "string"
      }
    }
  },
  "truncated": {
    "type": "boolean"
  }
},
"required": [
  "text",
  "bytes",
  "sha256",
  "truncated",
  "selection"
],
"type": "object"
},
"timed_out": {},
"timeout_seconds": {}
},
"required": [
  "ran",
  "command",
  "returncode",
  "stdout",
  "stderr",
  "timed_out",
  "timeout_seconds",
  "elapsed_ms",
  "process_group_cleanup",
  "required_counts",

```

```

  "required_skips_fail_closed",
  "optional_platform_tests_excluded",
  "marker_expression",
  "selection_expression",
  "junit_sha256",

```

```

    "passed",
    "scale_environment_configured",
    "raw_artifacts",
    "raw_artifact_manifest_digest",
    "installed_environment",
    "source_tree_shadowing_disabled"
  ],
  "type": "object"
},
"validation": {
  "additionalProperties": false,
  "properties": {
    "checks": {},
    "errors": {},
    "record_type": {},
    "root": {},
    "valid": {}
  },
  "required": [
    "checks",
    "errors",
    "record_type",
    "root",

```

```

    "valid"
  ],
  "type": "object"
}
},
"required": [
  "record_type",
  "status",
  "platform",
  "no_degradation",
  "source",
  "root",
  "artifact_binding",
  "candidate_binding_digest",
  "test_manifest",
  "install_mode",
  "required_predicates",
  "validation",
  "tests",
  "passed",
  "pass_credit",
  "acceptance_pass",
  "product_acceptance_pass",
  "product_public_approval",
  "scale_selection",
  "artifact_binding_unchanged",
  "standard_distribution_gate_pass",
  "phases",

```

```

  "execution_budget",
  "producer",
  "invocation",
  "raw_artifact_manifest_digest",
  "producer_attestation",
  "result_digest"
],
"type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/NoDegradationResult | Валідатор:
urn:promin:contracts:1#/\$defs/NoDegradationResult

61. OperationMetrics

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, claim_scope, authoritative, duration_ms, changed_records, event_write, projection_update, runtime_checkpoint, physical_payload_bytes, bytes_per_changed_record

Властивість	Обов'язков е	Структурне правило
authoritative	так	const=false
bytes_per_changed_record	так	type="number"; minimum=0
changed_records	так	type="integer"; minimum=0

Властивість	Обов'язков е	Структурне правило
claim_scope	так	const="single-command-observation-only"
duration_ms	так	type="number"; minimum=0
event_write	так	type="object"; additionalProperties=false
physical_payload_bytes	так	type="integer"; minimum=0
projection_update	так	oneOf[3]=type:object type:object type:object
record_type	так	const="OperationMetrics"
runtime_checkpoint	так	oneOf[2]=type:object type:object
<pre> { "additionalProperties": false, "properties": { "authoritative": { "const": false }, "bytes_per_changed_record": { "minimum": 0, "type": "number" }, "changed_records": { "minimum": 0, "type": "integer" }, "claim_scope": { "const": "single-command-observation-only" }, "duration_ms": { "minimum": 0, "type": "number" }, "event_write": { "additionalProperties": false, "properties": { "changed_records": { "minimum": 0, "type": "integer" }, </pre>		
<pre> "derived_index_bytes": { "minimum": 0, "type": "integer" }, "head_bytes": { "minimum": 0, "type": "integer" }, "journal_authority_bytes": { "minimum": 0, "type": "integer" }, "journal_checkpoint_bytes": { "minimum": 0, "type": "integer" }, "logical_final_bytes": { "minimum": 0, "type": "integer" }, "physical_payload_bytes": { "minimum": 0, "type": "integer" }, "temporary_staging_bytes": { "minimum": 0, "type": "integer" } } }, "required": ["changed_records", "journal_authority_bytes", "temporary_staging_bytes", "head_bytes", "derived_index_bytes", "journal_checkpoint_bytes", </pre>		

```

        "logical_final_bytes",
        "physical_payload_bytes"
    ],
    "type": "object"
},
"physical_payload_bytes": {
    "minimum": 0,
    "type": "integer"
},
"projection_update": {
    "oneOf": [
        {
            "additionalProperties": false,
            "properties": {
                "bytes_per_changed_record": {
                    "minimum": 0,
                    "type": "number"
                },
                "changed_records": {
                    "minimum": 0,

```

```

                    "type": "integer"
                },
                "changed_shards": {
                    "minimum": 0,
                    "type": "integer"
                },
                "physical_payload_bytes": {
                    "minimum": 0,
                    "type": "integer"
                },
                "projection_authoritative": {
                    "const": false
                },
                "status": {
                    "enum": [
                        "already-current",
                        "missing",
                        "stale-rebuild-required"
                    ]
                }
            },
            "required": [
                "status",
                "projection_authoritative",
                "changed_records",
                "changed_shards",
                "physical_payload_bytes",
                "bytes_per_changed_record"

```

```

        ],
        "type": "object"
    },
    {
        "additionalProperties": false,
        "properties": {
            "bytes_per_changed_record": {
                "minimum": 0,
                "type": "number"
            },
            "changed_records": {
                "minimum": 0,
                "type": "integer"
            },
            "changed_shards": {
                "minimum": 0,
                "type": "integer"
            },
            "error": {
                "maxLength": 4096,
                "minLength": 1,
                "type": "string"
            },
            "physical_payload_bytes": {
                "minimum": 0,
                "type": "integer"
            },
            "projection_authoritative": {

```

```

        "const": false
    },
    "status": {
        "const": "failed-rebuild-required"
    }
},
"required": [
    "status",
    "projection_authoritative",
    "changed_records",
    "changed_shards",
    "physical_payload_bytes",
    "bytes_per_changed_record",
    "error"
],
"type": "object"
},
{
    "additionalProperties": false,
    "properties": {
        "bytes_per_changed_record": {
            "minimum": 0,
            "type": "number"
        },
        "changed_records": {
            "minimum": 0,
            "type": "integer"
        },

```

```

        "changed_shards": {
            "minimum": 0,
            "type": "integer"
        },
        "head": {
            "additionalProperties": false,
            "properties": {
                "batch_digest": {
                    "oneOf": [
                        {
                            "$ref": "#/$defs/Digest"
                        },
                        {
                            "type": "null"
                        }
                    ]
                },
                "batch_id": {
                    "oneOf": [
                        {
                            "$ref": "#/$defs/Id"
                        },
                        {
                            "type": "null"
                        }
                    ]
                }
            },
            "sequence": {

```

```

                "minimum": 0,
                "type": "integer"
            }
        },
        "required": [
            "sequence",
            "batch_id",
            "batch_digest"
        ],
        "type": "object"
    },
    "logical_payload_bytes": {
        "minimum": 0,
        "type": "integer"
    },
    "physical_payload_bytes": {
        "minimum": 0,
        "type": "integer"
    },
    "projection_authoritative": {
        "const": false
    },

```

```

    "projection_growth_bytes": {
      "minimum": 0,
      "type": "integer"
    },
    "semantic_digest": {
      "$ref": "#/$defs/Digest"
    },
    "status": {
      "const": "updated"
    }
  },
  "required": [
    "status",
    "projection_authoritative",
    "changed_records",
    "changed_shards",
    "physical_payload_bytes",
    "bytes_per_changed_record",
    "head",
    "semantic_digest",
    "logical_payload_bytes",
    "projection_growth_bytes"
  ],
  "type": "object"
}
]
},
"record_type": {
  "const": "OperationMetrics"
},
"runtime_checkpoint": {
  "oneOf": [
    {
      "additionalProperties": false,

```

```

    "properties": {
      "authoritative": {
        "const": false
      },
      "batch_threshold": {
        "minimum": 1,
        "type": "integer"
      },
      "byte_threshold": {
        "minimum": 1,
        "type": "integer"
      },
      "checkpoint_bytes": {
        "minimum": 0,
        "type": "integer"
      },
      "checkpoint_count": {
        "minimum": 0,
        "type": "integer"
      },
      "tail_batches": {
        "minimum": 0,
        "type": "integer"
      },
      "tail_bytes": {
        "minimum": 0,
        "type": "integer"
      }
    },

```

```

    "written": {
      "const": false
    }
  },
  "required": [
    "authoritative",
    "batch_threshold",
    "byte_threshold",
    "checkpoint_bytes",
    "checkpoint_count",
    "tail_batches",
    "tail_bytes",
    "written"
  ],
  "type": "object"

```



```

    },
    {
      "additionalProperties": false,
      "properties": {
        "authoritative": {
          "const": false
        },
        "batch_threshold": {
          "minimum": 1,
          "type": "integer"
        },
        "byte_threshold": {
          "minimum": 1,
          "type": "integer"
        },
        "checkpoint_bytes": {
          "minimum": 0,
          "type": "integer"
        },
        "checkpoint_count": {
          "minimum": 0,
          "type": "integer"
        },
        "compacted_tail_batches": {
          "minimum": 0,
          "type": "integer"
        },
        "compacted_tail_bytes": {
          "minimum": 0,
          "type": "integer"
        },
        "tail_batches": {
          "minimum": 0,
          "type": "integer"
        },
        "tail_bytes": {
          "minimum": 0,
          "type": "integer"
        },
        "written": {
          "const": true
        }
      }
    },
    {
      "required": [
        "authoritative",
        "batch_threshold",
        "byte_threshold",
        "checkpoint_bytes",
        "checkpoint_count",
        "tail_batches",
        "tail_bytes",
        "written",
        "compacted_tail_batches",
        "compacted_tail_bytes"
      ],
      "type": "object"
    }
  ],
  "type": "object"
},
{
  "required": [
    "record_type",
    "claim_scope",
    "authoritative",
    "duration_ms",
    "changed_records",
    "event_write",
    "projection_update",
    "runtime_checkpoint",
    "physical_payload_bytes",
    "bytes_per_changed_record"
  ],
  "type": "object",
  "x-promin-persistence": "derived-live-only"
}

```

Власник: [core/contracts.schema.json#/\\$defs/OperationMetrics](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/OperationMetrics](#)

62. PersistentEntityDef

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: kind, purpose, admission_rule

Властивість	Обов'язков е	Структурне правило
admission_rule	так	type="string"; minLength=1; maxLength=256
kind	так	enum=["Завдання", "Candidate", "Artifact", "Run", "Finding", "Decision", "GateResult", "Grant", "Оренда"]
purpose	так	type="string"; minLength=1
<pre>{ "additionalProperties": false, "properties": { "admission_rule": { "maxLength": 256, "minLength": 1, "type": "string" }, "kind": { "enum": ["Завдання", "Candidate", "Artifact", "Run", "Finding", "Decision", "GateResult", "Grant", "Оренда"] }, "purpose": { "minLength": 1, "type": "string" } }, "required": ["kind", "purpose", "admission_rule"], "type": "object" }</pre>		

Власник: core/contracts.schema.json#/\$defs/PersistentEntityDef | Валідатор: urn:promin:contracts:1#/\$defs/PersistentEntityDef

63. PlanProposal

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, proposal_id, project_id, project_mode, goal, source_plan_digest, created_at, tasks, authority, pass_credit, proposal_digest

Властивість	Обов'язков е	Структурне правило
authority	так	const=false
created_at	так	\$ref="#/\$defs/Timestamp"
goal	так	type="string"; minLength=1; maxLength=8192
pass_credit	так	const=false
project_id	так	\$ref="#/\$defs/Id"
project_mode	так	enum=["greenfield", "existing-code", "hybrid"]
proposal_digest	так	\$ref="#/\$defs/Digest"
proposal_id	так	\$ref="#/\$defs/Id"
record_type	так	const="PlanProposal"
source_plan_digest	так	\$ref="#/\$defs/Digest"
tasks	так	type="array"; maxItems=1024

```

{
  "additionalProperties": false,
  "properties": {
    "authority": {
      "const": false
    },
    "created_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "goal": {
      "maxLength": 8192,
      "minLength": 1,
      "type": "string"
    },
    "pass_credit": {
      "const": false
    },
    "project_id": {
      "$ref": "#/$defs/Id"
    },
    "project_mode": {
      "enum": [
        "greenfield",
        "existing-code",
        "hybrid"
      ]
    },
    "proposal_digest": {
      "$ref": "#/$defs/Digest"
    },
    "proposal_id": {
      "$ref": "#/$defs/Id"
    },
    "record_type": {
      "const": "PlanProposal"
    },
    "source_plan_digest": {
      "$ref": "#/$defs/Digest"
    },
    "tasks": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "acceptance_predicate": {
            "maxLength": 8192,
            "minLength": 1,
            "type": "string"
          },
          "allowed_paths": {
            "items": {
              "maxLength": 4096,
              "minLength": 1,
              "type": "string"
            },
            "maxItems": 256,
            "minItems": 1,
            "type": "array",
            "uniqueItems": true
          },
          "authority": {
            "const": false
          },
          "depends_on": {
            "items": {
              "$ref": "#/$defs/Id"
            },
            "maxItems": 256,
            "type": "array",
            "uniqueItems": true
          },
          "operation": {
            "enum": [
              "read",
              "mutate"
            ]
          },
          "pass_credit": {
            "const": false
          }
        }
      }
    }
  }
}

```

```

    },
    "source_bindings": {
      "items": {
        "maxLength": 4096,
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 256,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "task_id": {
      "$ref": "#/$defs/Id"
    },
    "title": {
      "maxLength": 4096,
      "minLength": 1,
      "type": "string"
    }
  },
  "required": [
    "task_id",
    "title",
    "operation",
    "depends_on",
    "acceptance_predicate",
    "allowed_paths",
    "source_bindings",
    "authority"
  ],
  "type": "object"
},
"maxItems": 1024,
"type": "array"
}
},
"required": [
  "record_type",
  "proposal_id",
  "project_id",
  "project_mode",
  "goal",
  "source_plan_digest",
  "created_at",
  "tasks",
  "authority",
  "pass_credit",
  "proposal_digest"
],
"type": "object",
"x-promin-persistence": "transient-ingress-only"
}

```

Власник: [core/contracts.schema.json#/\\$defs/PlanProposal](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/PlanProposal](#)

64. PlatformVerificationResult

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, status, candidate_binding_digest, platform, platform_identity, exact_candidate_verified, archive_sha256, archive_bytes, portable_implementation_closure_digest, observed_environment_closure_digest, install_mode, installed_command_verified, installation, product_acceptance_pass, product_public_approval, producer, invocation, raw_artifact_manifest_digest, producer_attestation, result_digest

Властивість	Обов'язков е	Структурне правило
archive_bytes	так	type="integer"; minimum=1
archive_sha256	так	\$ref="#/\$defs/Digest"
candidate_binding_digest	так	\$ref="#/\$defs/Digest"
exact_candidate_verified	так	const=true
install_mode	так	const="online-clean"
installation	так	type="object"; additionalProperties=false

Властивість	Обов'язков е	Структурне правило
installed_command_verified	так	const=true
invocation	так	\$ref="#/\$defs/ReleaseEvidenceInvocation"
observed_environment_closure_digest	так	\$ref="#/\$defs/Digest"
platform	так	enum=["linux", "windows"]
platform_identity	так	type="object"; additionalProperties=false
portable_implementation_closure_digest	так	\$ref="#/\$defs/Digest"
producer	так	\$ref="#/\$defs/ReleaseEvidenceProducer"
producer_attestation	так	\$ref="#/\$defs/EvidenceProducerAttestation"
product_acceptance_pass	так	const=false
product_public_approval	так	const="not_approved"
raw_artifact_manifest_digest	так	\$ref="#/\$defs/Digest"
record_type	так	const="PlatformVerificationResult"
result_digest	так	\$ref="#/\$defs/Digest"
status	так	const="pass"
<pre> { "additionalProperties": false, "properties": { "archive_bytes": { "minimum": 1, "type": "integer" }, "archive_sha256": { "\$ref": "#/\$defs/Digest" }, "candidate_binding_digest": { "\$ref": "#/\$defs/Digest" }, "exact_candidate_verified": { "const": true }, "install_mode": { "const": "online-clean" }, "installation": { "additionalProperties": false, "properties": { "build_backend": { "additionalProperties": false, "properties": { "distribution": { "const": "setuptools" }, "name": { "const": "setuptools.build_meta" }, "record_sha256": { "\$ref": "#/\$defs/Digest" }, "version": { "minLength": 1, "type": "string" } }, "required": ["name", "distribution", "version", "record_sha256"], "type": "object" }, "build_dependency_source": { "const": "online-index-explicit" }, "command_invocations": { </pre>		

```

"items": {
  "additionalProperties": false,
  "properties": {
    "argv_digest": {
      "$ref": "#/$defs/Digest"

```

```

    },
    "command": {
      "maxLength": 128,
      "minLength": 1,
      "type": "string"
    },
    "returncode": {
      "const": 0
    },
    "stderr_bytes": {
      "maximum": 1048576,
      "minimum": 0,
      "type": "integer"
    },
    "stderr_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "stdout_bytes": {
      "maximum": 1048576,
      "minimum": 0,
      "type": "integer"
    },
    "stdout_sha256": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "command",

```

```

    "argv_digest",
    "returncode",
    "stdout_sha256",
    "stdout_bytes",
    "stderr_sha256",
    "stderr_bytes"
  ],
  "type": "object"
},
"maxItems": 7,
"minItems": 7,
"type": "array"
},
"console_script": {
  "additionalProperties": false,
  "properties": {
    "declared": {
      "const": true
    },
    "installed_wrapper_checked": {
      "const": true
    },
    "path": {
      "minLength": 1,
      "type": "string"
    },
    "posix_execute_bits": {
      "type": [

```

```

        "boolean",
        "null"
      ]
    },
    "present": {
      "const": true
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    },
    "size_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "source_module_invoked": {
      "const": false

```

```

    }
  },
  "required": [
    "present",
    "declared",
    "installed_wrapper_checked",
    "source_module_invoked",
    "posix_execute_bits",
    "path",
    "sha256",
    "size_bytes"
  ],

```

```

    "type": "object"
  },
  "declared_python_requirement": {
    "minLength": 1,
    "type": "string"
  },
  "dependency_closure": {
    "additionalProperties": false,
    "properties": {
      "closure_check": {
        "const": "pass"
      },
      "declared_build": {
        "items": {
          "type": "string"
        },
        "minItems": 1,
        "type": "array"
      },
      "declared_runtime": {
        "items": {
          "type": "string"
        },
        "minItems": 1,
        "type": "array"
      }
    },
    "dependency_artifact_digest": {
      "$ref": "#/$defs/Digest"
    }
  },

```

```

  },
  "digest": {
    "$ref": "#/$defs/Digest"
  },
  "resolved_runtime": {
    "additionalProperties": false,
    "properties": {
      "cryptography": {
        "additionalProperties": false,
        "properties": {
          "requirement": {
            "minLength": 1,
            "type": "string"
          },
          "version": {
            "minLength": 1,
            "type": "string"
          }
        }
      },
      "required": [
        "requirement",
        "version"
      ],
      "type": "object"
    },
    "jsonschema": {
      "additionalProperties": false,
      "properties": {

```

```

        "requirement": {
          "minLength": 1,
          "type": "string"
        },
        "version": {
          "minLength": 1,
          "type": "string"
        }
      }
    },
  },

```

```

        "required": [
            "requirement",
            "version"
        ],
        "type": "object"
    },
    "pypdf": {
        "additionalProperties": false,
        "properties": {
            "requirement": {
                "minLength": 1,
                "type": "string"
            },
            "version": {
                "minLength": 1,
                "type": "string"
            }
        }
    },
    "required": [
        "requirement",
        "version"
    ],
    "type": "object"
},
"required": [
    "cryptography",
    "jsonschema",
    "pypdf"
],
"type": "object"
},
"transitive_distribution_digest": {
    "$ref": "#/$defs/Digest"
},
"transitive_distributions": {
    "items": {
        "additionalProperties": false,
        "properties": {
            "license": {
                "maxLength": 2048,
                "minLength": 1,
                "type": "string"
            },
            "name": {
                "minLength": 1,
                "type": "string"
            }
        }
    },
    "record": {
        "additionalProperties": false,
        "properties": {
            "bytes": {
                "minimum": 1,
                "type": "integer"
            },
            "path": {
                "minLength": 1,
                "pattern": "\\\\.dist-info/RECORD$",
                "type": "string"
            },
            "sha256": {
                "$ref": "#/$defs/Digest"
            }
        }
    },
    "required": [
        "path",
        "sha256",
        "bytes"
    ],
    "type": "object"
},
"requires": {
    "items": {
        "minLength": 1,
        "type": "string"
    }
},
"type": "array"

```



```

        },
        "version": {
            "minLength": 1,
            "type": "string"
        }
    },
    "required": [
        "name",
        "version",
        "record",
        "requires",
        "license"
    ],
    "type": "object"
},
"maxItems": 256,
"minItems": 1,
"type": "array"
}
},
"required": [
    "declared_runtime",
    "declared_build",
    "resolved_runtime",
    "closure_check",
    "transitive_distributions",
    "transitive_distribution_digest",
    "dependency_artifact_digest",
    "digest"
],
"type": "object"
},
"environment": {
    "const": "clean-venv"
},
"installation_performed": {
    "const": true
},
"installed_distribution": {
    "additionalProperties": false,
    "properties": {
        "name": {
            "const": "promin"
        },
        "runtime_version": {
            "$ref": "#/$defs/SemVer"
        },
        "version": {
            "$ref": "#/$defs/SemVer"
        }
    }
},
"required": [
    "name",
    "version",
    "runtime_version"
],
"type": "object"
},
"installed_environment_observation": {
    "additionalProperties": false,
    "properties": {
        "build_backend": {
            "additionalProperties": false,
            "properties": {
                "distribution": {
                    "const": "setuptools"
                },
                "name": {
                    "const": "setuptools.build_meta"
                }
            },
            "record_sha256": {
                "$ref": "#/$defs/Digest"
            },
            "version": {
                "minLength": 1,
                "type": "string"
            }
        }
    }
},

```

```

    "required": [
      "name",
      "distribution",
      "version",
      "record_sha256"
    ],
    "type": "object"
  },
  "console_wrapper": {
    "additionalProperties": false,
    "properties": {
      "bytes": {
        "minimum": 1,
        "type": "integer"
      },
      "path": {
        "minLength": 1,
        "type": "string"
      },
      "sha256": {
        "$ref": "#/$defs/Digest"
      }
    }
  },
  "required": [
    "path",
    "sha256",
    "bytes"
  ],
  "type": "object"
},
"license_closure": {
  "additionalProperties": false,

```

```

    "properties": {
      "complete": {
        "const": true
      },
      "digest": {
        "$ref": "#/$defs/Digest"
      },
      "entries": {
        "items": {
          "additionalProperties": false,
          "properties": {
            "license": {
              "maxLength": 2048,
              "minLength": 1,
              "not": {
                "const": "UNKNOWN"
              },
              "type": "string"
            },
            "name": {
              "minLength": 1,
              "type": "string"
            },
            "version": {
              "minLength": 1,
              "type": "string"
            }
          }
        }
      }
    },

```

```

    "required": [
      "name",
      "version",
      "license"
    ],
    "type": "object"
  },
  "maxItems": 256,
  "minItems": 1,
  "type": "array"
}
},
"required": [
  "entries",
  "complete",
  "digest"
],

```

```

    "type": "object"
  },
  "observation_digest": {
    "$ref": "#/$defs/Digest"
  },
  "platform": {
    "additionalProperties": false,
    "properties": {
      "machine": {
        "minLength": 1,
        "type": "string"

```

```

      },
      "release": {
        "minLength": 1,
        "type": "string"
      },
      "sys_platform": {
        "minLength": 1,
        "type": "string"
      },
      "system": {
        "enum": [
          "linux",
          "windows"
        ]
      },
      "tags": {
        "items": {
          "minLength": 1,
          "type": "string"
        },
        "maxItems": 64,
        "minItems": 1,
        "type": "array"
      }
    },
    "required": [
      "system",
      "machine",

```

```

      "release",
      "sys_platform",
      "tags"
    ],
    "type": "object"
  },
  "promin": {
    "additionalProperties": false,
    "properties": {
      "installed_file_count": {
        "maximum": 4096,
        "minimum": 1,
        "type": "integer"
      },
      "installed_files": {
        "items": {
          "additionalProperties": false,
          "properties": {
            "bytes": {
              "minimum": 0,
              "type": "integer"
            },
            "path": {
              "minLength": 1,
              "type": "string"
            },
            "sha256": {
              "$ref": "#/$defs/Digest"

```

```

          }
        },
        "required": [
          "path",
          "sha256",
          "bytes"
        ],
        "type": "object"
      },
      "maxItems": 4096,

```

```

        "minItems": 1,
        "type": "array"
    },
    "installed_files_digest": {
        "$ref": "#/$defs/Digest"
    },
    "module_file": {
        "minLength": 1,
        "type": "string"
    },
    "module_sha256": {
        "$ref": "#/$defs/Digest"
    },
    "record": {
        "additionalProperties": false,
        "properties": {
            "bytes": {
                "minimum": 1,
                "type": "integer"
            },
            "path": {
                "minLength": 1,
                "pattern": "\\\\.dist-info/RECORD$",
                "type": "string"
            },
            "sha256": {
                "$ref": "#/$defs/Digest"
            }
        },
        "required": [
            "path",
            "sha256",
            "bytes"
        ],
        "type": "object"
    },
    "version": {
        "$ref": "#/$defs/SemVer"
    }
},
"required": [
    "version",
    "module_file",
    "module_sha256",
    "record",
    "installed_files",
    "installed_files_digest",
    "installed_file_count"
],
"type": "object"
},
"python": {
    "additionalProperties": false,
    "properties": {
        "abi_tag": {
            "minLength": 1,
            "type": "string"
        },
        "base_executable": {
            "minLength": 1,
            "type": "string"
        },
        "base_executable_sha256": {
            "$ref": "#/$defs/Digest"
        },
        "base_prefix": {
            "minLength": 1,
            "type": "string"
        },
        "executable": {
            "minLength": 1,
            "type": "string"
        },
        "executable_sha256": {
            "$ref": "#/$defs/Digest"
        },
        "implementation": {

```

```

        "minLength": 1,
        "type": "string"
    },
    "prefix": {
        "minLength": 1,
        "type": "string"
    },
    "version": {
        "minLength": 1,
        "type": "string"
    }
},
"required": [
    "executable",
    "executable_sha256",
    "base_executable",
    "base_executable_sha256",
    "implementation",
    "version",
    "abi_tag",
    "prefix",
    "base_prefix"
],
"type": "object"
},

```

```

"sbom": {
    "additionalProperties": false,
    "properties": {
        "components": {
            "items": {
                "additionalProperties": false,
                "properties": {
                    "name": {
                        "minLength": 1,
                        "type": "string"
                    },
                    "record_sha256": {
                        "$ref": "#/$defs/Digest"
                    },
                    "version": {
                        "minLength": 1,
                        "type": "string"
                    }
                },
                "required": [
                    "name",
                    "version",
                    "record_sha256"
                ],
                "type": "object"
            },
            "maxItems": 256,
            "minItems": 1,

```

```

            "type": "array"
        },
        "digest": {
            "$ref": "#/$defs/Digest"
        },
        "format": {
            "const": "promin-spx-lite-v1"
        }
    },
    "required": [
        "format",
        "components",
        "digest"
    ],
    "type": "object"
},
"site_packages": {
    "minLength": 1,
    "type": "string"
},
"sqlite_version": {
    "minLength": 1,
    "type": "string"
},
"transitive_distribution_digest": {

```

```

    "$ref": "#/$defs/Digest"
  },
  "transitive_distributions": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "license": {
          "maxLength": 2048,
          "minLength": 1,
          "type": "string"
        },
        "name": {
          "minLength": 1,
          "type": "string"
        },
        "record": {
          "additionalProperties": false,
          "properties": {
            "bytes": {
              "minimum": 1,
              "type": "integer"
            },
            "path": {
              "minLength": 1,
              "pattern": "\\\\.dist-info/RECORD$",
              "type": "string"
            },
            "sha256": {
              "$ref": "#/$defs/Digest"
            }
          }
        },
        "required": [
          "path",
          "sha256",
          "bytes"
        ],
        "type": "object"
      },
      "requires": {
        "items": {
          "minLength": 1,
          "type": "string"
        },
        "type": "array"
      },
      "version": {
        "minLength": 1,
        "type": "string"
      }
    },
    "required": [
      "name",
      "version",
      "record",
      "requires",
      "license"
    ],
    "type": "object"
  },
  "maxItems": 256,
  "minItems": 1,
  "type": "array"
}
},
"required": [
  "python",
  "platform",
  "sqlite_version",
  "site_packages",
  "promin",
  "console_wrapper",
  "transitive_distributions",
  "transitive_distribution_digest",
  "sbom",
  "license_closure",
  "build_backend",
  "observation_digest"

```

```

    ],
    "type": "object"
  },
  "license_closure": {
    "additionalProperties": false,
    "properties": {
      "complete": {
        "const": true
      },
      "digest": {
        "$ref": "#/$defs/Digest"
      },
      "entries": {
        "items": {
          "additionalProperties": false,
          "properties": {
            "license": {
              "maxLength": 2048,
              "minLength": 1,
              "not": {
                "const": "UNKNOWN"
              },
              "type": "string"
            },
            "name": {
              "minLength": 1,
              "type": "string"
            },
            "version": {
              "minLength": 1,
              "type": "string"
            }
          }
        },
        "required": [
          "name",
          "version",
          "license"
        ]
      },
      "type": "object"
    },
    "maxItems": 256,
    "minItems": 1,
    "type": "array"
  },
  "required": [
    "entries",
    "complete",
    "digest"
  ],
  "type": "object"
},
"nested_venv_created": {
  "const": true
},
"network_disabled": {
  "const": false
},
"pip_report": {
  "additionalProperties": false,
  "properties": {
    "artifact_digest": {
      "$ref": "#/$defs/Digest"
    },
    "report_count": {
      "const": 2
    }
  },
  "reports": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "artifact_digest": {
          "$ref": "#/$defs/Digest"
        },
        "artifacts": {
          "items": {
            "additionalProperties": false,

```

```

        "properties": {
            "hashes": {
                "items": {
                    "minLength": 3,
                    "type": "string"
                },
                "type": "array"
            },
            "is_direct": {
                "type": "boolean"
            },
            "name": {
                "minLength": 1,
                "type": "string"
            },
            "version": {
                "minLength": 1,

```

```

                "type": "string"
            }
        },
        "required": [
            "name",
            "version",
            "is_direct",
            "hashes"
        ],
        "type": "object"
    },
    "maxItems": 256,
    "minItems": 1,
    "type": "array"
},
"bytes": {
    "maximum": 16777216,
    "minimum": 1,
    "type": "integer"
},
"install_count": {
    "maximum": 256,
    "minimum": 1,
    "type": "integer"
},
"pip_version": {
    "minLength": 1,
    "type": "string"

```

```

},
"role": {
    "enum": [
        "build-requirements",
        "candidate-install"
    ]
},
"sha256": {
    "$ref": "#/$defs/Digest"
}
},
"required": [
    "role",
    "sha256",
    "bytes",
    "pip_version",
    "install_count",
    "artifacts",
    "artifact_digest"
],
"type": "object"
},
"maxItems": 2,
"minItems": 2,
"type": "array"
}
},
"required": [

```

```

    "reports",
    "report_count",
    "artifact_digest"
],

```



```

    "type": "object"
  },
  "runtime_dependency_source": {
    "const": "online-index"
  },
  "sbom": {
    "additionalProperties": false,
    "properties": {
      "components": {
        "items": {
          "additionalProperties": false,
          "properties": {
            "name": {
              "minLength": 1,
              "type": "string"
            },
            "record_sha256": {
              "$ref": "#/$defs/Digest"
            },
            "version": {
              "minLength": 1,
              "type": "string"
            }
          }
        }
      }
    }
  },

```

```

    "required": [
      "name",
      "version",
      "record_sha256"
    ],
    "type": "object"
  },
  "maxItems": 256,
  "minItems": 1,
  "type": "array"
},
"digest": {
  "$ref": "#/$defs/Digest"
},
"format": {
  "const": "promin-spdx-lite-v1"
}
},
"required": [
  "format",
  "components",
  "digest"
],
"type": "object"
}
},
"required": [
  "environment",

```

```

    "installation_performed",
    "nested_venv_created",
    "runtime_dependency_source",
    "build_dependency_source",
    "network_disabled",
    "declared_python_requirement",
    "installed_distribution",
    "console_script",
    "command_invocations",
    "dependency_closure",
    "installed_environment_observation",
    "pip_report",
    "sbom",
    "license_closure",
    "build_backend"
  ],
  "type": "object"
},
"installed_command_verified": {
  "const": true
},
"invocation": {
  "$ref": "#/$defs/ReleaseEvidenceInvocation"
},
"observed_environment_closure_digest": {
  "$ref": "#/$defs/Digest"
}

```

```

},
"platform": {
  "enum": [
    "linux",
    "windows"
  ]
},
"platform_identity": {
  "additionalProperties": false,
  "properties": {
    "machine": {
      "minLength": 1,
      "type": "string"
    },
    "platform": {
      "enum": [
        "linux",
        "windows"
      ]
    },
    "platform_release": {
      "minLength": 1,
      "type": "string"
    },
    "platform_tags": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 64,
      "minItems": 1,
      "type": "array"
    },
    "python_abi_tag": {
      "minLength": 1,
      "type": "string"
    },
    "python_executable": {
      "minLength": 1,
      "type": "string"
    },
    "python_executable_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "python_implementation": {
      "minLength": 1,
      "type": "string"
    },
    "python_version": {
      "minLength": 1,
      "type": "string"
    },
    "sys_platform": {
      "minLength": 1,
      "type": "string"
    }
  }
},
"required": [
  "platform",
  "machine",
  "platform_release",
  "sys_platform",
  "platform_tags",
  "python_version",
  "python_implementation",
  "python_executable",
  "python_executable_sha256",
  "python_abi_tag"
],
"type": "object"
},
"portable_implementation_closure_digest": {
  "$ref": "#/$defs/Digest"
},
"producer": {
  "$ref": "#/$defs/ReleaseEvidenceProducer"
},

```

```

"producer_attestation": {
  "$ref": "#/$defs/EvidenceProducerAttestation"
},
"product_acceptance_pass": {
  "const": false
},
"product_public_approval": {
  "const": "not_approved"
},

```

```

"raw_artifact_manifest_digest": {
  "$ref": "#/$defs/Digest"
},
"record_type": {
  "const": "PlatformVerificationResult"
},
"result_digest": {
  "$ref": "#/$defs/Digest"
},
"status": {
  "const": "pass"
}

```

```

},
"required": [
  "record_type",
  "status",
  "candidate_binding_digest",
  "platform",
  "platform_identity",
  "exact_candidate_verified",
  "archive_sha256",
  "archive_bytes",
  "portable_implementation_closure_digest",
  "observed_environment_closure_digest",
  "install_mode",
  "installed_command_verified",
  "installation",
  "product_acceptance_pass",

```

```

"product_public_approval",
"producer",
"invocation",
"raw_artifact_manifest_digest",
"producer_attestation",
"result_digest"
],
"type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/PlatformVerificationResult | Валідатор: urn:promin:contracts:1#/\$defs/PlatformVerificationResult

65. PolicyDef

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: id, name, validator_id, severity, statement

Властивість	Обов'язков е	Структурне правило
id	так	enum=[68 items]
name	так	\$ref="#/\$defs/Id"
severity	так	enum=["P0", "P1", "P2"]
statement	так	type="string"; minLength=1
validator_id	так	\$ref="#/\$defs/Id"

```

{
  "additionalProperties": false,
  "properties": {
    "id": {
      "enum": [
        "P-001",
        "P-002",
        "P-003",
        "P-004",
        "P-005",
        "P-006",
        "P-007",

```

```
"P-008",  
"P-009",  
"P-010",  
"P-011",  
"P-012",  
"P-013",  
"P-014",  
"P-015",  
"P-016",  
"P-017",  
"P-018",  
"P-019",  
"P-020",  
"P-021",  
"P-022",  
"P-023",
```

```
"P-024",  
"P-025",  
"P-026",  
"P-027",  
"P-028",  
"P-029",  
"P-030",  
"P-031",  
"P-032",  
"P-033",  
"P-034",  
"P-035",  
"P-036",  
"P-037",  
"P-038",  
"P-039",  
"P-040",  
"P-041",  
"P-042",  
"P-043",  
"P-044",  
"P-045",  
"P-046",  
"P-047",  
"P-048",  
"P-049",  
"P-050",  
"P-051",
```

```
"P-052",  
"P-053",  
"P-054",  
"P-055",  
"P-056",  
"P-057",  
"P-058",  
"P-059",  
"P-060",  
"P-061",  
"P-062",  
"P-063",  
"P-064",  
"P-065",  
"P-066",  
"P-067",  
"P-068"
```

```
]  
,  
"name": {  
  "$ref": "#/$defs/Id"  
},  
"severity": {  
  "enum": [  
    "P0",  
    "P1",  
    "P2"  
  ]  
}
```

```
,  
"statement": {  
  "minLength": 1,  
  "type": "string"  
},
```

```

    "validator_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "id",
    "name",
    "validator_id",
    "severity",
    "statement"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/PolicyDef | Валідатор: urn:promin:contracts:1#/\$defs/PolicyDef

66. PolicySet

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, schema_ref, document, version, function, validator_registry_version, optimization_gate, state_machines, policies, derived_result_contracts, gate_run_definition_contract, research_draft_sanitation, semantic_export_contract

Властивість	Обов'язков	Структурне правило
derived_result_contracts	так	const=schema(keys:DoctorResult,OperationMetrics,ReadyFrontier,RetrievalPage,WorkCardProjection)
document	так	const="promin.policy-set"
function	так	const="Єдиний власник executable cross-record invariants через стабільні validator identifiers без дублювання budget values."
gate_run_definition_contract	так	const=schema(keys:artifact_binding_fields,artifact_record_digest_rule,authorization_scope_contract,command_binding,definition_change,definition_digest_rule,definition_owner,definition_provenance_fields,evidence_class_purpose_pairs,expected_owner_fields,implementation_closure_binding_rule,inline_definition_submission,mixed_evidence_classes,payload_digest_as_artifact_record_binding,persistence,product_credit_required_rule,provider_binding_digest_rule,run_binding_rule,status_outcome_rules,submitted_artifacts_define_expectations,target_binding_rule,task_binding,task_owner_digest_rule)
optimization_gate	так	const=schema(keys:base_predicate,formula,stress_predicate)
policies	так	const=[68 items]
record_type	так	const="PolicySet"
research_draft_sanitation	так	const=schema(keys:allowed_license_disposition,canonical_bytes_max,citation_required_rule,claim_digest_rule,claim_kinds,evidence_reference_rule,forbidden_normative_claim_classes,forbidden_requested_flag_mutations,metadata_only_rule,migration_rule,normative_candidate_rule,persistence,result_flags,result_persistence,result_record_type,source_binding,source_license_rule,source_payload_bytes_max,source_payload_total_bytes_max,verification_statuses)
schema_ref	так	const="urn:promin:contracts:1#/\$defs/PolicySet"
semantic_export_contract	так	const=schema(keys:export_kind,input_digests_max,input_digests_rule,output_artifact_kind,output_artifact_record_digest_rule,output_bytes_max,output_bytes_rule,output_media_type,output_record_type,output_records_max,package_or_archive,persistence,product_flags,record_order,required_bindings)
state_machines	так	const=schema(keys:capacity_release,lease,task)
validator_registry_version	так	const=1
version	так	const="1.0.0-alpha.1"

```

{
  "additionalProperties": false,
  "properties": {
    "derived_result_contracts": {
      "const": {
        "DoctorResult": {
          "archival_diagnostics_effect": "none",
          "claim_scope": "runtime-health-only",
          "component_status_values": [
            "healthy",
            "degraded",
            "failed",
            "incomplete"
          ],
          "current_components": [
            "core",

```

```

    "init",
    "providers",
    "recovery",
    "replay",
    "projection"
  ],
  "forbidden_allocation_scopes": [
    "batch",
    "goal",
    "per-run",
    "session",
    "workspace"
  ]

```

```

  ],
  "forbidden_metric_families": [
    "cost",
    "token",
    "usage"
  ],
  "live_only": true,
  "metrics": [
    "core_artifacts_verified",
    "event_batches",
    "init_records_verified",
    "projection_entities",
    "projection_relations",
    "provider_healthchecks_executed"
  ],
  "pass_credit": false,
  "product_acceptance_pass": false,
  "product_public_approval": "not_approved",
  "product_tree_scans": 0,
  "report_authoritative": false,
  "rollup_precedence": [
    "failed",
    "incomplete",
    "degraded",
    "healthy"
  ],
  "source_scope": "installed-core-and-live-operational-state-only"
},

```

```

"OperationMetrics": {
  "authoritative": false,
  "claim_scope": "single-command-observation-only",
  "forbidden_allocation_scopes": [
    "batch",
    "goal",
    "per-run",
    "session",
    "workspace"
  ],
  "forbidden_metric_families": [
    "cost",
    "token",
    "usage"
  ],
  "live_only": true,
  "metric_groups": [
    "duration_ms",
    "changed_records",
    "event_write",
    "projection_update",
    "runtime_checkpoint",
    "physical_payload_bytes",
    "bytes_per_changed_record"
  ]
},
"ReadyFrontier": {
  "blocking_finding_statuses": [

```

```

    "OPEN"
  ],
  "cycle": "reject",
  "dependency_relation_kind": "DEPENDS_ON",
  "eligible_task_state": "READY",
  "gate_currentness_bindings": [
    "activation_digest",
    "candidate_digest",
    "definition_digest"
  ]

```

```

    ],
    "gate_outcome": "pass",
    "gate_pass_credit": true,
    "live_only": true,
    "ordering": [
        "created_at-ascending",
        "task_id-ascending"
    ],
    "projection_authoritative": false,
    "required_dependency_state": "COMPLETED",
    "silent_truncation": false
},
"RetrievalPage": {
    "canonical_cursor_fields": [
        "stream_cursor",
        "next_stream_cursor"
    ],
    "forbidden_compatibility_fields": [
        "seed_cursor",
        "next_seed_cursor"
    ],
    "live_only": true,
    "projection_authoritative": false,
    "query_bytes_max": 4096,
    "refinement_hints_max": 8,
    "silent_truncation": false
},
"WorkCardProjection": {
    "continuation_exposed": false,
    "live_only": true,
    "projection_authoritative": false,
    "selected_task_only": true
}
}
},
"document": {
    "const": "promin.policy-set"
},
"function": {
    "const": "Єдиний власник executable cross-record invariants через стабільні validator identifiers без дублювання budget values."
},
"gate_run_definition_contract": {
    "const": {
        "artifact_binding_fields": [
            "artifact_id",
            "artifact_record_digest",
            "run_id",
            "run_digest"
        ],
        "artifact_record_digest_rule": "sha256(canonical finalized record with exact artifact and authoritative commit_binding)",
        "authorization_scope_contract": {
            "all_selector": "reject",
            "allowed_extra_kinds": [
                "project",
                "task"
            ],
            "lease_bound_task_selector_required": true,
            "project_items_max": 1,
            "target_scope_relation": "subset",
            "task_items_max": 1,
            "task_value_rule": "GateResult.task_id",
            "unrelated_selector": "reject"
        },
        "command_binding": "CommandRequest.definition_digest equals the immutable GateRunDefinition digest and the payload definition_digest",
        "definition_change": "reject after Task creation; define a new Task identity for changed requirements",
        "definition_digest_rule": "sha256(canonical GateRunDefinition)",
        "definition_owner": "Task.gate_run_definitions",
        "definition_provenance_fields": [
            "owner_kind",
            "owner_digest",
            "defined_at_head_digest"
        ],
        "evidence_class_purpose_pairs": {

```

```

"diagnostic": [
  "validator"
],
"gate": [
  "validator",
  "product-execution"
],
"product": [
  "product-execution"
]
},
"expected_owner_fields": [
  "expected_evidence_class",
  "expected_evidence_purpose",
  "product_credit_required",
  "target_kind",
  "target_digest",
  "target_scope",
  "candidate_digest",
  "policy_digest",
  "tool_digest",
  "implementation_closure_digest",
  "provider_binding_digest",
  "input_digests"
],
"implementation_closure_binding_rule": "GateRunDefinition, Run, and every exact bound Artifact
EvidenceBinding carry the same implementation_closure_digest",
"inline_definition_submission": "reject",
"mixed_evidence_classes": "reject",
"payload_digest_as_artifact_record_binding": "forbidden",
"persistence": "immutable-definition-input",

"product_credit_required_rule": "true requires the product purpose and product-execution class;
false may use any declared evidence class-purpose pair",
"provider_binding_digest_rule": "sha256(canonical installed provider-binding evidence array);
actual ProviderInvocationEvidence records remain exact immutable Artifact inputs and are independently
validated against the active provider bindings when present",
"run_binding_rule": "GateResult.run_id and run_digest and every GateEvidenceArtifactBinding.run_id
and run_digest bind the exact canonical Run record",
"status_outcome_rules": {
  "blocked": {
    "evidence_binding": "обов'язкові",
    "outcome": "blocked",
    "pass_credit": false,
    "reason": "обов'язкові"
  },
  "fail": {
    "evidence_binding": "обов'язкові",
    "outcome": "fail",
    "pass_credit": false,
    "reason": "необов'язкові"
  },
  "pass": {
    "diagnostic_pass_with_no_credit": true,
    "evidence_binding": "обов'язкові",
    "outcome": "pass",
    "pass_credit_rule": "true only when definition product_credit_required is true and every exact
bound Artifact is fresh, resolved, passing, product-execution evidence"
  },
  "skipped": {
    "evidence_binding": "forbidden",
    "outcome": "skipped",
    "pass_credit": false,
    "reason": "обов'язкові"
  }
}
},
"submitted_artifacts_define_expectations": false,
"target_binding_rule": "candidate target_digest equals candidate_digest; finding target_digest is
the immutable exact OPEN Finding record digest captured at definition time and is not recomputed from
later mutable Finding state",
"task_binding": "GateResult.task_id and definition_digest resolve one exact embedded Task
definition",
"task_owner_digest_rule": "sha256(canonical immutable Task fields excluding state and
gate_run_definitions)"
}
},
"optimization_gate": {
  "const": {

```



```

    "base_predicate": "NetValue>0",
    "formula": "NetValue=3*RiskReduction+2*AutomationGain+1.5*QueryabilityGain+ReuseGain-1.5*Implementa
tionCost-2*MaintenanceCost-2*CognitiveNoise-1.5*ReferenceCost-MigrationCost",
    "stress_predicate": "NetValue залишається >0, коли вигоди зменшуються на 20 відсотків, а витрати
зростають на 20 відсотків"
  },
  "policies": {
    "const": [
      {
        "id": "P-001",
        "name": "no-false-success",
        "severity": "P0",
        "statement": "Gate pass credit дорівнює true лише тоді, коли normalized status є pass і кожний
обов'язковий gate проходить.",
        "validator_id": "no_false_success"
      },
      {
        "id": "P-002",
        "name": "activation-binding",
        "severity": "P0",
        "statement": "Кожний нормативний запис прив'язує точну активну конфігурацію Core, preset, init і
profile.",
        "validator_id": "activation_binding"
      },
      {
        "id": "P-003",
        "name": "grant-validity",
        "severity": "P0",
        "statement": "Subject, capability, conjunctive scope, interval, nonce, exact claim digest, proof
chain, revocation state і Activation повністю збігаються.",
        "validator_id": "grant_validity"
      },
      {
        "id": "P-004",
        "name": "lease-fence",
        "severity": "P0",
        "statement": "Кожна мутація використовує поточну активну generation Lease та монотонний fencing
token.",
        "validator_id": "lease_fence"
      },
      {
        "id": "P-005",
        "name": "candidate-evidence-binding",
        "severity": "P0",
        "statement": "Every evidence Artifact requires an explicit diagnostic, gate, or product purpose
and binds Activation, candidate, policy, provider/tool, and input digests inside the immutable event
payload.",
        "validator_id": "candidate_evidence_binding"
      },
      {
        "id": "P-006",
        "name": "atomic-event-commit",
        "severity": "P0",
        "statement": "Одна команда створює один immutable batch через lock, compare-head, sync, atomic
replace і recovery.",
        "validator_id": "atomic_event_commit"
      },
      {
        "id": "P-007",
        "name": "projection-is-derived",
        "severity": "P0",
        "statement": "Видалення проєкції та deterministic replay не можуть змінити нормативне значення.",
        "validator_id": "projection_derived"
      },
      {
        "id": "P-008",
        "name": "minimal-graph",
        "severity": "P1",
        "statement": "Each verified raw file creates exactly one non-authoritative derived Artifact
proxy. Raw inventory creates no Task, READS, or PRODUCES record; Tasks and typed relations require
explicit semantic content.",
        "validator_id": "minimal_graph"
      },
      {
        "id": "P-009",
        "name": "bounded-workcard",
        "severity": "P0",

```

```

    "statement": "Every WorkCard stays within both Core hard ceilings and its selected profile byte,
entity, relation, fanout, and top-k budgets; every truncation is explicit and losslessly resumable.",
    "validator_id": "bounded_workcard"
  },
  {
    "id": "P-010",
    "name": "size-independent-init",
    "severity": "P0",
    "statement": "Ініціалізація виконує нуль product-tree scans і створює рівно п'ять init records.",
    "validator_id": "zero_scan_init"
  },
  {
    "id": "P-011",
    "name": "explicit-external-bindings",
    "severity": "P1",
    "statement": "Зовнішні стандарти та providers існують лише в explicit init records, прив'язаних
до digest.",
    "validator_id": "explicit_bindings"
  },
  {
    "id": "P-012",
    "name": "canonical-name-only",
    "severity": "P0",
    "statement": "Канонічна ідентичність - промін; aliases, redirects і compatibility names
відхиляються.",
    "validator_id": "canonical_name"
  },
  {
    "id": "P-013",
    "name": "safe-export",
    "severity": "P0",
    "statement": "Export використовує allowlist, рекурсивно обмежений, безпечний щодо symlink і
secrets та очищує paths.",
    "validator_id": "safe_export"
  },
  {
    "id": "P-014",
    "name": "closed-acknowledgement",
    "severity": "P0",
    "statement": "Лише чинний перехід до CLOSED з явним підтвердженням закриття звільняє слот
мутації.",
    "validator_id": "closed_ack"
  },
  {
    "id": "P-015",
    "name": "fail-closed-critical-fields",
    "severity": "P0",
    "statement": "Відсутні, невідомі, stale, downgraded або unbound critical fields спричиняють
відхилення до мутації.",
    "validator_id": "critical_fields"
  },
  {
    "id": "P-016",
    "name": "preset-capability-ceiling",
    "severity": "P0",
    "statement": "Preset може налаштовувати потреби в провайдерах і бюджети, але не може надавати
повноваження на дії.",
    "validator_id": "preset_no_authority"
  },
  {
    "id": "P-017",
    "name": "single-scan-inventory",
    "severity": "P1",
    "statement": "Один запит inventory виконує не більше одного повного проходу product tree; усі
проекції походять із його stream.",
    "validator_id": "single_scan"
  },
  {
    "id": "P-018",
    "name": "idempotent-commands",
    "severity": "P0",
    "statement": "Повторний idempotency key у межах тієї самої Activation і subject повертає
початковий результат або конфлікт.",
    "validator_id": "idempotent_commands"
  },
  {
    "id": "P-019",

```

```

        "name": "core-preset-separation",
        "severity": "P1",
        "statement": "Ідентичність Core bundle не включає замінні presets; Activation прив'язує вибраний
preset окремо.",
        "validator_id": "core_preset_separation"
    },
    {
        "id": "P-020",
        "name": "acyclic-bootstrap-authority",
        "severity": "P0",
        "statement": "Init config визначає root ceilings, але ніколи не створює pre-activation runtime
Grants.",
        "validator_id": "bootstrap_authority"
    },
    {
        "id": "P-021",
        "name": "normalization-collision-rejection",
        "severity": "P1",
        "statement": "Повторні JSON keys і колізії path або ID після NFC/casefold normalization
відхиляються.",
        "validator_id": "normalization_collision"
    },
    {
        "id": "P-022",
        "name": "provider-substitution-binding",
        "severity": "P1",
        "statement": "A provider identity or implementation-closure change invalidates the Activation
and every dependent continuation, evidence Artifact, projection, replay result, and current release
closure; it never silently changes authority.",
        "validator_id": "provider_substitution"
    },
    {
        "id": "P-023",
        "name": "exact-init-keyset",
        "severity": "P1",
        "statement": "Activation прив'язує рівно project.json, standards.json, technologies.json і
authority.json.",
        "validator_id": "exact_init_keyset"
    },
    {
        "id": "P-024",
        "name": "stale-projection-rejection",
        "severity": "P1",
        "statement": "Генерація status, next і WorkCard відхиляє проєкцію, source HEAD або Activation
якої відрізняється.",
        "validator_id": "stale_projection"
    },
    {
        "id": "P-025",
        "name": "one-command-one-batch",
        "severity": "P0",
        "statement": "Кожний committed batch прив'язує рівно один subject, validated command digest,
command intent digest, authorization digest та idempotency key.",
        "validator_id": "one_command_one_batch"
    },
    {
        "id": "P-026",
        "name": "deterministic-retrieval",
        "severity": "P1",
        "statement": "Однакові ranking scores розв'язуються детерміновано, а acceptance-critical facts
додаються через typed closure, не fuzzy ranking.",
        "validator_id": "deterministic_retrieval"
    },
    {
        "id": "P-027",
        "name": "strict-time-order",
        "severity": "P1",
        "statement": "Timestamps використовують canonical UTC seconds; порядок issued, heartbeat,
expiry, finish і decision перевіряється семантично.",
        "validator_id": "strict_time"
    },
    {
        "id": "P-028",
        "name": "exact-standard-files",
        "severity": "P1",
        "statement": "Вхідні Core і вибраний preset відхиляють symlink та неочікувані shadow files.",
        "validator_id": "exact_standard_files"
    },

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{
  "id": "P-029",
  "name": "root-authorization-is-bootstrap-only",
  "severity": "P0",
  "statement": "Root command authorization підписує один command intent і може видати лише перший Grant authority.manage; вона не може виконувати звичайні команди проєкту.",
  "validator_id": "root_authorization_limit"
},
{
  "id": "P-030",
  "name": "configuration-exact-idempotency",
  "severity": "P1",
  "statement": "Повторна ініціалізація успішна лише тоді, коли запитаний canonical configuration digest точно дорівнює наявній Activation; інакше повертається конфлікт.",
  "validator_id": "config_exact_idempotency"
},
{
  "id": "P-031",
  "name": "truthful-observability",
  "severity": "P1",
  "statement": "Команда може повідомляти про scan, healthcheck, validation або durability action лише тоді, коли дію фактично виконано та прив'язано до доказів.",
  "validator_id": "truthful_observability"
},
{
  "id": "P-032",
  "name": "depth-aware-retrieval-budget",
  "severity": "P1",
  "statement": "Effective retrieval seed count reserves entity and relation capacity for typed closure at the requested dependency depth; top_k is a ceiling, byte/entity/relation/fanout budgets all remain effective, and no removed or unknown budget field may bypass a ceiling.",
  "validator_id": "depth_aware_retrieval"
},
{
  "id": "P-033",
  "name": "lease-only-for-mutation",
  "severity": "P1",
  "statement": "Mutation WorkCards потребують поточної Lease; read-only WorkCards не повинні отримувати або нести mutation Lease.",
  "validator_id": "lease_only_for_mutation"
},
{
  "id": "P-034",
  "name": "command-capability-scope",
  "severity": "P0",
  "statement": "Кожна команда визначає рівно одну capability, власником якої є Core; requested scope зберігає кожне authorization constraint і явно містить кожну ціль впливу команди.",
  "validator_id": "command_capability_scope"
},
{
  "id": "P-035",
  "name": "grant-issuance-chain",
  "severity": "P0",
  "statement": "Кожний доказ Grant прив'язує claim_digest і є або єдиним bootstrap root issuance, або перевіреним team threshold, або активним issuer Grant з authority.manage.",
  "validator_id": "grant_issuance_chain"
},
{
  "id": "P-036",
  "name": "verified-team-signatures",
  "severity": "P0",
  "statement": "Team-signature fields are never valid without threshold evaluation by the signature adapter resolved from the exact technologies binding; process-local injected adapters are explicitly non-reconstructable and cannot prove a cross-process CLI workflow.",
  "validator_id": "verified_team_signatures"
},
{
  "id": "P-037",
  "name": "provider-preflight-before-activation",
  "severity": "P1",
  "statement": "Кожна обов'язкова provider identity і healthcheck успішно перевіряється до commit ініціалізації; оголошений provider не є доказом доступності.",
  "validator_id": "provider_preflight"
},
{
  "id": "P-038",
  "name": "canonical-state-machines",
  "severity": "P0",

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"statement": "Переходи Task і Lease приймаються лише state machines, власником яких є policy; generated lifecycle views не можуть вигадувати переходи.",

```
"validator_id": "state_machine_transition"
},
{
  "id": "P-039",
  "name": "decision-target-binding",
  "severity": "P0",
  "statement": "Кожний Decision називає одну типізовану ціль, а decision kind, requested scope,
candidate і authority узгоджені з цією ціллю.",
  "validator_id": "decision_target_binding"
},
{
  "id": "P-040",
  "name": "credit-requires-evidence",
  "severity": "P0",
  "statement": "Pass, fail, and blocked gate results cite at least one exact immutable Artifact
record reference; skipped results never receive pass credit.",
  "validator_id": "credit_requires_evidence"
},
{
  "id": "P-041",
  "name": "command-event-correspondence",
  "severity": "P0",
  "statement": "Один command batch прив'язує точне authorization claim і містить рівно одну
primary event, рівну validated command payload; auxiliary events є лише типізованими відношеннями, а event
IDs унікальні.",
  "validator_id": "command_event_correspondence"
},
{
  "id": "P-042",
  "name": "continuation-authorization-and-completeness",
  "severity": "P0",
  "statement": "A continuation is authenticated by a random per-Activation secret and binds the
exact subject, projection.read Grant claim, capability, requested scope, revocation state, normalized
query, deterministic cursor, Activation, HEAD, projection, ranking, complete budgets, depth,
implementation closure, issue time, and expiry; its complete page union emits every reachable entity and
relation exactly once without loss.",
  "validator_id": "continuation_completeness"
},
{
  "id": "P-043",
  "name": "candidate-snapshot-consistency",
  "severity": "P0",
  "statement": "The candidate recipe is applied in the single inventory pass; promin v1 permits
only provider-proven immutable VCS trees for credit, while best-effort observations are explicit and
receive no pass credit.",
  "validator_id": "candidate_snapshot_consistency"
},
{
  "id": "P-044",
  "name": "provider-adapter-dispatch",
  "severity": "P0",
  "statement": "Every technologies.json binding resolves to the adapter actually invoked and
carries a recomputed capability-specific implementation closure for the promin runtime, Python
interpreter, jsonschema or SQLite component where used, and adapter. Unsupported or drifted tuples reject,
and dependent evidence binds the exact implementation_closure_digest.",
  "validator_id": "provider_adapter_dispatch"
},
{
  "id": "P-045",
  "name": "current-release-closure",
  "severity": "P0",
  "statement": "A project-operational release Decision is immutable history; current product
release eligibility is derived and becomes false after a blocker, failed gate, stale or missing evidence,
implementation drift, or Activation drift. It is separate from the external StandardReleaseDecision for
promin distribution.",
  "validator_id": "current_release_closure"
},
{
  "id": "P-046",
  "name": "finding-disposition-evidence",
  "severity": "P0",
  "statement": "Every Finding cites exact immutable Artifact record references. Every resolve or
waive Decision cites evidence bound to the exact canonical Finding digest; waiver also requires an
exception-policy digest, future expiry, and separation of duties.",
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    "validator_id": "finding_disposition_evidence"
  },
  {
    "id": "P-047",
    "name": "candidate-delta-scope",
    "severity": "P0",
    "statement": "A diff Artifact binds base and new Candidate digests, the exact WorkCard digest,
and the complete changed-path set; every changed path is contained by both Task allowed_paths and the
authorizing Grant effect scope.",
    "validator_id": "candidate_delta_scope"
  },
  {
    "id": "P-048",
    "name": "verified-state-checkpoint",
    "severity": "P1",
    "statement": "Events remain the sole authority. A normal command appends 0(batch), incrementally
updates changed projection rows/shards, and advances a verified bounded replay tail. Full derived-state
compaction occurs only after declared batch/byte thresholds or explicit recovery, publishes crash-safely
against exact HEAD, and missing, stale, or corrupt checkpoints require full verified fallback with equal
semantic digest.",
    "validator_id": "verified_state_checkpoint"
  },
  {
    "id": "P-049",
    "name": "profile-default-depth",
    "severity": "P1",
    "statement": "A profile default_dependency_depth applies only when depth is omitted; explicit
depth is accepted through the single Core dependency_depth_hard_max of 12 and remains subject to all
retrieval budgets.",
    "validator_id": "profile_default_depth"
  },
  {
    "id": "P-050",
    "name": "canonical-distribution-archive",
    "severity": "P0",
    "statement": "The sole package builder emits immutable candidate bytes and an external
StandardReleaseCandidateBinding. VERSION.json describes SemVer and package identity only; approval is
derived later and never changes the candidate ZIP.",

```

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    "validator_id": "canonical_distribution_archive"
  },
  {
    "id": "P-051",
    "name": "verified-inventory-provenance",
    "severity": "P0",
    "statement": "Public rebuild accepts only a verified InventoryResult or its persisted manifest;
caller provenance fields are reserved, data_class is forced to untrusted-source, and runtime injects the
exact path and digest.",
    "validator_id": "verified_inventory_provenance"
  },
  {
    "id": "P-052",
    "name": "supported-command-surface",
    "severity": "P1",
    "statement": "The v1 preset advertises exactly init, doctor, status, next, validate, and
continue; no optional or administrative command catalogue may claim an absent executable path.",
    "validator_id": "supported_command_surface"
  },
  {
    "id": "P-053",
    "name": "profile-bound-scale",
    "severity": "P0",
    "statement": "Physical 100k evidence binds content-addressed raw inventory, query, continuation,
process-sample, phase-log, and operation-metric artifacts for at least 198999 explicitly recorded
Core-valid typed Relations and a representative mixed workload. The verifier recomputes every summary and
fails closed on truthful process/invoke status, commit p95/p99, bytes per changed record, checkpoint
count, semantic-ingestion time, search latency, incremental inventory memory amplification, absolute peak
RSS, database bytes, projection amplification, semantic inflation, continuation completeness, or
traversal-pass ceiling.",
    "validator_id": "profile_bound_scale"
  },
  {
    "id": "P-054",
    "name": "cross-platform-release-evidence",
    "severity": "P0",
    "statement": "The exact candidate ZIP must pass authenticated Windows and Linux platform
observations, clean-install, package, handle-relative path/race, locking, crash, replay, physical scale,
offline installed no-degradation, online compatibility, dependency/SBOM/RECORD/license closure, and typed
PDF gates. One StandardReleaseEvidenceManifest physically resolves exactly seven unique authority roles,

```

the exact non-authoritative eight-lane CPython 3.13/3.14 online/offline matrix, all four CPython 3.14 supplemental records, and every nested physical record by path, SHA-256, and byte count. All producer attestations bind the same candidate and participate in one replay/SOD/chronology graph; matrix_authoritative, pass_credit, acceptance_pass, product_acceptance_pass, and public approval remain false.",

```

    "validator_id": "cross_platform_release"
  },
  {
    "id": "P-055",
    "name": "standard-release-separation",
    "severity": "P0",
    "statement": "An external StandardReleaseDecision has explicit approve or reject outcome, binds candidate and evidence manifest digests, follows the manifest-derived maximum evidence completion time, stays within 300 seconds of verifier clock skew, and verifies decider, standard.distribute capability, independently pinned trust root, Ed25519 provider, event-time key validity, nonce, signed claim digest, and signature. The signed decision remains immutable history; current eligibility is derived; product_acceptance_pass is false and product_public_approval is not_approved.",
    "validator_id": "standard_release_separation"
  },
  {
    "id": "P-056",
    "name": "required-gates-fail-closed",
    "severity": "P0",
    "statement": "No-degradation runs the complete required suite only through the clean installed interpreter outside the install source, with isolated mode, cleared PYTHONPATH, import-origin assertion, zero skipped tests, and full hash-bound JUnit/stdout/stderr bytes. Distribution acceptance executes every required positive, mutation, concurrency, crash, replay, path, package, platform, Python-minor, offline/online, scale, search, PDF, evidence-resolution, chronology, and signature gate; a skipped, missing, stale, blocked, failed, unsigned, unresolved, source-shadowed, or unpinned gate receives no pass credit.",
    "validator_id": "required_gates_fail_closed"
  },
  {
    "id": "P-057",
    "name": "research-draft-sanitation",
    "severity": "P0",
    "statement": "Research crosses into normative owners only through the non-persistent MIGRATE_FACTS_NOT_FILES intake: every cited source is digest- and license-plan-bound, citation-required claims cite allowed sources or remain blocked with a reason, and marketing, proof, distribution, acceptance, or public-approval claims cannot become normative text.",
    "validator_id": "research_draft_sanitation"
  },
  {
    "id": "P-058",
    "name": "doctor-result-is-live-non-authority",
    "severity": "P0",
    "statement": "DoctorResult is an exact live-only derived health report with report_authoritative=false, pass_credit=false, product_acceptance_pass=false, product_public_approval=not approved, zero product scans, and only its Core-owned operational metrics; token, cost, usage, and goal/session/batch/workspace/per-run allocation are forbidden.",
    "validator_id": "doctor_result_non_authority"
  },
  {
    "id": "P-059",
    "name": "semantic-inheritance-is-structural",
    "severity": "P0",
    "statement": "The compiled projection preserves the complete SemanticModel owner: admission, five independent granularities, promotion rules, exact entity purposes and admission rules, exact relation domain/range/rules, and structured provider forbidden boundaries.",
    "validator_id": "semantic_inheritance"
  },
  {
    "id": "P-060",
    "name": "authority-owner-is-structural",
    "severity": "P0",
    "statement": "AuthorityModel is the single owner of the capability catalogue, delegation depth, and exact separation-of-duty pairs; runtime policy may derive from that Activation-bound owner but may not substitute hardcoded authority facts.",
    "validator_id": "authority_owner"
  },
  {
    "id": "P-061",
    "name": "gate-run-definition-is-independent",
    "severity": "P0",
    "statement": "An immutable GateRunDefinition independently owns the expected evidence class, product-credit requirement, exact target scope, and candidate, policy, tool, provider, and input digests. A run or gate command binds its exact definition digest before submitted Artifacts are evaluated.",
    "validator_id": "gate_run_definition_integrity"
  }

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```

    },
    {
      "id": "P-062",
      "name": "authority-runtime-policy-is-derived-live-only",
      "severity": "P0",
      "statement": "AuthorityRuntimePolicy is rebuilt from the installed AuthorityModel for every
mutation and replay, binds the exact owner digest, ordered capability IDs, structured separation-of-duty
rules and delegation ceiling, and is never persisted or imported as authority.",
      "validator_id": "authority_runtime_policy_projection"
    },
    {
      "id": "P-063",
      "name": "semantic-export-is-export-only",
      "severity": "P0",
      "statement": "SemanticExport is a strict bounded export-only semantic-state record binding the
exact Candidate, Activation, HEAD, Core, preset, implementation closure, providers, inputs, and output. It
is neither a package nor archive and cannot carry product acceptance, pass credit, public approval, or
release approval.",
      "validator_id": "semantic_export_integrity"
    },
    {
      "id": "P-064",
      "name": "operation-metrics-are-live-observations",
      "severity": "P0",
      "statement": "OperationMetrics is a strict live-only non-authoritative observation for one
command. It permits only Core-owned duration, changed-record, write, projection, checkpoint, and
physical-byte fields and forbids token, cost, usage, and goal/session/batch/workspace/per-run allocation.",
      "validator_id": "operation_metrics_non_authority"
    },
    {
      "id": "P-065",
      "name": "workcard-projection-is-v2-live-only",
      "severity": "P0",
      "statement": "RetrievalPage is the strict bounded live-only v2 retrieval result and exclusively
owns stream_cursor, next_stream_cursor, and retrieval continuation. WorkCardProjection is internal
selected-Task materialization and exposes no continuation or cursor fields.",
      "validator_id": "workcard_projection_v2"
    },
    {
      "id": "P-066",
      "name": "event-store-policy-is-derived-live-only",
      "severity": "P0",
      "statement": "EventStorePolicy is rebuilt from the installed AuthorityModel before
normalization, mutation, and replay. It binds exact command and batch bounds, envelope and commitment
fields, lease-bound commands and states, primary-event mapping, and UTC-second timestamps, with no
defaults or nullable critical fields.",
      "validator_id": "event_store_policy_projection"
    },
    {
      "id": "P-067",
      "name": "depends-on-graph-is-acyclic",
      "severity": "P0",
      "statement": "Every command, import, replay, and rebuild rejects a DEPENDS_ON cycle. Relation
domain and range validity alone cannot admit a cyclic dependency graph.",
      "validator_id": "dependency_graph_acyclicity"
    },
    {
      "id": "P-068",
      "name": "ready-frontier-is-current-derived-state",
      "severity": "P0",
      "statement": "ReadyFrontier is derived live-only from the current acyclic DEPENDS_ON graph. A
READY Task is returned only after every dependency is currently COMPLETED, every required GateResult is
current passing credit, and no current OPEN Finding blocks it; ordering and continuation are deterministic
and truncation is explicit.",
      "validator_id": "ready_frontier_projection"
    }
  ],
  "record_type": {
    "const": "PolicySet"
  },
  "research_draft_sanitation": {
    "const": {
      "allowed_license_disposition": "allowed",
      "canonical_bytes_max": 1048576,
      "citation_required_rule": "source-refs-or-blocked-with-reason",
      "claim_digest_rule": "sha256(canonical complete claim excluding claim_digest and derived

```



```

normative_use_allowed)",
  "claim_kinds": [
    "research_question",
    "hypothesis",
    "assumption",
    "evidence_claim",
    "blocked_claim"
  ],
  "evidence_reference_rule": "citation_refs identify research sources; evidence_refs identify
independently resolved Artifact records; the sets are semantically distinct",
  "forbidden_normative_claim_classes": [
    "distribution",
    "marketing",
    "proof"
  ],
  "forbidden_requested_flag_mutations": [
    "acceptance_pass",
    "pass_credit",
    "product_acceptance_pass",
    "product_public_approval",
    "public_release_approved"
  ],
  "metadata_only_rule": "missing payload receipt is representable only by source_mode metadata-only
and every referencing claim remains blocked, non-authoritative, and normative_use_allowed=false",
  "migration_rule": "MIGRATE_FACTS_NOT_FILES",
  "normative_candidate_rule": "only a source-bound or verified evidence_claim classified fact with
eligible citations and resolved evidence_refs may become a sanitized normative candidate",
  "persistence": "transient-ingress-only",
  "result_flags": {
    "pass_credit": false,
    "product_acceptance_pass": false,
    "product_public_approval": "not_approved",
    "public_release_approved": false
  },
  "result_persistence": "derived-live-only",
  "result_record_type": "SanitizedResearchDraft",
  "source_binding": "source-id+trusted-source-locator+provenance-digest+artifact-id+artifact-record-d
igest+recomputed-content-digest+size+media-type+active-license-plan-digest",
  "source_license_rule": "allowed only when the active LicensesPlan binding review_state is reviewed
or source-verified",
  "source_payload_bytes_max": 16777216,
  "source_payload_total_bytes_max": 67108864,
  "verification_statuses": [
    "unverified",
    "source-bound",
    "verified",
    "blocked"
  ]
},
},
"schema_ref": {
  "const": "urn:promin:contracts:1#/$defs/PolicySet"
},
"semantic_export_contract": {
  "const": {
    "export_kind": "semantic-state",
    "input_digests_max": 128,
    "input_digests_rule": "sorted unique sha256(canonical exported record) values for the exact
replay-derived record set",
    "output_artifact_kind": "evidence",
    "output_artifact_record_digest_rule": "sha256(canonical finalized record with exact artifact and
authoritative commit_binding)",
    "output_bytes_max": 1073741824,
    "output_bytes_rule": "exact canonical SemanticStatePayload bytes stored in content-addressed
storage",
    "output_media_type": "application/vnd.promin.semantic-state+json",
    "output_record_type": "SemanticStatePayload",
    "output_records_max": 400000,
    "package_or_archive": false,
    "persistence": "export-only",
    "product_flags": {
      "pass_credit": false,
      "product_acceptance_pass": false,
      "product_public_approval": "not_approved",
      "public_release_approved": false
    },
    "record_order": "record-digest-ascending after exact digest deduplication",
  }
}

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```

"required_bindings": [
  "candidate_digest",
  "activation_digest",
  "head_event_id",
  "head_sequence",
  "head_digest",
  "core_bundle_digest",
  "preset_digest",
  "implementation_closure_digest",
  "provider_binding_digest",
  "input_digests",
  "output_artifact_id",
  "output_artifact_record_digest",
  "output_digest",
  "output_size_bytes",
  "output_media_type"
]

```

```

}
},
"state_machines": {
  "const": {
    "capacity_release": {
      "closed_release": {
        "requires_field": "close_ack",
        "slot_reusable": true,
        "state": "CLOSED"
      },
      "terminal_reconciliation": {
        "fencing_token_binding": "exact-lease-fencing-token",
        "field": "capacity_reconciliation",
        "generation_binding": "exact-lease-generation",
        "manager_capability": "lease.manage",
        "required_fields": [
          "reconciled_by",
          "grant_id",
          "grant_claim_digest",
          "reconciled_at",
          "generation",
          "fencing_token"
        ],
        "slot_reusable_after_record": true,
        "states": [
          "EXPIRED",
          "REVOKED"
        ]
      }
    }
  }
}

```

```

},
"unreconciled_slot_reusable": false
},
"lease": {
  "ACTIVE": [
    "CLOSING",
    "EXPIRED",
    "REVOKED"
  ],
  "CLOSED": [],
  "CLOSING": [
    "CLOSED",
    "EXPIRED",
    "REVOKED"
  ],
  "EXPIRED": [],
  "REVOKED": []
},
"task": {
  "BLOCKED": [
    "READY",
    "CANCELLED"
  ],
  "CANCELLED": [],
  "COMPLETED": [],
  "LEASED": [
    "RUNNING",
    "READY",
    "BLOCKED",
    "CANCELLED"
  ],
  "PLANNED": [

```

```

        "READY",
        "BLOCKED",
        "CANCELLED"
    ],
    "READY": [
        "LEASED",
        "BLOCKED",
        "CANCELLED"
    ],
    "RUNNING": [
        "COMPLETED",
        "BLOCKED",
        "CANCELLED"
    ]
  }
},
"validator_registry_version": {
  "const": 1
},
"version": {
  "const": "1.0.0-alpha.1"
}
},
"required": [
  "record_type",
  "schema_ref",
  "document",
  "version",
  "function",
  "validator_registry_version",
  "optimization_gate",
  "state_machines",
  "policies",
  "derived_result_contracts",
  "gate_run_definition_contract",
  "research_draft_sanitation",
  "semantic_export_contract"
],
"type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/PolicySet | Валідатор: urn:promin:contracts:1#/\$defs/PolicySet

67. Preset

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, schema_ref, document, version, function, preset_id, default_profile, base_user_commands, required_provider_capabilities, optional_provider_capabilities, profiles, semantic_programming, forbidden_implicit_behavior

Властивість	Обов'язков е	Структурне правило
base_user_commands	так	type="array"; const=["init", "doctor", "status", "next", "validate", "continue", "audit", "refresh", "context", "skills"]
default_profile	так	\$ref="#/\$defs/Id"
document	так	const="promin.preset"
forbidden_implicit_behavior	так	type="array"; minItems=1; maxItems=16; uniqueItems=true
function	так	type="string"; minLength=1
optional_provider_capabilities	так	type="array"; minItems=0; maxItems=16; uniqueItems=true
preset_id	так	\$ref="#/\$defs/Id"
profiles	так	type="object"; additionalProperties=schema(keys:\$ref)
record_type	так	const="Preset"
required_provider_capabilities	так	type="array"; minItems=1; maxItems=16; uniqueItems=true
schema_ref	так	const="urn:promin:contracts:1#/\$defs/Preset"
semantic_programming	так	type="object"
version	так	const="1.0.0-alpha.1"
{ "additionalProperties": false,		

```

"properties": {
  "base_user_commands": {
    "const": [
      "init",
      "doctor",
      "status",
      "next",
      "validate",
      "continue",
      "audit",
      "refresh",
      "context",
      "skills"
    ],
    "type": "array"
  },
  "default_profile": {
    "$ref": "#/$defs/Id"
  },
  "document": {
    "const": "promin.preset"
  },
  "forbidden_implicit_behavior": {
    "items": {
      "$ref": "#/$defs/Id"
    },

```

```

    "maxItems": 16,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "function": {
    "minLength": 1,
    "type": "string"
  },
  "optional_provider_capabilities": {
    "items": {
      "$ref": "#/$defs/Id"
    },
    "maxItems": 16,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "preset_id": {
    "$ref": "#/$defs/Id"
  },
  "profiles": {
    "additionalProperties": {
      "$ref": "#/$defs/Profile"
    },
    "minProperties": 1,
    "type": "object"
  },
},

```

```

"record_type": {
  "const": "Preset"
},
"required_provider_capabilities": {
  "items": {
    "$ref": "#/$defs/Id"
  },
  "maxItems": 16,
  "minItems": 1,
  "type": "array",
  "uniqueItems": true
},
"schema_ref": {
  "const": "urn:promin:contracts:1#/$defs/Preset"
},
"semantic_programming": {
  "minProperties": 2,
  "type": "object"
},
"version": {
  "const": "1.0.0-alpha.1"
}
},
"required": [

```

```
"record_type",
"schema_ref",
"document",
"version",

"function",
"preset_id",
"default_profile",
"base_user_commands",
"required_provider_capabilities",
"optional_provider_capabilities",
"profiles",
"semantic_programming",
"forbidden_implicit_behavior"
],
"type": "object"
}
```

Власник: core/contracts.schema.json#/\$defs/Preset | Валідатор: urn:promin:contracts:1#/\$defs/Preset

68. Profile

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: model_tier, max_context_bytes, max_entities, max_relations, max_fanout_per_entity, top_k, max_parallel_tasks, default_dependency_depth

Властивість	Обов'язков е	Структурне правило
default_dependency_depth	так	type="integer"; minimum=1; maximum=12
max_context_bytes	так	type="integer"; minimum=1024
max_entities	так	type="integer"; minimum=1
max_fanout_per_entity	так	type="integer"; minimum=1
max_parallel_tasks	так	type="integer"; minimum=1
max_relations	так	type="integer"; minimum=0
model_tier	так	enum=["weak-local", "capable", "strong"]
top_k	так	type="integer"; minimum=1

```
{
  "additionalProperties": false,
  "properties": {
    "default_dependency_depth": {
      "maximum": 12,
      "minimum": 1,
      "type": "integer"
    },
    "max_context_bytes": {
      "minimum": 1024,
      "type": "integer"
    },
    "max_entities": {
      "minimum": 1,
      "type": "integer"
    },
    "max_fanout_per_entity": {
      "minimum": 1,
      "type": "integer"
    },
    "max_parallel_tasks": {
      "minimum": 1,
      "type": "integer"
    },
    "max_relations": {
      "minimum": 0,
      "type": "integer"
    }
  },

```

```
  "model_tier": {
    "enum": [
      "weak-local",
      "capable",
      "strong"
    ]
  },
  "top_k": {
    "minimum": 1,

```

```

    "type": "integer"
  },
  "required": [
    "model_tier",
    "max_context_bytes",
    "max_entities",
    "max_relations",
    "max_fanout_per_entity",
    "top_k",
    "max_parallel_tasks",
    "default_dependency_depth"
  ],
  "type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/Profile](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/Profile](#)

69. ProjectInit

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, project_id, roots, candidate_recipe, preset_id, operating_profile, intent, resolved_profile, orchestration, telemetry

Властивість	Обов'язков е	Структурне правило
candidate_recipe	так	\$ref="#/\$defs/CandidateRecipe"
intent	так	type="object"; additionalProperties=false
operating_profile	так	\$ref="#/\$defs/Id"
orchestration	так	type="object"; additionalProperties=false
preset_id	так	\$ref="#/\$defs/Id"
project_id	так	\$ref="#/\$defs/Id"
record_type	так	const="ProjectInit"
resolved_profile	так	type="object"; additionalProperties=false
roots	так	type="array"; minItems=1; maxItems=16
telemetry	так	type="object"; additionalProperties=false

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_recipe": {
      "$ref": "#/$defs/CandidateRecipe"
    },
    "intent": {
      "additionalProperties": false,
      "properties": {
        "autonomy": {
          "enum": [
            "ask",
            "safe-auto",
            "unsafe-auto"
          ]
        }
      },
      "constraints": {
        "items": {
          "maxLength": 4096,
          "minLength": 1,
          "type": "string"
        },
        "maxItems": 256,
        "type": "array",
        "uniqueItems": true
      },
      "deliverables": {
        "items": {
          "maxLength": 4096,
          "minLength": 1,
          "type": "string"
        },
        "maxItems": 256,
        "type": "array",
        "uniqueItems": true
      }
    }
  }
}

```

```

},
"goal": {
  "maxLength": 8192,
  "minLength": 1,
  "type": "string"
},
"non_goals": {
  "items": {
    "maxLength": 4096,
    "minLength": 1,
    "type": "string"
  },
  "maxItems": 256,
  "type": "array",
  "uniqueItems": true
},
"project_mode": {
  "enum": [
    "greenfield",
    "existing-code",
    "hybrid"

```

```

]
},
"references": {
  "items": {
    "maxLength": 4096,
    "minLength": 1,
    "type": "string"
  },
  "maxItems": 256,
  "type": "array",
  "uniqueItems": true
},
"reporting_language": {
  "enum": [
    "uk",
    "en"
  ]
},
"success_criteria": {
  "items": {
    "maxLength": 4096,
    "minLength": 1,
    "type": "string"
  },
  "maxItems": 256,
  "type": "array",
  "uniqueItems": true
},

```

```

"work_sources": {
  "items": {
    "maxLength": 4096,
    "minLength": 1,
    "type": "string"
  },
  "maxItems": 256,
  "type": "array",
  "uniqueItems": true
}
},
"required": [
  "project_mode",
  "goal",
  "success_criteria",
  "constraints",
  "non_goals",
  "deliverables",
  "references",
  "work_sources",
  "reporting_language",
  "autonomy"
],
"type": "object"
},
"operating_profile": {
  "$ref": "#/$defs/Id"
},

```

```

"orchestration": {
  "additionalProperties": false,
  "properties": {
    "enabled": {
      "const": true
    },
    "enforcement": {
      "const": "обов'язкові"
    },
    "host_source": {
      "enum": [
        "generic-agent-envelope",
        "explicit-promin",
        "inherited-compatible-host"
      ]
    },
    "model_routing": {
      "const": "cheapest-adequate-with-evidence-escalation"
    },
    "unregistered_work": {
      "const": "external-proposal-only"
    }
  },
  "required": [
    "enabled",
    "enforcement",
    "unregistered_work",
    "model_routing",

```

```

    "host_source"
  ],
  "type": "object"
},
"preset_id": {
  "$ref": "#/$defs/Id"
},
"project_id": {
  "$ref": "#/$defs/Id"
},
"record_type": {
  "const": "ProjectInit"
},
"resolved_profile": {
  "additionalProperties": false,
  "properties": {
    "authority_effect": {
      "const": "none"
    }
  },
  "detected_technologies": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "confidence": {
          "maximum": 1,
          "minimum": 0,
          "type": "number"
        }
      }
    },

```

```

    "sources": {
      "items": {
        "maxLength": 4096,
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 64,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "technology": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "technology",
    "sources",
    "confidence"
  ],
  "type": "object"
},

```



```

    "maxItems": 128,
    "type": "array"
  },
  "layers": {
    "items": {
      "$ref": "#/$defs/Id"

```

```

    },
    "maxItems": 32,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "operation_profiles": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "model_tier": {
          "enum": [
            "tool-only",
            "micro",
            "standard",
            "strong",
            "critical-review"
          ]
        },
        "operation": {
          "$ref": "#/$defs/Id"
        },
        "reason": {
          "maxLength": 2048,
          "minLength": 1,
          "type": "string"
        }
      }
    },

```

```

    "required": [
      "operation",
      "model_tier",
      "reason"
    ],
    "type": "object"
  },
  "maxItems": 64,
  "type": "array"
},
"profile_digest": {
  "$ref": "#/$defs/Digest"
},
"resolution": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "confidence": {
        "maximum": 1,
        "minimum": 0,
        "type": "number"
      },
      "profile_id": {
        "$ref": "#/$defs/Id"
      },
      "reason": {
        "maxLength": 2048,
        "minLength": 1,

```

```

        "type": "string"
      }
    },
    "required": [
      "profile_id",
      "reason",
      "confidence"
    ],
    "type": "object"
  },
  "maxItems": 64,
  "minItems": 1,
  "type": "array"
},
"revision": {

```

```

        "minimum": 1,
        "type": "integer"
    },
    "required": [
        "revision",
        "layers",
        "detected_technologies",
        "operation_profiles",
        "resolution",
        "authority_effect",
        "profile_digest"
    ],

```

```

        "type": "object"
    },
    "roots": {
        "items": {
            "$ref": "#/$defs/ProjectRoot"
        },
        "maxItems": 16,
        "minItems": 1,
        "type": "array"
    },
    "telemetry": {
        "additionalProperties": false,
        "properties": {
            "enabled": {
                "type": "boolean"
            },
            "export_enabled": {
                "const": false
            },
            "local_only": {
                "const": true
            },
            "max_log_bytes": {
                "maximum": 67108864,
                "minimum": 65536,
                "type": "integer"
            },
            "retention_records": {

```

```

                "maximum": 100000,
                "minimum": 100,
                "type": "integer"
            },
            "store_full_prompts": {
                "const": false
            },
            "store_secrets_or_pii": {
                "const": false
            },
            "store_source_bodies": {
                "const": false
            }
        },
        "required": [
            "enabled",
            "local_only",
            "export_enabled",
            "max_log_bytes",
            "retention_records",
            "store_full_prompts",
            "store_source_bodies",
            "store_secrets_or_pii"
        ],
        "type": "object"
    },
    "required": [

```

```

        "record_type",
        "project_id",
        "roots",
        "candidate_recipe",
        "preset_id",
        "operating_profile",
        "intent",
        "resolved_profile",

```

```

    "orchestration",
    "telemetry"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/ProjectInit | Валідатор: urn:promin:contracts:1#/\$defs/ProjectInit

70. ProjectRoot

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: path, kind

Властивість	Обов'язков е	Структурне правило
kind	так	const="product"
path	так	\$ref="#/\$defs/RelativePath"
<pre> { "additionalProperties": false, "properties": { "kind": { "const": "product" }, "path": { "\$ref": "#/\$defs/RelativePath" } }, "required": ["path", "kind"], "type": "object" } </pre>		

Власник: core/contracts.schema.json#/\$defs/ProjectRoot | Валідатор: urn:promin:contracts:1#/\$defs/ProjectRoot

71. ProjectionLimits

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, token_version, ranking_algorithm_id, traversal_algorithm_id, dependency_depth_hard_max, continuation_ttl_seconds_max, selected_profile_id, selected_profile_digest, persistent_entity_types, default_budget, hard_budget, default_depth, depth_min, depth_max, default_ttl_seconds, ttl_min_seconds, ttl_max_seconds, max_token_bytes, max_continuation_state_bytes, max_query_bytes, min_result_bytes, max_resume_binding_fields, max_resume_binding_key_chars, max_resume_binding_value_bytes, max_resume_binding_bytes, required_resume_binding_fields, policy_digest

Властивість	Обов'язков е	Структурне правило
continuation_ttl_seconds_max	так	const=3600
default_budget	так	\$ref="#/\$defs/Budget"
default_depth	так	type="integer"; minimum=1; maximum=12
default_ttl_seconds	так	const=900
dependency_depth_hard_max	так	const=12
depth_max	так	const=12
depth_min	так	const=1
hard_budget	так	\$ref="#/\$defs/Budget"
max_continuation_state_bytes	так	const=16384
max_query_bytes	так	const=4096
max_resume_binding_bytes	так	const=4096
max_resume_binding_fields	так	const=8
max_resume_binding_key_chars	так	const=64
max_resume_binding_value_bytes	так	const=512
max_token_bytes	так	const=256
min_result_bytes	так	const=1024

Властивість	Обов'язков е	Структурне правило
persistent_entity_types	так	const=["Завдання", "Candidate", "Artifact", "Run", "Finding", "Decision", " GateResult", "Grant", "Опенда"]
policy_digest	так	\$ref="#/\$defs/Digest"
ranking_algorithm_id	так	const="bm25-v1"
record_type	так	const="ProjectionLimits"
required_resume_binding_fields	так	const=["activation_digest", "capability_id", "grant_claim_digest", "grant_id", " implementation_closure_digest", "requested_scope_digest", " revocation_epoch", "subject_id"]
selected_profile_digest	так	\$ref="#/\$defs/Digest"
selected_profile_id	так	\$ref="#/\$defs/Id"
token_version	так	const=2
traversal_algorithm_id	так	const="typed-bfs-v2"
ttl_max_seconds	так	const=3600
ttl_min_seconds	так	const=1
<pre> { "additionalProperties": false, "properties": { "continuation_ttl_seconds_max": { "const": 3600 }, "default_budget": { "\$ref": "#/\$defs/Budget" }, "default_depth": { "maximum": 12, "minimum": 1, "type": "integer" }, "default_ttl_seconds": { "const": 900 }, "dependency_depth_hard_max": { "const": 12 }, "depth_max": { "const": 12 }, "depth_min": { "const": 1 }, "hard_budget": { "\$ref": "#/\$defs/Budget" }, "max_continuation_state_bytes": { "const": 16384 }, "max_query_bytes": { "const": 4096 }, "max_resume_binding_bytes": { "const": 4096 }, "max_resume_binding_fields": { "const": 8 }, "max_resume_binding_key_chars": { "const": 64 }, "max_resume_binding_value_bytes": { "const": 512 }, "max_token_bytes": { "const": 256 }, "min_result_bytes": { "const": 1024 }, "persistent_entity_types": { </pre>		

```

    "const": [
      "Завдання",
      "Candidate",
      "Artifact",
      "Run",
      "Finding",
      "Decision",
      "GateResult",
      "Grant",
      "Оренда"
    ]
  },
  "policy_digest": {
    "$ref": "#/$defs/Digest"
  },
  "ranking_algorithm_id": {
    "const": "bm25-v1"
  },
  "record_type": {
    "const": "ProjectionLimits"
  },
  "required_resume_binding_fields": {
    "const": [
      "activation_digest",
      "capability_id",
      "grant_claim_digest",
      "grant_id",
      "implementation_closure_digest",
      "requested_scope_digest",
      "revocation_epoch",
      "subject_id"
    ]
  },
  "selected_profile_digest": {
    "$ref": "#/$defs/Digest"
  },
  "selected_profile_id": {
    "$ref": "#/$defs/Id"
  },
  "token_version": {
    "const": 2
  },
  "traversal_algorithm_id": {
    "const": "typed-bfs-v2"
  },
  "ttl_max_seconds": {
    "const": 3600
  },
  "ttl_min_seconds": {
    "const": 1
  }
},
"required": [
  "record_type",
  "token_version",
  "ranking_algorithm_id",
  "traversal_algorithm_id",
  "dependency_depth_hard_max",
  "continuation_ttl_seconds_max",
  "selected_profile_id",
  "selected_profile_digest",
  "persistent_entity_types",
  "default_budget",
  "hard_budget",
  "default_depth",
  "depth_min",
  "depth_max",
  "default_ttl_seconds",
  "ttl_min_seconds",
  "ttl_max_seconds",
  "max_token_bytes",
  "max_continuation_state_bytes",
  "max_query_bytes",
  "min_result_bytes",
  "max_resume_binding_fields",
  "max_resume_binding_key_chars",
  "max_resume_binding_value_bytes",

```

```

    "max_resume_binding_bytes",
    "required_resume_binding_fields",
    "policy_digest"
  ],
  "type": "object",
  "x-promin-ingress-operations": [
    "rebuild"
  ],
  "x-promin-persistence": "derived-live-only"
}

```

Власник: core/contracts.schema.json#/\$defs/ProjectionLimits | Валідатор: urn:promin:contracts:1#/\$defs/ProjectionLimits

72. ProminPlatformNoDegradationMatrix

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, standard_name, version, candidate_binding_digest, archive_sha256, archive_bytes, trust_configuration_sha256, lanes, lane_count, complete, semantic_validation_complete, unique_invocation_ids, unique_key_nonces, matrix_authoritative, pass_credit, acceptance_pass, product_acceptance_pass, product_public_approval, matrix_digest

Властивість	Обов'язков е	Структурне правило
acceptance_pass	так	const=false
archive_bytes	так	type="integer"; minimum=1
archive_sha256	так	\$ref="#/\$defs/Digest"
candidate_binding_digest	так	\$ref="#/\$defs/Digest"
complete	так	const=true
lane_count	так	const=8
lanes	так	type="array"; minItems=8; maxItems=8
matrix_authoritative	так	const=false
matrix_digest	так	\$ref="#/\$defs/Digest"
pass_credit	так	const=false
product_acceptance_pass	так	const=false
product_public_approval	так	const="not_approved"
record_type	так	const="ProminPlatformNoDegradationMatrix"
semantic_validation_complete	так	const=true
standard_name	так	const="promin"
trust_configuration_sha256	так	\$ref="#/\$defs/Digest"
unique_invocation_ids	так	const=true
unique_key_nonces	так	const=true
version	так	const="1.0.0-alpha.1"

```

{
  "additionalProperties": false,
  "properties": {
    "acceptance_pass": {
      "const": false
    },
    "archive_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "archive_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "complete": {
      "const": true
    },
    "lane_count": {
      "const": 8
    },
    "lanes": {
      "items": {

```

```
"additionalProperties": false,
"properties": {
  "attestation_digest": {
    "$ref": "#/$defs/Digest"
```

```
  },
  "evidence_role": {
    "enum": [
      "linux",
      "linux-no-degradation",
      "windows",
      "windows-no-degradation"
    ]
  },
  "install_mode": {
    "enum": [
      "offline-wheelhouse",
      "online-clean"
    ]
  },
  "invocation_id": {
    "$ref": "#/$defs/Id"
  },
  "key_id": {
    "$ref": "#/$defs/Id"
  },
  "lane_id": {
    "enum": [
      "linux-cp313-offline",
      "linux-cp313-online",
      "linux-cp314-offline",
      "linux-cp314-online",
      "windows-cp313-offline",
```

```
      "windows-cp313-online",
      "windows-cp314-offline",
      "windows-cp314-online"
    ]
  },
  "nonce": {
    "maxLength": 128,
    "minLength": 24,
    "type": "string"
  },
  "os": {
    "enum": [
      "linux",
      "windows"
    ]
  },
  "path": {
    "$ref": "#/$defs/RelativePath"
  },
  "python_abi": {
    "maxLength": 128,
    "minLength": 1,
    "type": "string"
  },
  "python_implementation": {
    "const": "cpython"
  },
  "python_version": {
```

```
    "maxLength": 128,
    "minLength": 1,
    "type": "string"
  },
  "record_type": {
    "enum": [
      "NoDegradationResult",
      "PlatformVerificationResult"
    ]
  },
  "result_digest": {
    "$ref": "#/$defs/Digest"
  },
  "semantic_valid": {
    "const": true
  },
  "sha256": {
```

```

    "$ref": "#/$defs/Digest"
  },
  "size_bytes": {
    "minimum": 1,
    "type": "integer"
  },
  "status": {
    "const": "pass"
  },
  "subject_id": {
    "$ref": "#/$defs/Id"
  }

```

```

  }
},
"required": [
  "lane_id",
  "path",
  "sha256",
  "size_bytes",
  "result_digest",
  "record_type",
  "status",
  "os",
  "python_version",
  "python_implementation",
  "python_abi",
  "install_mode",
  "evidence_role",
  "key_id",
  "subject_id",
  "attestation_digest",
  "invocation_id",
  "nonce",
  "semantic_valid"
],
"type": "object"
},
"maxItems": 8,
"minItems": 8,
"type": "array"

```

```

},
"matrix_authoritative": {
  "const": false
},
"matrix_digest": {
  "$ref": "#/$defs/Digest"
},
"pass_credit": {
  "const": false
},
"product_acceptance_pass": {
  "const": false
},
"product_public_approval": {
  "const": "not_approved"
},
"record_type": {
  "const": "ProminPlatformNoDegradationMatrix"
},
"semantic_validation_complete": {
  "const": true
},
"standard_name": {
  "const": "promin"
},
"trust_configuration_sha256": {
  "$ref": "#/$defs/Digest"
},

```

```

"unique_invocation_ids": {
  "const": true
},
"unique_key_nonces": {
  "const": true
},
"version": {
  "const": "1.0.0-alpha.1"
}
},

```



```

"required": [
  "record_type",
  "standard_name",
  "version",
  "candidate_binding_digest",
  "archive_sha256",
  "archive_bytes",
  "trust_configuration_sha256",
  "lanes",
  "lane_count",
  "complete",
  "semantic_validation_complete",
  "unique_invocation_ids",
  "unique_key_nonces",
  "matrix_authoritative",
  "pass_credit",
  "acceptance_pass",
  "product_acceptance_pass",
  "product_public_approval",
  "matrix_digest"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/ProminPlatformNoDegradationMatrix](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/ProminPlatformNoDegradationMatrix](#)

73. ProviderBinding

Опис типу: type="object"; additionalProperties=false; allOf[9]=keys:if,then,title | keys:if,then,title | keys:if,then,title | keys:if,then,title | keys:if,then,title | keys:if,then,title | keys:if,then,title | keys:if,then,title

Обов'язкові поля: capability_id, provider_id, version, invocation, purpose, required, healthcheck, license, identity, implementation_closure, dependency_receipt

Властивість	Обов'язков е	Структурне правило
capability_id	так	enum=["control-runtime", "shape-validation", "content-identity", "local-serialization", "query-projection", "filesystem-inventory", "export-scan", "signature", "build-dependency"]
dependency_receipt	так	\$ref="#/\$defs/ProviderDependencyReceipt"
healthcheck	так	\$ref="#/\$defs/Healthcheck"
identity	так	\$ref="#/\$defs/ProviderIdentity"
implementation_closure	так	\$ref="#/\$defs/ImplementationClosure"
invocation	так	\$ref="#/\$defs/Invocation"
license	так	\$ref="#/\$defs/LicenseBinding"
provider_id	так	\$ref="#/\$defs/Id"
purpose	так	type="string"; minLength=1
обов'язкові	так	type="boolean"
version	так	type="string"; minLength=1

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "if": {
        "properties": {
          "capability_id": {
            "const": "control-runtime"
          }
        },
        "required": [
          "capability_id"
        ]
      },
      "then": {
        "oneOf": [
          {
            "properties": {
              "identity": {
                "properties": {
                  "kind": {
                    "const": "file-digest"

```

```

    }
  },
  "required": [
    "kind"
  ]
},

```

```

  "implementation_closure": {
    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "adapter_id": {
              "const": "python-runtime-v1"
            }
          },
          "required": [
            "adapter_id"
          ]
        }
      }
    },
    "required": [
      "adapters"
    ]
  },
  "invocation": {
    "properties": {
      "kind": {
        "const": "python-runtime"
      }
    },
    "required": [
      "kind"
    ]
  }
}

```

```

    }
  }
},
],
"properties": {
  "implementation_closure": {
    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "capability_id": {
              "const": "control-runtime"
            },
            "protocol_id": {
              "const": "promin.control-runtime.v1"
            }
          },
          "required": [
            "capability_id",
            "protocol_id"
          ]
        }
      }
    },
    "required": [
      "adapters"
    ]
  }
}

```

```

    }
  },
  "title": "control-runtime binds its exact adapter protocol"
},
{
  "if": {
    "properties": {
      "capability_id": {
        "const": "shape-validation"
      }
    },
    "required": [
      "capability_id"
    ]
  },
}

```

```

"then": {
  "oneOf": [
    {
      "properties": {
        "identity": {
          "properties": {
            "kind": {
              "const": "file-digest"
            }
          }
        },
        "required": [
          "kind"
        ]
      }
    }
  ],
  "implementation_closure": {
    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "adapter_id": {
              "const": "python-runtime-v1"
            }
          }
        },
        "required": [
          "adapter_id"
        ]
      }
    },
    "required": [
      "adapters"
    ]
  },
  "invocation": {
    "properties": {
      "kind": {
        "const": "python-runtime"
      }
    },
    "required": [
      "kind"
    ]
  }
}

```

```

    },
    "required": [
      "adapters"
    ]
  },
  "invocation": {
    "properties": {
      "kind": {
        "const": "python-runtime"
      }
    },
    "required": [
      "kind"
    ]
  }
}

```

```

    }
  ],
  "properties": {
    "implementation_closure": {
      "properties": {
        "adapters": {
          "items": {
            "properties": {
              "capability_id": {
                "const": "shape-validation"
              },
              "protocol_id": {
                "const": "promin.shape-validation.v1"
              }
            },
            "required": [
              "capability_id",
              "protocol_id"
            ]
          }
        },
        "required": [
          "adapters"
        ]
      }
    },
    "required": [
      "adapters"
    ]
  }
}

```

```

    },
    "title": "shape-validation binds its exact adapter protocol"
  },
  {
    "if": {
      "properties": {

```

```

        "capability_id": {
            "const": "content-identity"
        }
    },
    "required": [
        "capability_id"
    ]
},
"then": {
    "oneOf": [
        {
            "properties": {
                "identity": {
                    "properties": {
                        "kind": {
                            "const": "file-digest"
                        }
                    }
                },
                "required": [
                    "kind"

```

```

                ]
            },
            "implementation_closure": {
                "properties": {
                    "adapters": {
                        "items": {
                            "properties": {
                                "adapter_id": {
                                    "const": "python-runtime-v1"
                                }
                            }
                        },
                        "required": [
                            "adapter_id"
                        ]
                    }
                },
                "required": [
                    "adapters"
                ]
            },
            "invocation": {
                "properties": {
                    "kind": {
                        "const": "python-runtime"
                    }
                },
                "required": [

```

```

                    "kind"
                ]
            }
        }
    ],
    "properties": {
        "implementation_closure": {
            "properties": {
                "adapters": {
                    "items": {
                        "properties": {
                            "capability_id": {
                                "const": "content-identity"
                            },
                            "protocol_id": {
                                "const": "promin.content-identity.v1"
                            }
                        },
                        "required": [
                            "capability_id",
                            "protocol_id"
                        ]
                    }
                },
                "required": [
                    "adapters"

```

```

    }
  }
},
"title": "content-identity binds its exact adapter protocol"
},
{
  "if": {
    "properties": {
      "capability_id": {
        "const": "local-serialization"
      }
    },
    "required": [
      "capability_id"
    ]
  },
  "then": {
    "oneOf": [
      {
        "properties": {
          "identity": {
            "properties": {
              "kind": {
                "const": "file-digest"
              }
            }
          },
          "required": [

```

```

            "kind"
          ]
        },
        "implementation_closure": {
          "properties": {
            "adapters": {
              "items": {
                "properties": {
                  "adapter_id": {
                    "const": "python-runtime-v1"
                  }
                },
                "required": [
                  "adapter_id"
                ]
              }
            },
            "required": [
              "adapters"
            ]
          },
          "invocation": {
            "properties": {
              "kind": {
                "const": "python-runtime"
              }
            }
          },
          "required": [

```

```

            "kind"
          ]
        }
      }
    ],
    "properties": {
      "implementation_closure": {
        "properties": {
          "adapters": {
            "items": {
              "properties": {
                "capability_id": {
                  "const": "local-serialization"
                },
                "protocol_id": {
                  "const": "promin.local-serialization.v1"
                }
              },
              "required": [
                "capability_id",

```

```

        "protocol_id"
      ]
    }
  },
  "required": [
    "adapters"
  ]
},
{
  "title": "local-serialization binds its exact adapter protocol"
},
{
  "if": {
    "properties": {
      "capability_id": {
        "const": "query-projection"
      }
    },
    "required": [
      "capability_id"
    ]
  },
  "then": {
    "oneOf": [
      {
        "properties": {
          "identity": {
            "properties": {
              "kind": {
                "const": "file-digest"
              }
            }
          }
        },
        "required": [
          "kind"
        ]
      }
    ]
  },
  "implementation_closure": {
    "properties": {
      "adapters": {
        "items": {
          "properties": {
            "adapter_id": {
              "const": "python-runtime-v1"
            }
          },
          "required": [
            "adapter_id"
          ]
        }
      }
    },
    "required": [
      "adapters"
    ]
  },
  "invocation": {
    "properties": {
      "kind": {
        "const": "python-runtime"
      }
    }
  },
  "required": [
    "kind"
  ]
}
},
{
  "properties": {
    "implementation_closure": {
      "properties": {
        "adapters": {
          "items": {
            "properties": {
              "capability_id": {

```

```
      "const": "query-projection"
    },
    "protocol_id": {
      "const": "promin.query-projection.v1"
    }
  ],
  "required": [
    "capability_id",
    "protocol_id"
  ]
}
},
{
  "required": [
    "adapters"
  ]
}
},
{
  "title": "query-projection binds its exact adapter protocol"
},
{
  "if": {
    "properties": {
      "capability_id": {
        "const": "filesystem-inventory"
      }
    },
    "required": [
      "capability_id"
    ]
  },
  "then": {
    "oneOf": [
      {
        "properties": {
          "identity": {
            "properties": {
              "kind": {
                "const": "file-digest"
              }
            }
          },
          "required": [
            "kind"
          ]
        },
        "implementation_closure": {
          "properties": {
            "adapters": {
              "items": {
                "properties": {
                  "adapter_id": {
                    "const": "git-executable-v1"
                  }
                },
                "required": [
                  "adapter_id"
                ]
              }
            }
          },
          "required": [
            "adapters"
          ]
        },
        "invocation": {
          "properties": {
            "kind": {
              "const": "executable"
            }
          },
          "required": [
            "kind"
          ]
        }
      }
    ]
  }
}
```

```

    ],
    "properties": {
      "implementation_closure": {
        "properties": {
          "adapters": {
            "items": {
              "properties": {
                "capability_id": {
                  "const": "filesystem-inventory"
                },
                "protocol_id": {
                  "const": "promin.filesystem-inventory.v1"
                }
              },
              "required": [
                "capability_id",
                "protocol_id"
              ]
            }
          }
        }
      },
      "required": [
        "adapters"
      ]
    }
  },
  "title": "filesystem-inventory binds its exact adapter protocol"
},
{
  "if": {
    "properties": {
      "capability_id": {
        "const": "export-scan"
      }
    },
    "required": [
      "capability_id"
    ]
  },
  "then": {
    "oneOf": [
      {
        "properties": {
          "identity": {
            "properties": {
              "kind": {
                "const": "file-digest"
              }
            }
          },
          "required": [
            "kind"
          ]
        },
        "implementation_closure": {
          "properties": {
            "adapters": {
              "items": {
                "properties": {
                  "adapter_id": {
                    "const": "executable-export-scan-v1"
                  }
                },
                "required": [
                  "adapter_id"
                ]
              }
            },
            "required": [
              "adapters"
            ]
          }
        },
        "invocation": {
          "properties": {
            "kind": {

```



```

        "const": "executable"
    }
},
"required": [
    "kind"
]
}
}
},
"properties": {
    "implementation_closure": {
        "properties": {
            "adapters": {
                "items": {
                    "properties": {
                        "capability_id": {
                            "const": "export-scan"
                        },
                        "protocol_id": {
                            "const": "promin.export-scan.v1"
                        }
                    },
                    "required": [
                        "capability_id",
                        "protocol_id"
                    ]
                }
            }
        }
    },
    "required": [
        "capability_id",
        "protocol_id"
    ]
}
}
},
},
"required": [
    "adapters"
]
}
},
},
"title": "export-scan binds its exact adapter protocol"
},
{
    "if": {
        "properties": {
            "capability_id": {
                "const": "signature"
            }
        },
        "required": [
            "capability_id"
        ]
    },
    "then": {
        "oneOf": [
            {
                "properties": {
                    "identity": {
                        "properties": {
                            "kind": {
                                "const": "file-digest"
                            }
                        }
                    },
                    "required": [
                        "kind"
                    ]
                },
                "implementation_closure": {
                    "properties": {
                        "adapters": {
                            "items": {
                                "properties": {
                                    "adapter_id": {
                                        "const": "executable-signature-v1"
                                    }
                                },
                                "required": [
                                    "adapter_id"
                                ]
                            }
                        }
                    }
                }
            }
        ]
    }
}
},

```



```

    }
  },
  "required": [
    "adapters"
  ]
}
},
"if": {

```

```

  "properties": {
    "capability_id": {
      "const": "build-dependency"
    }
  },
  "required": [
    "capability_id"
  ]
},
"then": {
  "oneOf": [
    {
      "properties": {
        "identity": {
          "properties": {
            "kind": {
              "const": "file-digest"
            }
          },
          "required": [
            "kind"
          ]
        },
        "implementation_closure": {
          "properties": {
            "adapters": {
              "items": {

```

```

                "adapter_id": {
                  "const": "executable-build-dependency-v1"
                }
              },
              "required": [
                "adapter_id"
              ]
            }
          },
          "required": [
            "adapters"
          ]
        },
        "invocation": {
          "properties": {
            "kind": {
              "const": "executable"
            }
          },
          "required": [
            "kind"
          ]
        }
      ]
    }
  ],
  "properties": {

```

```

    "implementation_closure": {
      "properties": {
        "adapters": {
          "items": {
            "properties": {
              "capability_id": {
                "const": "build-dependency"
              },

```

```

        "protocol_id": {
            "const": "promin.build-dependency.v1"
        },
        "required": [
            "capability_id",
            "protocol_id"
        ]
    },
    "required": [
        "adapters"
    ]
}
},
"title": "build-dependency binds its exact adapter protocol"
}
],

```

```

"properties": {
    "capability_id": {
        "enum": [
            "control-runtime",
            "shape-validation",
            "content-identity",
            "local-serialization",
            "query-projection",
            "filesystem-inventory",
            "export-scan",
            "signature",
            "build-dependency"
        ]
    },
    "dependency_receipt": {
        "$ref": "#/$defs/ProviderDependencyReceipt"
    },
    "healthcheck": {
        "$ref": "#/$defs/Healthcheck"
    },
    "identity": {
        "$ref": "#/$defs/ProviderIdentity"
    },
    "implementation_closure": {
        "$ref": "#/$defs/ImplementationClosure"
    },
    "invocation": {
        "$ref": "#/$defs/Invocation"
    },

```

```

},
    "license": {
        "$ref": "#/$defs/LicenseBinding"
    },
    "provider_id": {
        "$ref": "#/$defs/Id"
    },
    "purpose": {
        "minLength": 1,
        "type": "string"
    },
    "required": {
        "type": "boolean"
    },
    "version": {
        "minLength": 1,
        "type": "string"
    }
},
"required": [
    "capability_id",
    "provider_id",
    "version",
    "invocation",
    "purpose",
    "обов'язкові",
    "healthcheck",
    "license",

```

```

    "identity",
    "implementation_closure",
    "dependency_receipt"
  ],
  "type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/ProviderBinding](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/ProviderBinding](#)

74. ProviderDependencyReceipt

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: complete, provider_owned_scope, os_substrate_excluded, components, aggregate_digest

Властивість	Обов'язков е	Структурне правило
aggregate_digest	так	\$ref="#/\$defs/Digest"
complete	так	const=true
components	так	type="array"; minItems=1; maxItems=4096; uniqueItems=true
os_substrate_excluded	так	const=true
provider_owned_scope	так	const="configured-provider-root-and-runtime-components"

```

{
  "additionalProperties": false,
  "properties": {
    "aggregate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "complete": {
      "const": true
    },
    "components": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "component_id": {
            "$ref": "#/$defs/Id"
          },
          "component_kind": {
            "enum": [
              "native-runtime",
              "provider-file",
              "provider-tree",
              "python-distribution"
            ]
          },
          "digest": {
            "$ref": "#/$defs/Digest"
          },
          "size_bytes": {
            "minimum": 1,
            "type": "integer"
          },
          "source": {
            "maxLength": 4096,
            "minLength": 1,
            "type": "string"
          },
          "version": {
            "maxLength": 256,
            "minLength": 1,
            "type": "string"
          }
        },
        "required": [
          "component_id",
          "component_kind",
          "source",
          "version",
          "digest",
          "size_bytes"
        ],
        "type": "object"
      },
      "maxItems": 4096,

```

```

    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "os_substrate_excluded": {
    "const": true
  },
  "provider_owned_scope": {
    "const": "configured-provider-root-and-runtime-components"
  }
},
"required": [
  "complete",
  "provider_owned_scope",
  "os_substrate_excluded",
  "components",
  "aggregate_digest"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/ProviderDependencyReceipt](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/ProviderDependencyReceipt](#)

75. ProviderIdentity

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: kind, digest, source

Властивість	Обов'язков е	Структурне правило
digest	так	\$ref="#/\$defs/Digest"
kind	так	enum=["file-digest", "module-file-digest", "distribution-tree-digest", "module-closure-digest", "platform-attestation"]
source	так	type="string"; minLength=1

```

{
  "additionalProperties": false,
  "properties": {
    "digest": {
      "$ref": "#/$defs/Digest"
    },
    "kind": {
      "enum": [
        "file-digest",
        "module-file-digest",
        "distribution-tree-digest",
        "module-closure-digest",
        "platform-attestation"
      ]
    },
    "source": {
      "minLength": 1,
      "type": "string"
    }
  },
  "required": [
    "kind",
    "digest",
    "source"
  ],
  "type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/ProviderIdentity](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/ProviderIdentity](#)

76. ProviderInvocationEvidence

[illegible]

Обов'язкові поля: capability_id, provider_id, invocation_kind, identity_kind, identity_digest, adapter_id, protocol_id, operation, operation_contract_digest, dependency_receipt_digest, implementation_closure_digest, invoked, started_at, completed_at, outcome, exit_code, invocation_request_digest, input_digest, output_digest, output_size_bytes, output_size_ceiling_bytes, stdout_capture_digest, stdout_capture_size_bytes, stdout_capture_truncated, stderr_capture_digest, stderr_capture_size_bytes, stderr_capture_truncated, invocation_receipt_digest, authoritative, pass_credit

Властивість	Обов'язков е	Структурне правило
adapter_id	так	\$ref="#/\$defs/Id"
authoritative	так	const=false
capability_id	так	enum=["control-runtime", "shape-validation", "content-identity", "local-serialization", "query-projection", "filesystem-inventory", "export-scan", "signature", "build-dependency"]
completed_at	так	\$ref="#/\$defs/Timestamp"
dependency_receipt_digest	так	\$ref="#/\$defs/Digest"
exit_code	так	oneOf[2]=type:integer type:null
identity_digest	так	\$ref="#/\$defs/Digest"
identity_kind	так	enum=["file-digest", "module-file-digest", "distribution-tree-digest", "module-closure-digest", "platform-attestation"]
implementation_closure_digest	так	\$ref="#/\$defs/Digest"
input_digest	так	\$ref="#/\$defs/Digest"
invocation_kind	так	enum=["python-runtime", "python-module", "executable", "platform-service"]
invocation_receipt_digest	так	\$ref="#/\$defs/Digest"
invocation_request_digest	так	\$ref="#/\$defs/Digest"
invoked	так	const=true
operation	так	\$ref="#/\$defs/Id"
operation_contract_digest	так	\$ref="#/\$defs/Digest"
outcome	так	enum=["success", "failure", "blocked"]
output_digest	так	\$ref="#/\$defs/Digest"
output_size_bytes	так	type="integer"; minimum=0; maximum=1099511627776
output_size_ceiling_bytes	так	type="integer"; minimum=0; maximum=1099511627776
pass_credit	так	const=false
protocol_id	так	\$ref="#/\$defs/Id"
provider_id	так	\$ref="#/\$defs/Id"
started_at	так	\$ref="#/\$defs/Timestamp"
stderr_capture_digest	так	\$ref="#/\$defs/Digest"
stderr_capture_size_bytes	так	type="integer"; minimum=0; maximum=1048576
stderr_capture_truncated	так	type="boolean"
stdout_capture_digest	так	\$ref="#/\$defs/Digest"
stdout_capture_size_bytes	так	type="integer"; minimum=0; maximum=1048576
stdout_capture_truncated	так	type="boolean"
{ "additionalProperties": false, "oneOf": [{		

```

"properties": {
  "adapter_id": {
    "const": "python-runtime-v1"
  },
  "capability_id": {
    "const": "control-runtime"
  },
  "identity_kind": {
    "const": "file-digest"
  },
  "invocation_kind": {
    "const": "python-runtime"
  },
  "operation": {
    "const": "continue"
  },
  "operation_contract_digest": {
    "const": "a8a728c5c39a2c00083fc3b6463288e8b3797d871675d3d2e27f7feb1be7ff74"
  },
  "protocol_id": {
    "const": "promin.control-runtime.v1"
  }
}
},

```

```

{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "control-runtime"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "doctor"
    },
    "operation_contract_digest": {
      "const": "dcf2e435cf7954edc17fd35fc7928951f9a607d2ee91343d1788c114d23cdead"
    },
    "protocol_id": {
      "const": "promin.control-runtime.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {

```

```

      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "control-runtime"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
    },
    "protocol_id": {
      "const": "promin.control-runtime.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"

```



```

    },
    "capability_id": {
        "const": "control-runtime"
    },
    "identity_kind": {
        "const": "file-digest"
    },
    "invocation_kind": {
        "const": "python-runtime"
    },
    "operation": {
        "const": "init"
    },
    "operation_contract_digest": {
        "const": "9c2e26f6706abd39bef62d2120fcd9785f4f1cf39cdae92c3e4fa8f20ff9dd12"
    },
    "protocol_id": {
        "const": "promin.control-runtime.v1"
    }
}
},
{
    "properties": {
        "adapter_id": {
            "const": "python-runtime-v1"
        },
        "capability_id": {
            "const": "control-runtime"
        },
        "identity_kind": {
            "const": "file-digest"
        },
        "invocation_kind": {
            "const": "python-runtime"
        },
        "operation": {
            "const": "next"
        },
        "operation_contract_digest": {
            "const": "b75143a53c7129f54c62fc5c0ebd97927a7732bb8fc6f91918d5858a9578fe5d"
        },
        "protocol_id": {
            "const": "promin.control-runtime.v1"
        }
    }
},
{
    "properties": {
        "adapter_id": {
            "const": "python-runtime-v1"
        },
        "capability_id": {
            "const": "control-runtime"
        },
        "identity_kind": {
            "const": "file-digest"
        },
        "invocation_kind": {
            "const": "python-runtime"
        },
        "operation": {
            "const": "provider-binding"
        },
        "operation_contract_digest": {
            "const": "135f8b5d81f1cec32ec18a36aledcb3455cd186266ed1a72a5b1ecb638102f6f"
        },
        "protocol_id": {
            "const": "promin.control-runtime.v1"
        }
    }
},
{
    "properties": {
        "adapter_id": {
            "const": "python-runtime-v1"
        },
        "capability_id": {

```

```

    "const": "control-runtime"
  },
  "identity_kind": {
    "const": "file-digest"
  },
  "invocation_kind": {
    "const": "python-runtime"
  },
  "operation": {
    "const": "status"
  },
  "operation_contract_digest": {
    "const": "0449eeb95febf5e15ffba4657cdb6cd606c0a12a148bd444384120faa5bc5eb1"
  },
  "protocol_id": {
    "const": "promin.control-runtime.v1"
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "control-runtime"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "validate"
    },
    "operation_contract_digest": {
      "const": "6cab313c41c9cd37de52a75d0ba0676cfd9fb0a3fc53eb3f225521a02dcd8cf3"
    },
    "protocol_id": {
      "const": "promin.control-runtime.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "shape-validation"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "command"
    },
    "operation_contract_digest": {
      "const": "5ba4690e0a2adc6d06b74673ab39b0b714215a8543c188a7bed695b71a54c2bf"
    },
    "protocol_id": {
      "const": "promin.shape-validation.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "shape-validation"
    }
  },

```

```

    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "continue"
    },
    "operation_contract_digest": {
      "const": "5ba4690e0a2adc6d06b74673ab39b0b714215a8543c188a7bed695b71a54c2bf"
    },
    "protocol_id": {
      "const": "promin.shape-validation.v1"
    }
  }
}

```

```

},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "shape-validation"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "export"
    },
    "operation_contract_digest": {
      "const": "5ba4690e0a2adc6d06b74673ab39b0b714215a8543c188a7bed695b71a54c2bf"
    },
    "protocol_id": {
      "const": "promin.shape-validation.v1"
    }
  }
},
{
  "properties": {

```

```

    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "shape-validation"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
    },
    "protocol_id": {
      "const": "promin.shape-validation.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },

```

```

    "capability_id": {
      "const": "shape-validation"
    },
    "identity_kind": {
      "const": "file-digest"
    }
  }
}

```

```

    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    },
    "operation": {
      "const": "import"
    },
    },
    "operation_contract_digest": {
      "const": "5ba4690e0a2adc6d06b74673ab39b0b714215a8543c188a7bed695b71a54c2bf"
    },
    },
    "protocol_id": {
      "const": "promin.shape-validation.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    },
    "capability_id": {
      "const": "shape-validation"
    },
    },
  },

```

```

    "identity_kind": {
      "const": "file-digest"
    },
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    },
    "operation": {
      "const": "provider-binding"
    },
    },
    "operation_contract_digest": {
      "const": "135f8b5d81f1cec32ec18a36a1edcb3455cd186266ed1a72a5b1ecb638102f6f"
    },
    },
    "protocol_id": {
      "const": "promin.shape-validation.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    },
    "capability_id": {
      "const": "shape-validation"
    },
    },
    "identity_kind": {
      "const": "file-digest"
    },
    },
  },

```

```

    "invocation_kind": {
      "const": "python-runtime"
    },
    },
    "operation": {
      "const": "rebuild"
    },
    },
    "operation_contract_digest": {
      "const": "5ba4690e0a2adc6d06b74673ab39b0b714215a8543c188a7bed695b71a54c2bf"
    },
    },
    "protocol_id": {
      "const": "promin.shape-validation.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    },
    "capability_id": {
      "const": "shape-validation"
    },
    },
    "identity_kind": {
      "const": "file-digest"
    },
    },
    "invocation_kind": {

```

```

    "const": "python-runtime"
  },
  "operation": {
    "const": "replay"
  },
  "operation_contract_digest": {
    "const": "5ba4690e0a2adc6d06b74673ab39b0b714215a8543c188a7bed695b71a54c2bf"
  },
  "protocol_id": {
    "const": "promin.shape-validation.v1"
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "content-identity"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "digest-bytes"
    }
  },
  "operation_contract_digest": {
    "const": "1898e33042a5a6c55d8c1687c07f53f3f8df020c5475a57e882ca54c9adcebfd"
  },
  "protocol_id": {
    "const": "promin.content-identity.v1"
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "content-identity"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "digest-file"
    },
    "operation_contract_digest": {
      "const": "3876865e4a21f0047d92daaca9b6d7177769130a97c781e90ce130ced513f60b"
    }
  },
  "protocol_id": {
    "const": "promin.content-identity.v1"
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "content-identity"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    }
  },

```

```

"operation": {
  "const": "digest-value"
},
"operation_contract_digest": {
  "const": "b0a54275c219f51585f519a4d9c1814478bf072dbeff050a92a80780eb492a39"
},
"protocol_id": {
  "const": "promin.content-identity.v1"
}

```

```

}
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "content-identity"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
    },
    "protocol_id": {
      "const": "promin.content-identity.v1"
    }
  }
},
{

```

```

"properties": {
  "adapter_id": {
    "const": "python-runtime-v1"
  },
  "capability_id": {
    "const": "content-identity"
  },
  "identity_kind": {
    "const": "file-digest"
  },
  "invocation_kind": {
    "const": "python-runtime"
  },
  "operation": {
    "const": "provider-binding"
  },
  "operation_contract_digest": {
    "const": "135f8b5d81f1cec32ec18a36a1edcb3455cd186266ed1a72a5b1ecb638102f6f"
  },
  "protocol_id": {
    "const": "promin.content-identity.v1"
  }
}
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"

```

```

},
"capability_id": {
  "const": "local-serialization"
},
"identity_kind": {
  "const": "file-digest"
},
"invocation_kind": {
  "const": "python-runtime"
},
"operation": {
  "const": "atomic-replace"

```

```

    },
    "operation_contract_digest": {
      "const": "afab188552442a9aa0a271a4021e25f96a34655eaa687bc4f82b934a04652706"
    },
    "protocol_id": {
      "const": "promin.local-serialization.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "local-serialization"
    }
  }
}

```

```

    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "directory-sync"
    },
    "operation_contract_digest": {
      "const": "83842a439d25cc9e85bd51afec2381576f2ff42390d364672b4ee0bd415b1b23"
    },
    "protocol_id": {
      "const": "promin.local-serialization.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "local-serialization"
    },
    "identity_kind": {
      "const": "file-digest"
    }
  }
}

```

```

    },
    "invocation_kind": {
      "const": "python-runtime"
    },
    "operation": {
      "const": "file-sync"
    },
    "operation_contract_digest": {
      "const": "3d757d7af8c2fcfb955b76db89a8917262aec19de8419305c6617a6f668cbefd"
    },
    "protocol_id": {
      "const": "promin.local-serialization.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-runtime-v1"
    },
    "capability_id": {
      "const": "local-serialization"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "python-runtime"
    }
  }
}

```

```

    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {

```

```

        "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
    },
    "protocol_id": {
        "const": "promin.local-serialization.v1"
    }
},
{
    "properties": {
        "adapter_id": {
            "const": "python-runtime-v1"
        },
        "capability_id": {
            "const": "local-serialization"
        },
        "identity_kind": {
            "const": "file-digest"
        },
        "invocation_kind": {
            "const": "python-runtime"
        },
        "operation": {
            "const": "local-writer-lock"
        }
    }
},
{
    "operation_contract_digest": {
        "const": "ad8ca42303233f79dfc5f72b097602947e2a3df3b24fa2b5d452e374505820c1"
    },
    "protocol_id": {
        "const": "promin.local-serialization.v1"
    }
},
{
    "properties": {
        "adapter_id": {
            "const": "python-runtime-v1"
        },
        "capability_id": {
            "const": "local-serialization"
        },
        "identity_kind": {
            "const": "file-digest"
        },
        "invocation_kind": {
            "const": "python-runtime"
        },
        "operation": {
            "const": "provider-binding"
        },
        "operation_contract_digest": {
            "const": "135f8b5d81f1cec32ec18a36a1edcb3455cd186266ed1a72a5b1ecb638102f6f"
        }
    }
},
{
    "protocol_id": {
        "const": "promin.local-serialization.v1"
    }
},
{
    "properties": {
        "adapter_id": {
            "const": "python-runtime-v1"
        },
        "capability_id": {
            "const": "local-serialization"
        },
        "identity_kind": {
            "const": "file-digest"
        },
        "invocation_kind": {
            "const": "python-runtime"
        },
        "operation": {
            "const": "recover"
        },
        "operation_contract_digest": {
            "const": "148d9b508acd481c22a9e74e9021bd7741bbbb4557bcaa0e863e8ab6b9481f98"
        }
    }
},

```



```

    "protocol_id": {
      "const": "promin.local-serialization.v1"
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "query-projection"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "continue"
      },
      "operation_contract_digest": {
        "const": "84f9a7260289ac24fea9192b7f97982539b86b489aba2f347c4d77a0b1121d1c"
      },
      "protocol_id": {
        "const": "promin.query-projection.v1"
      }
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "query-projection"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "healthcheck"
      },
      "operation_contract_digest": {
        "const": "b8579e40a20a9617d82182e8775d1d3ace71566bfac3c6d6a5745754c3f812e0"
      },
      "protocol_id": {
        "const": "promin.query-projection.v1"
      }
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "query-projection"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "next"
      },
      "operation_contract_digest": {
        "const": "f9994e06188b1391ccdd2b1d592e9a1709db1874a5f437ac8bfebf4de349106"
      },
      "protocol_id": {
        "const": "promin.query-projection.v1"
      }
    }
  }
}

```

```

    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "query-projection"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "provider-binding"
      },
      "operation_contract_digest": {
        "const": "135f8b5d81f1cec32ec18a36aledcb3455cd186266ed1a72a5b1ecb638102f6f"
      },
      "protocol_id": {
        "const": "promin.query-projection.v1"
      }
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "query-projection"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "rebuild"
      },
      "operation_contract_digest": {
        "const": "825da07c8c0c880475d1e292bbc0ff059f10e3d47197d93da1cf1fc8c23d58e9"
      },
      "protocol_id": {
        "const": "promin.query-projection.v1"
      }
    }
  },
  {
    "properties": {
      "adapter_id": {
        "const": "python-runtime-v1"
      },
      "capability_id": {
        "const": "query-projection"
      },
      "identity_kind": {
        "const": "file-digest"
      },
      "invocation_kind": {
        "const": "python-runtime"
      },
      "operation": {
        "const": "status"
      },
      "operation_contract_digest": {
        "const": "6a48bc71d6f0876bde29d4267ebecf27c694e0652fd0f807aea49810a6d8b527"
      },
      "protocol_id": {
        "const": "promin.query-projection.v1"
      }
    }
  }
}

```

```

    },
    {
      "properties": {
        "adapter_id": {
          "const": "git-executable-v1"
        },
        "capability_id": {
          "const": "filesystem-inventory"
        },
        "identity_kind": {
          "const": "file-digest"
        },
        "invocation_kind": {
          "const": "executable"
        },
        "operation": {
          "const": "healthcheck"
        },
        "operation_contract_digest": {
          "const": "053c4181d6d35bd4ca5a13b8a1f18c19571c21b3426644bfa6bda99066dce5b3"
        },
        "protocol_id": {
          "const": "promin.filesystem-inventory.v1"
        }
      }
    },
    {
      "properties": {
        "adapter_id": {
          "const": "git-executable-v1"
        },
        "capability_id": {
          "const": "filesystem-inventory"
        },
        "identity_kind": {
          "const": "file-digest"
        },
        "invocation_kind": {
          "const": "executable"
        },
        "operation": {
          "const": "immutable-tree-object"
        },
        "operation_contract_digest": {
          "const": "7b9234db382b1728299f568503507fd04df4e1f19433c0e378c21378caad3241"
        },
        "protocol_id": {
          "const": "promin.filesystem-inventory.v1"
        }
      }
    },
    {
      "properties": {
        "adapter_id": {
          "const": "git-executable-v1"
        },
        "capability_id": {
          "const": "filesystem-inventory"
        },
        "identity_kind": {
          "const": "file-digest"
        },
        "invocation_kind": {
          "const": "executable"
        },
        "operation": {
          "const": "immutable-tree-stream"
        },
        "operation_contract_digest": {
          "const": "a286c3c566c2af8bb74011affe25071137b9ec9452a472f0e4997c4dc1a1038e"
        },
        "protocol_id": {
          "const": "promin.filesystem-inventory.v1"
        }
      }
    }
  ],
  {

```

```

"properties": {
  "adapter_id": {
    "const": "git-executable-v1"
  },
  "capability_id": {
    "const": "filesystem-inventory"
  },
  "identity_kind": {
    "const": "file-digest"
  },
  "invocation_kind": {
    "const": "executable"
  },
  "operation": {
    "const": "provider-binding"
  },
  "operation_contract_digest": {
    "const": "561e1b0729e20cd9586ee796ce58ac2f2759bcdff6cd5e1c31524f89aa4cbd97"
  },
  "protocol_id": {
    "const": "promin.filesystem-inventory.v1"
  }
}
}

```

```

},
{
  "properties": {
    "adapter_id": {
      "const": "executable-export-scan-v1"
    },
    "capability_id": {
      "const": "export-scan"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "bounded-export-scan"
    },
    "operation_contract_digest": {
      "const": "7e0326ca387c256798bc2c37c1d1c3e0fa17440d775c9d0c7deafa7ada33d1c3"
    },
    "protocol_id": {
      "const": "promin.export-scan.v1"
    }
  }
},
{
  "properties": {

```

```

    "adapter_id": {
      "const": "executable-export-scan-v1"
    },
    "capability_id": {
      "const": "export-scan"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "healthcheck"
    },
    "operation_contract_digest": {
      "const": "116e5e8a519592f63585f0fae7b90f86863ba3369f0b006913a0e4f7d51eaf2e"
    },
    "protocol_id": {
      "const": "promin.export-scan.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {

```

```

    "const": "executable-export-scan-v1"
  },
  "capability_id": {
    "const": "export-scan"
  },
  "identity_kind": {
    "const": "file-digest"
  },
  "invocation_kind": {
    "const": "executable"
  },
  "operation": {
    "const": "provider-binding"
  },
  "operation_contract_digest": {
    "const": "561e1b0729e20cd9586ee796ce58ac2f2759bcdff6cd5e1c31524f89aa4cbd97"
  },
  "protocol_id": {
    "const": "promin.export-scan.v1"
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
  },
  "identity_kind": {
    "const": "file-digest"
  },
  "invocation_kind": {
    "const": "executable"
  },
  "operation": {
    "const": "healthcheck"
  },
  "operation_contract_digest": {
    "const": "0e2delaec47aec67eacbec7838d7f7f19f457bcff9c3fae57c9e88a72b6bb6be"
  },
  "protocol_id": {
    "const": "promin.signature.v1"
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-module-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "module-file-digest"
    },
  },
  "invocation_kind": {
    "const": "python-module"
  },
  "operation": {
    "const": "healthcheck"
  },
  "operation_contract_digest": {
    "const": "0e2delaec47aec67eacbec7838d7f7f19f457bcff9c3fae57c9e88a72b6bb6be"
  },
  "protocol_id": {
    "const": "promin.signature.v1"
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-signature-v1"
    },
  },

```

```

"capability_id": {
  "const": "signature"
},
"identity_kind": {
  "const": "file-digest"
},
"invocation_kind": {
  "const": "executable"
},

```

```

"operation": {
  "const": "provider-binding"
},
"operation_contract_digest": {
  "const": "340c605d4d232f122703a2d47141fd524232107e7f69a8968a53a4d4e3903a1d"
},
"protocol_id": {
  "const": "promin.signature.v1"
}
}
},
{
  "properties": {
    "adapter_id": {
      "const": "python-module-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "module-file-digest"
    },
    "invocation_kind": {
      "const": "python-module"
    },
    "operation": {
      "const": "provider-binding"
    },
  },

```

```

"operation_contract_digest": {
  "const": "340c605d4d232f122703a2d47141fd524232107e7f69a8968a53a4d4e3903a1d"
},
"protocol_id": {
  "const": "promin.signature.v1"
}
}
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "sign-digest"
    },
    "operation_contract_digest": {
      "const": "ba5393e6098a6e47715d7eab7df317b837c087766dd6dc82aa750da9fa9e3237"
    },
  },

```

```

"protocol_id": {
  "const": "promin.signature.v1"
}
}
},
{
  "properties": {
    "adapter_id": {
      "const": "python-module-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
  },

```

```

    },
    "identity_kind": {
      "const": "module-file-digest"
    },
    "invocation_kind": {
      "const": "python-module"
    },
    "operation": {
      "const": "sign-digest"
    },
    "operation_contract_digest": {
      "const": "ba5393e6098a6e47715d7eab7df317b837c087766dd6dc82aa750da9fa9e3237"
    },
    "protocol_id": {
      "const": "promin.signature.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "file-digest"
    },
    "invocation_kind": {
      "const": "executable"
    },
    "operation": {
      "const": "verify-digest"
    },
    "operation_contract_digest": {
      "const": "7985bee569d1fbd30f73edc3878027a7e2a388a86c02c7dad6894b0ef9d4e67b"
    },
    "protocol_id": {
      "const": "promin.signature.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "python-module-signature-v1"
    },
    "capability_id": {
      "const": "signature"
    },
    "identity_kind": {
      "const": "module-file-digest"
    },
    "invocation_kind": {
      "const": "python-module"
    },
    "operation": {
      "const": "verify-digest"
    },
    "operation_contract_digest": {
      "const": "7985bee569d1fbd30f73edc3878027a7e2a388a86c02c7dad6894b0ef9d4e67b"
    },
    "protocol_id": {
      "const": "promin.signature.v1"
    }
  }
},
{
  "properties": {
    "adapter_id": {
      "const": "executable-build-dependency-v1"

```

```

    },
    "capability_id": {
      "const": "build-dependency"
    },
    "identity_kind": {

```

```

        "const": "file-digest"
    },
    "invocation_kind": {
        "const": "executable"
    },
    "operation": {
        "const": "healthcheck"
    },
    "operation_contract_digest": {
        "const": "116e5e8a519592f63585f0fae7b90f86863ba3369f0b006913a0e4f7d51eaf2e"
    },
    "protocol_id": {
        "const": "promin.build-dependency.v1"
    }
}
},
{
    "properties": {
        "adapter_id": {
            "const": "executable-build-dependency-v1"
        },
        "capability_id": {
            "const": "build-dependency"
        }
    },
    "identity_kind": {
        "const": "file-digest"
    },
    "invocation_kind": {
        "const": "executable"
    },
    "operation": {
        "const": "import-build-facts"
    },
    "operation_contract_digest": {
        "const": "25063736ddf272a51989981c79fcbb8e9cf2decf040522d3e050b13f0e2457db"
    },
    "protocol_id": {
        "const": "promin.build-dependency.v1"
    }
}
},
{
    "properties": {
        "adapter_id": {
            "const": "executable-build-dependency-v1"
        },
        "capability_id": {
            "const": "build-dependency"
        },
        "identity_kind": {
            "const": "file-digest"
        }
    },
    "invocation_kind": {
        "const": "executable"
    },
    "operation": {
        "const": "provider-binding"
    },
    "operation_contract_digest": {
        "const": "561e1b0729e20cd9586ee796ce58ac2f2759bcdff6cd5e1c31524f89aa4cbd97"
    },
    "protocol_id": {
        "const": "promin.build-dependency.v1"
    }
}
},
{
    "properties": {
        "adapter_id": {
            "$ref": "#/$defs/Id"
        },
        "authoritative": {
            "const": false
        },
        "capability_id": {
            "enum": [
                "control-runtime",

```



```

"shape-validation",
"content-identity",

"local-serialization",
"query-projection",
"filesystem-inventory",
"export-scan",
"signature",
"build-dependency"
]
},
"completed_at": {
  "$ref": "#/$defs/Timestamp"
},
"dependency_receipt_digest": {
  "$ref": "#/$defs/Digest"
},
"exit_code": {
  "oneOf": [
    {
      "type": "integer"
    },
    {
      "type": "null"
    }
  ]
},
"identity_digest": {
  "$ref": "#/$defs/Digest"
},
"identity_kind": {

```

```

  "enum": [
    "file-digest",
    "module-file-digest",
    "distribution-tree-digest",
    "module-closure-digest",
    "platform-attestation"
  ]
},
"implementation_closure_digest": {
  "$ref": "#/$defs/Digest"
},
"input_digest": {
  "$ref": "#/$defs/Digest"
},
"invocation_kind": {
  "enum": [
    "python-runtime",
    "python-module",
    "executable",
    "platform-service"
  ]
},
"invocation_receipt_digest": {
  "$ref": "#/$defs/Digest"
},
"invocation_request_digest": {
  "$ref": "#/$defs/Digest"
},

```

```

"invoked": {
  "const": true
},
"operation": {
  "$ref": "#/$defs/Id"
},
"operation_contract_digest": {
  "$ref": "#/$defs/Digest"
},
"outcome": {
  "enum": [
    "success",
    "failure",
    "blocked"
  ]
},
"output_digest": {
  "$ref": "#/$defs/Digest"
},

```

```

"output_size_bytes": {
  "maximum": 1099511627776,
  "minimum": 0,
  "type": "integer"
},
"output_size_ceiling_bytes": {
  "maximum": 1099511627776,
  "minimum": 0,
  "type": "integer"
},
"pass_credit": {
  "const": false
},
"protocol_id": {
  "$ref": "#/$defs/Id"
},
"provider_id": {
  "$ref": "#/$defs/Id"
},
"started_at": {
  "$ref": "#/$defs/Timestamp"
},
"stderr_capture_digest": {
  "$ref": "#/$defs/Digest"
},
"stderr_capture_size_bytes": {
  "maximum": 1048576,
  "minimum": 0,
  "type": "integer"
},
"stderr_capture_truncated": {
  "type": "boolean"
},
"stdout_capture_digest": {
  "$ref": "#/$defs/Digest"
},
"stdout_capture_size_bytes": {
  "maximum": 1048576,
  "minimum": 0,
  "type": "integer"
},
"stdout_capture_truncated": {
  "type": "boolean"
}
},
"required": [
  "capability_id",
  "provider_id",
  "invocation_kind",
  "identity_kind",
  "identity_digest",
  "adapter_id",
  "protocol_id",
  "operation",
  "operation_contract_digest",
  "dependency_receipt_digest",
  "implementation_closure_digest",
  "invoked",
  "started_at",
  "completed_at",
  "outcome",
  "exit_code",
  "invocation_request_digest",
  "input_digest",
  "output_digest",
  "output_size_bytes",
  "output_size_ceiling_bytes",
  "stdout_capture_digest",
  "stdout_capture_size_bytes",
  "stdout_capture_truncated",
  "stderr_capture_digest",
  "stderr_capture_size_bytes",
  "stderr_capture_truncated",
  "invocation_receipt_digest",
  "authoritative",
  "pass_credit"
],

```

```
"type": "object"
}
```

Власник: [core/contracts.schema.json#/\\$defs/ProviderInvocationEvidence](#) | Валідатор:
[urn:promin:contracts:1#/\\$defs/ProviderInvocationEvidence](#)

77. ReadyFrontier

Опис типу: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then

Обов'язкові поля: record_type, activation_digest, head_digest, projection_digest, ordering, ready_tasks, evaluated_task_count, blocked_task_count, cycle_count, truncated, continuation, silent_truncation, projection_authoritative

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
blocked_task_count	так	type="integer"; minimum=0
continuation	так	oneOf[2]=type:null type:object
cycle_count	так	type="integer"; const=0
evaluated_task_count	так	type="integer"; minimum=0
head_digest	так	oneOf[2]=#/\$defs/Digest type:null
ordering	так	const=["created_at-ascending", "task_id-ascending"]
projection_authoritative	так	const=false
projection_digest	так	\$ref="#/\$defs/Digest"
ready_tasks	так	type="array"; minItems=0; maxItems=12; uniqueItems=true
record_type	так	const="ReadyFrontier"
silent_truncation	так	const=false
truncated	так	type="boolean"

```
{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "properties": {
          "continuation": {
            "type": "null"
          }
        }
      },
      "if": {
        "properties": {
          "truncated": {
            "const": true
          }
        }
      },
      "required": [
        "truncated"
      ]
    },
    {
      "then": {
        "properties": {
          "continuation": {
            "additionalProperties": false,
            "properties": {
              "activation_digest": {
                "$ref": "#/$defs/Digest"
              }
            }
          },
          "budget_digest": {
            "$ref": "#/$defs/Digest"
          },
          "cursor": {
            "minimum": 0,
            "type": "integer"
          },
          "depth": {
            "maximum": 12,
            "minimum": 1,
            "type": "integer"
          }
        }
      }
    }
  ]
}
```

```

        "expiry": {
            "$ref": "#/$defs/Timestamp"
        },
        "head_digest": {
            "oneOf": [
                {
                    "$ref": "#/$defs/Digest"
                },
                {
                    "type": "null"
                }
            ]
        },
        "implementation_closure_digest": {
            "$ref": "#/$defs/Digest"
        },
        "projection_digest": {
            "$ref": "#/$defs/Digest"
        },
        "ranking": {
            "const": "bm25-v1"
        },
        "resume_binding_digest": {
            "$ref": "#/$defs/Digest"
        },
        "token": {
            "maxLength": 256,
            "minLength": 1,
            "pattern": "^promin-v2\\.\\.[A-Za-z0-9_-]{22}\\\\.\\.[0-9a-f]{64}\\\\.\\.[A-Za-z0-9_-]{43}$",
            "type": "string"
        },
        "traversal": {
            "const": "typed-bfs-v2"
        },
        "version": {
            "const": 2
        }
    },
    "required": [
        "version",
        "traversal",
        "token",
        "cursor",
        "expiry",
        "activation_digest",
        "head_digest",
        "projection_digest",
        "implementation_closure_digest",
        "ranking",
        "depth",
        "budget_digest",
        "resume_binding_digest"
    ],
    "type": "object"
}
}
}
},
"properties": {
    "activation_digest": {
        "$ref": "#/$defs/Digest"
    },
    "blocked_task_count": {
        "minimum": 0,
        "type": "integer"
    },
    "continuation": {
        "oneOf": [
            {
                "type": "null"
            },
            {
                "additionalProperties": false,
                "properties": {
                    "activation_digest": {
                        "$ref": "#/$defs/Digest"
                    }
                }
            }
        ]
    }
}

```

```

    },
    "budget_digest": {
      "$ref": "#/$defs/Digest"
    },
    "cursor": {
      "minimum": 0,
      "type": "integer"
    },
    "depth": {
      "maximum": 12,
      "minimum": 1,
      "type": "integer"
    },
    "expiry": {
      "$ref": "#/$defs/Timestamp"
    },
    "head_digest": {
      "oneOf": [
        {
          "$ref": "#/$defs/Digest"
        },
        {

```

```

          "type": "null"
        }
      ]
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "projection_digest": {
      "$ref": "#/$defs/Digest"
    },
    "ranking": {
      "const": "bm25-v1"
    },
    "resume_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "token": {
      "maxLength": 256,
      "minLength": 1,
      "pattern": "^promin-v2\\.\\.[A-Za-z0-9_-]{22}\\\\.\\.[0-9a-f]{64}\\\\.\\.[A-Za-z0-9_-]{43}$",
      "type": "string"
    },
    "traversal": {
      "const": "typed-bfs-v2"
    },
    "version": {
      "const": 2
    }
  }

```

```

},
"required": [
  "version",
  "traversal",
  "token",
  "cursor",
  "expiry",
  "activation_digest",
  "head_digest",
  "projection_digest",
  "implementation_closure_digest",
  "ranking",
  "depth",
  "budget_digest",
  "resume_binding_digest"
],
"type": "object"
}
]
},
"cycle_count": {
  "const": 0,
  "type": "integer"
},
"evaluated_task_count": {
  "minimum": 0,
  "type": "integer"
},

```

```

"head_digest": {
  "oneOf": [
    {
      "$ref": "#/$defs/Digest"
    },
    {
      "type": "null"
    }
  ]
},
"ordering": {
  "const": [
    "created_at-ascending",
    "task_id-ascending"
  ]
},
"projection_authoritative": {
  "const": false
},
"projection_digest": {
  "$ref": "#/$defs/Digest"
},
"ready_tasks": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "acceptance_predicate": {
        "maxLength": 256,

```

```

        "minLength": 1,
        "type": "string"
      },
      "candidate_digest": {
        "$ref": "#/$defs/Digest"
      },
      "created_at": {
        "$ref": "#/$defs/Timestamp"
      },
      "dependency_task_ids": {
        "items": {
          "$ref": "#/$defs/Id"
        },
        "maxItems": 32,
        "minItems": 0,
        "type": "array",
        "uniqueItems": true
      },
      "gate_result_digests": {
        "items": {
          "$ref": "#/$defs/Digest"
        },
        "maxItems": 64,
        "minItems": 0,
        "type": "array",
        "uniqueItems": true
      },
      "task_digest": {

```

```

        "$ref": "#/$defs/Digest"
      },
      "task_id": {
        "$ref": "#/$defs/Id"
      }
    },
    "required": [
      "task_id",
      "task_digest",
      "created_at",
      "candidate_digest",
      "acceptance_predicate",
      "dependency_task_ids",
      "gate_result_digests"
    ],
    "type": "object"
  },
  "maxItems": 12,
  "minItems": 0,
  "type": "array",
  "uniqueItems": true
},

```

```

    "record_type": {
      "const": "ReadyFrontier"
    },
    "silent_truncation": {
      "const": false
    },
  },
  "truncated": {
    "type": "boolean"
  }
},
"required": [
  "record_type",
  "activation_digest",
  "head_digest",
  "projection_digest",
  "ordering",
  "ready_tasks",
  "evaluated_task_count",
  "blocked_task_count",
  "cycle_count",
  "truncated",
  "continuation",
  "silent_truncation",
  "projection_authoritative"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Власник: [core/contracts.schema.json#/\\$defs/ReadyFrontier](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/ReadyFrontier](#)

78. Relation

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, relation_id, kind, source_type, source_id, target_type, target_id, activation_digest, created_at

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
created_at	так	\$ref="#/\$defs/Timestamp"
kind	так	enum=["DEPENDS_ON", "READS", "PRODUCES", "VALIDATES", "CITES", "BLOCKS"]
record_type	так	const="Відношення"
relation_id	так	\$ref="#/\$defs/Id"
source_id	так	\$ref="#/\$defs/Id"
source_type	так	\$ref="#/\$defs/Id"
target_id	так	\$ref="#/\$defs/Id"
target_type	так	\$ref="#/\$defs/Id"

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "created_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "kind": {
      "enum": [
        "DEPENDS_ON",
        "READS",
        "PRODUCES",
        "VALIDATES",
        "CITES",
        "BLOCKS"
      ]
    },
    "record_type": {
      "const": "Відношення"
    },
    "relation_id": {

```

```

    "$ref": "#/$defs/Id"
  },
  "source_id": {
    "$ref": "#/$defs/Id"
  },
  "source_type": {
    "$ref": "#/$defs/Id"
  },
  "target_id": {
    "$ref": "#/$defs/Id"
  },
  "target_type": {
    "$ref": "#/$defs/Id"
  }
},
"required": [
  "record_type",
  "relation_id",
  "kind",
  "source_type",
  "source_id",
  "target_type",
  "target_id",
  "activation_digest",
  "created_at"
],
"type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/Relation | Валідатор: urn:promin:contracts:1#/\$defs/Relation

79. RelationDef

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: kind, source, target, rules

Властивість	Обов'язков е	Структурне правило
kind	так	enum=["DEPENDS_ON", "READS", "PRODUCES", "VALIDATES", "CITES", "BLOCKS"]
rules	так	type="array"; minItems=1; maxItems=8; uniqueItems=true
source	так	type="array"; minItems=1; maxItems=4; uniqueItems=true
target	так	type="array"; minItems=1; maxItems=4; uniqueItems=true

```

{
  "additionalProperties": false,
  "properties": {
    "kind": {
      "enum": [
        "DEPENDS_ON",
        "READS",
        "PRODUCES",
        "VALIDATES",
        "CITES",
        "BLOCKS"
      ]
    },
    "rules": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 8,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "source": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 4,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "target": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 4,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    }
  }
}

```



```

    },
    "target": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 4,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    }
  },
  "required": [
    "kind",
    "source",
    "target",
    "rules"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/RelationDef | Валідатор: urn:promin:contracts:1#/\$defs/RelationDef

80. RelativePath

Опис типу: type="string"; pattern="^(?!/)(?![A-Za-z:])(?!.*\\\\)(?!.*(?:^|/|\\\\.\\.(?:/|\$))(?!.*//).+\$"; minLength=1; maxLength=4096

Обов'язкові поля: -

```

{
  "maxLength": 4096,
  "minLength": 1,
  "pattern": "^(?!/)(?![A-Za-z:])(?!.*\\\\\\)(?!.*(?:^|/|\\\\.\\.(?:/|$))(?!.*//).+$",
  "type": "string"
}

```

Власник: core/contracts.schema.json#/\$defs/RelativePath | Валідатор: urn:promin:contracts:1#/\$defs/RelativePath

81. ReleaseEvidenceInvocation

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: invocation_id, operation, arguments_digest, started_at, completed_at, exit_code, platform_binding_digest

Властивість	Обов'язков е	Структурне правило
arguments_digest	так	\$ref="#/\$defs/Digest"
completed_at	так	\$ref="#/\$defs/Timestamp"
exit_code	так	type="integer"
invocation_id	так	\$ref="#/\$defs/Id"
operation	так	\$ref="#/\$defs/Id"
platform_binding_digest	так	\$ref="#/\$defs/Digest"
started_at	так	\$ref="#/\$defs/Timestamp"

```

{
  "additionalProperties": false,
  "properties": {
    "arguments_digest": {
      "$ref": "#/$defs/Digest"
    },
    "completed_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "exit_code": {
      "type": "integer"
    },
    "invocation_id": {
      "$ref": "#/$defs/Id"
    },
    "operation": {
      "$ref": "#/$defs/Id"
    },
    "platform_binding_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
}

```

```

    "started_at": {
      "$ref": "#/$defs/Timestamp"
    }
  },
  "required": [
    "invocation_id",
    "operation",
    "arguments_digest",
    "started_at",
    "completed_at",
    "exit_code",
    "platform_binding_digest"
  ],
  "type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/ReleaseEvidenceInvocation](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/ReleaseEvidenceInvocation](#)

82. ReleaseEvidenceProducer

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: tool_path, tool_sha256, tool_version

Властивість	Обов'язков е	Структурне правило
tool_path	так	enum=["tools/generate_human.py", "tools/promin_no_degradation.py", "tools/promin_package.py", "tools/promin_saturation.py", "tools/promin_saturation_audit.py", "tools/promin_validate.py"]
tool_sha256	так	\$ref="#/\$defs/Digest"
tool_version	так	\$ref="#/\$defs/SemVer"

```

{
  "additionalProperties": false,
  "properties": {
    "tool_path": {
      "enum": [
        "tools/generate_human.py",
        "tools/promin_no_degradation.py",
        "tools/promin_package.py",
        "tools/promin_saturation.py",
        "tools/promin_saturation_audit.py",
        "tools/promin_validate.py"
      ]
    },
    "tool_sha256": {
      "$ref": "#/$defs/Digest"
    },
    "tool_version": {
      "$ref": "#/$defs/SemVer"
    }
  },
  "required": [
    "tool_path",
    "tool_sha256",
    "tool_version"
  ],
  "type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/ReleaseEvidenceProducer](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/ReleaseEvidenceProducer](#)

83. ResearchDraftClaim

Опис типу: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then

Обов'язкові поля: claim_id, claim_kind, claim_class, claim_digest, text, citation_required, citation_refs, evidence_refs, verification_status, normative_candidate, normative_use_allowed

Властивість	Обов'язков е	Структурне правило
blocked_reason	ні	type="string"; minLength=1; maxLength=2048
citation_refs	так	type="array"; maxItems=32; uniqueItems=true
citation_required	так	type="boolean"

Властивість	Обов'язков е	Структурне правило
claim_class	так	enum=["distribution", "fact", "marketing", "proof"]
claim_digest	так	\$ref="#/\$defs/Digest"
claim_id	так	\$ref="#/\$defs/Id"
claim_kind	так	enum=["research_question", "hypothesis", "assumption", "evidence_claim", "blocked_claim"]
evidence_refs	так	type="array"; maxItems=32; uniqueItems=true
normative_candidate	так	type="boolean"
normative_use_allowed	так	const=false
text	так	type="string"; minLength=1; maxLength=8192
verification_status	так	enum=["unverified", "source-bound", "verified", "blocked"]

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [
            "blocked_reason"
          ]
        }
      },
      "if": {
        "anyOf": [
          {
            "properties": {
              "claim_kind": {
                "const": "blocked_claim"
              }
            },
            "required": [
              "claim_kind"
            ]
          },
          {
            "properties": {
              "verification_status": {
                "const": "blocked"
              }
            }
          }
        ]
      },
      "required": [
        "verification_status"
      ]
    }
  ],
  "then": {
    "required": [
      "blocked_reason"
    ]
  }
},
{
  "properties": {
    "blocked_reason": {
      "maxLength": 2048,
      "minLength": 1,
      "type": "string"
    },
    "citation_refs": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 32,
      "type": "array",
      "uniqueItems": true
    },
    "citation_required": {
      "type": "boolean"
    }
  }
}

```

```

"claim_class": {
  "enum": [
    "distribution",
    "fact",
    "marketing",
    "proof"
  ]
},
"claim_digest": {
  "$ref": "#/$defs/Digest"
},
"claim_id": {
  "$ref": "#/$defs/Id"
},
"claim_kind": {
  "enum": [
    "research_question",
    "hypothesis",
    "assumption",
    "evidence_claim",
    "blocked_claim"
  ]
},
"evidence_refs": {
  "items": {

```

```

    "additionalProperties": false,
    "properties": {
      "artifact_id": {
        "$ref": "#/$defs/Id"
      },
      "artifact_record_digest": {
        "$ref": "#/$defs/Digest"
      }
    },
    "required": [
      "artifact_id",
      "artifact_record_digest"
    ],
    "type": "object"
  },
  "maxItems": 32,
  "type": "array",
  "uniqueItems": true
},
"normative_candidate": {
  "type": "boolean"
},
"normative_use_allowed": {
  "const": false
},
"text": {
  "maxLength": 8192,
  "minLength": 1,

```

```

    "type": "string"
  },
  "verification_status": {
    "enum": [
      "unverified",
      "source-bound",
      "verified",
      "blocked"
    ]
  }
},
"required": [
  "claim_id",
  "claim_kind",
  "claim_class",
  "claim_digest",
  "text",
  "citation_required",
  "citation_refs",
  "evidence_refs",
  "verification_status",
  "normative_candidate",
  "normative_use_allowed"
],

```

```
"type": "object"
}
```

Власник: core/contracts.schema.json#/\$defs/ResearchDraftClaim | Валідатор: urn:promin:contracts:1#/\$defs/ResearchDraftClaim

84. ResearchDraftIntake

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, migration_rule, license_plan_digest, sources, claims, requested_flag_mutations

Властивість	Обов'язков е	Структурне правило
claims	так	type="array"; minItems=1; maxItems=256
license_plan_digest	так	\$ref="#/\$defs/Digest"
migration_rule	так	const="MIGRATE_FACTS_NOT_FILES"
record_type	так	const="ResearchDraftIntake"
requested_flag_mutations	так	type="array"; const=[]
sources	так	type="array"; minItems=1; maxItems=128

```
{
  "additionalProperties": false,
  "properties": {
    "claims": {
      "items": {
        "$ref": "#/$defs/ResearchDraftClaim"
      },
      "maxItems": 256,
      "minItems": 1,
      "type": "array"
    },
    "license_plan_digest": {
      "$ref": "#/$defs/Digest"
    },
    "migration_rule": {
      "const": "MIGRATE_FACTS_NOT_FILES"
    },
    "record_type": {
      "const": "ResearchDraftIntake"
    },
    "requested_flag_mutations": {
      "const": [],
      "type": "array"
    },
    "sources": {
      "items": {
        "$ref": "#/$defs/ResearchSourceIntake"
      },
      "maxItems": 128,
      "minItems": 1,
      "type": "array"
    }
  },
  "required": [
    "record_type",
    "migration_rule",
    "license_plan_digest",
    "sources",
    "claims",
    "requested_flag_mutations"
  ],
  "type": "object",
  "x-promin-ingress-operations": [
    "transient"
  ],
  "x-promin-max-canonical-bytes": 1048576,
  "x-promin-persistence": "transient-ingress-only"
}
```

Власник: core/contracts.schema.json#/\$defs/ResearchDraftIntake | Валідатор: urn:promin:contracts:1#/\$defs/ResearchDraftIntake

85. ResearchSourceIntake

Опис типу: type="object"; additionalProperties=false; allOf[1]=keys:else,if,then

Обов'язкові поля: source_id, source_mode, source_locator, source_provenance_digest, media_type, license_provider_id, license_expression, license_plan_digest

Властивість	Обов'язков е	Структурне правило
artifact_id	ні	\$ref="#/\$defs/Id"
artifact_record_digest	ні	\$ref="#/\$defs/Digest"
content_digest	ні	\$ref="#/\$defs/Digest"
license_expression	так	type="string"; minLength=1; maxLength=512
license_plan_digest	так	\$ref="#/\$defs/Digest"
license_provider_id	так	\$ref="#/\$defs/Id"
media_type	так	type="string"; minLength=1; maxLength=256
missing_reason	ні	type="string"; minLength=1; maxLength=2048
size_bytes	ні	type="integer"; minimum=0
source_id	так	\$ref="#/\$defs/Id"
source_locator	так	type="string"; minLength=1; maxLength=2048
source_mode	так	enum=["captured-payload", "metadata-only"]
source_provenance_digest	так	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "anyOf": [
            {
              "required": [
                "artifact_id"
              ]
            },
            {
              "required": [
                "artifact_record_digest"
              ]
            },
            {
              "required": [
                "content_digest"
              ]
            },
            {
              "required": [
                "size_bytes"
              ]
            }
          ]
        }
      }
    },
    {
      "required": [
        "missing_reason"
      ]
    },
    {
      "if": {
        "properties": {
          "source_mode": {
            "const": "captured-payload"
          }
        }
      },
      "required": [
        "source_mode"
      ]
    },
    {
      "then": {
        "not": {
          "required": [
            "missing_reason"
          ]
        }
      },
      "required": [
        "artifact_id",
        "artifact_record_digest",

```

```

        "content_digest",
        "size_bytes"
    ]
}

},
"properties": {
    "artifact_id": {
        "$ref": "#/$defs/Id"
    },
    "artifact_record_digest": {
        "$ref": "#/$defs/Digest"
    },
    "content_digest": {
        "$ref": "#/$defs/Digest"
    },
    "license_expression": {
        "maxLength": 512,
        "minLength": 1,
        "type": "string"
    },
    "license_plan_digest": {
        "$ref": "#/$defs/Digest"
    },
    "license_provider_id": {
        "$ref": "#/$defs/Id"
    },
    "media_type": {
        "maxLength": 256,
        "minLength": 1,
        "type": "string"
    },
},

```

```

    "missing_reason": {
        "maxLength": 2048,
        "minLength": 1,
        "type": "string"
    },
    "size_bytes": {
        "minimum": 0,
        "type": "integer"
    },
    "source_id": {
        "$ref": "#/$defs/Id"
    },
    "source_locator": {
        "maxLength": 2048,
        "minLength": 1,
        "type": "string"
    },
    "source_mode": {
        "enum": [
            "captured-payload",
            "metadata-only"
        ]
    },
    "source_provenance_digest": {
        "$ref": "#/$defs/Digest"
    }
},
"required": [

```

```

    "source_id",
    "source_mode",
    "source_locator",
    "source_provenance_digest",
    "media_type",
    "license_provider_id",
    "license_expression",
    "license_plan_digest"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/ResearchSourceIntake](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/ResearchSourceIntake](#)

86. RetrievalPage

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, query, activation_digest, depth, ranking, budget, head_digest, projection_digest, entities, relations, evidence, truncated, continuation, continuation_version, stream_cursor, next_stream_cursor, effective_top_k, selected_seed_count, refinement_required, refinement_hints, unselected_matches_traversable, selected_closure_complete, silent_truncation, projection_authoritative

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
budget	так	\$ref="#/\$defs/Budget"
continuation	так	oneOf[2]=type:null type:object
continuation_version	так	const=2
depth	так	type="integer"; minimum=1; maximum=12
effective_top_k	так	type="integer"; minimum=1; maximum=12
entities	так	type="array"; maxItems=32
evidence	так	type="array"; maxItems=32
head_digest	так	oneOf[2]=#/\$defs/Digest type:null
next_stream_cursor	так	type="integer"; minimum=0
projection_authoritative	так	const=false
projection_digest	так	\$ref="#/\$defs/Digest"
query	так	type="string"; minLength=1; maxLength=4096
ranking	так	const="bm25-v1"
record_type	так	const="RetrievalPage"
refinement_hints	так	type="array"; maxItems=8; uniqueItems=true
refinement_required	так	type="boolean"
relations	так	type="array"; maxItems=48
selected_closure_complete	так	type="boolean"
selected_seed_count	так	type="integer"; minimum=0
silent_truncation	так	const=false
stream_cursor	так	type="integer"; minimum=0
truncated	так	type="boolean"
unselected_matches_traversable	так	const=false

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "budget": {
      "$ref": "#/$defs/Budget"
    },
    "continuation": {
      "oneOf": [
        {
          "type": "null"
        },
        {
          "additionalProperties": false,
          "properties": {
            "activation_digest": {
              "$ref": "#/$defs/Digest"
            },
            "budget_digest": {
              "$ref": "#/$defs/Digest"
            },
            "cursor": {
              "minimum": 0,
              "type": "integer"
            },
            "depth": {

```



```

        "maximum": 12,
        "minimum": 1,
        "type": "integer"
    },
    "expiry": {
        "$ref": "#/$defs/Timestamp"
    },
    "head_digest": {
        "oneOf": [
            {
                "$ref": "#/$defs/Digest"
            },
            {
                "type": "null"
            }
        ]
    },
    "implementation_closure_digest": {
        "$ref": "#/$defs/Digest"
    },
    "projection_digest": {
        "$ref": "#/$defs/Digest"
    },
    "ranking": {
        "const": "bm25-v1"
    },
    "resume_binding_digest": {
        "$ref": "#/$defs/Digest"
    }

```

```

    },
    "token": {
        "maxLength": 256,
        "minLength": 1,
        "pattern": "^promin-v2\\. [A-Za-z0-9_-]{22}\\.[0-9a-f]{64}\\.[A-Za-z0-9_-]{43}$",
        "type": "string"
    },
    "traversal": {
        "const": "typed-bfs-v2"
    },
    "version": {
        "const": 2
    }
},
"required": [
    "version",
    "traversal",
    "token",
    "cursor",
    "expiry",
    "activation_digest",
    "head_digest",
    "projection_digest",
    "implementation_closure_digest",
    "ranking",
    "depth",
    "budget_digest",
    "resume_binding_digest"
]

```

```

    ],
    "type": "object"
}
]
},
"continuation_version": {
    "const": 2
},
"depth": {
    "maximum": 12,
    "minimum": 1,
    "type": "integer"
},
"effective_top_k": {
    "maximum": 12,
    "minimum": 1,
    "type": "integer"
},
"entities": {
    "items": {
        "additionalProperties": false,
        "properties": {

```

```

    "data_class": {
      "enum": [
        "operational-record",
        "untrusted-source"
      ]
    },

```

```

    "entity_type": {
      "enum": [
        "Завдання",
        "Candidate",
        "Artifact",
        "Run",
        "Finding",
        "Decision",
        "GateResult",
        "Grant",
        "Оренда"
      ]
    },
    "id": {
      "$ref": "#/$defs/Id"
    },
    "payload": {
      "type": "object"
    },
    "source_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "id",
    "entity_type",
    "data_class",
    "payload",

```

```

    "source_digest"
  ],
  "type": "object"
},
"maxItems": 32,
"type": "array"
},
"evidence": {
  "items": {
    "$ref": "#/$defs/Artifact"
  },
  "maxItems": 32,
  "type": "array"
},
"head_digest": {
  "oneOf": [
    {
      "$ref": "#/$defs/Digest"
    },
    {
      "type": "null"
    }
  ]
},
"next_stream_cursor": {
  "minimum": 0,
  "type": "integer"
},

```

```

"projection_authoritative": {
  "const": false
},
"projection_digest": {
  "$ref": "#/$defs/Digest"
},
"query": {
  "maxLength": 4096,
  "minLength": 1,
  "type": "string"
},
"ranking": {
  "const": "bm25-v1"
},
"record_type": {

```

```

    "const": "RetrievalPage"
  },
  "refinement_hints": {
    "items": {
      "maxLength": 512,
      "minLength": 1,
      "type": "string"
    },
    "maxItems": 8,
    "type": "array",
    "uniqueItems": true
  },
  "refinement_required": {
    "type": "boolean"
  },
  "relations": {
    "items": {
      "$ref": "#/$defs/Relation"
    },
    "maxItems": 48,
    "type": "array"
  },
  "selected_closure_complete": {
    "type": "boolean"
  },
  "selected_seed_count": {
    "minimum": 0,
    "type": "integer"
  },
  "silent_truncation": {
    "const": false
  },
  "stream_cursor": {
    "minimum": 0,
    "type": "integer"
  },
  "truncated": {
    "type": "boolean"
  },
  "unselected_matches_traversable": {
    "const": false
  }
},
"required": [
  "record_type",
  "query",
  "activation_digest",
  "depth",
  "ranking",
  "budget",
  "head_digest",
  "projection_digest",
  "entities",
  "relations",
  "evidence",
  "truncated",
  "continuation",
  "continuation_version",
  "stream_cursor",
  "next_stream_cursor",
  "effective_top_k",
  "selected_seed_count",
  "refinement_required",
  "refinement_hints",
  "unselected_matches_traversable",
  "selected_closure_complete",
  "silent_truncation",
  "projection_authoritative"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Власник: [core/contracts.schema.json#/\\$defs/RetrievalPage](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/RetrievalPage](#)

87. RootAuthority

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: subject_id, capability_ceiling, scope

Властивість	Обов'язков е	Структурне правило
capability_ceiling	так	type="array"; minItems=1; maxItems=32; uniqueItems=true
scope	так	type="array"; minItems=1; maxItems=32
subject_id	так	\$ref="#/\$defs/Id"

```
{
  "additionalProperties": false,
  "properties": {
    "capability_ceiling": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "scope": {
      "items": {
        "$ref": "#/$defs/ScopeSelector"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array"
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "subject_id",
    "capability_ceiling",
    "scope"
  ],
  "type": "object"
}
```

Власник: core/contracts.schema.json#/\$defs/RootAuthority | Валідатор: urn:promin:contracts:1#/\$defs/RootAuthority

88. RootAuthorization

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: kind, subject_id, proofs

Властивість	Обов'язков е	Структурне правило
kind	так	const="root"
proofs	так	type="array"; minItems=1; maxItems=32; uniqueItems=true
subject_id	так	\$ref="#/\$defs/Id"

```
{
  "additionalProperties": false,
  "properties": {
    "kind": {
      "const": "root"
    },
    "proofs": {
      "items": {
        "$ref": "#/$defs/RootCommandProof"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    }
  }
}
```

```

},
"required": [
  "kind",
  "subject_id",
  "proofs"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/RootAuthorization](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/RootAuthorization](#)

89. RootCommandProof

Опис типу: [oneOf\[2\]=#/\\$defs/LocalRootCommandProof](#) | [#/\\$defs/SignatureRootCommandProof](#)

Обов'язкові поля: -

```

{
  "oneOf": [
    {
      "$ref": "#/$defs/LocalRootCommandProof"
    },
    {
      "$ref": "#/$defs/SignatureRootCommandProof"
    }
  ]
}

```

Власник: [core/contracts.schema.json#/\\$defs/RootCommandProof](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/RootCommandProof](#)

90. Run

Опис типу: `type="object"; additionalProperties=false`

Обов'язкові поля: `record_type`, `run_id`, `run_kind`, `task_id`, `candidate_digest`, `policy_digest`, `tool_digest`, `input_digests`, `status`, `started_at`, `finished_at`, `activation_digest`, `definition_digest`, `implementation_closure_digest`, `provider_binding_digest`

Властивість	Обов'язков е	Структурне правило
<code>activation_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>candidate_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>definition_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>finished_at</code>	так	<code>\$ref="#/\$defs/Timestamp"</code>
<code>implementation_closure_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>input_digests</code>	так	<code>type="array"; minItems=0; maxItems=128; uniqueItems=true</code>
<code>policy_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>provider_binding_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>record_type</code>	так	<code>const="Run"</code>
<code>run_id</code>	так	<code>\$ref="#/\$defs/Id"</code>
<code>run_kind</code>	так	<code>enum=["execution", "validation"]</code>
<code>started_at</code>	так	<code>\$ref="#/\$defs/Timestamp"</code>
<code>status</code>	так	<code>enum=["pass", "fail", "blocked", "skipped", "error"]</code>
<code>task_id</code>	так	<code>\$ref="#/\$defs/Id"</code>
<code>tool_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "definition_digest": {
      "$ref": "#/$defs/Digest"
    },
    "finished_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "implementation_closure_digest": {

```

```

    "$ref": "#/$defs/Digest"
  },
  "input_digests": {
    "items": {
      "$ref": "#/$defs/Digest"
    },
    "maxItems": 128,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "policy_digest": {

```

```

    "$ref": "#/$defs/Digest"
  },
  "provider_binding_digest": {
    "$ref": "#/$defs/Digest"
  },
  "record_type": {
    "const": "Run"
  },
  "run_id": {
    "$ref": "#/$defs/Id"
  },
  "run_kind": {
    "enum": [
      "execution",
      "validation"
    ]
  },
  "started_at": {
    "$ref": "#/$defs/Timestamp"
  },
  "status": {
    "enum": [
      "pass",
      "fail",
      "blocked",
      "skipped",
      "error"
    ]
  }
]

```

```

  },
  "task_id": {
    "$ref": "#/$defs/Id"
  },
  "tool_digest": {
    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "record_type",
  "run_id",
  "run_kind",
  "task_id",
  "candidate_digest",
  "policy_digest",
  "tool_digest",
  "input_digests",
  "status",
  "started_at",
  "finished_at",
  "activation_digest",
  "definition_digest",
  "implementation_closure_digest",
  "provider_binding_digest"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/Run](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/Run](#)

91. SanitizedResearchDraft

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, migration_rule, license_plan_digest, sources, normative_candidates, excluded_claims, requested_flag_mutations, pass_credit, product_acceptance_pass, product_public_approval, public_release_approved

Властивість	Обов'язков	Структурне правило
excluded_claims	так	type="array"; minItems=0; maxItems=256; uniqueItems=true
license_plan_digest	так	\$ref="#/\$defs/Digest"
migration_rule	так	const="MIGRATE_FACTS_NOT_FILES"
normative_candidates	так	type="array"; minItems=0; maxItems=256; uniqueItems=true
pass_credit	так	const=false
product_acceptance_pass	так	const=false
product_public_approval	так	const="not_approved"
public_release_approved	так	const=false
record_type	так	const="SanitizedResearchDraft"
requested_flag_mutations	так	type="array"; const=[]
sources	так	type="array"; minItems=1; maxItems=128; uniqueItems=true

```
{
  "additionalProperties": false,
  "properties": {
    "excluded_claims": {
      "items": {
        "additionalProperties": false,
        "allOf": [
          {
            "else": {
              "not": {
                "required": [
                  "blocked_reason"
                ]
              }
            },
            "if": {
              "anyOf": [
                {
                  "properties": {
                    "claim_kind": {
                      "const": "blocked_claim"
                    }
                  },
                  "required": [
                    "claim_kind"
                  ]
                }
              ],
            },
          ]
        ],
      },
    },
  },
}
```

```
    "properties": {
      "verification_status": {
        "const": "blocked"
      },
      "required": [
        "verification_status"
      ]
    },
    "then": {
      "required": [
        "blocked_reason"
      ]
    }
  },
  "properties": {
    "blocked_reason": {
      "maxLength": 2048,
      "minLength": 1,
      "type": "string"
    },
    "citation_refs": {
      "items": {
        "$ref": "#/$defs/Id"
      },
    },
  },
}
```

```

        "maxItems": 32,
        "type": "array",
        "uniqueItems": true
    },
    "citation_required": {
        "type": "boolean"
    },
    "claim_class": {
        "enum": [
            "distribution",
            "fact",
            "marketing",
            "proof"
        ]
    },
    "claim_digest": {
        "$ref": "#/$defs/Digest"
    },
    "claim_id": {
        "$ref": "#/$defs/Id"
    },
    "claim_kind": {
        "enum": [
            "research_question",
            "hypothesis",
            "assumption",
            "evidence_claim",
            "blocked_claim"
        ]
    },
    "evidence_refs": {
        "items": {
            "additionalProperties": false,
            "properties": {
                "artifact_id": {
                    "$ref": "#/$defs/Id"
                },
                "artifact_record_digest": {
                    "$ref": "#/$defs/Digest"
                }
            },
            "required": [
                "artifact_id",
                "artifact_record_digest"
            ],
            "type": "object"
        },
        "maxItems": 32,
        "type": "array",
        "uniqueItems": true
    },
    "normative_candidate": {
        "type": "boolean"
    },
    "normative_use_allowed": {
        "const": false
    },
    "reason": {
        "maxLength": 2048,
        "minLength": 1,
        "type": "string"
    },
    "text": {
        "maxLength": 8192,
        "minLength": 1,
        "type": "string"
    },
    "verification_status": {
        "enum": [
            "unverified",
            "source-bound",
            "verified",
            "blocked"
        ]
    },
    "required": [
        "claim_id",

```



```
"claim_kind",
"claim_class",
"claim_digest",
"text",
"citation_required",
"citation_refs",
```

```
"evidence_refs",
"verification_status",
"normative_candidate",
"normative_use_allowed",
"reason"
```

```
],
"type": "object"
},
"maxItems": 256,
"minItems": 0,
"type": "array",
"uniqueItems": true
},
"license_plan_digest": {
"$ref": "#/$defs/Digest"
},
"migration_rule": {
"const": "MIGRATE_FACTS_NOT_FILES"
},
"normative_candidates": {
"items": {
"additionalProperties": false,
"properties": {
"citation_refs": {
"items": {
"$ref": "#/$defs/Id"
},
"maxItems": 32,
```

```
"minItems": 1,
"type": "array",
"uniqueItems": true
},
"citation_required": {
"const": true
},
"claim_class": {
"const": "fact"
},
"claim_digest": {
"$ref": "#/$defs/Digest"
},
"claim_id": {
"$ref": "#/$defs/Id"
},
"claim_kind": {
"const": "evidence_claim"
},
"evidence_refs": {
"items": {
"additionalProperties": false,
"properties": {
"artifact_id": {
"$ref": "#/$defs/Id"
},
"artifact_record_digest": {
"$ref": "#/$defs/Digest"
```

```

}
},
"required": [
"artifact_id",
"artifact_record_digest"
],
"type": "object"
},
"maxItems": 32,
"minItems": 1,
"type": "array",
"uniqueItems": true
},
"normative_candidate": {
"const": true
```

```

},
"normative_use_allowed": {
  "const": true
},
"source_bindings": {
  "items": {
    "additionalProperties": false,
    "properties": {
      "artifact_id": {
        "$ref": "#/$defs/Id"
      },
      "artifact_record_digest": {
        "$ref": "#/$defs/Digest"

```

```

      },
      "content_digest": {
        "$ref": "#/$defs/Digest"
      },
      "license_expression": {
        "maxLength": 512,
        "minLength": 1,
        "type": "string"
      },
      "license_plan_digest": {
        "$ref": "#/$defs/Digest"
      },
      "license_provider_id": {
        "$ref": "#/$defs/Id"
      },
      "media_type": {
        "maxLength": 256,
        "minLength": 1,
        "type": "string"
      },
      "size_bytes": {
        "minimum": 0,
        "type": "integer"
      },
      "source_id": {
        "$ref": "#/$defs/Id"
      },
      "source_locator": {

```

```

        "maxLength": 2048,
        "minLength": 1,
        "type": "string"
      },
      "source_mode": {
        "enum": [
          "captured-payload",
          "metadata-only"
        ]
      },
      "source_provenance_digest": {
        "$ref": "#/$defs/Digest"
      }
    },
    "required": [
      "source_id",
      "source_mode",
      "source_locator",
      "source_provenance_digest",
      "artifact_id",
      "artifact_record_digest",
      "content_digest",
      "size_bytes",
      "media_type",
      "license_provider_id",
      "license_expression",
      "license_plan_digest"
    ],

```

```

    "type": "object"
  },
  "maxItems": 32,
  "minItems": 1,
  "type": "array",
  "uniqueItems": true
},
"text": {

```

```

        "maxLength": 8192,
        "minLength": 1,
        "type": "string"
    },
    "verification_status": {
        "enum": [
            "source-bound",
            "verified"
        ]
    }
},
"required": [
    "claim_id",
    "claim_kind",
    "claim_class",
    "claim_digest",
    "text",
    "citation_required",
    "citation_refs",
    "evidence_refs",

```

```

        "verification_status",
        "normative_candidate",
        "normative_use_allowed",
        "source_bindings"
    ],
    "type": "object"
},
"maxItems": 256,
"minItems": 0,
"type": "array",
"uniqueItems": true
},
"pass_credit": {
    "const": false
},
"product_acceptance_pass": {
    "const": false
},
"product_public_approval": {
    "const": "not_approved"
},
"public_release_approved": {
    "const": false
},
"record_type": {
    "const": "SanitizedResearchDraft"
},
"requested_flag_mutations": {

```

```

    "const": [],
    "type": "array"
},
"sources": {
    "items": {
        "additionalProperties": false,
        "allOf": [
            {
                "else": {
                    "not": {
                        "anyOf": [
                            {
                                "required": [
                                    "artifact_id"
                                ]
                            },
                            {
                                "required": [
                                    "artifact_record_digest"
                                ]
                            },
                            {
                                "required": [
                                    "content_digest"
                                ]
                            }
                        ],
                    },
                }
            },
            {
                "required": [

```

```

        "size_bytes"
      ]
    }
  ],
  "required": [
    "missing_reason"
  ],
  "if": {
    "properties": {
      "source_mode": {
        "const": "captured-payload"
      }
    },
    "required": [
      "source_mode"
    ]
  },
  "then": {
    "not": {
      "required": [
        "missing_reason"
      ]
    },
    "required": [
      "artifact_id",
      "artifact_record_digest",

```

```

      "content_digest",
      "size_bytes"
    ]
  }
},
"properties": {
  "artifact_id": {
    "$ref": "#/$defs/Id"
  },
  "artifact_record_digest": {
    "$ref": "#/$defs/Digest"
  },
  "content_digest": {
    "$ref": "#/$defs/Digest"
  },
  "license_expression": {
    "maxLength": 512,
    "minLength": 1,
    "type": "string"
  },
  "license_plan_digest": {
    "$ref": "#/$defs/Digest"
  },
  "license_provider_id": {
    "$ref": "#/$defs/Id"
  },
  "media_type": {

```

```

    "maxLength": 256,
    "minLength": 1,
    "type": "string"
  },
  "missing_reason": {
    "maxLength": 2048,
    "minLength": 1,
    "type": "string"
  },
  "size_bytes": {
    "minimum": 0,
    "type": "integer"
  },
  "source_id": {
    "$ref": "#/$defs/Id"
  },
  "source_locator": {
    "maxLength": 2048,
    "minLength": 1,
    "type": "string"
  },
  "source_mode": {

```

```

        "enum": [
            "captured-payload",
            "metadata-only"
        ],
        "source_provenance_digest": {
            "$ref": "#/$defs/Digest"
        },
        "required": [
            "source_id",
            "source_mode",
            "source_locator",
            "source_provenance_digest",
            "media_type",
            "license_provider_id",
            "license_expression",
            "license_plan_digest"
        ],
        "type": "object"
    },
    "maxItems": 128,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
},
"required": [
    "record_type",
    "migration_rule",
    "license_plan_digest",
    "sources",
    "normative_candidates",
    "excluded_claims",
    "requested_flag_mutations",
    "pass_credit",
    "product_acceptance_pass",
    "product_public_approval",
    "public_release_approved"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Власник: [core/contracts.schema.json#/\\$defs/SanitizedResearchDraft](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/SanitizedResearchDraft](#)

92. SaturationAudit

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, status, candidate_binding_digest, zero_new_iterations, new_findings, pass_credit, artifact_binding, families, collection, iterations, consecutive_full_zero_new, requirements, predeclared_zero_new, acceptance_pass, product_acceptance_pass, producer, invocation, raw_artifact_manifest_digest, producer_attestation, result_digest

Властивість	Обов'язков е	Структурне правило
acceptance_pass	так	const=false
artifact_binding	так	type="object"
candidate_binding_digest	так	\$ref="#/\$defs/Digest"
collection	так	type="object"; additionalProperties=false
consecutive_full_zero_new	так	const=3
families	так	type="array"; minItems=1; uniqueItems=true
invocation	так	\$ref="#/\$defs/ReleaseEvidenceInvocation"
iterations	так	type="array"; minItems=3; maxItems=18
new_findings	так	const=0
pass_credit	так	const=false
predeclared_zero_new	так	const=false
producer	так	\$ref="#/\$defs/ReleaseEvidenceProducer"

Властивість	Обов'язков е	Структурне правило
producer_attestation	так	\$ref="#/\$defs/EvidenceProducerAttestation"
product_acceptance_pass	так	const=false
raw_artifact_manifest_digest	так	\$ref="#/\$defs/Digest"
record_type	так	const="SaturationAudit"
requirements	так	type="object"; additionalProperties=false
result_digest	так	\$ref="#/\$defs/Digest"
status	так	const="pass"
zero_new_iterations	так	const=3

```

{
  "additionalProperties": false,
  "properties": {
    "acceptance_pass": {
      "const": false
    },
    "artifact_binding": {
      "type": "object"
    },
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "collection": {
      "additionalProperties": false,
      "properties": {
        "catalogue_digest": {},
        "families_discovered": {},
        "production_probes_discovered": {}
      },
      "required": [
        "catalogue_digest",
        "families_discovered",
        "production_probes_discovered"
      ],
      "type": "object"
    },
    "consecutive_full_zero_new": {
      "const": 3
    }
  },
  "families": {
    "items": {
      "type": "string"
    },
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "invocation": {
    "$ref": "#/$defs/ReleaseEvidenceInvocation"
  },
  "iterations": {
    "items": {
      "additionalProperties": false,
      "properties": {
        "finding_fingerprints": {},
        "focused_and_integration_tests": {
          "additionalProperties": false,
          "properties": {
            "cache_provider_disabled": {},
            "elapsed_ms": {},
            "exit_code": {},
            "log_bytes": {},
            "log_path": {},
            "log_sha256": {},
            "marker_expression": {},
            "output_digest": {}
          }
        },
        "pytest_only_sentinel_executed": {},
        "pytest_only_sentinel_token_digest": {},
        "runner": {},
        "sentinel_bytes": {},
        "sentinel_path": {},
        "sentinel_sha256": {}
      }
    }
  }
}

```

```

    },
    "required": [
        "exit_code",
        "elapsed_ms",
        "output_digest",
        "runner",
        "marker_expression",
        "cache_provider_disabled",
        "pytest_only_sentinel_executed",
        "pytest_only_sentinel_token_digest",
        "log_path",
        "log_sha256",
        "log_bytes",
        "sentinel_path",
        "sentinel_sha256",
        "sentinel_bytes"
    ],
    "type": "object"
},
"full_system_corpus_rerun": {},
"iteration": {},
"mutation_families_run": {},

```

```

"mutation_seed": {},
"new_finding_classes": {},
"physical_runtime_saturation": {
    "additionalProperties": false,
    "properties": {
        "artifact_binding_digest": {},
        "current_release_regression": {},
        "elapsed_ms": {},
        "execution_route": {},
        "exit_code": {},
        "files": {},
        "forced_continuation_chains": {},
        "forced_union_matches": {},
        "fresh_control_state": {
            "additionalProperties": false,
            "properties": {
                "activation_digest": {
                    "$ref": "#/$defs/Digest"
                },
                "core_bundle_digest": {
                    "$ref": "#/$defs/Digest"
                },
                "implementation_closure_digest": {
                    "$ref": "#/$defs/Digest"
                },
                "init_record_count": {
                    "const": 5
                }
            }
        }
    }
},

```

```

"physical_corpus_recipe": {
    "additionalProperties": false,
    "properties": {
        "bytes": {
            "minimum": 1,
            "type": "integer"
        },
        "file_count": {
            "const": 100000
        },
        "sha256": {
            "$ref": "#/$defs/Digest"
        }
    }
},
"required": [
    "sha256",
    "bytes",
    "file_count"
],
"type": "object"
},
"preset_digest": {
    "$ref": "#/$defs/Digest"
},
"product_tree_reused": {
    "type": "boolean"
},
"product_tree_scans": {

```

```

        "const": 0
    },
    "semantic_state_reused": {
        "const": false
    },
    "status": {
        "const": "created"
    }
},
"required": [
    "status",
    "product_tree_scans",
    "init_record_count",
    "activation_digest",
    "implementation_closure_digest",
    "core_bundle_digest",
    "preset_digest",
    "semantic_state_reused",
    "product_tree_reused",
    "physical_corpus_recipe"
],
"type": "object"
},
"immutable_snapshot_creditable": {},
"invocation_id": {},
"log_bytes": {},
"log_path": {},
"log_sha256": {},

```

```

"metrics": {},
"output_digest": {},
"performance_profile": {},
"platform_binding_digest": {},
"pytest_marker": {},
"pytest_selection_command": {},
"pytest_selection_command_executed": {},
"queries": {},
"record_bytes": {
    "minimum": 1,
    "type": "integer"
},
"record_path": {},
"record_sha256": {},
"result_digest": {}
},
"required": [
    "exit_code",
    "elapsed_ms",
    "output_digest",
    "log_path",
    "log_sha256",
    "log_bytes",
    "files",
    "queries",
    "performance_profile",
    "pytest_marker",
    "pytest_selection_command",

```

```

    "pytest_selection_command_executed",
    "execution_route",
    "fresh_control_state",
    "metrics",
    "artifact_binding_digest",
    "immutable_snapshot_creditable",
    "platform_binding_digest",
    "record_path",
    "record_sha256",
    "record_bytes",
    "result_digest",
    "invocation_id",
    "forced_continuation_chains",
    "forced_union_matches",
    "current_release_regression"
],
"type": "object"
},
"silent_truncations": {},
"status": {},
"streak_reset_reason": {},
"unexpected_accepts": {},

```



```

    "unexpected_crashes": {},
    "unexpected_rejects": {},
    "zero_new_streak_after": {},
    "zero_new_streak_before": {}
  },
  "required": [
    "finding_fingerprints",
    "focused_and_integration_tests",
    "full_system_corpus_rerun",
    "iteration",
    "mutation_families_run",
    "mutation_seed",
    "new_finding_classes",
    "physical_runtime_saturation",
    "silent_truncations",
    "status",
    "streak_reset_reason",
    "unexpected_accepts",
    "unexpected_crashes",
    "unexpected_rejects",
    "zero_new_streak_after",
    "zero_new_streak_before"
  ],
  "type": "object"
},
"maxItems": 18,
"minItems": 3,
"type": "array"
},
"new_findings": {
  "const": 0
},
"pass_credit": {
  "const": false
},
"predeclared_zero_new": {
  "const": false
},
"producer": {
  "$ref": "#/$defs/ReleaseEvidenceProducer"
},
"producer_attestation": {
  "$ref": "#/$defs/EvidenceProducerAttestation"
},
"product_acceptance_pass": {
  "const": false
},
"raw_artifact_manifest_digest": {
  "$ref": "#/$defs/Digest"
},
"record_type": {
  "const": "SaturationAudit"
},
"requirements": {
  "additionalProperties": false,
  "properties": {
    "actual_runtime_queries": {},
    "configured_iteration_limit": {},
    "core_valid_relations": {},
    "depths": {},
    "focused_pytest_marker": {},
    "fresh_control_state_per_iteration": {
      "const": true
    },
    "init_product_tree_scans": {
      "const": 0
    },
    "maximum_iterations": {},
    "minimum_consecutive_full_zero_new": {},
    "missing_scale_workspace_fails": {},
    "mixed_query_classes": {},
    "performance_profile": {},
    "physical_files": {},
    "physical_pytest_command": {},
    "physical_pytest_marker": {},
    "profile_bound_fail_closed": {},
    "raw_file_proxy_ratio": {}
  }
}

```

```

    "semantic_corpus_relations": {},
    "semantic_corpus_tasks": {},
    "semantic_state_reused": {
      "const": false
    },
    "synthetic_task_ratio": {}
  },
  "required": [
    "minimum_consecutive_full_zero_new",
    "maximum_iterations",
    "configured_iteration_limit",
    "physical_files",
    "core_valid_relations",

```

```

    "actual_runtime_queries",
    "depths",
    "raw_file_proxy_ratio",
    "synthetic_task_ratio",
    "semantic_corpus_tasks",
    "semantic_corpus_relations",
    "mixed_query_classes",
    "performance_profile",
    "profile_bound_fail_closed",
    "focused_pytest_marker",
    "physical_pytest_marker",
    "physical_pytest_command",
    "missing_scale_workspace_fails",
    "fresh_control_state_per_iteration",
    "semantic_state_reused",
    "init_product_tree_scans"
  ],
  "type": "object"
},
"result_digest": {
  "$ref": "#/$defs/Digest"
},
"status": {
  "const": "pass"
},
"zero_new_iterations": {
  "const": 3
}
}

```

```

},
"required": [
  "record_type",
  "status",
  "candidate_binding_digest",
  "zero_new_iterations",
  "new_findings",
  "pass_credit",
  "artifact_binding",
  "families",
  "collection",
  "iterations",
  "consecutive_full_zero_new",
  "requirements",
  "predeclared_zero_new",
  "acceptance_pass",
  "product_acceptance_pass",
  "producer",
  "invocation",
  "raw_artifact_manifest_digest",
  "producer_attestation",
  "result_digest"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/SaturationAudit](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/SaturationAudit](#)

93. SaturationEvidence

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, status, candidate_binding_digest, artifact_binding, artifact_binding_unchanged, workspace_initialization, runtime_binding, physical_files, core_valid_relations, runtime_queries, silent_truncations, selected_closure_union_completeness, memory_amplification_at_most_32, core_valid_relations_exact_198999, broad_query_refinement_required, high_cardinality_terms_verified, content_search_verified, miss_behavior_verified,

hostile_proxy_content_verified, exact_artifact_search_verified,
mixed_query_classes_complete, continuation_token_bytes_at_most_256,
continuation_state_bytes_at_most_16384, continuation_token_overhead_at_most_10_percent, pass_credit, physical,
inventory, projection, query_authorization, search, resources, performance, contract_predicates,
raw_artifact_manifest, current_release_regression, claim_scope, acceptance_pass, product_acceptance_pass,
public_release_approved, producer, invocation, raw_artifact_manifest_digest, producer_attestation, result_digest

Властивість	Обов'язков е	Структурне правило
acceptance_pass	так	const=false
artifact_binding	так	type="object"
artifact_binding_unchanged	так	const=true
broad_query_refinement_required	так	type="boolean"
candidate_binding_digest	так	\$ref="#/\$defs/Digest"
claim_scope	так	type="string"; minLength=1
content_search_verified	так	type="boolean"
continuation_state_bytes_at_most_16384	так	type="boolean"
continuation_token_bytes_at_most_256	так	type="boolean"
continuation_token_overhead_at_most_10_percent	так	type="boolean"
contract_predicates	так	type="object"; additionalProperties=false
core_valid_relations	так	const=198999
core_valid_relations_exact_198999	так	const=true
current_release_regression	так	type="object"; additionalProperties=false
exact_artifact_search_verified	так	type="boolean"
high_cardinality_terms_verified	так	type="boolean"
hostile_proxy_content_verified	так	type="boolean"
inventory	так	type="object"; additionalProperties=false
invocation	так	\$ref="#/\$defs/ReleaseEvidenceInvocation"
memory_amplification_at_most_32	так	type="boolean"
miss_behavior_verified	так	type="boolean"
mixed_query_classes_complete	так	type="boolean"
pass_credit	так	const=false
performance	так	type="object"; additionalProperties=false
physical	так	type="object"; additionalProperties=false
physical_files	так	const=100000
producer	так	\$ref="#/\$defs/ReleaseEvidenceProducer"
producer_attestation	так	\$ref="#/\$defs/EvidenceProducerAttestation"
product_acceptance_pass	так	const=false
projection	так	type="object"; additionalProperties=false
public_release_approved	так	const=false
query_authorization	так	type="object"; additionalProperties=false
raw_artifact_manifest	так	type="object"; additionalProperties=false
raw_artifact_manifest_digest	так	\$ref="#/\$defs/Digest"
record_type	так	const="SaturationEvidence"
resources	так	type="object"; additionalProperties=false
result_digest	так	\$ref="#/\$defs/Digest"
runtime_binding	так	type="object"; additionalProperties=false
runtime_queries	так	const=600
search	так	type="object"; additionalProperties=false

Властивість	Обов'язков е	Структурне правило
selected_closure_union_completeness	так	type="number"; minimum=0.0; maximum=1.0
silent_truncations	так	const=0
status	так	enum=["pass", "fail"]
workspace_initialization	так	type="object"; additionalProperties=false
<pre> { "additionalProperties": false, "properties": { "acceptance_pass": { "const": false }, "artifact_binding": { "type": "object" }, "artifact_binding_unchanged": { "const": true }, "broad_query_refinement_required": { "type": "boolean" }, "candidate_binding_digest": { "\$ref": "#/\$defs/Digest" }, "claim_scope": { "minLength": 1, "type": "string" }, "content_search_verified": { "type": "boolean" }, "continuation_state_bytes_at_most_16384": { "type": "boolean" }, "continuation_token_bytes_at_most_256": { "type": "boolean" }, "continuation_token_overhead_at_most_10_percent": { "type": "boolean" }, "contract_predicates": { "additionalProperties": false, "properties": { "broad_query_refinement_required": {}, "content_search_verified": {}, "continuation_state_bytes_at_most_16384": {}, "continuation_token_bytes_at_most_256": {}, "continuation_token_overhead_at_most_10_percent": {}, "continuation_union_complete": {}, "core_valid_relations_exact": { "const": true }, "exact_artifact_binding_unchanged": {}, "exact_artifact_search_verified": {}, "high_cardinality_terms_verified": {}, "hostile_proxy_content_verified": {}, "inventory_incremental_memory_amplification_at_most_32": {}, "inventory_passes_exact": {}, "miss_behavior_verified": {}, "mixed_query_classes_complete": {}, "physical_relation_artifact_coverage_complete": {}, "raw_file_proxy_ratio_exact": {}, "rebuild_digest_equal": {}, "rebuild_product_passes_zero": {}, "runtime_depths_1_through_12": {}, "runtime_queries_exact": { "const": true }, "runtime_query_budget_bounded": {}, "selected_closure_union_complete": {}, "semantic_commit_count_exact": {}, "silent_truncations_zero": {}, "synthetic_task_ratio_zero": {} } } } } </pre>		

```

},
"required": [
  "broad_query_refinement_required",
  "content_search_verified",
  "continuation_state_bytes_at_most_16384",
  "continuation_token_bytes_at_most_256",
  "continuation_token_overhead_at_most_10_percent",
  "continuation_union_complete",
  "core_valid_relations_exact",
  "exact_artifact_binding_unchanged",
  "exact_artifact_search_verified",
  "high_cardinality_terms_verified",
  "hostile_proxy_content_verified",
  "inventory_incremental_memory_amplification_at_most_32",
  "inventory_passes_exact",
  "miss_behavior_verified",
  "mixed_query_classes_complete",

```

```

  "physical_relation_artifact_coverage_complete",
  "raw_file_proxy_ratio_exact",
  "rebuild_digest_equal",
  "rebuild_product_passes_zero",
  "runtime_depths_1_through_12",
  "runtime_queries_exact",
  "runtime_query_budget_bounded",
  "selected_closure_union_complete",
  "semantic_commit_count_exact",
  "silent_truncations_zero",
  "synthetic_task_ratio_zero"
],
"type": "object"
},
"core_valid_relations": {
  "const": 198999
},
"core_valid_relations_exact_198999": {
  "const": true
},
"current_release_regression": {
  "additionalProperties": false,
  "properties": {
    "after": {},
    "before": {},
    "eligibility_remained_false": {},
    "historical_decision_preserved": {}
  }
},

```

```


```

```

"required": [
  "before",
  "after",
  "eligibility_remained_false",
  "historical_decision_preserved"
],

```

```

"type": "object"
},

```

```

"exact_artifact_search_verified": {
  "type": "boolean"
},

```

```

"high_cardinality_terms_verified": {
  "type": "boolean"
},

```

```

"hostile_proxy_content_verified": {
  "type": "boolean"
},

```

```

"inventory": {
  "additionalProperties": false,
  "properties": {
    "candidate_digest": {},
    "elapsed_ms": {},
    "entries": {},
    "passes": {},
    "snapshot": {},
    "stream_bytes": {}
  }
},

```

```

"required": [

```

```

  "candidate_digest",
  "elapsed_ms",
  "entries",
  "passes",

```

```

        "snapshot",
        "stream_bytes"
    ],
    "type": "object"
},
"invocation": {
    "$ref": "#/$defs/ReleaseEvidenceInvocation"
},
"memory_amplification_at_most_32": {
    "type": "boolean"
},
"miss_behavior_verified": {
    "type": "boolean"
},
"mixed_query_classes_complete": {
    "type": "boolean"
},
"pass_credit": {
    "const": false
},
"performance": {
    "additionalProperties": false,
    "properties": {
        "all_within_profile": {},

```

```

        "compatible_platforms": {
            "const": [
                "linux",
                "windows"
            ]
        },
        "observed": {},
        "platform_binding": {
            "additionalProperties": false,
            "properties": {
                "binding_digest": {
                    "pattern": "^sha256:[0-9a-f]{64}$",
                    "type": "string"
                },
                "machine": {
                    "minLength": 1,
                    "type": "string"
                },
                "profile_key": {
                    "minLength": 1,
                    "type": "string"
                },
                "python_executable_sha256": {
                    "pattern": "^sha256:[0-9a-f]{64}$",
                    "type": "string"
                },
                "python_implementation": {
                    "const": "CPython"

```

```

            },
            "python_version": {
                "pattern": "^[0-9]+\\.?[0-9]+\\.?[0-9]+$",
                "type": "string"
            },
            "release": {
                "minLength": 1,
                "type": "string"
            },
            "sqlite_version": {
                "pattern": "^[0-9]+\\.?[0-9]+\\.?[0-9]+$",
                "type": "string"
            },
            "system": {
                "enum": [
                    "darwin",
                    "linux",
                    "windows"
                ]
            },
            "required": [
                "binding_digest",
                "system",
                "release",
                "machine",

```

```

        "python_implementation",
        "python_version",

        "python_executable_sha256",
        "sqlite_version",
        "profile_key"
    ],
    "type": "object"
},
"predicates": {},
"profile_id": {
    "const": "portable-local-v1"
},
"requires_same_runner_no_degradation": {
    "const": true
},
"thresholds": {
    "additionalProperties": false,
    "properties": {
        "commit_bytes_per_changed_record_max": {
            "const": 32768
        },
        "commit_p95_ms_max": {
            "const": 500
        },
        "commit_p99_ms_max": {
            "const": 1000
        },
        "database_bytes_max": {
            "const": 402653184
        }
    },

    "p50_ms_max": {
        "const": 100
    },
    "p95_ms_max": {
        "const": 150
    },
    "p99_ms_max": {
        "const": 500
    },
    "peak_rss_bytes_max": {
        "const": 805306368
    },
    "projection_amplification_max": {
        "const": 20
    },
    "runtime_checkpoint_count_max": {
        "const": 24
    },
    "semantic_inflation_max": {
        "const": 3.25
    },
    "semantic_ingestion_seconds_max": {
        "const": 600
    }
},
"required": [
    "p50_ms_max",
    "p95_ms_max",
    "p99_ms_max",
    "peak_rss_bytes_max",
    "database_bytes_max",
    "projection_amplification_max",
    "semantic_inflation_max",
    "commit_p95_ms_max",
    "commit_p99_ms_max",
    "commit_bytes_per_changed_record_max",
    "runtime_checkpoint_count_max",
    "semantic_ingestion_seconds_max"
],
"type": "object"
}
},
"required": [
    "all_within_profile",
    "compatible_platforms",
    "observed",
    "platform_binding",

```

```

    "predicates",
    "profile_id",
    "requires_same_runner_no_degradation",
    "thresholds"
  ],
  "type": "object"
},
"physical": {
  "additionalProperties": false,

```

```

    "properties": {
      "explicit_semantic_corpus": {
        "additionalProperties": false,
        "properties": {
          "conflicting_exact_id_text": {},
          "continuation_query_ids": {},
          "depths": {},
          "generation": {},
          "harness_generated": {
            "const": true
          },
          "high_fanout": {},
          "physical_relation_fixture": {
            "additionalProperties": false,
            "properties": {
              "artifact_target_count": {},
              "artifact_target_coverage": {},
              "relation_count": {},
              "relation_kind": {},
              "relations_per_atomic_batch_max": {},
              "target_type": {},
              "task_count": {}
            },
            "required": [
              "task_count",
              "relation_count",
              "relation_kind",
              "target_type",

```

```

              "artifact_target_count",
              "artifact_target_coverage",
              "relations_per_atomic_batch_max"
            ],
            "type": "object"
          },
          "physical_relation_fixture_reused": {
            "const": false
          },
          "product_acceptance_credit": {
            "const": false
          },
          "query_ids": {},
          "record_type": {
            "const": "SaturationSemanticCorpus"
          },
          "relation_count": {},
          "reused": {
            "const": false
          },
          "search_fixture": {
            "additionalProperties": false,
            "properties": {
              "depths": {},
              "high_fanout": {},
              "relation_count": {},
              "task_count": {}
            },

```

```

            "required": [
              "task_count",
              "relation_count",
              "depths",
              "high_fanout"
            ],
            "type": "object"
          },
          "search_fixture_reused": {
            "const": false
          },
          "task_count": {}

```



```

    },
    "required": [
      "record_type",
      "generation",
      "harness_generated",
      "product_acceptance_credit",
      "task_count",
      "relation_count",
      "depths",
      "high_fanout",
      "conflicting_exact_id_text",
      "query_ids",
      "continuation_query_ids",
      "reused",
      "search_fixture",
      "physical_relation_fixture",
      "search_fixture_reused",
      "physical_relation_fixture_reused"
    ],
    "type": "object"
  },
  "files": {},
  "generation_elapsed_ms": {},
  "inventory_relations": {},
  "raw_file_proxies": {},
  "raw_file_proxy_ratio": {},
  "raw_files": {},
  "relations": {},
  "reused_product": {},
  "semantic_proxies": {},
  "synthetic_task_count": {},
  "synthetic_task_ratio": {},
  "synthetic_tasks": {},
  "vcs_commit": {},
  "vcs_provider_version": {},
  "vcs_tree_digest": {},
  "vcs_tree_files": {}
},
"required": [
  "explicit_semantic_corpus",
  "files",
  "generation_elapsed_ms",
  "inventory_relations",
  "raw_file_proxies",
  "raw_file_proxy_ratio",
  "raw_files",
  "relations",
  "reused_product",
  "semantic_proxies",
  "synthetic_task_count",
  "synthetic_task_ratio",
  "synthetic_tasks",
  "vcs_commit",
  "vcs_provider_version",
  "vcs_tree_digest",
  "vcs_tree_files"
],
"type": "object"
},
"physical_files": {
  "const": 100000
},
"producer": {
  "$ref": "#/$defs/ReleaseEvidenceProducer"
},
"producer_attestation": {
  "$ref": "#/$defs/EvidenceProducerAttestation"
},
"product_acceptance_pass": {
  "const": false
},
"projection": {
  "additionalProperties": false,
  "properties": {
    "database_bytes": {},
    "elapsed_ms": {},
    "entity_count": {},

```

```

    "equal_semantic_digest": {},
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "initial_inventory_passes": {},
    "initial_product_passes": {},
    "inventory_integrity": {},
    "inventory_projection_amplification": {},
    "projection_amplification": {},
    "rebuild_inventory_passes": {},
    "rebuild_product_passes": {},
    "relation_count": {},
    "semantic_digest": {},
    "semantic_inflation": {}
  },
  "required": [
    "database_bytes",
    "elapsed_ms",
    "entity_count",
    "equal_semantic_digest",
    "implementation_closure_digest",
    "initial_inventory_passes",
    "initial_product_passes",

```

```

    "inventory_integrity",
    "inventory_projection_amplification",
    "projection_amplification",
    "rebuild_inventory_passes",
    "rebuild_product_passes",
    "relation_count",
    "semantic_digest",
    "semantic_inflation"
  ],
  "type": "object"
},
"public_release_approved": {
  "const": false
},
"query_authorization": {
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "capability_id": {
      "const": "projection.read"
    },
    "claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "expires_at": {
      "$ref": "#/$defs/Timestamp"
    }
  }
},

```

```

  },
  "grant_id": {
    "$ref": "#/$defs/Id"
  },
  "scope": {
    "items": {
      "type": "object"
    },
    "maxItems": 128,
    "minItems": 1,
    "type": "array"
  },
  "subject_id": {
    "$ref": "#/$defs/Id"
  }
},
"required": [
  "activation_digest",
  "capability_id",
  "claim_digest",
  "expires_at",
  "grant_id",
  "scope",
  "subject_id"
],
"type": "object"

```

```

},
"raw_artifact_manifest": {
  "additionalProperties": false,
  "properties": {
    "artifact_count": {
      "const": 6
    },
    "artifacts": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "bytes": {
            "maximum": 268435456,
            "minimum": 1,
            "type": "integer"
          },
          "media_type": {
            "enum": [
              "application/json",
              "application/x-ndjson"
            ]
          },
          "path": {
            "pattern": "^raw/",
            "type": "string"
          },
          "records": {
            "maximum": 100000,
            "minimum": 1,
            "type": "integer"
          }
        }
      },
      "role": {
        "enum": [
          "inventory-stream",
          "query-results",
          "process-samples",
          "continuation-state-manifest",
          "phase-log",
          "operation-metrics"
        ]
      },
      "sha256": {
        "$ref": "#/$defs/Digest"
      }
    },
    "required": [
      "role",
      "path",
      "media_type",
      "sha256",
      "bytes",
      "records"
    ],
    "type": "object"
  },
  "maxItems": 6,
  "minItems": 6,
  "type": "array"
},
"evidence_class": {
  "const": "harness_generated"
},
"inventory_identity_digest": {
  "$ref": "#/$defs/Digest"
},
"inventory_stream_digest": {
  "$ref": "#/$defs/Digest"
},
"manifest_digest": {
  "$ref": "#/$defs/Digest"
},
"path_scope": {
  "const": "saturation-result-directory"
},
"product_acceptance_credit": {
  "const": false
},

```

```

    "record_type": {
      "const": "SaturationRawArtifactManifest"
    }
  },
  "required": [
    "record_type",
    "path_scope",
    "evidence_class",
    "product_acceptance_credit",
    "artifacts",
    "artifact_count",
    "inventory_stream_digest",
    "inventory_identity_digest",
    "manifest_digest"
  ],
  "type": "object"
},
"raw_artifact_manifest_digest": {
  "$ref": "#/$defs/Digest"
},
"record_type": {
  "const": "SaturationEvidence"
},
"resources": {
  "additionalProperties": false,
  "properties": {
    "inventory_absolute_rss_amplification": {},
    "inventory_incremental_memory_amplification": {},
    "inventory_pipeline_incremental_peak_bytes": {},
    "inventory_pipeline_peak_rss_bytes": {},
    "inventory_stage_rss": {},
    "memory_amplification_metric": {},
    "peak_rss_bytes": {},
    "projection_stage_rss": {}
  }
},
"required": [
  "inventory_absolute_rss_amplification",
  "inventory_incremental_memory_amplification",
  "inventory_pipeline_incremental_peak_bytes",
  "inventory_pipeline_peak_rss_bytes",
  "inventory_stage_rss",
  "memory_amplification_metric",
  "peak_rss_bytes",
  "projection_stage_rss"
],
"type": "object"
},
"result_digest": {
  "$ref": "#/$defs/Digest"
},
"runtime_binding": {
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "implementation_closure_digest": {
      "$ref": "#/$defs/Digest"
    },
    "platform_binding_digest": {
      "pattern": "^sha256:[0-9a-f]{64}$",
      "type": "string"
    }
  }
},
"required": [
  "activation_digest",
  "implementation_closure_digest",
  "platform_binding_digest"
],
"type": "object"
},
"runtime_queries": {
  "const": 600
},
"search": {
  "additionalProperties": false,
  "properties": {

```

```

"actual_runtime_queries": {},
"broad_query_refinement_required": {},
"content_search_verified": {},
"continuation_state": {},
"continuation_token_overhead_at_most_10_percent": {},
"continuation_union_complete": {},
"continuation_union_completeness": {},
"continuations_checked": {},
"depth_counts": {},
"depth_max": {},
"depth_min": {},
"exact_artifact_search_verified": {},
"explicit_truncations": {},
"forced_continuation_chains": {},
"forced_depths": {},
"forced_query_budget": {},

```

```

"forced_union_matches": {},
"high_cardinality_terms_verified": {},
"hostile_proxy_content_verified": {},
"maximum_continuation_token_bytes": {},
"miss_behavior_verified": {},
"mixed_query_classes_complete": {},
"p50_ms": {},
"p95_ms": {},
"p99_ms": {},
"pages_observed": {},
"promin_service_continuation_calls": {},
"promin_service_search_calls": {},
"query_class_depths": {},
"query_class_latency_ms": {},
"query_mix": {},
"result_digest": {},
"runtime_ingress": {},
"runtime_query_budget": {},
"selected_closure_chains": {},
"selected_closure_union_completeness": {},
"silent_truncations": {}
},

```

```

"required": [
  "actual_runtime_queries",
  "broad_query_refinement_required",
  "content_search_verified",
  "continuation_state",
  "continuation_token_overhead_at_most_10_percent",

```

```

  "continuation_union_complete",
  "continuation_union_completeness",
  "continuations_checked",
  "depth_counts",
  "depth_max",
  "depth_min",
  "exact_artifact_search_verified",
  "explicit_truncations",
  "forced_continuation_chains",
  "forced_depths",
  "forced_query_budget",
  "forced_union_matches",
  "high_cardinality_terms_verified",
  "hostile_proxy_content_verified",
  "maximum_continuation_token_bytes",
  "miss_behavior_verified",
  "mixed_query_classes_complete",
  "p50_ms",
  "p95_ms",
  "p99_ms",
  "pages_observed",
  "promin_service_continuation_calls",
  "promin_service_search_calls",
  "query_class_depths",
  "query_class_latency_ms",
  "query_mix",
  "result_digest",
  "runtime_ingress",

```

```

  "runtime_query_budget",
  "selected_closure_chains",
  "selected_closure_union_completeness",
  "silent_truncations"
],

```

```

    "type": "object"
  },
  "selected_closure_union_completeness": {
    "maximum": 1.0,
    "minimum": 0.0,
    "type": "number"
  },
  "silent_truncations": {
    "const": 0
  },
  "status": {
    "enum": [
      "pass",
      "fail"
    ]
  },
  "workspace_initialization": {
    "additionalProperties": false,
    "properties": {
      "activation_digest": {
        "$ref": "#/$defs/Digest"
      },
      "implementation_closure_digest": {
        "$ref": "#/$defs/Digest"
      },
      "init_record_count": {
        "const": 5
      },
      "init_records": {
        "const": [
          "activation.json",
          "authority.json",
          "project.json",
          "standards.json",
          "technologies.json"
        ]
      },
      "product_tree_scans": {
        "const": 0
      },
      "project_id": {
        "const": "promin-physical-saturation"
      },
      "record_type": {
        "const": "SaturationWorkspaceInitialization"
      },
      "snapshot_provider_id": {
        "const": "git-filessystem-inventory"
      },
      "snapshot_provider_version": {
        "minLength": 1,
        "type": "string"
      },
      "status": {
        "enum": [
          "created",
          "reused"
        ]
      }
    }
  },
  "required": [
    "activation_digest",
    "implementation_closure_digest",
    "init_record_count",
    "init_records",
    "product_tree_scans",
    "project_id",
    "record_type",
    "snapshot_provider_id",
    "snapshot_provider_version",
    "status"
  ],
  "type": "object"
}
},
"required": [
  "record_type",

```

```

"status",
"candidate_binding_digest",

"artifact_binding",
"artifact_binding_unchanged",
"workspace_initialization",
"runtime_binding",
"physical_files",
"core_valid_relations",
"runtime_queries",
"silent_truncations",
"selected_closure_union_completeness",
"memory_amplification_at_most_32",
"core_valid_relations_exact_198999",
"broad_query_refinement_required",
"high_cardinality_terms_verified",
"content_search_verified",
"miss_behavior_verified",
"hostile_proxy_content_verified",
"exact_artifact_search_verified",
"mixed_query_classes_complete",
"continuation_token_bytes_at_most_256",
"continuation_state_bytes_at_most_16384",
"continuation_token_overhead_at_most_10_percent",
"pass_credit",
"physical",
"inventory",
"projection",
"query_authorization",
"search",
"resources",

"performance",
"contract_predicates",
"raw_artifact_manifest",
"current_release_regression",
"claim_scope",
"acceptance_pass",
"product_acceptance_pass",
"public_release_approved",
"producer",
"invocation",
"raw_artifact_manifest_digest",
"producer_attestation",
"result_digest"
],
"type": "object"
}

```

Власник: `core/contracts.schema.json#/$defs/SaturationEvidence` | Валідатор: `urn:promin:contracts:1#/$defs/SaturationEvidence`

94. ScopeSelector

Опис типу: `type="object"; additionalProperties=false`

Обов'язкові поля: `kind`, `value`

Властивість	Обов'язков е	Структурне правило
<code>kind</code>	так	<code>enum=["project", "path", "task", "candidate", "artifact", "finding", "all"]</code>
<code>value</code>	так	<code>type="string"; minLength=1; maxLength=4096</code>

```

{
  "additionalProperties": false,
  "properties": {
    "kind": {
      "enum": [
        "project",
        "path",
        "task",
        "candidate",
        "artifact",
        "finding",
        "all"
      ]
    },
    "value": {
      "maxLength": 4096,
      "minLength": 1,

```

```

    "type": "string"
  },
  "required": [
    "kind",
    "value"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/ScopeSelector | Валідатор: urn:promin:contracts:1#/\$defs/ScopeSelector

95. SemVer

Опис типу: type="string"; pattern="^(0|[1-9][0-9]*)\\. (0|[1-9][0-9]*)\\. (0|[1-9][0-9]*) (?:(?:0|[1-9][0-9]*[0-9]*[A-Za-z-][0-9A-Za-z-]*)?(?:\\.(?:0|[1-9][0-9]*[0-9]*[A-Za-z-][0-9A-Za-z-]*)?))?(?:\\+([0-9A-Za-z-]+(?:\\. [0-9A-Za-z-]+)*)?)?\$"; maxLength=128

Обов'язкові поля: -

```

{
  "maxLength": 128,
  "pattern": "^(0|[1-9][0-9]*)\\. (0|[1-9][0-9]*)\\. (0|[1-9][0-9]*) (?:(?:0|[1-9][0-9]*[0-9]*[A-Za-z-][0-9A-Za-z-]*)?(?:\\.(?:0|[1-9][0-9]*[0-9]*[A-Za-z-][0-9A-Za-z-]*)?))?(?:\\+([0-9A-Za-z-]+(?:\\. [0-9A-Za-z-]+)*)?)?$",
  "type": "string"
}

```

Власник: core/contracts.schema.json#/\$defs/SemVer | Валідатор: urn:promin:contracts:1#/\$defs/SemVer

96. SemanticExport

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, export_kind, package_or_archive, candidate_digest, activation_digest, head_event_id, head_sequence, head_digest, core_bundle_digest, preset_digest, implementation_closure_digest, provider_binding_digest, input_digests, output_artifact_id, output_artifact_record_digest, output_digest, output_size_bytes, output_media_type, pass_credit, product_acceptance_pass, product_public_approval, public_release_approved

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
candidate_digest	так	\$ref="#/\$defs/Digest"
core_bundle_digest	так	\$ref="#/\$defs/Digest"
export_kind	так	const="semantic-state"
head_digest	так	oneOf[2]=#/\$defs/Digest type:null
head_event_id	так	oneOf[2]=#/\$defs/Id type:null
head_sequence	так	type="integer"; minimum=0
implementation_closure_digest	так	\$ref="#/\$defs/Digest"
input_digests	так	type="array"; minItems=0; maxItems=128; uniqueItems=true
output_artifact_id	так	\$ref="#/\$defs/Id"
output_artifact_record_digest	так	\$ref="#/\$defs/Digest"
output_digest	так	\$ref="#/\$defs/Digest"
output_media_type	так	const="application/vnd.promin.semantic-state+json"
output_size_bytes	так	type="integer"; minimum=1; maximum=1073741824
package_or_archive	так	const=false
pass_credit	так	const=false
preset_digest	так	\$ref="#/\$defs/Digest"
product_acceptance_pass	так	const=false
product_public_approval	так	const="not_approved"
provider_binding_digest	так	\$ref="#/\$defs/Digest"
public_release_approved	так	const=false
record_type	так	const="SemanticExport"
<pre> { "additionalProperties": false, </pre>		


```

"properties": {
  "activation_digest": {
    "$ref": "#/$defs/Digest"
  },
  "candidate_digest": {
    "$ref": "#/$defs/Digest"
  },
  "core_bundle_digest": {
    "$ref": "#/$defs/Digest"
  },
  "export_kind": {
    "const": "semantic-state"
  },
  "head_digest": {
    "oneOf": [
      {
        "$ref": "#/$defs/Digest"
      },
      {
        "type": "null"
      }
    ]
  },
  "head_event_id": {
    "oneOf": [
      {

```

```

        "$ref": "#/$defs/Id"
      },
      {
        "type": "null"
      }
    ]
  },
  "head_sequence": {
    "minimum": 0,
    "type": "integer"
  },
  "implementation_closure_digest": {
    "$ref": "#/$defs/Digest"
  },
  "input_digests": {
    "items": {
      "$ref": "#/$defs/Digest"
    },
    "maxItems": 128,
    "minItems": 0,
    "type": "array",
    "uniqueItems": true
  },
  "output_artifact_id": {
    "$ref": "#/$defs/Id"
  },
  "output_artifact_record_digest": {
    "$ref": "#/$defs/Digest"
  },

```

```

  },
  "output_digest": {
    "$ref": "#/$defs/Digest"
  },
  "output_media_type": {
    "const": "application/vnd.promin.semantic-state+json"
  },
  "output_size_bytes": {
    "maximum": 1073741824,
    "minimum": 1,
    "type": "integer"
  },
  "package_or_archive": {
    "const": false
  },
  "pass_credit": {
    "const": false
  },
  "preset_digest": {
    "$ref": "#/$defs/Digest"
  },
  "product_acceptance_pass": {
    "const": false
  },
  },

```

```

"product_public_approval": {
  "const": "not_approved"
},
"provider_binding_digest": {
  "$ref": "#/$defs/Digest"
},
"public_release_approved": {
  "const": false
},
"record_type": {
  "const": "SemanticExport"
}
},
"required": [
  "record_type",
  "export_kind",
  "package_or_archive",
  "candidate_digest",
  "activation_digest",
  "head_event_id",
  "head_sequence",
  "head_digest",
  "core_bundle_digest",
  "preset_digest",
  "implementation_closure_digest",
  "provider_binding_digest",
  "input_digests",
  "output_artifact_id",
  "output_artifact_record_digest",
  "output_digest",
  "output_size_bytes",
  "output_media_type",
  "pass_credit",
  "product_acceptance_pass",
  "product_public_approval",
  "public_release_approved"
],
"type": "object",
"x-promin-ingress-operations": [
  "export"
],
"x-promin-persistence": "export-only"
}

```

Власник: core/contracts.schema.json#/\$defs/SemanticExport | Валідатор: urn:promin:contracts:1#/\$defs/SemanticExport

97. SemanticModel

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, schema_ref, document, version, function, persistent_entities, relations, derived_projection_kinds, value_objects, design_rules, technology_capabilities, admission

Властивість	Обов'язков е	Структурне правило
admission	так	const=schema(keys:derived_by_default,entity,relation)
derived_projection_kinds	так	const=[16 items]
design_rules	так	const=schema(keys:artifact_subtypes,candidate_consistency,granularities_are_i ndependent,implementation_closure,inventory_projection,promotion_test,provid er_invocation_evidence,semantic_objects_default)
document	так	const="promin.semantic-model"
function	так	const="Єдиний власник мінімального persistent vocabulary, relation grammar, межі derived projection і provider capability contracts."
persistent_entities	так	const=[9 items]; maxItems=9
record_type	так	const="SemanticModel"
relations	так	const=[6 items]; maxItems=6
schema_ref	так	const="urn:promin:contracts:1#/\$defs/SemanticModel"
technology_capabilities	так	const=[9 items]

Властивість	Обов'язков е	Структурне правило
value_objects	так	const=schema(keys:CandidateDelta,Command,Comparator,ContinuationToken,CurrentReleaseStatus,ImplementationClosure,Operation,PlanProposal,Policy,Predicate,ProviderInvocationEvidence,Query,StandardReleaseDecision,StateCheckpoint,WorkCard)
version	так	const="1.0.0-alpha.1"
<pre> { "additionalProperties": false, "properties": { "admission": { "const": { "derived_by_default": "Each raw inventory file yields exactly one derived Artifact proxy. Raw-file proxies never create a persistent Task or relation; persistence still requires an independent invariant.", "entity": "EntityUtility=QueriesEnabled+RulesEnabled+AutomationEnabled-StorageCost-MaintenanceCost-NavigationNoise > 0", "relation": "RelationUtility=QueriesEnabled+RulesEnabled+SchedulingGain-WriteCost-FanoutNoise-Ambiguity > 0" } }, "derived_projection_kinds": { "const": ["InventoryEntry", "SearchDocument", "InventoryProjectionRow", "ProjectionLimits", "RetrievalPage", "ReadyFrontier", "WorkCardProjection", "WorkCard", "NextResult", "ContinueResult", "NavigationView", "StateCheckpoint", "CurrentReleaseStatus", "DoctorResult", "OperationMetrics", "Report"] }, "design_rules": { "const": { "artifact_subtypes": { "candidate-delta": "An Artifact with kind diff binds one base Candidate, one new Candidate, one WorkCard, and the complete normalized changed-path set.", "evidence": "An evidence Artifact binds exact Candidate, policy, tool, inputs, and every invoked provider adapter; Finding disposition evidence additionally binds the canonical Finding digest." }, "candidate_consistency": { "creditable": "Only an immutable-vcs-tree provider may produce a creditable Candidate in promin v1.", "observational": "A best-effort filesystem observation is explicit, non-creditable, and cannot receive product or release pass credit.", "recipe_binding": "Candidate identity binds the exact candidate_recipe digest, including include, exclude, path identity, collision, and symlink rules." }, "granularities_are_independent": ["semantic", "public", "compilation", "runtime-dispatch", "documentation"], "implementation_closure": { "activation_binding": "Activation binds the aggregate digest of every capability-specific implementation closure.", "binding_shape": "Each TechnologiesInit binding carries runtime, interpreter, zero or more capability components, exactly one matching adapter, and a recomputed closure_digest.", "required_components": { "control-runtime": ["promin-runtime", "python-interpreter"], "query-projection": ["promin-runtime", "python-interpreter", </pre>		

```

        "sqlite"
    ],
    "shape-validation": [
        "promin-runtime",
        "python-interpreter",
        "jsonschema"
    ]
},
"stale_rule": "A runtime, interpreter, component, adapter, or closure digest change invalidates
the Activation and every dependent evidence Artifact, projection, continuation, and current release
closure."
},
"inventory_projection": {
    "current_inventory_persistence": "derived-rebuild-only",
    "current_inventory_record_type": "CurrentInventory",
    "data_class": "untrusted-source",
    "derived_entity_type": "Artifact",
    "input": "Only a verified InventoryResult or its persisted manifest may enter the public rebuild
path.",
    "input_manifest_persistence": "transient-ingress-only",
    "input_manifest_record_type": "InventoryInputManifest",
    "jsonl_row_fields": [
        "path",
        "digest",
        "size",
        "search_text"
    ],
    "one_file_rule": "Each verified raw file produces exactly one InventoryProjectionRow and one
deterministic derived Artifact proxy.",
    "prohibited_derivations": [
        "Завдання",
        "READS",
        "PRODUCES"
    ],
    "reserved_provenance": [
        "data_class",
        "inventory_path",
        "inventory_digest"
    ],
    "runtime_owned_fields": [
        "path",
        "digest",
        "size",
        "inventory_path",
        "inventory_digest"
    ],
    "search_text_rule": "exact NFC-normalized canonical path, UTF-8 bounded by Conformance
scale_contracts.inventory.search_text_bytes_max",
    "stream_artifact_kind": "report",
    "stream_artifact_record_digest_rule": "sha256(canonical finalized record with exact artifact and
authoritative commit_binding)",
    "stream_media_type": "application/x-ndjson"
},
"promotion_test": [
    "independent-invariant",
    "independent-policy",
    "replaceability",
    "measurable-algorithm",
    "failure-boundary",
    "reuse",
    "domain-significance"
],
"provider_invocation_evidence": {
    "diagnostic_capture_bytes_max": 1048576,
    "diagnostic_capture_fields": [
        "stdout_capture_digest",
        "stdout_capture_size_bytes",
        "stdout_capture_truncated",
        "stderr_capture_digest",
        "stderr_capture_size_bytes",
        "stderr_capture_truncated"
    ],
    "full_output_bytes_hard_max": 1099511627776,
    "full_output_fields": [
        "output_digest",
        "output_size_bytes",
        "output_size_ceiling_bytes"
    ],

```

```

    "full_output_rule": "output_digest and output_size_bytes cover the complete provider operation
output stream; they are never derived from the bounded diagnostic captures",
    "immutable_tree_stream_rule": "filesystem-inventory immutable-tree-stream digests and counts the
complete raw archive stream while stdout/stderr captures remain bounded diagnostics",
    "invocation_receipt_digest_rule": "sha256(canonical ProviderInvocationEvidence without
invocation_receipt_digest)",
    "invocation_request_digest_rule": "sha256(canonical pre-invocation request receipt binding exact
argv or module call, configured cwd, stdin digest, selected output ceiling, environment allowlist,
provider identity, dependency receipt, and implementation closure)",
    "selected_ceiling_rule": "output_size_ceiling_bytes is the exact operation or
configured-workspace ceiling selected before invocation, does not exceed full_output_bytes_hard_max, and
rejects output_size_bytes above it"
  },
  "semantic_objects_default": [
    "private",
    "final",

    "immutable-or-stateless",
    "no-runtime-dispatch",
    "no-separate-build-root",
    "bounded-context-local"
  ]
},
"document": {
  "const": "promin.semantic-model"
},
"function": {
  "const": "Єдиний власник мінімального persistent vocabulary, relation grammar, межі derived
projection і provider capability contracts."
},
"persistent_entities": {
  "const": [
    {
      "admission_rule": "independent-scheduling-and-acceptance-boundary",
      "kind": "Завдання",
      "purpose": "Представляти одну плановану одиницю з одним acceptance predicate та обмеженою
mutation scope."
    },
    {
      "admission_rule": "independent-candidate-identity-and-consistency-boundary",
      "kind": "Candidate",
      "purpose": "Identify either one provider-proven immutable product snapshot or one explicitly
non-credible observation, independently from control state and bound to the exact candidate recipe."
    },
    {
      "admission_rule": "independent-content-identity-provenance-and-retention-boundary",
      "kind": "Artifact",

      "purpose": "Identify immutable product, evidence, log, report, or diff bytes with provenance and
retention metadata; raw-file inventory produces only derived Artifact proxies, while a diff may carry the
canonical Candidate-delta change manifest without creating another persistent entity."
    },
    {
      "admission_rule": "independent-execution-attempt-and-evidence-boundary",
      "kind": "Run",
      "purpose": "Записувати одну спробу execution або validation щодо точних candidate, policy, tool
та input digests."
    },
    {
      "admission_rule": "independent-defect-lifecycle-and-blocking-boundary",
      "kind": "Finding",
      "purpose": "Записувати спостережений дефект або порушення policy з явною effective-blocking
semantics."
    },
    {
      "admission_rule": "independent-human-judgement-and-target-binding-boundary",
      "kind": "Decision",
      "purpose": "Record a project-operational acceptance, rejection, resolution, waiver, promotion,
or release judgement. It is distinct from the external StandardReleaseDecision for the standard
distribution."
    },
    {
      "admission_rule": "immutable-task-owned-gate-definition-plus-current-run-and-finalized-artifact-r
ecord-bindings",
      "kind": "GateResult",
      "purpose": "Record one gate outcome against the exact immutable Task-owned GateRunDefinition,
current Run, and finalized evidence Artifact records; skipped, blocked, failed, stale, or unresolved
evidence never receives pass credit."
    }
  ]
},

```

<pre> { "admission_rule": "independent-capability-claim-and-revocation-boundary", "kind": "Grant", "purpose": "Доводити одну capability subject у межах однієї Activation, time interval, nonce і scope." }, { </pre>	
<pre> "admission_rule": "independent-mutation-ownership-and-fencing-boundary", "kind": "Оренда", "purpose": "Координувати тимчасове володіння мутацією через generation, fencing, expiry, heartbeat і підтвердження закриття." }], "maxItems": 9 }, "record_type": { "const": "SemanticModel" }, "relations": { "const": [{ "kind": "DEPENDS_ON", "rules": ["acyclic-for-ready-frontier", "accepted-current-outcome-required"], "source": ["Завдання"], "target": ["Завдання"] }], { "kind": "READS", "rules": [</pre>	
<pre> "read-does-not-imply-authority"], "source": ["Завдання", "Run"], "target": ["Candidate", "Artifact"] }, { "kind": "PRODUCES", "rules": ["content-or-record-identity-required"], "source": ["Завдання", "Run"], "target": ["Candidate", "Artifact", "Finding", "Decision"] }, { </pre>	
<pre> "kind": "VALIDATES", "rules": ["candidate-policy-tool-input-binding-required"], "source": ["Run"], "target": ["Candidate", "Artifact", "Завдання"] }, { </pre>	

```

{
  "kind": "CITES",
  "rules": [
    "selector-and-content-resolution-required"
  ],
  "source": [
    "Finding",
    "Decision"
  ],
  "target": [
    "Artifact",
    "Run",
    "Finding"
  ]
},
{
  "kind": "BLOCKS",
  "rules": [
    "effective-only-while-open-and-blocking"
  ],
  "source": [
    "Finding"
  ],
  "target": [
    "Завдання",
    "Candidate",
    "Decision"
  ]
}
],
"maxItems": 6
},
"schema_ref": {
  "const": "urn:promin:contracts:1#/$defs/SemanticModel"
},
"technology_capabilities": {
  "const": [
    {
      "adapter_protocol": {
        "dependency_receipt_rule": {
          "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
          "complete": true,
          "component_fields": [
            "component_id",
            "component_kind",
            "source",
            "version",
            "digest",
            "size_bytes"
          ],
          "component_order": "component-id-ascending",
          "os_substrate_excluded": true,
          "provider_owned_scope": "configured-provider-root-and-runtime-components",
          "unresolved_dependencies": "reject"
        },
        "operation_contracts": {
          "continue": {
            "request": "validated-continuation-token",
            "response": "WorkCardProjection"
          },
          "doctor": {
            "request": "installed-activation",
            "response": "DoctorResult"
          },
          "healthcheck": {
            "request": "configured-healthcheck",
            "response": "bounded-process-observation"
          },
          "init": {
            "request": "explicit-init-plans",
            "response": "InitResult"
          }
        },
        "next": {
          "request": "authorized-bounded-query",
          "response": "WorkCardProjection"
        },
        "provider_binding": {

```

```

        "request": "configured-provider-binding",
        "response": "content-addressed-receipt"
    },
    "status": {
        "request": "installed-activation",
        "response": "StatusResult"
    },
    "validate": {
        "request": "installed-activation",
        "response": "ValidationResult"
    }
},
"protocol_id": "promin.control-runtime.v1",
"receipt_persistence": "content-addressed-receipt",
"supported_bindings": [
    {
        "adapter_id": "python-runtime-v1",
        "identity_kind": "file-digest",
        "invocation_kind": "python-runtime"
    }
],
"supported_commands": [
    "continue",
    "doctor",
    "healthcheck",
    "init",
    "next",
    "provider-binding",
    "status",
    "validate"
],
"unknown_binding": "reject"
},
"forbidden_boundary": {
    "effect": "reject",
    "forbids": [
        "authority-invention",
        "policy-invention",
        "provider-binding-invention"
    ],
    "normative_authority": false
},
"id": "control-runtime",
"task": "Розбирати explicit configuration, канонізувати JSON і виконувати local CLI orchestration."
},
{
    "adapter_protocol": {
        "dependency_receipt_rule": {
            "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
            "complete": true,

```



```

    "healthcheck": {
      "request": "configured-healthcheck",
      "response": "bounded-process-observation"
    },
    "import": {
      "request": "strict-json-record",
      "response": "Draft-2020-12-plus-semantic-result"
    },
    "provider-binding": {
      "request": "configured-provider-binding",
      "response": "content-addressed-receipt"
    },
    "rebuild": {
      "request": "strict-json-record",
      "response": "Draft-2020-12-plus-semantic-result"
    },
    "replay": {
      "request": "strict-json-record",
      "response": "Draft-2020-12-plus-semantic-result"
    }
  },
  "protocol_id": "promin.shape-validation.v1",
  "receipt_persistence": "content-addressed-receipt",
  "supported_bindings": [
    {
      "adapter_id": "python-runtime-v1",
      "identity_kind": "file-digest",
      "invocation_kind": "python-runtime"
    }
  ],
  "supported_commands": [
    "command",
    "continue",
    "export",
    "healthcheck",
    "import",
    "provider-binding",
    "rebuild",
    "replay"
  ],
  "unknown_binding": "reject"
},
"forbidden_boundary": {
  "effect": "reject",
  "forbids": [
    "authority-decision",
    "cross-record-semantic-decision"
  ],
  "normative_authority": false
},
"id": "shape-validation",
"task": "Перевіряти єдиний contract bundle Draft 2020-12 з обов'язковими formats і local
references."
},
{
  "adapter_protocol": {
    "dependency_receipt_rule": {
      "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
      "complete": true,
      "component_fields": [
        "component_id",
        "component_kind",
        "source",
        "version",
        "digest",
        "size_bytes"
      ],
      "component_order": "component-id-ascending",
      "os_substrate_excluded": true,
      "provider_owned_scope": "configured-provider-root-and-runtime-components",
      "unresolved_dependencies": "reject"
    },
    "operation_contracts": {
      "digest-bytes": {
        "request": "bounded-bytes",
        "response": "lowercase-sha256"
      }
    }
  }
}

```

```

    "digest-file": {
      "request": "no-follow-regular-file",
      "response": "lowercase-sha256"
    },
    "digest-value": {
      "request": "canonical-json-value",
      "response": "lowercase-sha256"
    },
    "healthcheck": {
      "request": "configured-healthcheck",
      "response": "bounded-process-observation"
    },
    "provider-binding": {
      "request": "configured-provider-binding",
      "response": "content-addressed-receipt"
    }
  },
  "protocol_id": "promin.content-identity.v1",
  "receipt_persistence": "content-addressed-receipt",
  "supported_bindings": [
    {
      "adapter_id": "python-runtime-v1",
      "identity_kind": "file-digest",
      "invocation_kind": "python-runtime"
    }
  ],
  "supported_commands": [
    "digest-bytes",
    "digest-file",
    "digest-value",
    "healthcheck",
    "provider-binding"
  ],
  "unknown_binding": "reject"
},
"forbidden_boundary": {
  "effect": "reject",
  "forbids": [
    "digest-as-action-authority"
  ],
  "normative_authority": false
},
"id": "content-identity",
"task": "Обчислювати content digests та ідентичності content-addressed objects."
},
{
  "adapter_protocol": {
    "dependency_receipt_rule": {
      "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
      "complete": true,
      "component_fields": [
        "component_id",
        "component_kind",
        "source",
        "version",
        "digest",
        "size_bytes"
      ],
      "component_order": "component-id-ascending",
      "os_substrate_excluded": true,
      "provider_owned_scope": "configured-provider-root-and-runtime-components",
      "unresolved_dependencies": "reject"
    },
    "operation_contracts": {
      "atomic-replace": {
        "request": "same-directory-staged-bytes",
        "response": "durable-replacement"
      },
      "directory-sync": {
        "request": "owning-directory",
        "response": "durability-observation"
      },
      "file-sync": {
        "request": "open-regular-file",
        "response": "durability-observation"
      },
      "healthcheck": {

```

```

    "request": "configured-healthcheck",
    "response": "bounded-process-observation"
  },
  "local-writer-lock": {
    "request": "activation-bound-lock",
    "response": "exclusive-lease"
  },
  "provider-binding": {
    "request": "configured-provider-binding",
    "response": "content-addressed-receipt"
  },
  "recover": {
    "request": "durable-journal-state",
    "response": "verified-head"
  }
},
"protocol_id": "promin.local-serialization.v1",

```

```

"receipt_persistence": "content-addressed-receipt",
"supported_bindings": [
  {
    "adapter_id": "python-runtime-v1",
    "identity_kind": "file-digest",
    "invocation_kind": "python-runtime"
  }
],
"supported_commands": [
  "atomic-replace",
  "directory-sync",
  "file-sync",
  "healthcheck",
  "local-writer-lock",
  "provider-binding",
  "recover"
],
"unknown_binding": "reject"
},
"forbidden_boundary": {
  "effect": "reject",
  "forbids": [
    "distributed-consensus-claim"
  ],
  "normative_authority": false
},
"id": "local-serialization",
"task": "Надавати один local-writer lock, atomic replace, file sync, directory sync i recovery primitives."

```

```

},
{
  "adapter_protocol": {
    "dependency_receipt_rule": {
      "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
      "complete": true,
      "component_fields": [
        "component_id",
        "component_kind",
        "source",
        "version",
        "digest",
        "size_bytes"
      ],
      "component_order": "component-id-ascending",
      "os_substrate_excluded": true,
      "provider_owned_scope": "configured-provider-root-and-runtime-components",
      "unresolved_dependencies": "reject"
    },
    "operation_contracts": {
      "continue": {
        "request": "authenticated-ready-frontier-continuation",
        "response": "ContinueResult"
      },
      "healthcheck": {
        "request": "configured-healthcheck",
        "response": "bounded-process-observation"
      }
    },
    "next": {
      "request": "authorized-bounded-ready-query",
      "response": "NextResult"
    }
  }
}

```

```

    },
    "provider-binding": {
      "request": "configured-provider-binding",
      "response": "content-addressed-receipt"
    },
    "rebuild": {
      "request": "verified-event-and-inventory-input",
      "response": "current-projection-state"
    },
    "status": {
      "request": "projection-file",
      "response": "current-projection-state"
    }
  },
  "protocol_id": "promin.query-projection.v1",
  "receipt_persistence": "content-addressed-receipt",
  "supported_bindings": [
    {
      "adapter_id": "python-runtime-v1",
      "identity_kind": "file-digest",
      "invocation_kind": "python-runtime"
    }
  ],
  "supported_commands": [
    "continue",

```

```

    "healthcheck",
    "next",
    "provider-binding",
    "rebuild",
    "status"
  ],
  "unknown_binding": "reject"
},
"forbidden_boundary": {
  "effect": "reject",
  "forbids": [
    "projection-as-normative-authority"
  ],
  "normative_authority": false
},
"id": "query-projection",
"task": "Build disposable status and bounded typed-retrieval projections; derive ReadyFrontier,
materialize one exact WorkCardProjection, and return Core-owned NextResult or ContinueResult envelopes for
the six supported CLI commands."
},
{

```

```

  "adapter_protocol": {
    "dependency_receipt_rule": {
      "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
      "complete": true,
      "component_fields": [
        "component_id",
        "component_kind",
        "source",
        "version",
        "digest",
        "size_bytes"
      ],
      "component_order": "component-id-ascending",
      "os_substrate_excluded": true,
      "provider_owned_scope": "configured-provider-root-and-runtime-components",
      "unresolved_dependencies": "reject"
    },
    "operation_contracts": {
      "healthcheck": {
        "request": "configured-bounded-git-healthcheck",
        "response": "exit-zero-captured-stdout-stderr"
      },
      "immutable-tree-object": {
        "request": "argv:rev-parse --verify HEAD^{tree};cwd:configured-root;stdin:none",
        "response": "exit-zero;stdout:one-lowercase-hex-object-id-plus-LF;stderr:captured"
      },
      "immutable-tree-stream": {
        "request": "argv:archive --format=tar <verified-tree-object>;cwd:configured-root;stdin:none",
        "response": "exit-zero;stdout:raw-git-tar-stream;stderr:captured"
      }
    },
    "provider-binding": {

```

```

        "request": "configured-executable-file-digest",
        "response": "content-addressed-receipt"
    }
},
"protocol_id": "promin.filesystem-inventory.v1",
"receipt_persistence": "content-addressed-receipt",
"supported_bindings": [
    {
        "adapter_id": "git-executable-v1",
        "identity_kind": "file-digest",
        "invocation_kind": "executable"
    }
],
"supported_commands": [
    "healthcheck",
    "immutable-tree-object",
    "immutable-tree-stream",
    "provider-binding"
],
"unknown_binding": "reject"
},
"forbidden_boundary": {
    "effect": "reject",
    "forbids": [
        "implicit-initialization-scan"
    ],
    "normative_authority": false
},
"id": "filesystem-inventory",
"task": "Apply the exact candidate recipe in one product pass, use no-follow reads, and publish
either a provider-proven immutable VCS tree stream or an explicit non-creditable observation."
},
{
    "adapter_protocol": {
        "dependency_receipt_rule": {
            "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
            "complete": true,
            "component_fields": [
                "component_id",
                "component_kind",
                "source",
                "version",
                "digest",
                "size_bytes"
            ],
            "component_order": "component-id-ascending",
            "os_substrate_excluded": true,
            "provider_owned_scope": "configured-provider-root-and-runtime-components",
            "unresolved_dependencies": "reject"
        },
        "operation_contracts": {
            "bounded-export-scan": {
                "request": "canonical-json-stdin:{operation,root,allowlist_digest,max_files,max_bytes}",
                "response": "exit-zero;canonical-json-stdout:{status,files_scanned,bytes_scanned,violations,
output_digest};stderr:captured"
            },
            "healthcheck": {
                "request": "configured-bounded-healthcheck",
                "response": "exit-zero-captured-stdout-stderr"
            },
            "provider-binding": {
                "request": "configured-executable-file-digest",
                "response": "content-addressed-receipt"
            }
        }
    },
    "protocol_id": "promin.export-scan.v1",
    "receipt_persistence": "content-addressed-receipt",
    "supported_bindings": [
        {
            "adapter_id": "executable-export-scan-v1",
            "identity_kind": "file-digest",
            "invocation_kind": "executable"
        }
    ],
    "supported_commands": [
        "bounded-export-scan",
        "healthcheck",

```

```

        "provider-binding"
    ],
    "unknown_binding": "reject"
},
"forbidden_boundary": {
    "effect": "reject",
    "forbids": [
        "package-as-authorization-boundary"
    ],
    "normative_authority": false
},
"id": "export-scan",
"task": "Виконувати bounded recursive export inspection, allowlisting, secret classification i
відхилення symlink."
},
{

```

```

    "adapter_protocol": {
        "dependency_receipt_rule": {
            "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
            "complete": true,
            "component_fields": [
                "component_id",
                "component_kind",
                "source",
                "version",
                "digest",
                "size_bytes"
            ],
            "component_order": "component-id-ascending",
            "os_substrate_excluded": true,
            "provider_owned_scope": "configured-provider-root-and-runtime-components",
            "unresolved_dependencies": "reject"
        },
        "operation_contracts": {
            "healthcheck": {
                "request": "configured-bounded-signature-healthcheck",
                "response": "exit-zero-captured-stdout-stderr"
            },
            "provider-binding": {
                "request": "configured-module-or-executable-digest",
                "response": "content-addressed-receipt"
            },
            "sign-digest": {
                "request": "domain-separated-canonical-claim-digest",

```

```

                "response": "configured-adapter-signature"
            },
            "verify-digest": {
                "request": "domain-separated-canonical-claim-digest-plus-signature",
                "response": "strict-boolean-verification"
            }
        },
        "protocol_id": "promin.signature.v1",
        "receipt_persistence": "content-addressed-receipt",
        "supported_bindings": [
            {
                "adapter_id": "executable-signature-v1",
                "identity_kind": "file-digest",
                "invocation_kind": "executable"
            },
            {
                "adapter_id": "python-module-signature-v1",
                "identity_kind": "module-file-digest",
                "invocation_kind": "python-module"
            }
        ],
        "supported_commands": [
            "healthcheck",
            "provider-binding",
            "sign-digest",
            "verify-digest"
        ],
        "unknown_binding": "reject"
    },

```

```

    },
    "forbidden_boundary": {
        "effect": "reject",
        "forbids": [
            "core-signer-implementation-lock-in"

```

```

    },
    "normative_authority": false
  },
  "id": "signature",
  "task": "Dispatch the exact configured signature adapter to sign or verify activation, Grant,
and Decision digests for team trust mode, and evidence the invoked adapter identity."
},
{
  "adapter_protocol": {
    "dependency_receipt_rule": {
      "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
      "complete": true,
      "component_fields": [
        "component_id",
        "component_kind",
        "source",
        "version",
        "digest",
        "size_bytes"
      ],
      "component_order": "component-id-ascending",
      "os_substrate_excluded": true,
      "provider_owned_scope": "configured-provider-root-and-runtime-components",
      "unresolved_dependencies": "reject"
    },
    "operation_contracts": {
      "healthcheck": {
        "request": "configured-bounded-healthcheck",
        "response": "exit-zero-captured-stdout-stderr"
      },
      "import-build-facts": {
        "request": "canonical-json-stdin:{operation,project_root,input_digests,max_files,max_bytes}"
      },
      "response": "exit-zero;canonical-json-stdout:{status,facts,input_digests,output_digest};std
err:captured"
    },
    "provider_binding": {
      "request": "configured-executable-file-digest",
      "response": "content-addressed-receipt"
    }
  },
  "protocol_id": "promin.build-dependency.v1",
  "receipt_persistence": "content-addressed-receipt",
  "supported_bindings": [
    {
      "adapter_id": "executable-build-dependency-v1",
      "identity_kind": "file-digest",
      "invocation_kind": "executable"
    }
  ],
  "supported_commands": [
    "healthcheck",
    "import-build-facts",
    "provider-binding"
  ],
  "unknown_binding": "reject"
},
{
  "forbidden_boundary": {
    "effect": "reject",
    "forbids": [
      "semantic-model-redefinition"
    ],
    "normative_authority": false
  },
  "id": "build-dependency",
  "task": "Імпортувати project-specific build і dependency facts як похідні проєкції."
}
]
},
"value_objects": {
  "const": {
    "CandidateDelta": "A diff Artifact payload binding base and new Candidate digests, WorkCard
digest, and the complete normalized changed-path set.",
    "Command": "Запит на зміну стану, перевірений до commit.",
    "Comparator": "Чисте правило впорядкування або рівності.",
    "ContinuationToken": "An authenticated bounded runtime value signed with the random per-Activation
secret. It binds subject, exact projection.read Grant claim, requested scope, revocation state, normalized
query, cursor, Activation, HEAD, projection, ranking, complete budgets, depth, implementation closure,

```

```

issue time, and expiry; it never grants authority by itself.",
  "CurrentReleaseStatus": "A derived evaluation of an immutable historical release Decision against
the current closure, blockers, evidence, providers, and Activation.",
  "ImplementationClosure": "A canonical capability-specific binding of the promin runtime, Python
interpreter, jsonschema or SQLite component where used, exactly one adapter, and the recomputed aggregate
closure digest.",
  "Operation": "Один завершений семантичний крок із явними семантиками input, output, failure i
evidence.",
  "PlanProposal": "A non-authoritative bounded proposal that derives candidate Tasks and relations
from project intent, repository facts, backlog, and references; it cannot mutate operational state or
grant authority.",
  "Policy": "Машинно-адресований інваріант, власником якого є policy-set.json.",
  "Predicate": "Чисте правило ухвалення рішення.",
  "ProviderInvocationEvidence": "The exact capability, provider, invocation kind, identity kind and
digest, and adapter ID actually used by one operation.",
  "Query": "Read-only запит, який не може змінювати нормативний стан.",
  "StandardReleaseDecision": "An external human-reviewable distribution record bound to the exact
promin ZIP, Core, preset, tools, implementation closure, tests, platform results, saturation, PDFs, and
evidence manifest. It never grants product acceptance.",
  "StateCheckpoint": "A disposable self-digested state or journal index bound to an exact
authoritative HEAD; it cannot grant authority or acceptance.",
  "WorkCard": "A bounded generated context package for one Task that separately binds the
projection.read query Grant claim and the task.execute holder Grant; neither binding implies acceptance."
},
"version": {
  "const": "1.0.0-alpha.1"
},
"required": [
  "record_type",
  "schema_ref",
  "document",
  "version",
  "function",
  "persistent_entities",
  "relations",
  "derived_projection_kinds",
  "value_objects",
  "design_rules",
  "technology_capabilities",
  "admission"
],
"type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/SemanticModel | Валідатор: urn:promin:contracts:1#/\$defs/SemanticModel

98. SemanticStatePayload

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, candidate_digest, activation_digest, head_event_id, head_sequence, head_digest, records

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
candidate_digest	так	\$ref="#/\$defs/Digest"
head_digest	так	oneOf[2]=#/\$defs/Digest type:null
head_event_id	так	oneOf[2]=#/\$defs/Id type:null
head_sequence	так	type="integer"; minimum=0
record_type	так	const="SemanticStatePayload"
records	так	type="array"; minItems=0; maxItems=400000

```

{
  "additionalProperties": false,
  "properties": {
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "head_digest": {

```



```

    "oneOf": [
      {
        "$ref": "#/$defs/Digest"
      },
      {
        "type": "null"
      }
    ]
  },
  "head_event_id": {
    "oneOf": [
      {
        "$ref": "#/$defs/Id"
      },
      {
        "type": "null"
      }
    ]
  }
]

```

```

},
"head_sequence": {
  "minimum": 0,
  "type": "integer"
},
"record_type": {
  "const": "SemanticStatePayload"
},
"records": {
  "items": {
    "oneOf": [
      {
        "$ref": "#/$defs/Activation"
      },
      {
        "$ref": "#/$defs/Task"
      },
      {
        "$ref": "#/$defs/Candidate"
      },
      {
        "$ref": "#/$defs/Artifact"
      },
      {
        "$ref": "#/$defs/Run"
      },
      {
        "$ref": "#/$defs/Finding"

```

```

      },
      {
        "$ref": "#/$defs/Decision"
      },
      {
        "$ref": "#/$defs/Grant"
      },
      {
        "$ref": "#/$defs/Lease"
      },
      {
        "$ref": "#/$defs/GateResult"
      },
      {
        "$ref": "#/$defs/GrantRevocation"
      },
      {
        "$ref": "#/$defs/Relation"
      }
    ]
  },
  "maxItems": 400000,
  "minItems": 0,
  "type": "array"
}
},
"required": [
  "record_type",

```

```

  "candidate_digest",
  "activation_digest",
  "head_event_id",

```

```

    "head_sequence",
    "head_digest",
    "records"
  ],
  "type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/SemanticStatePayload](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/SemanticStatePayload](#)

99. SignatureActivationProof

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: kind, key_id, algorithm, signed_digest, signature

Властивість	Обов'язков е	Структурне правило
algorithm	так	type="string"; minLength=1
key_id	так	\$ref="#/\$defs/Id"
kind	так	const="signature"
signature	так	type="string"; minLength=1
signed_digest	так	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "properties": {
    "algorithm": {
      "minLength": 1,
      "type": "string"
    },
    "key_id": {
      "$ref": "#/$defs/Id"
    },
    "kind": {
      "const": "signature"
    },
    "signature": {
      "minLength": 1,
      "type": "string"
    },
    "signed_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "kind",
    "key_id",
    "algorithm",
    "signed_digest",
    "signature"
  ],
  "type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/SignatureActivationProof](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/SignatureActivationProof](#)

100. SignatureRootCommandProof

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: kind, key_id, algorithm, signed_intent_digest, signature

Властивість	Обов'язков е	Структурне правило
algorithm	так	type="string"; minLength=1
key_id	так	\$ref="#/\$defs/Id"
kind	так	const="signature-command"
signature	так	type="string"; minLength=1
signed_intent_digest	так	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,

```

```

"properties": {
  "algorithm": {
    "minLength": 1,
    "type": "string"
  },
  "key_id": {
    "$ref": "#/$defs/Id"
  },
  "kind": {
    "const": "signature-command"
  },
  "signature": {
    "minLength": 1,
    "type": "string"
  },
  "signed_intent_digest": {
    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "kind",
  "key_id",
  "algorithm",
  "signed_intent_digest",
  "signature"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/SignatureRootCommandProof](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/SignatureRootCommandProof](#)

101. SignatureTrustProof

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: kind, key_id, algorithm, signed_digest, signature

Властивість	Обов'язков е	Структурне правило
algorithm	так	type="string"; minLength=1; maxLength=128
key_id	так	\$ref="#/\$defs/Id"
kind	так	const="signature"
signature	так	type="string"; minLength=1; maxLength=8192
signed_digest	так	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "properties": {
    "algorithm": {
      "maxLength": 128,
      "minLength": 1,
      "type": "string"
    },
    "key_id": {
      "$ref": "#/$defs/Id"
    },
    "kind": {
      "const": "signature"
    },
    "signature": {
      "maxLength": 8192,
      "minLength": 1,
      "type": "string"
    },
    "signed_digest": {
      "$ref": "#/$defs/Digest"
    }
  },
  "required": [
    "kind",
    "key_id",
    "algorithm",
    "signed_digest",
    "signature"
  ],
}

```

```
"type": "object"
}
```

Власник: core/contracts.schema.json#/\$defs/SignatureTrustProof | Валідатор: urn:promin:contracts:1#/\$defs/SignatureTrustProof

102. StandardBinding

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: standard_id, version, source_uri, source_digest, purpose

Властивість	Обов'язков е	Структурне правило
purpose	так	type="string"; minLength=1
source_digest	так	\$ref="#/\$defs/Digest"
source_uri	так	type="string"; minLength=1
standard_id	так	\$ref="#/\$defs/Id"
version	так	type="string"; minLength=1

```
{
  "additionalProperties": false,
  "properties": {
    "purpose": {
      "minLength": 1,
      "type": "string"
    },
    "source_digest": {
      "$ref": "#/$defs/Digest"
    },
    "source_uri": {
      "minLength": 1,
      "type": "string"
    },
    "standard_id": {
      "$ref": "#/$defs/Id"
    },
    "version": {
      "minLength": 1,
      "type": "string"
    }
  },
  "required": [
    "standard_id",
    "version",
    "source_uri",
    "source_digest",
    "purpose"
  ],
  "type": "object"
}
```

Власник: core/contracts.schema.json#/\$defs/StandardBinding | Валідатор: urn:promin:contracts:1#/\$defs/StandardBinding

103. StandardManifest

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, schema_ref, canonical_name, aliases, canonical_name_only, version, function, core_components, bundle_digest, core_artifact_count, schema_definition_count, selected_preset_count, persistent_entity_kind_count, relation_kind_count, technology_capability_count, action_capability_count, policy_count, acceptance_predicate_count, mutation_family_count

Властивість	Обов'язков е	Структурне правило
acceptance_predicate_count	так	type="integer"; const=102
action_capability_count	так	type="integer"; const=16
aliases	так	type="array"; maxItems=0
bundle_digest	так	\$ref="#/\$defs/Digest"
canonical_name	так	const="promin"
canonical_name_only	так	const=true
core_artifact_count	так	type="integer"; const=6

Властивість	Обов'язков е	Структурне правило
core_components	так	type="array"; minItems=5; maxItems=5
function	так	type="string"; minLength=1
mutation_family_count	так	type="integer"; const=77
persistent_entity_kind_count	так	type="integer"; const=9
policy_count	так	type="integer"; const=68
record_type	так	const="StandardManifest"
relation_kind_count	так	type="integer"; const=6
schema_definition_count	так	type="integer"; const=127
schema_ref	так	const="urn:promin:contracts:1#/\$defs/StandardManifest"
selected_preset_count	так	type="integer"; const=1
technology_capability_count	так	type="integer"; const=9
version	так	const="1.0.0-alpha.1"

```

{
  "additionalProperties": false,
  "properties": {
    "acceptance_predicate_count": {
      "const": 102,
      "type": "integer"
    },
    "action_capability_count": {
      "const": 16,
      "type": "integer"
    },
    "aliases": {
      "maxItems": 0,
      "type": "array"
    },
    "bundle_digest": {
      "$ref": "#/$defs/Digest"
    },
    "canonical_name": {
      "const": "promin"
    },
    "canonical_name_only": {
      "const": true
    },
    "core_artifact_count": {
      "const": 6,
      "type": "integer"
    },
    "core_components": {
      "items": {
        "$ref": "#/$defs/CoreComponent"
      },
      "maxItems": 5,
      "minItems": 5,
      "type": "array"
    },
    "function": {
      "minLength": 1,
      "type": "string"
    },
    "mutation_family_count": {
      "const": 77,
      "type": "integer"
    },
    "persistent_entity_kind_count": {
      "const": 9,
      "type": "integer"
    },
    "policy_count": {
      "const": 68,
      "type": "integer"
    },
    "record_type": {
      "const": "StandardManifest"
    },
    "relation_kind_count": {

```

```

    "const": 6,
    "type": "integer"
  },
  "schema_definition_count": {
    "const": 127,
    "type": "integer"
  },
  "schema_ref": {
    "const": "urn:promin:contracts:1#/$defs/StandardManifest"
  },
  "selected_preset_count": {
    "const": 1,
    "type": "integer"
  },
  "technology_capability_count": {
    "const": 9,
    "type": "integer"
  },
  "version": {
    "const": "1.0.0-alpha.1"
  }
},
"required": [
  "record_type",
  "schema_ref",
  "canonical_name",
  "aliases",
  "canonical_name_only",
  "version",
  "function",
  "core_components",
  "bundle_digest",
  "core_artifact_count",
  "schema_definition_count",
  "selected_preset_count",
  "persistent_entity_kind_count",
  "relation_kind_count",
  "technology_capability_count",
  "action_capability_count",
  "policy_count",
  "acceptance_predicate_count",
  "mutation_family_count"
],
"type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/StandardManifest | Валідатор: urn:promin:contracts:1#/\$defs/StandardManifest

104. StandardReleaseCandidateBinding

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, standard_name, version, archive_sha256, archive_bytes, archive_member_manifest_digest, package_manifest_digest, checksums_digest, core_bundle_digest, preset_digest, package_tool_digest, validator_digest, test_manifest_digest, portable_implementation_closure_digest, evidence_tool_digests, candidate_binding_digest

Властивість	Обов'язков е	Структурне правило
archive_bytes	так	type="integer"; minimum=1
archive_member_manifest_digest	так	\$ref="#/\$defs/Digest"
archive_sha256	так	\$ref="#/\$defs/Digest"
candidate_binding_digest	так	\$ref="#/\$defs/Digest"
checksums_digest	так	\$ref="#/\$defs/Digest"
core_bundle_digest	так	\$ref="#/\$defs/Digest"
evidence_tool_digests	так	type="object"; additionalProperties=false
package_manifest_digest	так	\$ref="#/\$defs/Digest"
package_tool_digest	так	\$ref="#/\$defs/Digest"
portable_implementation_closure_digest	так	\$ref="#/\$defs/Digest"
preset_digest	так	\$ref="#/\$defs/Digest"

Властивість	Обов'язков е	Структурне правило
record_type	так	const="StandardReleaseCandidateBinding"
standard_name	так	const="promin"
test_manifest_digest	так	\$ref="#/\$defs/Digest"
validator_digest	так	\$ref="#/\$defs/Digest"
version	так	\$ref="#/\$defs/SemVer"
<pre> { "additionalProperties": false, "properties": { "archive_bytes": { "minimum": 1, "type": "integer" }, "archive_member_manifest_digest": { "\$ref": "#/\$defs/Digest" }, "archive_sha256": { "\$ref": "#/\$defs/Digest" }, "candidate_binding_digest": { "\$ref": "#/\$defs/Digest" }, "checksums_digest": { "\$ref": "#/\$defs/Digest" }, "core_bundle_digest": { "\$ref": "#/\$defs/Digest" }, "evidence_tool_digests": { "additionalProperties": false, "properties": { "tools/generate_human.py": { "\$ref": "#/\$defs/Digest" }, "tools/promin_no_degradation.py": { "\$ref": "#/\$defs/Digest" }, "tools/promin_package.py": { "\$ref": "#/\$defs/Digest" }, "tools/promin_saturation.py": { "\$ref": "#/\$defs/Digest" }, "tools/promin_saturation_audit.py": { "\$ref": "#/\$defs/Digest" }, "tools/promin_validate.py": { "\$ref": "#/\$defs/Digest" } }, "required": ["tools/generate_human.py", "tools/promin_no_degradation.py", "tools/promin_package.py", "tools/promin_saturation.py", "tools/promin_saturation_audit.py", "tools/promin_validate.py"], "type": "object" }, "package_manifest_digest": { "\$ref": "#/\$defs/Digest" }, "package_tool_digest": { "\$ref": "#/\$defs/Digest" }, "portable_implementation_closure_digest": { "\$ref": "#/\$defs/Digest" }, "preset_digest": { "\$ref": "#/\$defs/Digest" }, "record_type": { </pre>		

```

    "const": "StandardReleaseCandidateBinding"
  },
  "standard_name": {
    "const": "promin"
  },
  "test_manifest_digest": {
    "$ref": "#/$defs/Digest"
  },
  "validator_digest": {
    "$ref": "#/$defs/Digest"
  },
  "version": {
    "$ref": "#/$defs/SemVer"
  }
},
"required": [
  "record_type",

  "standard_name",
  "version",
  "archive_sha256",
  "archive_bytes",
  "archive_member_manifest_digest",
  "package_manifest_digest",
  "checksums_digest",
  "core_bundle_digest",
  "preset_digest",
  "package_tool_digest",
  "validator_digest",
  "test_manifest_digest",
  "portable_implementation_closure_digest",
  "evidence_tool_digests",
  "candidate_binding_digest"
],
"type": "object"
}

```

Власник: `core/contracts.schema.json#/$defs/StandardReleaseCandidateBinding` | Валідатор: `urn:promin:contracts:1#/$defs/StandardReleaseCandidateBinding`

105. StandardReleaseDecision

Опис типу: `type="object"; additionalProperties=false`

Обов'язкові поля: `record_type`, `decision_id`, `standard_name`, `version`, `candidate_binding_digest`, `evidence_manifest_digest`, `outcome`, `decider_id`, `release_capability`, `trust_root_id`, `signature_provider_id`, `key_id`, `nonce`, `decided_at`, `signed_claim_digest`, `signature`

Властивість	Обов'язков е	Структурне правило
<code>candidate_binding_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>decided_at</code>	так	<code>\$ref="#/\$defs/Timestamp"</code>
<code>decider_id</code>	так	<code>\$ref="#/\$defs/Id"</code>
<code>decision_id</code>	так	<code>\$ref="#/\$defs/Id"</code>
<code>evidence_manifest_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>key_id</code>	так	<code>\$ref="#/\$defs/Id"</code>
<code>nonce</code>	так	<code>type="string"; minLength=24; maxLength=128</code>
<code>outcome</code>	так	<code>enum=["approve", "reject"]</code>
<code>record_type</code>	так	<code>const="StandardReleaseDecision"</code>
<code>release_capability</code>	так	<code>const="standard.distribute"</code>
<code>signature</code>	так	<code>type="string"; minLength=80; maxLength=128</code>
<code>signature_provider_id</code>	так	<code>const="cryptography-ed25519-v1"</code>
<code>signed_claim_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>standard_name</code>	так	<code>const="promin"</code>
<code>trust_root_id</code>	так	<code>\$ref="#/\$defs/Id"</code>
<code>version</code>	так	<code>\$ref="#/\$defs/SemVer"</code>
<pre> { "additionalProperties": false, "properties": { </pre>		


```

    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "decided_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "decider_id": {
      "$ref": "#/$defs/Id"
    },
    "decision_id": {
      "$ref": "#/$defs/Id"
    },
    "evidence_manifest_digest": {
      "$ref": "#/$defs/Digest"
    },
    "key_id": {
      "$ref": "#/$defs/Id"
    },
    "nonce": {
      "maxLength": 128,
      "minLength": 24,
      "type": "string"
    },
    "outcome": {
      "enum": [
        "approve",
        "reject"
      ]
    },
    "record_type": {
      "const": "StandardReleaseDecision"
    },
    "release_capability": {
      "const": "standard.distribute"
    },
    "signature": {
      "maxLength": 128,
      "minLength": 80,
      "type": "string"
    },
    "signature_provider_id": {
      "const": "cryptography-ed25519-v1"
    },
    "signed_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    "standard_name": {
      "const": "promin"
    },
    "trust_root_id": {
      "$ref": "#/$defs/Id"
    },
    "version": {
      "$ref": "#/$defs/SemVer"
    }
  },
  "required": [
    "record_type",
    "decision_id",
    "standard_name",
    "version",
    "candidate_binding_digest",
    "evidence_manifest_digest",
    "outcome",
    "decider_id",
    "release_capability",
    "trust_root_id",
    "signature_provider_id",
    "key_id",
    "nonce",
    "decided_at",
    "signed_claim_digest",
    "signature"
  ],
  "type": "object"
}

```

106. StandardReleaseEvidenceEntry

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: evidence_id, evidence_role, path, sha256, size_bytes, record_type, status, candidate_binding_digest, predicates

Властивість	Обов'язков е	Структурне правило
candidate_binding_digest	так	\$ref="#/\$defs/Digest"
evidence_id	так	\$ref="#/\$defs/Id"
evidence_role	так	enum=["linux", "windows", "physical-scale", "saturation-audit", "human-documents", "linux-no-degradation", "windows-no-degradation"]
path	так	\$ref="#/\$defs/RelativePath"
predicates	так	type="array"; minItems=1; maxItems=64
record_type	так	\$ref="#/\$defs/Id"
sha256	так	\$ref="#/\$defs/Digest"
size_bytes	так	type="integer"; minimum=1
status	так	const="pass"

```
{
  "additionalProperties": false,
  "properties": {
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "evidence_id": {
      "$ref": "#/$defs/Id"
    },
    "evidence_role": {
      "enum": [
        "linux",
        "windows",
        "physical-scale",
        "saturation-audit",
        "human-documents",
        "linux-no-degradation",
        "windows-no-degradation"
      ]
    },
    "path": {
      "$ref": "#/$defs/RelativePath"
    },
    "predicates": {
      "items": {
        "$ref": "#/$defs/StandardReleaseEvidencePredicate"
      },
      "maxItems": 64,
      "minItems": 1,
      "type": "array"
    },
    "record_type": {
      "$ref": "#/$defs/Id"
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    },
    "size_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "status": {
      "const": "pass"
    }
  },
  "required": [
    "evidence_id",
    "evidence_role",
    "path",
    "sha256",
    "size_bytes",

```

```

    "record_type",
    "status",
    "candidate_binding_digest",
    "predicates"
  ],
  "type": "object"
}

```

Власник: `core/contracts.schema.json#/$defs/StandardReleaseEvidenceEntry` | Валідатор: `urn:promin:contracts:1#/$defs/StandardReleaseEvidenceEntry`

107. StandardReleaseEvidenceManifest

Опис типу: `type="object"; additionalProperties=false`

Обов'язкові поля: `record_type`, `standard_name`, `version`, `candidate_binding_digest`, `max_evidence_completed_at`, `entries`, `matrix_aggregate`, `supplemental_lanes`, `evidence_manifest_digest`

Властивість	Обов'язков е	Структурне правило
<code>candidate_binding_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>entries</code>	так	<code>type="array"; minItems=7; maxItems=7</code>
<code>evidence_manifest_digest</code>	так	<code>\$ref="#/\$defs/Digest"</code>
<code>matrix_aggregate</code>	так	<code>\$ref="#/\$defs/StandardReleaseMatrixBinding"</code>
<code>max_evidence_completed_at</code>	так	<code>\$ref="#/\$defs/Timestamp"</code>
<code>record_type</code>	так	<code>const="StandardReleaseEvidenceManifest"</code>
<code>standard_name</code>	так	<code>const="promin"</code>
<code>supplemental_lanes</code>	так	<code>type="array"; minItems=4; maxItems=4</code>
<code>version</code>	так	<code>\$ref="#/\$defs/SemVer"</code>

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "entries": {
      "items": {
        "$ref": "#/$defs/StandardReleaseEvidenceEntry"
      },
      "maxItems": 7,
      "minItems": 7,
      "type": "array"
    },
    "evidence_manifest_digest": {
      "$ref": "#/$defs/Digest"
    },
    "matrix_aggregate": {
      "$ref": "#/$defs/StandardReleaseMatrixBinding"
    },
    "max_evidence_completed_at": {
      "$ref": "#/$defs/Timestamp"
    },
    "record_type": {
      "const": "StandardReleaseEvidenceManifest"
    },
    "standard_name": {
      "const": "promin"
    },
    "supplemental_lanes": {
      "items": {
        "$ref": "#/$defs/StandardReleaseSupplementalLane"
      },
      "maxItems": 4,
      "minItems": 4,
      "type": "array"
    },
    "version": {
      "$ref": "#/$defs/SemVer"
    }
  },
  "required": [
    "record_type",

```

```

    "standard_name",
    "version",
    "candidate_binding_digest",
    "max_evidence_completed_at",
    "entries",
    "matrix_aggregate",
    "supplemental_lanes",
    "evidence_manifest_digest"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/StandardReleaseEvidenceManifest | Валідатор:
urn:promin:contracts:1#/\$defs/StandardReleaseEvidenceManifest

108. StandardReleaseEvidencePredicate

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: pointer, equals

Властивість	Обов'язков е	Структурне правило
equals	так	складене структурне визначення
pointer	так	type="string"; pattern="^/"; maxLength=1024

```

{
  "additionalProperties": false,
  "properties": {
    "equals": {},
    "pointer": {
      "maxLength": 1024,
      "pattern": "^/",
      "type": "string"
    }
  },
  "required": [
    "pointer",
    "equals"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/StandardReleaseEvidencePredicate | Валідатор:
urn:promin:contracts:1#/\$defs/StandardReleaseEvidencePredicate

109. StandardReleaseMatrixBinding

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: path, sha256, size_bytes, record_type, matrix_digest, lane_count, matrix_authoritative, pass_credit, acceptance_pass, product_acceptance_pass, product_public_approval

Властивість	Обов'язков е	Структурне правило
acceptance_pass	так	const=false
lane_count	так	const=8
matrix_authoritative	так	const=false
matrix_digest	так	\$ref="#/\$defs/Digest"
pass_credit	так	const=false
path	так	const="matrix-current/platform-no-degradation-matrix.json"
product_acceptance_pass	так	const=false
product_public_approval	так	const="not_approved"
record_type	так	const="ProminPlatformNoDegradationMatrix"
sha256	так	\$ref="#/\$defs/Digest"
size_bytes	так	type="integer"; minimum=1

```

{
  "additionalProperties": false,
  "properties": {
    "acceptance_pass": {
      "const": false
    },

```

```

"lane_count": {
  "const": 8
},
"matrix_authoritative": {
  "const": false
},
"matrix_digest": {
  "$ref": "#/$defs/Digest"
},
"pass_credit": {
  "const": false
},
"path": {
  "const": "matrix-current/platform-no-degradation-matrix.json"
},
"product_acceptance_pass": {
  "const": false
},
"product_public_approval": {
  "const": "not_approved"
},
"record_type": {
  "const": "ProminPlatformNoDegradationMatrix"
},
"sha256": {
  "$ref": "#/$defs/Digest"
},
"size_bytes": {
  "minimum": 1,
  "type": "integer"
}
},
"required": [
  "path",
  "sha256",
  "size_bytes",
  "record_type",
  "matrix_digest",
  "lane_count",
  "matrix_authoritative",
  "pass_credit",
  "acceptance_pass",
  "product_acceptance_pass",
  "product_public_approval"
],
"type": "object",
"x-promin-persistence": "derived-live-only"
}

```

Власник: [core/contracts.schema.json#/\\$defs/StandardReleaseMatrixBinding](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/StandardReleaseMatrixBinding](#)

110. StandardReleaseSupplementalLane

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: lane_id, path, sha256, size_bytes, record_type, status, candidate_binding_digest, result_digest, evidence_role, producer_attestation_digest

Властивість	Обов'язков е	Структурне правило
candidate_binding_digest	так	\$ref="#/\$defs/Digest"
evidence_role	так	enum=["linux", "linux-no-degradation", "windows", "windows-no-degradation"]
lane_id	так	enum=["linux-cp314-offline", "linux-cp314-online", "windows-cp314-offline", "windows-cp314-online"]
path	так	\$ref="#/\$defs/RelativePath"
producer_attestation_digest	так	\$ref="#/\$defs/Digest"
record_type	так	enum=["NoDegradationResult", "PlatformVerificationResult"]
result_digest	так	\$ref="#/\$defs/Digest"
sha256	так	\$ref="#/\$defs/Digest"
size_bytes	так	type="integer"; minimum=1
status	так	const="pass"

```

{
  "additionalProperties": false,
  "properties": {
    "candidate_binding_digest": {
      "$ref": "#/$defs/Digest"
    },
    "evidence_role": {
      "enum": [
        "linux",
        "linux-no-degradation",
        "windows",
        "windows-no-degradation"
      ]
    },
    "lane_id": {
      "enum": [
        "linux-cp314-offline",
        "linux-cp314-online",
        "windows-cp314-offline",
        "windows-cp314-online"
      ]
    },
    "path": {
      "$ref": "#/$defs/RelativePath"
    },
    "producer_attestation_digest": {
      "$ref": "#/$defs/Digest"
    }
  },

```

```

    "record_type": {
      "enum": [
        "NoDegradationResult",
        "PlatformVerificationResult"
      ]
    },
    "result_digest": {
      "$ref": "#/$defs/Digest"
    },
    "sha256": {
      "$ref": "#/$defs/Digest"
    },
    "size_bytes": {
      "minimum": 1,
      "type": "integer"
    },
    "status": {
      "const": "pass"
    }
  },

```

```

  "required": [
    "lane_id",
    "path",
    "sha256",
    "size_bytes",
    "record_type",
    "status",
    "candidate_binding_digest",

```

```

    "result_digest",
    "evidence_role",
    "producer_attestation_digest"
  ],
  "type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/StandardReleaseSupplementalLane](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/StandardReleaseSupplementalLane](#)

111. StandardReleaseTrustConfiguration

Опис типу: type="object"; additionalProperties=false; allOf[7]=keys:properties | keys:properties | keys:properties | keys:properties | keys:properties | keys:properties | keys:properties | keys:properties

Обов'язкові поля: record_type, trust_root_id, signature_provider_id, algorithm, keys

Властивість	Обов'язков е	Структурне правило
algorithm	так	const="Ed25519"
keys	так	type="array"; minItems=1; maxItems=32; uniqueItems=true

Властивість	Обов'язков е	Структурне правило
record_type	так	const="StandardReleaseTrustConfiguration"
signature_provider_id	так	const="cryptography-ed25519-v1"
trust_root_id	так	\$ref="#/\$defs/Id"
<pre> { "additionalProperties": false, "allOf": [{ "properties": { "keys": { "contains": { "properties": { "evidence_roles": { "contains": { "const": "human-documents" } } }, "required": ["evidence_roles"] }, "maxContains": 1, "minContains": 1 } }, "properties": { "keys": { "contains": { "properties": { "evidence_roles": { "contains": { "const": "linux" } } }, "required": ["evidence_roles"] }, "maxContains": 1, "minContains": 1 } }, "properties": { "keys": { "contains": { "properties": { "evidence_roles": { "contains": { "const": "linux-no-degradation" } } }, "required": ["evidence_roles"] }, "maxContains": 1, "minContains": 1 } }, "properties": { "keys": { "contains": { "properties": { "evidence_roles": { "contains": { </pre>		

```

        "const": "physical-scale"
      }
    },
    "required": [
      "evidence_roles"
    ]
  },
  "maxContains": 1,
  "minContains": 1
}
},
{

```

```

  "properties": {
    "keys": {
      "contains": {
        "properties": {
          "evidence_roles": {
            "contains": {
              "const": "saturation-audit"
            }
          }
        },
        "required": [
          "evidence_roles"
        ]
      },
      "maxContains": 1,
      "minContains": 1
    }
  },
{
  "properties": {
    "keys": {
      "contains": {
        "properties": {
          "evidence_roles": {
            "contains": {
              "const": "windows"
            }
          }
        }
      }
    }
  }

```

```

    },
    "required": [
      "evidence_roles"
    ]
  },
  "maxContains": 1,
  "minContains": 1
}
},
{
  "properties": {
    "keys": {
      "contains": {
        "properties": {
          "evidence_roles": {
            "contains": {
              "const": "windows-no-degradation"
            }
          }
        },
        "required": [
          "evidence_roles"
        ]
      },
      "maxContains": 1,
      "minContains": 1
    }
  }

```

```

    }
  },
  "properties": {
    "algorithm": {
      "const": "Ed25519"
    }
  }
}

```



```

},
"keys": {
  "items": {
    "additionalProperties": false,
    "allOf": [
      {
        "if": {
          "properties": {
            "capabilities": {
              "const": [
                "evidence.produce"
              ]
            }
          }
        },
        "then": {
          "properties": {
            "evidence_roles": {
              "maxItems": 1,
              "minItems": 1
            }
          },
          "platforms": {
            "minItems": 1
          }
        }
      }
    ],
    "if": {
      "properties": {
        "capabilities": {
          "const": [
            "standard.distribute"
          ]
        }
      },
      "then": {
        "properties": {
          "evidence_roles": {
            "maxItems": 0
          },
          "platforms": {
            "maxItems": 0
          }
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "evidence_roles": {
          "const": [
            "human-documents"
          ]
        }
      },
      "then": {
        "properties": {
          "platforms": {
            "const": [
              "windows"
            ]
          }
        }
      }
    },
    {
      "if": {
        "properties": {
          "evidence_roles": {
            "const": [
              "linux"
            ]
          }
        }
      }
    }
  }
}

```

```

    "if": {
      "properties": {
        "evidence_roles": {
          "const": [
            "human-documents"
          ]
        }
      },
      "then": {
        "properties": {
          "platforms": {
            "const": [
              "windows"
            ]
          }
        }
      }
    },
    {
      "if": {
        "properties": {
          "evidence_roles": {
            "const": [
              "linux"
            ]
          }
        }
      }
    }
  }
}

```

```

    },
    "then": {
      "properties": {
        "platforms": {
          "const": [
            "linux"
          ]
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "evidence_roles": {
          "const": [
            "linux-no-degradation"
          ]
        }
      }
    },
    "then": {
      "properties": {
        "platforms": {
          "const": [
            "linux"
          ]
        }
      }
    }
  }
}

```

```

    }
  },
  {
    "if": {
      "properties": {
        "evidence_roles": {
          "const": [
            "physical-scale"
          ]
        }
      }
    },
    "then": {
      "properties": {
        "platforms": {
          "const": [
            "windows"
          ]
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "evidence_roles": {
          "const": [

```

```

            "saturation-audit"
          ]
        }
      }
    },
    "then": {
      "properties": {
        "platforms": {
          "const": [
            "windows"
          ]
        }
      }
    }
  },
  {
    "if": {
      "properties": {
        "evidence_roles": {
          "const": [
            "windows"
          ]

```

```

    }
  },
  "then": {
    "properties": {
      "platforms": {
        "const": [
          "windows"
        ]
      }
    }
  },
  },
  {
    "if": {
      "properties": {
        "evidence_roles": {
          "const": [
            "windows-no-degradation"
          ]
        }
      }
    },
    "then": {
      "properties": {
        "platforms": {
          "const": [
            "windows"
          ]
        }
      }
    }
  }
},
],

```

```

"properties": {
  "capabilities": {
    "items": {
      "enum": [
        "evidence.produce",
        "standard.distribute"
      ]
    },
    "maxItems": 1,
    "minItems": 1,
    "type": "array",
    "uniqueItems": true
  },
  "evidence_roles": {
    "items": {
      "enum": [
        "human-documents",
        "linux",
        "linux-no-degradation",
        "physical-scale",
        "saturation-audit",
        "windows",
        "windows-no-degradation"
      ]
    },
    "maxItems": 1,
    "type": "array",
    "uniqueItems": true
  }
},

```

```

},
"key_id": {
  "$ref": "#/$defs/Id"
},
"not_after": {
  "$ref": "#/$defs/Timestamp"
},
"not_before": {
  "$ref": "#/$defs/Timestamp"
},
"platforms": {
  "items": {
    "enum": [
      "cross-platform",
      "darwin",

```

```

        "linux",
        "windows"
    ],
    "maxItems": 4,
    "type": "array",
    "uniqueItems": true
},
"public_key": {
    "maxLength": 44,
    "minLength": 44,
    "pattern": "^[A-Za-z0-9+/{43}=$",
    "type": "string"

```

```

    },
    "revoked": {
        "type": "boolean"
    },
    "subject_id": {
        "$ref": "#/$defs/Id"
    }
},
"required": [
    "key_id",
    "subject_id",
    "capabilities",
    "evidence_roles",
    "platforms",
    "public_key",
    "not_before",
    "not_after",
    "revoked"
],
"type": "object"
},
"maxItems": 32,
"minItems": 1,
"type": "array",
"uniqueItems": true
},
"record_type": {
    "const": "StandardReleaseTrustConfiguration"

```

```

    },
    "signature_provider_id": {
        "const": "cryptography-ed25519-v1"
    },
    "trust_root_id": {
        "$ref": "#/$defs/Id"
    }
},
"required": [
    "record_type",
    "trust_root_id",
    "signature_provider_id",
    "algorithm",
    "keys"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/StandardReleaseTrustConfiguration](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/StandardReleaseTrustConfiguration](#)

112. StandardsInit

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, bindings

Властивість	Обов'язков е	Структурне правило
bindings	так	type="array"; minItems=0; maxItems=64
record_type	так	const="StandardsInit"

```

{
  "additionalProperties": false,
  "properties": {
    "bindings": {

```

```

    "items": {
      "$ref": "#/$defs/StandardBinding"
    },
    "maxItems": 64,
    "minItems": 0,
    "type": "array"
  },
  "record_type": {
    "const": "StandardsInit"
  }
},
"required": [
  "record_type",
  "bindings"
],
"type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/StandardsInit | Валідатор: urn:promin:contracts:1#/\$defs/StandardsInit

113. StateBindingUpdate

Опис типу: type="object"; additionalProperties=false; oneOf[2]=keys:properties,required | keys:not,properties

Обов'язкові поля: leaf_type, leaf_id, operation

Властивість	Обов'язков е	Структурне правило
leaf_id	так	\$ref="#/\$defs/Id"
leaf_type	так	enum=["Activation", "Завдання", "Candidate", "Artifact", "Run", "Finding", "Decision", "GateResult", "Grant", "Оренда", "Відношення"]
operation	так	enum=["delete", "set"]
value_digest	ні	\$ref="#/\$defs/Digest"

```

{
  "additionalProperties": false,
  "oneOf": [
    {
      "properties": {
        "operation": {
          "const": "set"
        }
      },
      "required": [
        "value_digest"
      ]
    },
    {
      "not": {
        "required": [
          "value_digest"
        ]
      },
      "properties": {
        "operation": {
          "const": "delete"
        }
      }
    }
  ],
  "properties": {
    "leaf_id": {
      "$ref": "#/$defs/Id"
    },
    "leaf_type": {
      "enum": [
        "Activation",
        "Завдання",
        "Candidate",
        "Artifact",
        "Run",
        "Finding",
        "Decision",
        "GateResult",
        "Grant",
        "Оренда",
        "Відношення"
      ]
    }
  }
}

```

```

    ]
  },
  "operation": {
    "enum": [
      "delete",
      "set"
    ]
  },
  "value_digest": {
    "$ref": "#/$defs/Digest"
  }
},
"required": [
  "leaf_type",
  "leaf_id",
  "operation"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/StateBindingUpdate](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/StateBindingUpdate](#)

114. StateMachines

Опис типу: `const=schema(keys:capacity_release,lease,task)`

Обов'язкові поля: -

```

{
  "const": {
    "capacity_release": {
      "closed_release": {
        "requires_field": "close_ack",
        "slot_reusable": true,
        "state": "CLOSED"
      },
      "terminal_reconciliation": {
        "fencing_token_binding": "exact-lease-fencing-token",
        "field": "capacity_reconciliation",
        "generation_binding": "exact-lease-generation",
        "manager_capability": "lease.manage",
        "required_fields": [
          "reconciled_by",
          "grant_id",
          "grant_claim_digest",
          "reconciled_at",
          "generation",
          "fencing_token"
        ],
        "slot_reusable_after_record": true,
        "states": [
          "EXPIRED",
          "REVOKED"
        ]
      }
    },
    "unreconciled_slot_reusable": false
  },
  "lease": {
    "ACTIVE": [
      "CLOSING",
      "EXPIRED",
      "REVOKED"
    ],
    "CLOSED": [],
    "CLOSING": [
      "CLOSED",
      "EXPIRED",
      "REVOKED"
    ],
    "EXPIRED": [],
    "REVOKED": []
  },
  "task": {
    "BLOCKED": [
      "READY",
      "CANCELLED"
    ],
    "CANCELLED": [],

```

```

"COMPLETED": [],
"LEASED": [
  "RUNNING",
  "READY",
  "BLOCKED",
  "CANCELLED"
],
"PLANNED": [
  "READY",
  "BLOCKED",
  "CANCELLED"
],
"READY": [
  "LEASED",
  "BLOCKED",
  "CANCELLED"
],
"RUNNING": [
  "COMPLETED",
  "BLOCKED",
  "CANCELLED"
]
}
}
}

```

Власник: core/contracts.schema.json#/\$defs/StateMachines | Валідатор: urn:promin:contracts:1#/\$defs/StateMachines

115. StructuralBudgets

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: core_normative_artifacts, selected_presets, authority_bearing_init_records, persistent_entity_kinds_max, relation_kinds_max, base_user_commands_max

Властивість	Обов'язков е	Структурне правило
authority_bearing_init_records	так	type="integer"; minimum=1
base_user_commands_max	так	type="integer"; minimum=1
core_normative_artifacts	так	type="integer"; minimum=1
persistent_entity_kinds_max	так	type="integer"; minimum=1
relation_kinds_max	так	type="integer"; minimum=1
selected_presets	так	type="integer"; minimum=1

```

{
  "additionalProperties": false,
  "properties": {
    "authority_bearing_init_records": {
      "minimum": 1,
      "type": "integer"
    },
    "base_user_commands_max": {
      "minimum": 1,
      "type": "integer"
    },
    "core_normative_artifacts": {
      "minimum": 1,
      "type": "integer"
    },
    "persistent_entity_kinds_max": {
      "minimum": 1,
      "type": "integer"
    },
    "relation_kinds_max": {
      "minimum": 1,
      "type": "integer"
    },
    "selected_presets": {
      "minimum": 1,
      "type": "integer"
    }
  },
  "required": [
    "core_normative_artifacts",
    "selected_presets",

```

```

    "authority_bearing_init_records",
    "persistent_entity_kinds_max",
    "relation_kinds_max",
    "base_user_commands_max"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/StructuralBudgets | Валідатор: urn:promin:contracts:1#/\$defs/StructuralBudgets

116. SubjectInit

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: subject_id, kind, display_name

Властивість	Обов'язков е	Структурне правило
display_name	так	type="string"; minLength=1
kind	так	enum=["human", "service", "agent"]
subject_id	так	\$ref="#/\$defs/Id"

```

{
  "additionalProperties": false,
  "properties": {
    "display_name": {
      "minLength": 1,
      "type": "string"
    },
    "kind": {
      "enum": [
        "human",
        "service",
        "agent"
      ]
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "subject_id",
    "kind",
    "display_name"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/SubjectInit | Валідатор: urn:promin:contracts:1#/\$defs/SubjectInit

117. Task

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, task_id, state, required_capability, acceptance_predicate, allowed_paths, activation_digest, candidate_digest, created_at, gate_run_definitions

Властивість	Обов'язков е	Структурне правило
acceptance_predicate	так	type="string"; minLength=1
activation_digest	так	\$ref="#/\$defs/Digest"
allowed_paths	так	type="array"; minItems=0; maxItems=64; uniqueItems=true
candidate_digest	так	\$ref="#/\$defs/Digest"
created_at	так	\$ref="#/\$defs/Timestamp"
gate_run_definitions	так	type="array"; minItems=1; maxItems=64
operation_profile_id	ні	\$ref="#/\$defs/Id"
orchestration_required	ні	const=true
recommended_model_tier	ні	enum=["tool-only", "micro", "standard", "strong", "critical-review"]
record_type	так	const="Завдання"
required_capability	так	\$ref="#/\$defs/Id"
state	так	\$ref="#/\$defs/TaskState"

Властивість	Обов'язков е	Структурне правило
task_id	так	\$ref="#/\$defs/Id"
<pre> { "additionalProperties": false, "properties": { "acceptance_predicate": { "minLength": 1, "type": "string" }, "activation_digest": { "\$ref": "#/\$defs/Digest" }, "allowed_paths": { "items": { "\$ref": "#/\$defs/RelativePath" }, "maxItems": 64, "minItems": 0, "type": "array", "uniqueItems": true }, "candidate_digest": { "\$ref": "#/\$defs/Digest" }, "created_at": { "\$ref": "#/\$defs/Timestamp" }, "gate_run_definitions": { "items": { "additionalProperties": false, </pre>		
<pre> "properties": { "definition": { "\$ref": "#/\$defs/GateRunDefinition" }, "definition_digest": { "\$ref": "#/\$defs/Digest" } }, "required": ["definition_digest", "definition"], "type": "object" }, "maxItems": 64, "minItems": 1, "type": "array" }, "operation_profile_id": { "\$ref": "#/\$defs/Id" }, "orchestration_required": { "const": true }, "recommended_model_tier": { "enum": ["tool-only", "micro", </pre>		
<pre> "standard", "strong", "critical-review"] }, "record_type": { "const": "Завдання" }, "required_capability": { "\$ref": "#/\$defs/Id" }, "state": { "\$ref": "#/\$defs/TaskState" }, "task_id": { "\$ref": "#/\$defs/Id" } } } </pre>		

```

},
"required": [
  "record_type",
  "task_id",
  "state",
  "required_capability",
  "acceptance_predicate",
  "allowed_paths",
  "activation_digest",
  "candidate_digest",
  "created_at",

  "gate_run_definitions"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/Task](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/Task](#)

118. TaskState

Опис типу: enum=["BLOCKED", "CANCELLED", "COMPLETED", "LEASED", "PLANNED", "READY", "RUNNING"]

Обов'язкові поля: -

```

{
  "enum": [
    "BLOCKED",
    "CANCELLED",
    "COMPLETED",
    "LEASED",
    "PLANNED",
    "READY",
    "RUNNING"
  ]
}

```

Власник: [core/contracts.schema.json#/\\$defs/TaskState](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/TaskState](#)

119. TaskStateMachine

Опис типу: const=schema(keys:BLOCKED,CANCELLED,COMPLETED,LEASED,PLANNED,READY,RUNNING)

Обов'язкові поля: -

```

{
  "const": {
    "BLOCKED": [
      "READY",
      "CANCELLED"
    ],
    "CANCELLED": [],
    "COMPLETED": [],
    "LEASED": [
      "RUNNING",
      "READY",
      "BLOCKED",
      "CANCELLED"
    ],
    "PLANNED": [
      "READY",
      "BLOCKED",
      "CANCELLED"
    ],
    "READY": [
      "LEASED",
      "BLOCKED",
      "CANCELLED"
    ],
    "RUNNING": [
      "COMPLETED",
      "BLOCKED",
      "CANCELLED"
    ]
  }
}

```

Власник: [core/contracts.schema.json#/\\$defs/TaskStateMachine](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/TaskStateMachine](#)

120. TaskTransition

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: task_id, from_state, to_state, reason

Властивість	Обов'язков е	Структурне правило
from_state	так	\$ref="#/\$defs/TaskState"
reason	так	type="string"; minLength=1
task_id	так	\$ref="#/\$defs/Id"
to_state	так	\$ref="#/\$defs/TaskState"

```
{
  "additionalProperties": false,
  "properties": {
    "from_state": {
      "$ref": "#/$defs/TaskState"
    },
    "reason": {
      "minLength": 1,
      "type": "string"
    },
    "task_id": {
      "$ref": "#/$defs/Id"
    },
    "to_state": {
      "$ref": "#/$defs/TaskState"
    }
  },
  "required": [
    "task_id",
    "from_state",
    "to_state",
    "reason"
  ],
  "type": "object"
}
```

Власник: core/contracts.schema.json#/\$defs/TaskTransition | Валідатор: urn:promin:contracts:1#/\$defs/TaskTransition

121. TeamPolicy

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: threshold, key_ids, signature_provider_capability

Властивість	Обов'язков е	Структурне правило
key_ids	так	type="array"; minItems=1; maxItems=32; uniqueItems=true
signature_provider_capability	так	const="signature"
threshold	так	type="integer"; minimum=1

```
{
  "additionalProperties": false,
  "properties": {
    "key_ids": {
      "items": {
        "$ref": "#/$defs/Id"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "signature_provider_capability": {
      "const": "signature"
    },
    "threshold": {
      "minimum": 1,
      "type": "integer"
    }
  },
  "required": [
    "threshold",
    "key_ids",
    "signature_provider_capability"
  ]
}
```

```

    ],
    "type": "object"
  }

```

Власник: *core/contracts.schema.json#/\$defs/TeamPolicy* | Валідатор: *urn:pomin:contracts:1#/\$defs/TeamPolicy*

122. TechnologiesInit

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, bindings

Властивість	Обов'язков е	Структурне правило
bindings	так	type="array"; minItems=1; maxItems=32
record_type	так	const="TechnologiesInit"

```

{
  "additionalProperties": false,
  "properties": {
    "bindings": {
      "items": {
        "$ref": "#/$defs/ProviderBinding"
      },
      "maxItems": 32,
      "minItems": 1,
      "type": "array"
    },
    "record_type": {
      "const": "TechnologiesInit"
    }
  },
  "required": [
    "record_type",
    "bindings"
  ],
  "type": "object"
}

```

Власник: *core/contracts.schema.json#/\$defs/TechnologiesInit* | Валідатор: *urn:pomin:contracts:1#/\$defs/TechnologiesInit*

123. TechnologyCapabilityDef

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: id, task, forbidden_boundary, adapter_protocol

Властивість	Обов'язков е	Структурне правило
adapter_protocol	так	type="object"; additionalProperties=false
forbidden_boundary	так	type="object"; additionalProperties=false
id	так	enum=["control-runtime", "shape-validation", "content-identity", "local-serialization", "query-projection", "filesystem-inventory", "export-scan", "signature", "build-dependency"]
task	так	type="string"; minLength=1

```

{
  "additionalProperties": false,
  "properties": {
    "adapter_protocol": {
      "additionalProperties": false,
      "properties": {
        "dependency_receipt_rule": {
          "const": {
            "aggregate_digest_rule": "sha256(canonical receipt without aggregate_digest)",
            "complete": true,
            "component_fields": [
              "component_id",
              "component_kind",
              "source",
              "version",
              "digest",
              "size_bytes"
            ],
            "component_order": "component-id-ascending",
            "os_substrate_excluded": true,
            "provider_owned_scope": "configured-provider-root-and-runtime-components",
            "unresolved_dependencies": "reject"
          }
        }
      }
    }
  }
}

```

```

    },
    "operation_contracts": {
      "additionalProperties": {
        "additionalProperties": false,
        "properties": {
          "request": {
            "maxLength": 1024,
            "minLength": 1,
            "type": "string"
          },
          "response": {
            "maxLength": 1024,
            "minLength": 1,
            "type": "string"
          }
        },
        "required": [
          "request",
          "response"
        ],
        "type": "object"
      },
      "maxProperties": 32,
      "minProperties": 1,
      "propertyNames": {
        "$ref": "#/$defs/Id"
      },
      "type": "object"
    },
    "protocol_id": {
      "$ref": "#/$defs/Id"
    },
    "receipt_persistence": {
      "const": "content-addressed-receipt"
    },
    "supported_bindings": {
      "items": {
        "additionalProperties": false,
        "properties": {
          "adapter_id": {
            "$ref": "#/$defs/Id"
          },
          "identity_kind": {
            "enum": [
              "file-digest",
              "module-file-digest",
              "distribution-tree-digest",
              "module-closure-digest",
              "platform-attestation"
            ]
          },
          "invocation_kind": {
            "enum": [
              "python-runtime",
              "python-module",
              "executable",
              "platform-service"
            ]
          }
        },
        "required": [
          "adapter_id",
          "identity_kind",
          "invocation_kind"
        ],
        "type": "object"
      },
      "maxItems": 8,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    "supported_commands": {
      "items": {
        "$ref": "#/$defs/Id"
      },

```

```

        "maxItems": 32,
        "minItems": 1,
        "type": "array",
        "uniqueItems": true
      },
      "unknown_binding": {
        "const": "reject"
      }
    },
    "required": [
      "dependency_receipt_rule",
      "operation_contracts",
      "protocol_id",

```

```

      "receipt_persistence",
      "supported_bindings",
      "supported_commands",
      "unknown_binding"
    ],
    "type": "object"
  },
  "forbidden_boundary": {
    "additionalProperties": false,
    "properties": {
      "effect": {
        "const": "reject"
      },
      "forbids": {
        "items": {
          "maxLength": 128,
          "minLength": 1,
          "type": "string"
        },
        "maxItems": 16,
        "minItems": 1,
        "type": "array",
        "uniqueItems": true
      },
      "normative_authority": {
        "const": false
      }
    }
  },

```

```

    "required": [
      "effect",
      "forbids",
      "normative_authority"
    ],
    "type": "object"
  },
  "id": {
    "enum": [
      "control-runtime",
      "shape-validation",
      "content-identity",
      "local-serialization",
      "query-projection",
      "filesystem-inventory",
      "export-scan",
      "signature",
      "build-dependency"
    ]
  },
  "task": {
    "minLength": 1,
    "type": "string"
  }
},
"required": [
  "id",
  "task",

```

```

    "forbidden_boundary",
    "adapter_protocol"
  ],
  "type": "object"
}

```

Власник: core/contracts.schema.json#/\$defs/TechnologyCapabilityDef | Валідатор: urn:promin:contracts:1#/\$defs/TechnologyCapabilityDef

124. Timestamp

Опис типу: type="string"; format="date-time"; pattern="^\d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2}Z\$"

Обов'язкові поля: -

```
{
  "format": "date-time",
  "pattern": "^\d{4}-\d{2}-\d{2}T\d{2}:\d{2}:\d{2}Z$",
  "type": "string"
}
```

Власник: core/contracts.schema.json#/\$defs/Timestamp | Валідатор: urn:promin:contracts:1#/\$defs/Timestamp

125. TrustKey

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: key_id, subject_id, algorithm, public_key, fingerprint

Властивість	Обов'язков е	Структурне правило
algorithm	так	type="string"; minLength=1
fingerprint	так	\$ref="#/\$defs/Digest"
key_id	так	\$ref="#/\$defs/Id"
public_key	так	type="string"; minLength=1
subject_id	так	\$ref="#/\$defs/Id"

```
{
  "additionalProperties": false,
  "properties": {
    "algorithm": {
      "minLength": 1,
      "type": "string"
    },
    "fingerprint": {
      "$ref": "#/$defs/Digest"
    },
    "key_id": {
      "$ref": "#/$defs/Id"
    },
    "public_key": {
      "minLength": 1,
      "type": "string"
    },
    "subject_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "key_id",
    "subject_id",
    "algorithm",
    "public_key",
    "fingerprint"
  ],
  "type": "object"
}
```

Власник: core/contracts.schema.json#/\$defs/TrustKey | Валідатор: urn:promin:contracts:1#/\$defs/TrustKey

126. WorkCard

Опис типу: type="object"; additionalProperties=false; allOf[2]=keys:else,if,then | keys:else,if,then,title

Обов'язкові поля: record_type, task_id, operation_mode, operation, acceptance_predicate, allowed_paths, holder_grant_id, candidate_digest, activation_digest, context_digest, stop_conditions, budget, truncated, query_grant_id, query_grant_claim_digest

Властивість	Обов'язков е	Структурне правило
acceptance_predicate	так	type="string"; minLength=1
activation_digest	так	\$ref="#/\$defs/Digest"
allowed_paths	так	type="array"; minItems=0; maxItems=64; uniqueItems=true

Властивість	Обов'язков е	Структурне правило
budget	так	\$ref="#/\$defs/Budget"
candidate_digest	так	\$ref="#/\$defs/Digest"
context_digest	так	\$ref="#/\$defs/Digest"
continuation_query	ні	type="string"; minLength=1
fencing_token	ні	type="integer"; minimum=1
holder_grant_id	так	\$ref="#/\$defs/Id"
lease_generation	ні	type="integer"; minimum=1
lease_id	ні	\$ref="#/\$defs/Id"
operation	так	type="string"; minLength=1
operation_mode	так	enum=["read", "mutate"]
operation_profile_id	ні	\$ref="#/\$defs/Id"
orchestration_required	ні	const=true
query_grant_claim_digest	так	\$ref="#/\$defs/Digest"
query_grant_id	так	\$ref="#/\$defs/Id"
recommended_model_tier	ні	enum=["tool-only", "micro", "standard", "strong", "critical-review"]
record_type	так	const="WorkCard"
stop_conditions	так	type="array"; minItems=1; maxItems=16; uniqueItems=true
task_id	так	\$ref="#/\$defs/Id"
truncated	так	type="boolean"

```

{
  "additionalProperties": false,
  "allOf": [
    {
      "else": {
        "not": {
          "required": [
            "continuation_query"
          ]
        }
      },
      "if": {
        "properties": {
          "truncated": {
            "const": true
          }
        }
      },
      "then": {
        "required": [
          "continuation_query"
        ]
      }
    },
    {
      "else": {
        "not": {
          "anyOf": [
            {
              "required": [
                "lease_id"
              ]
            },
            {
              "required": [
                "lease_generation"
              ]
            },
            {
              "required": [
                "fencing_token"
              ]
            }
          ]
        }
      }
    }
  ]
}

```



```

    },
    "if": {
      "properties": {
        "operation_mode": {
          "const": "mutate"
        }
      },
      "required": [
        "operation_mode"
      ]
    },
    "then": {
      "required": [
        "lease_id",
        "lease_generation",
        "fencing_token"
      ]
    },
    "title": "вимагати поточні generation і fence Lease для мутації"
  },
  "properties": {
    "acceptance_predicate": {
      "minLength": 1,
      "type": "string"
    },
    "activation_digest": {
      "$ref": "#/$defs/Digest"
    },
    "allowed_paths": {
      "items": {
        "$ref": "#/$defs/RelativePath"
      },
      "maxItems": 64,
      "minItems": 0,
      "type": "array",
      "uniqueItems": true
    },
    "budget": {
      "$ref": "#/$defs/Budget"
    },
    "candidate_digest": {
      "$ref": "#/$defs/Digest"
    },
    "context_digest": {
      "$ref": "#/$defs/Digest"
    },
    "continuation_query": {
      "minLength": 1,
      "type": "string"
    },
    "fencing_token": {
      "minimum": 1,
      "type": "integer"
    },
    "holder_grant_id": {
      "$ref": "#/$defs/Id"
    },
    "lease_generation": {
      "minimum": 1,
      "type": "integer"
    },
    "lease_id": {
      "$ref": "#/$defs/Id"
    },
    "operation": {
      "minLength": 1,
      "type": "string"
    },
    "operation_mode": {
      "enum": [
        "read",
        "mutate"
      ]
    },
    "operation_profile_id": {
      "$ref": "#/$defs/Id"
    }
  }
}

```

```

    },
    "orchestration_required": {
      "const": true
    },
    },
    "query_grant_claim_digest": {
      "$ref": "#/$defs/Digest"
    },
    },
    "query_grant_id": {
      "$ref": "#/$defs/Id"
    },
    },
    "recommended_model_tier": {
      "enum": [
        "tool-only",
        "micro",
        "standard",
        "strong",
        "critical-review"
      ]
    }
  ]
}

```

```

    },
    "record_type": {
      "const": "WorkCard"
    },
    },
    "stop_conditions": {
      "items": {
        "minLength": 1,
        "type": "string"
      },
      "maxItems": 16,
      "minItems": 1,
      "type": "array",
      "uniqueItems": true
    },
    },
    "task_id": {
      "$ref": "#/$defs/Id"
    },
    },
    "truncated": {
      "type": "boolean"
    }
  }
},
"required": [
  "record_type",
  "task_id",
  "operation_mode",
  "operation",
  "acceptance_predicate",
  "allowed_paths",
  "holder_grant_id",
  "candidate_digest",
  "activation_digest",
  "context_digest",
  "stop_conditions",
  "budget",
  "truncated",
  "query_grant_id",
  "query_grant_claim_digest"
],
"type": "object"
}

```

Власник: [core/contracts.schema.json#/\\$defs/WorkCard](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/WorkCard](#)

127. WorkCardProjection

Опис типу: type="object"; additionalProperties=false

Обов'язкові поля: record_type, activation_digest, head_digest, projection_digest, task_id, task_digest, task, budget, projection_authoritative

Властивість	Обов'язков е	Структурне правило
activation_digest	так	\$ref="#/\$defs/Digest"
budget	так	\$ref="#/\$defs/Budget"
head_digest	так	oneOf[2]=#/\$defs/Digest type:null
projection_authoritative	так	const=false
projection_digest	так	\$ref="#/\$defs/Digest"

Властивість	Обов'язков е	Структурне правило
record_type	так	const="WorkCardProjection"
task	так	type="object"; additionalProperties=false
task_digest	так	\$ref="#/\$defs/Digest"
task_id	так	\$ref="#/\$defs/Id"
<pre> { "additionalProperties": false, "properties": { "activation_digest": { "\$ref": "#/\$defs/Digest" }, "budget": { "\$ref": "#/\$defs/Budget" }, "head_digest": { "oneOf": [{ "\$ref": "#/\$defs/Digest" }, { "type": "null" }] }, "projection_authoritative": { "const": false }, "projection_digest": { "\$ref": "#/\$defs/Digest" }, "record_type": { "const": "WorkCardProjection" } }, "task": { "additionalProperties": false, "properties": { "data_class": { "enum": ["operational-record", "untrusted-source"] }, "entity_type": { "enum": ["Завдання", "Candidate", "Artifact", "Run", "Finding", "Decision", "GateResult", "Grant", "Оренда"] }, "id": { "\$ref": "#/\$defs/Id" }, "payload": { "type": "object" } }, "source_digest": { "\$ref": "#/\$defs/Digest" } }, "required": ["id", "entity_type", "data_class", "payload", "source_digest"], "type": "object" </pre>		

```

    },
    "task_digest": {
      "$ref": "#/$defs/Digest"
    },
    "task_id": {
      "$ref": "#/$defs/Id"
    }
  },
  "required": [
    "record_type",
    "activation_digest",
    "head_digest",
    "projection_digest",
    "task_id",
    "task_digest",
    "task",
    "budget",
    "projection_authoritative"
  ],
  "type": "object",
  "x-promin-persistence": "derived-live-only"
}

```

Власник: [core/contracts.schema.json#/\\$defs/WorkCardProjection](#) | Валідатор: [urn:promin:contracts:1#/\\$defs/WorkCardProjection](#)

Реєстр політик - 68

ID	Критичність	Назва	Твердження	ID валідатора	Власник
P-001	P0	no-false-success	Gate pass credit дорівнює true лише тоді, коли normalized status є pass і кожний обов'язковий gate проходить.	no_false_success	core/policy-set.json
P-002	P0	activation-binding	Кожний нормативний запис прив'язує точну активну конфігурацію Core, preset, init і profile.	activation_binding	core/policy-set.json
P-003	P0	grant-validity	Subject, capability, conjunctive scope, interval, nonce, exact claim digest, proof chain, revocation state і Activation повністю збігаються.	grant_validity	core/policy-set.json
P-004	P0	lease-fence	Кожна мутація використовує поточну активну generation Lease та монотонний fencing token.	lease_fence	core/policy-set.json
P-005	P0	candidate-evidence-binding	Every evidence Artifact requires an explicit diagnostic, gate, or product purpose and binds Activation, candidate, policy, provider/tool, and input digests inside the immutable event payload.	candidate_evidence_binding	core/policy-set.json
P-006	P0	atomic-event-commit	Одна команда створює один immutable batch через lock, compare-head, sync, atomic replace і recovery.	atomic_event_commit	core/policy-set.json
P-007	P0	projection-is-derived	Видалення проєкції та deterministic replay не можуть змінити нормативне значення.	projection_derived	core/policy-set.json
P-008	P1	minimal-graph	Each verified raw file creates exactly one non-authoritative derived Artifact proxy. Raw inventory creates no Task, READS, or PRODUCES record; Tasks and typed relations require explicit semantic content.	minimal_graph	core/policy-set.json
P-009	P0	bounded-workcard	Every WorkCard stays within both Core hard ceilings and its selected profile byte, entity, relation, fanout, and top-k budgets; every truncation is explicit and losslessly resumable.	bounded_workcard	core/policy-set.json
P-010	P0	size-independent-init	Ініціалізація виконує нуль product-tree scans і створює рівно п'ять init records.	zero_scan_init	core/policy-set.json
P-011	P1	explicit-external-bindings	Зовнішні стандарти та providers існують лише в explicit init records, прив'язаних до digest.	explicit_bindings	core/policy-set.json
P-012	P0	canonical-name-only	Канонічна ідентичність - promin; aliases, redirects і compatibility names відхиляються.	canonical_name	core/policy-set.json
P-013	P0	safe-export	Export використовує allowlist, рекурсивно обмежений, безпечний щодо symlink і secrets та очищує paths.	safe_export	core/policy-set.json
P-014	P0	closed-acknowledgment	Лише чинний перехід до CLOSED з явним підтвердженням закриття звільняє слот мутації.	closed_ack	core/policy-set.json
P-015	P0	fail-closed-critical-fields	Відсутні, невідомі, stale, downgraded або unbound critical fields спричиняють відхилення до мутації.	critical_fields	core/policy-set.json
P-016	P0	preset-capability-ceiling	Preset може налаштовувати потреби в провайдерах і бюджети, але не може надавати повноваження на дії.	preset_no_authority	core/policy-set.json
P-017	P1	single-scan-inventory	Один запит inventory виконує не більше одного повного проходу product tree; усі проєкції походять із його stream.	single_scan	core/policy-set.json
P-018	P0	idempotent-commands	Повторний idempotency key у межах тієї самої Activation і subject повертає початковий результат або конфлікт.	idempotent_commands	core/policy-set.json
P-019	P1	core-preset-separation	Ідентичність Core bundle не включає заміни presets; Activation прив'язує вибраний preset окремо.	core_preset_separation	core/policy-set.json

ID	Критичність	Назва	Твердження	ID валідатора	Власник
P-020	P0	acyclic-bootstrap-authority	Init config визначає root ceilings, але ніколи не створює pre-activation runtime Grants.	bootstrap_authority	core/policy-set.json
P-021	P1	normalization-collision-rejection	Повторні JSON keys і колізії path або ID після NFC/casefold normalization відхиляються.	normalization_collision	core/policy-set.json
P-022	P1	provider-substitution-binding	A provider identity or implementation-closure change invalidates the Activation and every dependent continuation, evidence Artifact, projection, replay result, and current release closure; it never silently changes authority.	provider_substitution	core/policy-set.json
P-023	P1	exact-init-keyset	Activation прив'язує рівно project.json, standards.json, technologies.json і authority.json.	exact_init_keyset	core/policy-set.json
P-024	P1	stale-projection-rejection	Генерація status, next і WorkCard відхиляє проєкцію, source HEAD або Activation якої відрізняється.	stale_projection	core/policy-set.json
P-025	P0	one-command-one-batch	Кожний committed batch прив'язує рівно один subject, validated command digest, command intent digest, authorization digest та idempotency key.	one_command_one_batch	core/policy-set.json
P-026	P1	deterministic-retrieval	Однакові ranking scores розв'язуються детерміновано, а acceptance-critical facts додаються через typed closure, не fuzzy ranking.	deterministic_retrieval	core/policy-set.json
P-027	P1	strict-time-order	Timestamps використовують canonical UTC seconds; порядок issued, heartbeat, expiry, finish і decision перевіряється семантично.	strict_time	core/policy-set.json
P-028	P1	exact-standard-files	Вхідні Core і вибраний preset відхиляють symlink та неочікувані shadow files.	exact_standard_files	core/policy-set.json
P-029	P0	root-authorization-is-bootstrap-only	Root command authorization підписує один command intent і може видати лише перший Grant authority.manage; вона не може виконувати звичайні команди проєкту.	root_authorization_limit	core/policy-set.json
P-030	P1	configuration-exact-idempotency	Повторна ініціалізація успішна лише тоді, коли запитаний canonical configuration digest точно дорівнює наявній Activation; інакше повертається конфлікт.	config_exact_idempotency	core/policy-set.json
P-031	P1	truthful-observability	Команда може повідомляти про scan, healthcheck, validation або durability action лише тоді, коли дію фактично виконано та прив'язано до доказів.	truthful_observability	core/policy-set.json
P-032	P1	depth-aware-retrieval-budget	Effective retrieval seed count reserves entity and relation capacity for typed closure at the requested dependency depth; top_k is a ceiling, byte/entity/relation/fanout budgets all remain effective, and no removed or unknown budget field may bypass a ceiling.	depth_aware_retrieval	core/policy-set.json
P-033	P1	lease-only-for-mutation	Mutation WorkCards потребують поточної Lease; read-only WorkCards не повинні отримувати або нести mutation Lease.	lease_only_for_mutation	core/policy-set.json
P-034	P0	command-capability-scope	Кожна команда визначає рівно одну capability, власником якої є Core; requested score зберігає кожне authorization constraint і явно містить кожну ціль впливу команди.	command_capability_scope	core/policy-set.json
P-035	P0	grant-issuance-chain	Кожний доказ Grant прив'язує claim_digest і є або єдиним bootstrap root issuance, або перевіреним team threshold, або активним issuer Grant з authority.manage.	grant_issuance_chain	core/policy-set.json
P-036	P0	verified-team-signatures	Team-signature fields are never valid without threshold evaluation by the signature adapter resolved from the exact technologies binding; process-local injected adapters are explicitly non-reconstructable and cannot prove a cross-process CLI workflow.	verified_team_signatures	core/policy-set.json

ID	Критичність	Назва	Твердження	ID валідатора	Власник
P-037	P1	provider-preflight-before-activation	Кожна обов'язкова provider identity і healthcheck успішно перевіряється до commit ініціалізації; оголошений provider не є доказом доступності.	provider_preflight	core/policy-set.json
P-038	P0	canonical-state-machines	Переходи Task і Lease приймаються лише state machines, власником яких є policy; generated lifecycle views не можуть вигадувати переходи.	state_machine_transition	core/policy-set.json
P-039	P0	decision-target-binding	Кожний Decision називає одну типізовану ціль, а decision kind, requested scope, candidate і authority узгоджені з цією ціллю.	decision_target_binding	core/policy-set.json
P-040	P0	credit-requires-evidence	Pass, fail, and blocked gate results cite at least one exact immutable Artifact record reference; skipped results never receive pass credit.	credit_requires_evidence	core/policy-set.json
P-041	P0	command-event-correspondence	Один command batch прив'язує точне authorization claim і містить рівно одну primary event, рівну validated command payload; auxiliary events є лише типізованими відношеннями, а event IDs унікальні.	command_event_correspondence	core/policy-set.json
P-042	P0	continuation-authorization-and-completeness	A continuation is authenticated by a random per-Activation secret and binds the exact subject, projection.read Grant claim, capability, requested scope, revocation state, normalized query, deterministic cursor, Activation, HEAD, projection, ranking, complete budgets, depth, implementation closure, issue time, and expiry; its complete page union emits every reachable entity and relation exactly once without loss.	continuation_completeness	core/policy-set.json
P-043	P0	candidate-snapshot-consistency	The candidate recipe is applied in the single inventory pass; promin v1 permits only provider-proven immutable VCS trees for credit, while best-effort observations are explicit and receive no pass credit.	candidate_snapshot_consistency	core/policy-set.json
P-044	P0	provider-adapter-dispatch	Every technologies.json binding resolves to the adapter actually invoked and carries a recomputed capability-specific implementation closure for the promin runtime, Python interpreter, jsonschema or SQLite component where used, and adapter. Unsupported or drifted tuples reject, and dependent evidence binds the exact implementation_closure_digest.	provider_adapter_dispatch	core/policy-set.json
P-045	P0	current-release-closure	A project-operational release Decision is immutable history; current product release eligibility is derived and becomes false after a blocker, failed gate, stale or missing evidence, implementation drift, or Activation drift. It is separate from the external StandardReleaseDecision for promin distribution.	current_release_closure	core/policy-set.json
P-046	P0	finding-disposition-evidence	Every Finding cites exact immutable Artifact record references. Every resolve or waive Decision cites evidence bound to the exact canonical Finding digest; waiver also requires an exception-policy digest, future expiry, and separation of duties.	finding_disposition_evidence	core/policy-set.json
P-047	P0	candidate-delta-scope	A diff Artifact binds base and new Candidate digests, the exact WorkCard digest, and the complete changed-path set; every changed path is contained by both Task allowed_paths and the authorizing Grant effect scope.	candidate_delta_scope	core/policy-set.json

ID	Критичність	Назва	Твердження	ID валідатора	Власник
P-048	P1	verified-state-checkpoint	Events remain the sole authority. A normal command appends O(batch), incrementally updates changed projection rows/shards, and advances a verified bounded replay tail. Full derived-state compaction occurs only after declared batch/byte thresholds or explicit recovery, publishes crash-safely against exact HEAD, and missing, stale, or corrupt checkpoints require full verified fallback with equal semantic digest.	verified_state_checkpoint	core/policy-set.json
P-049	P1	profile-default-depth	A profile default_dependency_depth applies only when depth is omitted; explicit depth is accepted through the single Core dependency_depth_hard_max of 12 and remains subject to all retrieval budgets.	profile_default_depth	core/policy-set.json
P-050	P0	canonical-distribution-archive	The sole package builder emits immutable candidate bytes and an external StandardReleaseCandidateBinding. VERSION.json describes SemVer and package identity only; approval is derived later and never changes the candidate ZIP.	canonical_distribution_archive	core/policy-set.json
P-051	P0	verified-inventory-provenance	Public rebuild accepts only a verified InventoryResult or its persisted manifest; caller provenance fields are reserved, data_class is forced to untrusted-source, and runtime injects the exact path and digest.	verified_inventory_provenance	core/policy-set.json
P-052	P1	supported-command-surface	The v1 preset advertises exactly init, doctor, status, next, validate, and continue; no optional or administrative command catalogue may claim an absent executable path.	supported_command_surface	core/policy-set.json
P-053	P0	profile-bound-scale	Physical 100k evidence binds content-addressed raw inventory, query, continuation, process-sample, phase-log, and operation-metric artifacts for at least 198999 explicitly recorded Core-valid typed Relations and a representative mixed workload. The verifier recomputes every summary and fails closed on truthful process/invoke status, commit p95/p99, bytes per changed record, checkpoint count, semantic-ingestion time, search latency, incremental inventory memory amplification, absolute peak RSS, database bytes, projection amplification, semantic inflation, continuation completeness, or traversal-pass ceiling.	profile_bound_scale	core/policy-set.json
P-054	P0	cross-platform-release-evidence	The exact candidate ZIP must pass authenticated Windows and Linux platform observations, clean-install, package, handle-relative path/race, locking, crash, replay, physical scale, offline installed no-degradation, online compatibility, dependency/SBOM/RECORD/license closure, and typed PDF gates. One StandardReleaseEvidenceManifest physically resolves exactly seven unique authority roles, the exact non-authoritative eight-lane CPython 3.13/3.14 online/offline matrix, all four CPython 3.14 supplemental records, and every nested physical record by path, SHA-256, and byte count. All producer attestations bind the same candidate and participate in one replay/SOD/chronology graph; matrix_authoritative, pass_credit, acceptance_pass, product_acceptance_pass, and public approval remain false.	cross_platform_release	core/policy-set.json

ID	Критичність	Назва	Твердження	ID валідатора	Власник
P-055	P0	standard-release-separation	An external StandardReleaseDecision has explicit approve or reject outcome, binds candidate and evidence manifest digests, follows the manifest-derived maximum evidence completion time, stays within 300 seconds of verifier clock skew, and verifies decider, standard.distribute capability, independently pinned trust root, Ed25519 provider, event-time key validity, nonce, signed claim digest, and signature. The signed decision remains immutable history; current eligibility is derived; product_acceptance_pass is false and product_public_approval is not_approved.	standard_release_separation	core/policy-set.json
P-056	P0	required-gates-fail-closed	No-degradation runs the complete required suite only through the clean installed interpreter outside the install source, with isolated mode, cleared PYTHONPATH, import-origin assertion, zero skipped tests, and full hash-bound JUnit/stdout/stderr bytes. Distribution acceptance executes every required positive, mutation, concurrency, crash, replay, path, package, platform, Python-minor, offline/online, scale, search, PDF, evidence-resolution, chronology, and signature gate; a skipped, missing, stale, blocked, failed, unsigned, unresolved, source-shadowed, or unpinned gate receives no pass credit.	required_gates_fail_closed	core/policy-set.json
P-057	P0	research-draft-sanitation	Research crosses into normative owners only through the non-persistent MIGRATE_FACTS_NOT_FILES intake: every cited source is digest- and license-plan-bound, citation-required claims cite allowed sources or remain blocked with a reason, and marketing, proof, distribution, acceptance, or public-approval claims cannot become normative text.	research_draft_sanitation	core/policy-set.json
P-058	P0	doctor-result-is-live-non-authority	DoctorResult is an exact live-only derived health report with report_authoritative=false, pass_credit=false, product_acceptance_pass=false, product_public_approval=not_approved, zero product scans, and only its Core-owned operational metrics; token, cost, usage, and goal/session/batch/workspace/per-run allocation are forbidden.	doctor_result_non_authority	core/policy-set.json
P-059	P0	semantic-inheritance-is-structural	The compiled projection preserves the complete SemanticModel owner: admission, five independent granularities, promotion rules, exact entity purposes and admission rules, exact relation domain/range/rules, and structured provider forbidden boundaries.	semantic_inheritance	core/policy-set.json
P-060	P0	authority-owner-is-structural	AuthorityModel is the single owner of the capability catalogue, delegation depth, and exact separation-of-duty pairs; runtime policy may derive from that Activation-bound owner but may not substitute hardcoded authority facts.	authority_owner	core/policy-set.json
P-061	P0	gate-run-definition-is-independent	An immutable GateRunDefinition independently owns the expected evidence class, product-credit requirement, exact target scope, and candidate, policy, tool, provider, and input digests. A run or gate command binds its exact definition digest before submitted Artifacts are evaluated.	gate_run_definition_integrity	core/policy-set.json
P-062	P0	authority-runtime-policy-is-derived-live-only	AuthorityRuntimePolicy is rebuilt from the installed AuthorityModel for every mutation and replay, binds the exact owner digest, ordered capability IDs, structured separation-of-duty rules and delegation ceiling, and is never persisted or imported as authority.	authority_runtime_policy_projection	core/policy-set.json

ID	Критичність	Назва	Твердження	ID валідатора	Власник
P-063	P0	semantic-export-is-export-only	SemanticExport is a strict bounded export-only semantic-state record binding the exact Candidate, Activation, HEAD, Core, preset, implementation closure, providers, inputs, and output. It is neither a package nor archive and cannot carry product acceptance, pass credit, public approval, or release approval.	semantic_export_integrity	core/policy-set.json
P-064	P0	operation-metrics-a-re-live-observations	OperationMetrics is a strict live-only non-authoritative observation for one command. It permits only Core-owned duration, changed-record, write, projection, checkpoint, and physical-byte fields and forbids token, cost, usage, and goal/session/batch/workspace/per-run allocation.	operation_metrics_non_authority	core/policy-set.json
P-065	P0	workcard-projection-is-v2-live-only	RetrievalPage is the strict bounded live-only v2 retrieval result and exclusively owns stream_cursor, next_stream_cursor, and retrieval continuation. WorkCardProjection is internal selected-Task materialization and exposes no continuation or cursor fields.	workcard_projection_v2	core/policy-set.json
P-066	P0	event-store-policy-is-derived-live-only	EventStorePolicy is rebuilt from the installed AuthorityModel before normalization, mutation, and replay. It binds exact command and batch bounds, envelope and commitment fields, lease-bound commands and states, primary-event mapping, and UTC-second timestamps, with no defaults or nullable critical fields.	event_store_policy_projection	core/policy-set.json
P-067	P0	depends-on-graph-is-acyclic	Every command, import, replay, and rebuild rejects a DEPENDS_ON cycle. Relation domain and range validity alone cannot admit a cyclic dependency graph.	dependency_graph_acyclicity	core/policy-set.json
P-068	P0	ready-frontier-is-current-derived-state	ReadyFrontier is derived live-only from the current acyclic DEPENDS_ON graph. A READY Task is returned only after every dependency is currently COMPLETED, every required GateResult is current passing credit, and no current OPEN Finding blocks it; ordering and continuation are deterministic and truncation is explicit.	ready_frontier_projection	core/policy-set.json

Власник: core/policy-set.json | Валідатор: validator_registry_version=1

Індекс виконуваних валідаторів

P-001 | no_false_success | core/policy-set.json

P-002 | activation_binding | core/policy-set.json

P-003 | grant_validity | core/policy-set.json

P-004 | lease_fence | core/policy-set.json

P-005 | candidate_evidence_binding | core/policy-set.json

P-006 | atomic_event_commit | core/policy-set.json

P-007 | projection_derived | core/policy-set.json

P-008 | minimal_graph | core/policy-set.json

P-009 | bounded_workcard | core/policy-set.json

P-010 | zero_scan_init | core/policy-set.json

P-011 | explicit_bindings | core/policy-set.json

P-012 | canonical_name | core/policy-set.json

P-013 | safe_export | core/policy-set.json

P-014 | closed_ack | core/policy-set.json

P-015 | critical_fields | core/policy-set.json

P-016 | preset_no_authority | core/policy-set.json

P-017 | single_scan | core/policy-set.json
P-018 | idempotent_commands | core/policy-set.json
P-019 | core_preset_separation | core/policy-set.json
P-020 | bootstrap_authority | core/policy-set.json
P-021 | normalization_collision | core/policy-set.json
P-022 | provider_substitution | core/policy-set.json
P-023 | exact_init_keyset | core/policy-set.json
P-024 | stale_projection | core/policy-set.json
P-025 | one_command_one_batch | core/policy-set.json
P-026 | deterministic_retrieval | core/policy-set.json
P-027 | strict_time | core/policy-set.json
P-028 | exact_standard_files | core/policy-set.json
P-029 | root_authorization_limit | core/policy-set.json
P-030 | config_exact_idempotency | core/policy-set.json
P-031 | truthful_observability | core/policy-set.json
P-032 | depth_aware_retrieval | core/policy-set.json
P-033 | lease_only_for_mutation | core/policy-set.json
P-034 | command_capability_scope | core/policy-set.json
P-035 | grant_issuance_chain | core/policy-set.json
P-036 | verified_team_signatures | core/policy-set.json
P-037 | provider_preflight | core/policy-set.json
P-038 | state_machine_transition | core/policy-set.json
P-039 | decision_target_binding | core/policy-set.json
P-040 | credit_requires_evidence | core/policy-set.json
P-041 | command_event_correspondence | core/policy-set.json
P-042 | continuation_completeness | core/policy-set.json
P-043 | candidate_snapshot_consistency | core/policy-set.json
P-044 | provider_adapter_dispatch | core/policy-set.json
P-045 | current_release_closure | core/policy-set.json
P-046 | finding_disposition_evidence | core/policy-set.json
P-047 | candidate_delta_scope | core/policy-set.json
P-048 | verified_state_checkpoint | core/policy-set.json
P-049 | profile_default_depth | core/policy-set.json
P-050 | canonical_distribution_archive | core/policy-set.json
P-051 | verified_inventory_provenance | core/policy-set.json
P-052 | supported_command_surface | core/policy-set.json
P-053 | profile_bound_scale | core/policy-set.json
P-054 | cross_platform_release | core/policy-set.json
P-055 | standard_release_separation | core/policy-set.json
P-056 | required_gates_fail_closed | core/policy-set.json
P-057 | research_draft_sanitation | core/policy-set.json
P-058 | doctor_result_non_authority | core/policy-set.json
P-059 | semantic_inheritance | core/policy-set.json
P-060 | authority_owner | core/policy-set.json
P-061 | gate_run_definition_integrity | core/policy-set.json

P-062 | authority_runtime_policy_projection | core/policy-set.json

P-063 | semantic_export_integrity | core/policy-set.json

P-064 | operation_metrics_non_authority | core/policy-set.json

P-065 | workcard_projection_v2 | core/policy-set.json

P-066 | event_store_policy_projection | core/policy-set.json

P-067 | dependency_graph_acyclicity | core/policy-set.json

P-068 | readv_frontier_projection | core/policy-set.json

Власник: core/policy-set.json | Валідатор: policies[].validator_id

Предикати прийняття - 102

#	ID предиката	ID валідатора / предиката	Власник
1	canonical-name-only	canonical-name-only	core/conformance.json
2	exact-six-core-artifacts	exact-six-core-artifacts	core/conformance.json
3	preset-outside-core-digest	preset-outside-core-digest	core/conformance.json
4	exact-one-selected-preset	exact-one-selected-preset	core/conformance.json
5	exact-five-init-records	exact-five-init-records	core/conformance.json
6	zero-product-scan-during-init	zero-product-scan-during-init	core/conformance.json
7	single-pass-inventory	single-pass-inventory	core/conformance.json
8	no-false-success	no-false-success	core/conformance.json
9	strict-critical-fields	strict-critical-fields	core/conformance.json
10	activation-digest-binding	activation-digest-binding	core/conformance.json
11	root-ceiling-without-bootstrap-grant-cycle	root-ceiling-without-bootstrap-grant-cycle	core/conformance.json
12	grant-scope-expiry-nonce-proof	grant-scope-expiry-nonce-proof	core/conformance.json
13	lease-generation-fence-and-close-ack	lease-generation-fence-and-close-ack	core/conformance.json
14	candidate-evidence-provider-binding	candidate-evidence-provider-binding	core/conformance.json
15	one-command-one-atomic-batch	one-command-one-atomic-batch	core/conformance.json
16	deterministic-replay	deterministic-replay	core/conformance.json
17	projection-disposable-and-current	projection-disposable-and-current	core/conformance.json
18	bounded-workcard-through-depth-twelve	bounded-workcard-through-depth-twelve	core/conformance.json
19	typed-reference-domain-range-closure	typed-reference-domain-range-closure	core/conformance.json
20	dependency-graph-acyclic-and-current-ready-frontier	dependency-graph-acyclic-and-current-ready-frontier	core/conformance.json
21	safe-recursive-export	safe-recursive-export	core/conformance.json
22	idempotent-state-commands	idempotent-state-commands	core/conformance.json
23	normalization-collision-rejection	normalization-collision-rejection	core/conformance.json
24	provider-substitution-invalidates-derived-state	provider-substitution-invalidates-derived-state	core/conformance.json
25	rollback-and-recutover-equivalence	rollback-and-recutover-equivalence	core/conformance.json
26	zero-forbidden-name-occurrences	zero-forbidden-name-occurrences	core/conformance.json
27	zero-unowned-normative-facts	zero-unowned-normative-facts	core/conformance.json
28	zero-core-provider-lock-in	zero-core-provider-lock-in	core/conformance.json
29	zero-compiled-cache-files	zero-compiled-cache-files	core/conformance.json
30	physical-100k-one-artifact-proxy-per-file	physical-100k-one-artifact-proxy-per-file	core/conformance.json
31	external-standard-release-decision-exact-binding	external-standard-release-decision-exact-binding	core/conformance.json
32	root-bootstrap-command-limited	root-bootstrap-command-limited	core/conformance.json
33	deterministic-activation-identity	deterministic-activation-identity	core/conformance.json
34	configuration-exact-idempotency	configuration-exact-idempotency	core/conformance.json
35	truthful-operation-reporting	truthful-operation-reporting	core/conformance.json
36	depth-aware-seed-budget	depth-aware-seed-budget	core/conformance.json
37	lossless-continuation-in-reference-100k-scenario	lossless-continuation-in-reference-100k-scenario	core/conformance.json
38	read-workcard-without-lease	read-workcard-without-lease	core/conformance.json
39	mutation-workcard-requires-current-lease	mutation-workcard-requires-current-lease	core/conformance.json
40	command-capability-and-scope-resolution	command-capability-and-scope-resolution	core/conformance.json
41	grant-issuance-chain-closure	grant-issuance-chain-closure	core/conformance.json
42	team-signatures-cryptographically-verified	team-signatures-cryptographically-verified	core/conformance.json
43	required-provider-preflight-before-commit	required-provider-preflight-before-commit	core/conformance.json

#	ID предиката	ID валідатора / предиката	Власник
44	policy-owned-task-and-lease-state-machines	policy-owned-task-and-lease-state-machines	core/conformance.json
45	decision-kind-target-scope-binding	decision-kind-target-scope-binding	core/conformance.json
46	non-skipped-credit-result-has-evidence	non-skipped-credit-result-has-evidence	core/conformance.json
47	exact-command-primary-event-correspondence	exact-command-primary-event-correspondence	core/conformance.json
48	conjunctive-scope-containment	conjunctive-scope-containment	core/conformance.json
49	authorization-binds-exact-grant-claim	authorization-binds-exact-grant-claim	core/conformance.json
50	command-effect-target-in-requested-scope	command-effect-target-in-requested-scope	core/conformance.json
51	lease-holder-has-task-execute-grant	lease-holder-has-task-execute-grant	core/conformance.json
52	decision-grant-matches-command-authorization	decision-grant-matches-command-authorization	core/conformance.json
53	artifact-kind-resolves-correct-capability	artifact-kind-resolves-correct-capability	core/conformance.json
54	continuation-v2-complete-page-union	continuation-v2-complete-page-union	core/conformance.json
55	candidate-recipe-applied-once	candidate-recipe-applied-once	core/conformance.json
56	creditable-candidate-immutable-vcs-tree	creditable-candidate-immutable-vcs-tree	core/conformance.json
57	selected-provider-adapter-evidence	selected-provider-adapter-evidence	core/conformance.json
58	team-signed-cli-end-to-end	team-signed-cli-end-to-end	core/conformance.json
59	current-release-closure-revalidation	current-release-closure-revalidation	core/conformance.json
60	finding-bound-resolution-and-waiver-evidence	finding-bound-resolution-and-waiver-evidence	core/conformance.json
61	candidate-delta-within-task-and-grant-scope	candidate-delta-within-task-and-grant-scope	core/conformance.json
62	verified-head-checkpoint-delta-replay	verified-head-checkpoint-delta-replay	core/conformance.json
63	profile-default-with-core-depth-twelve	profile-default-with-core-depth-twelve	core/conformance.json
64	canonical-exact-zip-distribution	canonical-exact-zip-distribution	core/conformance.json
65	current-interpreter-and-clean-install-modes	current-interpreter-and-clean-install-modes	core/conformance.json
66	required-no-degradation-tests-fail-closed	required-no-degradation-tests-fail-closed	core/conformance.json
67	continuation-secret-subject-grant-binding	continuation-secret-subject-grant-binding	core/conformance.json
68	implementation-closure-bound-and-current	implementation-closure-bound-and-current	core/conformance.json
69	verified-inventory-result-provenance	verified-inventory-result-provenance	core/conformance.json
70	exact-six-base-command-surface	exact-six-base-command-surface	core/conformance.json
71	zero-dead-evidence-budget-fields	zero-dead-evidence-budget-fields	core/conformance.json
72	profile-bound-physical-100k-performance	profile-bound-physical-100k-performance	core/conformance.json
73	windows-linux-exact-zip-matrix	windows-linux-exact-zip-matrix	core/conformance.json
74	fresh-semantic-control-state-per-saturation-iteration	fresh-semantic-control-state-per-saturation-iteration	core/conformance.json
75	three-zero-new-saturation-iterations	three-zero-new-saturation-iterations	core/conformance.json
76	standard-distribution-separate-from-product-acceptance	standard-distribution-separate-from-product-acceptance	core/conformance.json
77	candidate-evidence-decision-acyclic-chain	candidate-evidence-decision-acyclic-chain	core/conformance.json
78	evidence-manifest-physical-resolution	evidence-manifest-physical-resolution	core/conformance.json
79	evidence-manifest-exact-eight-lane-matrix-resolution	evidence-manifest-exact-eight-lane-matrix-resolution	core/conformance.json
80	evidence-manifest-four-cp314-supplemental-records	evidence-manifest-four-cp314-supplemental-records	core/conformance.json
81	evidence-manifest-path-digest-size-closure	evidence-manifest-path-digest-size-closure	core/conformance.json
82	evidence-manifest-nested-and-supplemental-chronology	evidence-manifest-nested-and-supplemental-chronology	core/conformance.json
83	seven-distinct-role-producer-scopes	seven-distinct-role-producer-scopes	core/conformance.json
84	signed-standard-decision-authority-proof	signed-standard-decision-authority-proof	core/conformance.json
85	semver-from-candidate-binding	semver-from-candidate-binding	core/conformance.json

#	ID предиката	ID валідатора / предиката	Власник
86	typed-human-document-verification	typed-human-document-verification	core/conformance.json
87	broad-query-bounded-seed-refinement	broad-query-bounded-seed-refinement	core/conformance.json
88	continuation-token-bytes-at-most-256	continuation-token-bytes-at-most-256	core/conformance.json
89	inventory-stream-memory-amplification-at-most-32	inventory-stream-memory-amplification-at-most-32	core/conformance.json
90	explicit-scale-marker-selection	explicit-scale-marker-selection	core/conformance.json
91	authenticated-role-platform-evidence-attestations	authenticated-role-platform-evidence-attestations	core/conformance.json
92	installed-platform-observation-closure	installed-platform-observation-closure	core/conformance.json
93	handle-relative-evidence-root-safety	handle-relative-evidence-root-safety	core/conformance.json
94	raw-scale-summary-recomputation	raw-scale-summary-recomputation	core/conformance.json
95	bounded-incremental-commit-and-compaction	bounded-incremental-commit-and-compaction	core/conformance.json
96	offline-installed-no-degradation	offline-installed-no-degradation	core/conformance.json
97	offline-release-online-compatibility	offline-release-online-compatibility	core/conformance.json
98	decision-after-evidence-with-bounded-skew	decision-after-evidence-with-bounded-skew	core/conformance.json
99	single-derived-platform-closure-owner	single-derived-platform-closure-owner	core/conformance.json
100	truthful-process-invocation-status	truthful-process-invocation-status	core/conformance.json
101	product-public-approval-separation	product-public-approval-separation	core/conformance.json
102	historical-decision-current-eligibility-separation	historical-decision-current-eligibility-separation	core/conformance.json

Standard distribution success never grants product acceptance. Live product credit separately requires a creditable immutable-vcs-tree Candidate, all P0/P1 findings closed with exact target-bound evidence, current project release closure eligibility, rollback and recutover equivalence, and an authorized human project Decision. `product_acceptance_pass` remains false for promin v1 distribution evidence.

Власник: `core/conformance.json` | Валідатор: `required_acceptance`

Каталог мутацій - 77 сімейств

#	Сімейство мутацій	ID мутаційної перевірки	Власник
1	status-credit-contradiction	status-credit-contradiction	core/conformance.json
2	missing-critical-field	missing-critical-field	core/conformance.json
3	unknown-critical-field	unknown-critical-field	core/conformance.json
4	activation-drift	activation-drift	core/conformance.json
5	grant-replay	grant-replay	core/conformance.json
6	scope-escalation	scope-escalation	core/conformance.json
7	expired-grant	expired-grant	core/conformance.json
8	revoked-grant	revoked-grant	core/conformance.json
9	stale-fence	stale-fence	core/conformance.json
10	candidate-drift	candidate-drift	core/conformance.json
11	provider-substitution	provider-substitution	core/conformance.json
12	event-fork	event-fork	core/conformance.json
13	batch-command-mismatch	batch-command-mismatch	core/conformance.json
14	duplicate-idempotency-key	duplicate-idempotency-key	core/conformance.json
15	projection-corruption	projection-corruption	core/conformance.json
16	projection-staleness	projection-staleness	core/conformance.json
17	reference-break	reference-break	core/conformance.json
18	relation-domain-break	relation-domain-break	core/conformance.json
19	context-flood	context-flood	core/conformance.json
20	symlink-core-input	symlink-core-input	core/conformance.json
21	shadow-core-file	shadow-core-file	core/conformance.json
22	symlink-product-entry	symlink-product-entry	core/conformance.json
23	secret-export	secret-export	core/conformance.json
24	archive-expansion	archive-expansion	core/conformance.json
25	preset-authority-escalation	preset-authority-escalation	core/conformance.json
26	normalization-key-collision	normalization-key-collision	core/conformance.json
27	path-casefold-collision	path-casefold-collision	core/conformance.json
28	forbidden-name-reintroduction	forbidden-name-reintroduction	core/conformance.json
29	reinit-configuration-conflict	reinit-configuration-conflict	core/conformance.json
30	clock-order-violation	clock-order-violation	core/conformance.json
31	capability-confusion	capability-confusion	core/conformance.json
32	command-scope-escalation	command-scope-escalation	core/conformance.json
33	grant-issuer-chain-break	grant-issuer-chain-break	core/conformance.json
34	unverified-signature-acceptance	unverified-signature-acceptance	core/conformance.json
35	provider-declared-but-unavailable	provider-declared-but-unavailable	core/conformance.json
36	illegal-task-transition	illegal-task-transition	core/conformance.json
37	illegal-lease-transition	illegal-lease-transition	core/conformance.json
38	decision-target-confusion	decision-target-confusion	core/conformance.json
39	credit-without-evidence	credit-without-evidence	core/conformance.json
40	command-primary-event-mismatch	command-primary-event-mismatch	core/conformance.json
41	duplicate-event-id	duplicate-event-id	core/conformance.json
42	scope-selector-drop-broadening	scope-selector-drop-broadening	core/conformance.json
43	authorization-grant-claim-substitution	authorization-grant-claim-substitution	core/conformance.json
44	command-effect-outside-requested-scope	command-effect-outside-requested-scope	core/conformance.json
45	lease-holder-grant-mismatch	lease-holder-grant-mismatch	core/conformance.json
46	decision-grant-provenance-mismatch	decision-grant-provenance-mismatch	core/conformance.json

#	Сімейство мутацій	ID мутаційної перевірки	Власник
47	artifact-kind-capability-confusion	artifact-kind-capability-confusion	core/conformance.json
48	continuation-frontier-loss	continuation-frontier-loss	core/conformance.json
49	continuation-duplicate-emission	continuation-duplicate-emission	core/conformance.json
50	candidate-recipe-bypass	candidate-recipe-bypass	core/conformance.json
51	observational-candidate-pass-credit	observational-candidate-pass-credit	core/conformance.json
52	snapshot-provider-substitution	snapshot-provider-substitution	core/conformance.json
53	unsupported-provider-adapter	unsupported-provider-adapter	core/conformance.json
54	stale-release-closure-credit	stale-release-closure-credit	core/conformance.json
55	finding-evidence-target-mismatch	finding-evidence-target-mismatch	core/conformance.json
56	candidate-delta-scope-escape	candidate-delta-scope-escape	core/conformance.json
57	checkpoint-head-mismatch	checkpoint-head-mismatch	core/conformance.json
58	checkpoint-corruption-credit	checkpoint-corruption-credit	core/conformance.json
59	normal-command-full-replay	normal-command-full-replay	core/conformance.json
60	profile-default-depth-bypass	profile-default-depth-bypass	core/conformance.json
61	continuation-public-material-mismatch	continuation-public-material-mismatch	core/conformance.json
62	inventory-provenance-override	inventory-provenance-override	core/conformance.json
63	implementation-closure-drift	implementation-closure-drift	core/conformance.json
64	noncanonical-archive-layout	noncanonical-archive-layout	core/conformance.json
65	inventory-graph-inflation	inventory-graph-inflation	core/conformance.json
66	workcard-budget-bypass	workcard-budget-bypass	core/conformance.json
67	fabricated-evidence-digest	fabricated-evidence-digest	core/conformance.json
68	unsigned-standard-decision	unsigned-standard-decision	core/conformance.json
69	unconfigured-standard-decision-key	unconfigured-standard-decision-key	core/conformance.json
70	standard-decision-version-replay	standard-decision-version-replay	core/conformance.json
71	candidate-decision-cycle	candidate-decision-cycle	core/conformance.json
72	broad-query-corpus-pagination	broad-query-corpus-pagination	core/conformance.json
73	continuation-token-context-inflation	continuation-token-context-inflation	core/conformance.json
74	inventory-stream-materialization	inventory-stream-materialization	core/conformance.json
75	pdf-header-only-credit	pdf-header-only-credit	core/conformance.json
76	scale-environment-skip-credit	scale-environment-skip-credit	core/conformance.json
77	content-search-omission	content-search-omission	core/conformance.json

Власник: core/conformance.json | Валідатор: mutation_families

Автомати станів

Task

Стан	Дозволений наступний стан
BLOCKED	READY, CANCELLED
CANCELLED	-
COMPLETED	-
LEASED	RUNNING, READY, BLOCKED, CANCELLED
PLANNED	READY, BLOCKED, CANCELLED
READY	LEASED, BLOCKED, CANCELLED
RUNNING	COMPLETED, BLOCKED, CANCELLED

Власник: core/policy-set.json | Валідатор: state_machines.task + state_machine_transition

Lease

Стан	Дозволений наступний стан
ACTIVE	CLOSING, EXPIRED, REVOKED
CLOSED	-
CLOSING	CLOSED, EXPIRED, REVOKED
EXPIRED	-
REVOKED	-

Власник: `core/policy-set.json` | Валідатор: `state_machines.lease + state_machine_transition`

```
{'closed_release': {'requires_field': 'close_ack', 'slot_reusable': True, 'state': 'CLOSED'}, '
terminal_reconciliation': {'fencing_token_binding': 'exact-lease-fencing-token', 'field': '
capacity_reconciliation', 'generation_binding': 'exact-lease-generation', 'manager_capability': '
lease.manage', 'required_fields': ['reconciled_by', 'grant_id', 'grant_claim_digest', 'reconciled_at', '
generation', 'fencing_token'], 'slot_reusable_after_record': True, 'states': ['EXPIRED', 'REVOKED']}, '
unreconciled_slot_reusable': False}
```

Власник: `core/policy-set.json` | Валідатор: `state_machines.capacity_release`

Можливості й авторизація команд

Можливість	Призначення	Власник	Валідатор
standard.activate	Активувати одну конфігурацію репозиторію, прив'язану до digest.	core/authority-model.json	command_capability_scope
authority.manage	Видавати або відкликати runtime Grants у межах налаштованого root ceiling.	core/authority-model.json	command_capability_scope
task.plan	Створювати, ділити, впорядковувати або блокувати Tasks.	core/authority-model.json	command_capability_scope
task.execute	Виконувати один Task у межах його WorkCard і чинної Lease.	core/authority-model.json	command_capability_scope
lease.manage	Отримувати, підтверджувати heartbeat, закривати, завершувати строк або відкликати Lease.	core/authority-model.json	command_capability_scope
evidence.publish	Публікувати evidence Artifacts, прив'язані до candidate.	core/authority-model.json	command_capability_scope
validation.evaluate	Оцінювати gates без повноважень на promotion або release.	core/authority-model.json	command_capability_scope
finding.record	Записувати Findings.	core/authority-model.json	command_capability_scope
finding.resolve	Resolve one Finding only with evidence bound to that Finding's canonical digest.	core/authority-model.json	command_capability_scope
finding.waive	Waive one Finding only through an expiring exception-policy Decision, Finding-bound evidence, and separation of duties.	core/authority-model.json	command_capability_scope
candidate.promote	Просувати перевірений Candidate.	core/authority-model.json	command_capability_scope
release.decide	Record an immutable historical release judgement bound to an exact acceptance-closure and provider-binding digest; current eligibility remains derived.	core/authority-model.json	command_capability_scope
export.create	Створювати очищений зовнішній пакет.	core/authority-model.json	command_capability_scope
migration.manage	Виконувати cutover, rollback і deterministic recutover.	core/authority-model.json	command_capability_scope
projection.rebuild	Перебудовувати або перевіряти ненормативні проєкції.	core/authority-model.json	command_capability_scope
projection.read	Read one bounded current projection or resume its continuation under the exact current Grant claim; it grants no mutation, validation, promotion, product acceptance, or standard distribution authority.	core/authority-model.json	command_capability_scope

Відображення команди у можливість

Тип команди	Визначення можливості	Власник	Валідатор
task.record	task.plan	core/authority-model.json	command_capability_scope
task.transition	task.plan	core/authority-model.json	command_capability_scope
candidate.record	task.execute	core/authority-model.json	command_capability_scope
artifact.record	{ "capability_by_value": { "diff": "task.execute", "evidence": "evidence.publish", "log": "task.execute", "product": "task.execute", "report": "projection.rebuild", "payload_field": "artifact_kind" }	core/authority-model.json	command_capability_scope
run.record	{ "capability_by_value": { "execution": "task.execute", "validation": "validation.evaluate", "payload_field": "run_kind" }	core/authority-model.json	command_capability_scope
gate.record	validation.evaluate	core/authority-model.json	command_capability_scope
finding.record	finding.record	core/authority-model.json	command_capability_scope
decision.record	{ "capability_by_value": { "accept": "validation.evaluate", "promote": "candidate.promote", "reject": "validation.evaluate", "release": "release.decide", "resolve": "finding.resolve", "revoke": "authority.manage", "waive": "finding.waive", "payload_field": "decision_kind" }	core/authority-model.json	command_capability_scope
grant.issue	authority.manage	core/authority-model.json	command_capability_scope
grant.revoke	authority.manage	core/authority-model.json	command_capability_scope
lease.record	lease.manage	core/authority-model.json	command_capability_scope

Область дії команд

Тип команди	Режим	Правило цілі	Власник	Валідатор
task.record	payload-id	{ "id_field": "task_id", "kind": "task" }	core/authority-model.json	command_capability_scope
task.transition	payload-id	{ "id_field": "task_id", "kind": "task" }	core/authority-model.json	command_capability_scope
candidate.record	payload-id	{ "id_field": "candidate_id", "kind": "candidate" }	core/authority-model.json	command_capability_scope
artifact.record	payload-id	{ "id_field": "artifact_id", "kind": "artifact" }	core/authority-model.json	command_capability_scope
run.record	payload-multi-id	{ "targets": [{ "id_field": "task_id", "kind": "task" }, { "id_field": "candidate_digest", "kind": "candidate" }] }	core/authority-model.json	command_capability_scope
gate.record	payload-id	{ "id_field": "candidate_digest", "kind": "candidate" }	core/authority-model.json	command_capability_scope
finding.record	payload-id	{ "id_field": "finding_id", "kind": "finding" }	core/authority-model.json	command_capability_scope
decision.record	payload-dynamic-id-or-referenced-grant-scope	{ "grant_target_type": "Grant", "id_field": "target_id", "kind_field": "target_type", "kind_map": { "Candidate": "candidate", "Finding": "finding", "Task": "task" } }	core/authority-model.json	command_capability_scope
grant.issue	payload-scope	{ "scope_field": "scope" }	core/authority-model.json	command_capability_scope
grant.revoke	referenced-grant-scope	{ "grant_id_field": "grant_id" }	core/authority-model.json	command_capability_scope
lease.record	payload-id	{ "id_field": "task_id", "kind": "task" }	core/authority-model.json	command_capability_scope

Прив'язка мутаційної команди до оренди

Команда, прив'язана до Lease, допустима лише тоді, коли authorization доводить точну capability команди, holder_authorization незалежно доводить task.execute, обидва точні твердження Grant прив'язують той самий subject, Activation і conjunctive effect scope, зазначена Lease є поточною й ACTIVE у момент issued_at, а task, generation, fence, candidate, context і digests WorkCard збігаються точно.

Компонент твердження	Канонічне значення	Власник	Валідатор
command_authorization_source	CommandRequest.authorization.grant_id + grant_claim_digest	core/authority-model.js on	lease_fence + command_capability_scope
grant_separation_rule	Grant авторизації команди та holder Grant визначаються й перевіряються незалежно; вони можуть бути одним записом лише тоді, коли обов'язкова capability команди є task.execute.	core/authority-model.js on	lease_fence + command_capability_scope
holder_grant_capability	task.execute	core/authority-model.js on	lease_fence + command_capability_scope
holder_grant_field	holder_authorization.grant_id	core/authority-model.js on	lease_fence + command_capability_scope
lease_bound_command_kinds	["artifact.record", "run.record", "gate.record", "finding.record"]	core/authority-model.js on	lease_fence + command_capability_scope
lease_bound_task_transition_states	["LEASED", "RUNNING", "COMPLETED"]	core/authority-model.js on	lease_fence + command_capability_scope
lease_required_state	ACTIVE	core/authority-model.js on	lease_fence + command_capability_scope
required_command_fields	["workcard_task_id", "holder_authorization", "lease_id", "lease_generation", "fencing_token", "workcard_digest", "context_digest"]	core/authority-model.js on	lease_fence + command_capability_scope
required_workcard_fields	["holder_grant_id", "lease_id", "lease_generation", "fencing_token", "context_digest"]	core/authority-model.js on	lease_fence + command_capability_scope
stale_or_unresolved	reject	core/authority-model.js on	lease_fence + command_capability_scope
Рівність прив'язки		Власник	Валідатор
CommandRequest.workcard_task_id == WorkCard.task_id == Lease.task_id		core/authority-model.json	lease_fence + command_capability_scope
CommandRequest.subject_id == Lease.holder_subject_id		core/authority-model.json	lease_fence + command_capability_scope
CommandRequest.holder_authorization.grant_id == WorkCard.holder_grant_id == Lease.holder_grant_id		core/authority-model.json	lease_fence + command_capability_scope
CommandRequest.lease_id == WorkCard.lease_id == Lease.lease_id		core/authority-model.json	lease_fence + command_capability_scope
CommandRequest.lease_generation == WorkCard.lease_generation == Lease.generation		core/authority-model.json	lease_fence + command_capability_scope
CommandRequest.fencing_token == WorkCard.fencing_token == Lease.fencing_token		core/authority-model.json	lease_fence + command_capability_scope
CommandRequest.context_digest == WorkCard.context_digest		core/authority-model.json	lease_fence + command_capability_scope
CommandRequest.workcard_digest == sha256(canonical WorkCard)		core/authority-model.json	lease_fence + command_capability_scope

```

{
  "binding_equalities": [
    "CommandRequest.workcard_task_id == WorkCard.task_id == Lease.task_id",
    "CommandRequest.subject_id == Lease.holder_subject_id",
    "CommandRequest.holder_authorization.grant_id == WorkCard.holder_grant_id == Lease.holder_grant_id",
    "CommandRequest.lease_id == WorkCard.lease_id == Lease.lease_id",
    "CommandRequest.lease_generation == WorkCard.lease_generation == Lease.generation",
    "CommandRequest.fencing_token == WorkCard.fencing_token == Lease.fencing_token",
    "CommandRequest.context_digest == WorkCard.context_digest",
    "CommandRequest.workcard_digest == sha256(canonical WorkCard)"
  ],
  "claim_rule": "Команда, прив'язана до Lease, допустима лише тоді, коли authorization доводить точну capability команди, holder_authorization незалежно доводить task.execute, обидва точні твердження Grant прив'язують той самий subject, Activation і conjunctive effect scope, зазначена Lease є поточною й ACTIVE у момент issued_at, а task, generation, fence, candidate, context і digests WorkCard збігаються точно.",
  "command_authorization_source": "CommandRequest.authorization.grant_id + grant_claim_digest",
  "grant_separation_rule": "Grant авторизації команди та holder Grant визначаються й перевіряються незалежно; вони можуть бути одним записом лише тоді, коли обов'язкова capability команди є task.execute.",
  "holder_grant_capability": "task.execute",
  "holder_grant_field": "holder_authorization.grant_id",
  "lease_bound_command_kinds": [
    "artifact.record",
    "run.record",
    "gate.record",
    "finding.record"
  ],
  "lease_bound_task_transition_states": [
    "LEASED",
    "RUNNING",
    "COMPLETED"
  ],
  "lease_required_state": "ACTIVE",

  "required_command_fields": [
    "workcard_task_id",
    "holder_authorization",
    "lease_id",
    "lease_generation",
    "fencing_token",
    "workcard_digest",
    "context_digest"
  ],
  "required_workcard_fields": [
    "holder_grant_id",
    "lease_id",
    "lease_generation",
    "fencing_token",
    "context_digest"
  ],
  "stale_or_unresolved": "reject"
}

```

Власник: core/authority-model.json; core/policy-set.json | Валідатор: command_mutation_claim_rule + lease_fence + command_capability_scope

Повні для відтворення Evidence і CAS

Поле evidence	Правило	Схема
Artifact.required	["record_type", "artifact_id", "artifact_kind", "digest", "media_type", "size_bytes", "retention_class", "created_at"]	Artifact
EvidenceBinding.required	["activation_digest", "implementation_closure_digest", "candidate_digest", "policy_digest", "tool_digest", "input_digests"]	EvidenceBinding
outcome	["pass", "fail", "blocked", "skipped", "error"]	Artifact
evidence_class	["product-execution", "validator", "harness-generated", "migration"]	Artifact
product_credit_eligible	true лише для свіжого, вирішеного, успішного evidence класу product-execution	Artifact.allOf
GateResult.evidence_artifacts	exact artifact_id + artifact_record_digest for pass, fail, or blocked	GateResult.oneOf

Власник: core/contracts.schema.json; core/semantic-model.json; core/policy-set.json | Валідатор: \$defs/Artifact + \$defs/EvidenceBinding + \$defs/GateResult + candidate_evidence_binding + credit_requires_evidence

Розділення обов'язків

ID	Правило	Обмеження	Власник	Валідатор
SOD-001	lease-is-not-acceptance	{"forbid_lease_derived_capabilities": ["validation.evaluate", "finding.resolve", "finding.waive", "candidate.promote", "release.decide"], "scope_kind": "lease"}	core/authority-model.json	SOD-001
SOD-002	executor-not-release-decider	{"forbid_same_subject_for_same_candidate": ["task.execute", "release.decide"], "scope_kind": "candidate"}	core/authority-model.json	SOD-002
SOD-003	validator-not-self-promoter	{"forbid_same_subject_for_same_candidate": ["validation.evaluate", "candidate.promote"], "scope_kind": "candidate"}	core/authority-model.json	SOD-003
SOD-004	finder-not-sole-waiver	{"forbid_same_subject_for_same_finding": ["finding.record", "finding.waive"], "scope_kind": "finding"}	core/authority-model.json	SOD-004
SOD-005	evidence-producer-not-distribution-decider	{"forbid_same_key_capabilities": ["evidence.produce", "standard.distribute"], "forbid_same_subject_for_same_candidate": ["evidence.produce", "standard.distribute"], "scope_kind": "candidate-and-key"}	core/authority-model.json	SOD-005

Власник: core/authority-model.json | Валідатор: capabilities + command_capability_rules + command_effect_scope_rules + separation_of_duties

Довіра, початкова авторизація і presets ролей

authority.json configures root ceilings and Activation only. Root command proofs bind one command intent and may issue only the first authority.manage Grant. Every Grant proof binds claim_digest; later Grants cite an active authority.manage issuer Grant or team signatures verified by the exact signature adapter selected in technologies.json.

Режим довіри	Визначення	Власник	Валідатор
local-owner	{"boundary": "root authority однієї машини під контролем власника", "signer_required": false}	core/authority-model.json	grant_validity
team-signed	{"boundary": "кілька учасників або віддалений транспорт", "minimum_threshold": 1, "provider_capability": "signature", "provider_resolution": "Resolve the exact technologies.json signature binding for init and every ActivationGuard, service, CLI, replay, and mutation path. A process-local injected adapter is explicitly non-reconstructable and valid only while bound to the same project, Activation, provider, and identity; cross-process CLI workflows require a reconstructable configured adapter.", "signer_required": true, "unsupported_binding": "reject-before-workflow"}	core/authority-model.json	grant_validity
Preset ролі	Можливості	Статус повноважень	Власник
auditor	projection.read, finding.record, finding.resolve, export.create	ненормативний	core/authority-model.json
root-owner	standard.activate, authority.manage, task.plan, lease.manage, finding.resolve, finding.waive, candidate.promote, release.decide, export.create, migration.manage, projection.rebuild, projection.read	ненормативний	core/authority-model.json
validator	projection.read, validation.evaluate, evidence.publish, finding.record	ненормативний	core/authority-model.json
worker	projection.read, task.execute, evidence.publish	ненормативний	core/authority-model.json

Власник: core/authority-model.json | Валідатор: bootstrap_rule + trust_modes + role_presets_non_authoritative

Семантичний словник

Сутність	Призначення		Власник	Валідатор
Завдання	Представляти одну плановану одиницю з одним acceptance predicate та обмеженою mutation score.		core/semantic-model.json	admission.entity
Candidate	Identify either one provider-proven immutable product snapshot or one explicitly non-creditable observation, independently from control state and bound to the exact candidate recipe.		core/semantic-model.json	admission.entity
Artifact	Identify immutable product, evidence, log, report, or diff bytes with provenance and retention metadata; raw-file inventory produces only derived Artifact proxies, while a diff may carry the canonical Candidate-delta change manifest without creating another persistent entity.		core/semantic-model.json	admission.entity
Run	Записувати одну спробу execution або validation щодо точних candidate, policy, tool та input digests.		core/semantic-model.json	admission.entity
Finding	Записувати спостережений дефект або порушення policy з явною effective-blocking semantics.		core/semantic-model.json	admission.entity
Decision	Record a project-operational acceptance, rejection, resolution, waiver, promotion, or release judgement. It is distinct from the external StandardReleaseDecision for the standard distribution.		core/semantic-model.json	admission.entity
GateResult	Record one gate outcome against the exact immutable Task-owned GateRunDefinition, current Run, and finalized evidence Artifact records; skipped, blocked, failed, stale, or unresolved evidence never receives pass credit.		core/semantic-model.json	admission.entity
Grant	Доводити одну capability subject у межах однієї Activation, time interval, nonce і scope.		core/semantic-model.json	admission.entity
Оренда	Координувати тимчасове володіння мутацією через generation, fencing, expiry, heartbeat і підтвердження закриття.		core/semantic-model.json	admission.entity
Відношення	Джерело	Ціль	Правила	Валідатор
DEPENDS_ON	Завдання	Завдання	acyclic-for-ready-frontier, accepted-current-outcome-required	admission.relation
READS	Task, Run	Candidate, Artifact	read-does-not-imply-authority	admission.relation
PRODUCES	Task, Run	Candidate, Artifact, Finding, Decision	content-or-record-identity-required	admission.relation
VALIDATES	Run	Candidate, Artifact, Task	candidate-policy-tool-input-binding-required	admission.relation
CITES	Finding, Decision	Artifact, Run, Finding	selector-and-content-resolution-required	admission.relation
BLOCKS	Finding	Task, Candidate, Decision	effective-only-while-open-and-blocking	admission.relation
Об'єкт-значення	Значення		Власник	Валідатор
CandidateDelta	A diff Artifact payload binding base and new Candidate digests, WorkCard digest, and the complete normalized changed-path set.		core/semantic-model.json	design_rules
Команда	Запит на зміну стану, перевірений до commit.		core/semantic-model.json	design_rules
Comparator	Чисте правило впорядкування або рівності.		core/semantic-model.json	design_rules
ContinuationToken	An authenticated bounded runtime value signed with the random per-Activation secret. It binds subject, exact projection.read Grant claim, requested scope, revocation state, normalized query, cursor, Activation, HEAD, projection, ranking, complete budgets, depth, implementation closure, issue time, and expiry; it never grants authority by itself.		core/semantic-model.json	design_rules
CurrentReleaseStatus	A derived evaluation of an immutable historical release Decision against the current closure, blockers, evidence, providers, and Activation.		core/semantic-model.json	design_rules

Об'єкт-значення	Значення	Власник	Валідатор
ImplementationClosure	A canonical capability-specific binding of the promin runtime, Python interpreter, jsonschema or SQLite component where used, exactly one adapter, and the recomputed aggregate closure digest.	core/semantic-model.json	design_rules
Operation	Один завершений семантичний крок із явними семантиками input, output, failure і evidence.	core/semantic-model.json	design_rules
PlanProposal	A non-authoritative bounded proposal that derives candidate Tasks and relations from project intent, repository facts, backlog, and references; it cannot mutate operational state or grant authority.	core/semantic-model.json	design_rules
Policy	Машинно-адресований інваріант, власником якого є policy-set.json.	core/semantic-model.json	design_rules
Predicate	Чисте правило ухвалення рішення.	core/semantic-model.json	design_rules
ProviderInvocationEvidence	The exact capability, provider, invocation kind, identity kind and digest, and adapter ID actually used by one operation.	core/semantic-model.json	design_rules
Query	Read-only запит, який не може змінювати нормативний стан.	core/semantic-model.json	design_rules
StandardReleaseDecision	An external human-reviewable distribution record bound to the exact promin ZIP, Core, preset, tools, implementation closure, tests, platform results, saturation, PDFs, and evidence manifest. It never grants product acceptance.	core/semantic-model.json	design_rules
StateCheckpoint	A disposable self-digested state or journal index bound to an exact authoritative HEAD; it cannot grant authority or acceptance.	core/semantic-model.json	design_rules
WorkCard	A bounded generated context package for one Task that separately binds the projection.read query Grant claim and the task.execute holder Grant; neither binding implies acceptance.	core/semantic-model.json	design_rules

Похідна проєкція	Повноваження	Власник	Валідатор
InventoryEntry	похідна та ненормативна	core/semantic-model.json	derived_projection_kinds
SearchDocument	похідна та ненормативна	core/semantic-model.json	derived_projection_kinds
InventoryProjectionRow	похідна та ненормативна	core/semantic-model.json	derived_projection_kinds
ProjectionLimits	похідна та ненормативна	core/semantic-model.json	derived_projection_kinds
RetrievalPage	похідна та ненормативна	core/semantic-model.json	derived_projection_kinds
ReadyFrontier	похідна та ненормативна	core/semantic-model.json	derived_projection_kinds
WorkCardProjection	похідна та ненормативна	core/semantic-model.json	derived_projection_kinds
WorkCard	похідна та ненормативна	core/semantic-model.json	derived_projection_kinds
NextResult	похідна та ненормативна	core/semantic-model.json	derived_projection_kinds
ContinueResult	похідна та ненормативна	core/semantic-model.json	derived_projection_kinds
NavigationView	похідна та ненормативна	core/semantic-model.json	derived_projection_kinds
StateCheckpoint	похідна та ненормативна	core/semantic-model.json	derived_projection_kinds
CurrentReleaseStatus	похідна та ненормативна	core/semantic-model.json	derived_projection_kinds
DoctorResult	похідна та ненормативна	core/semantic-model.json	derived_projection_kinds
OperationMetrics	похідна та ненормативна	core/semantic-model.json	derived_projection_kinds
Report	похідна та ненормативна	core/semantic-model.json	derived_projection_kinds

Власник: core/semantic-model.json | Валідатор: admission + design_rules + persistent_entities + relations + value_objects + derived_projection_kinds

Технологічні межі

Можливість	Завдання	Заборонена межа	Власник	Валідатор
control-runtime	Розбирати explicit configuration, канонізувати JSON і виконувати local CLI orchestration.	{"effect": "reject", "forbids": ["authority-invention", "policy-invention", "provider-binding-invention"], "normative_authority": false}	core/semantic-model.json	technology_capabilities
shape-validation	Перевіряти єдиний contract bundle Draft 2020-12 з обов'язковими formats і local references.	{"effect": "reject", "forbids": ["authority-decision", "cross-record-semantic-decision"], "normative_authority": false}	core/semantic-model.json	technology_capabilities
content-identity	Обчислювати content digests та ідентичності content-addressed objects.	{"effect": "reject", "forbids": ["digest-as-action-authority"], "normative_authority": false}	core/semantic-model.json	technology_capabilities
local-serialization	Надавати один local-writer lock, atomic replace, file sync, directory sync і recovery primitives.	{"effect": "reject", "forbids": ["distributed-consensus-claim"], "normative_authority": false}	core/semantic-model.json	technology_capabilities
query-projection	Build disposable status and bounded typed-retrieval projections; derive ReadyFrontier, materialize one exact WorkCardProjection, and return Core-owned NextResult or ContinueResult envelopes for the six supported CLI commands.	{"effect": "reject", "forbids": ["projection-as-normative-authority"], "normative_authority": false}	core/semantic-model.json	technology_capabilities
filesystem-inventory	Apply the exact candidate recipe in one product pass, use no-follow reads, and publish either a provider-proven immutable VCS tree stream or an explicit non-creditable observation.	{"effect": "reject", "forbids": ["implicit-initialization-scan"], "normative_authority": false}	core/semantic-model.json	technology_capabilities
export-scan	Виконувати bounded recursive export inspection, allowlisting, secret classification і відхилення symlink.	{"effect": "reject", "forbids": ["package-as-authorization-boundary"], "normative_authority": false}	core/semantic-model.json	technology_capabilities
signature	Dispatch the exact configured signature adapter to sign or verify activation, Grant, and Decision digests for team trust mode, and evidence the invoked adapter identity.	{"effect": "reject", "forbids": ["core-signer-implementation-lock-in"], "normative_authority": false}	core/semantic-model.json	technology_capabilities
build-dependency	Імпортувати project-specific build і dependency facts як похідні проєкції.	{"effect": "reject", "forbids": ["semantic-model-redefinition"], "normative_authority": false}	core/semantic-model.json	technology_capabilities

Власник: core/semantic-model.json | Валідатор: technology_capabilities

Бюджети відповідності й масштабування

Фізична перевірка використовує 100 000 гав-файлів і рівно один похідний Artifact проху на файл. Raw inventory не створює Task або Relation семантики. Окремий авторизований harness corpus доводить проєкцію щонайменше до 198 999 Core-valid типізованих Relations. Щонайменше 600 фактичних runtime-запитів поєднують exact identities, content-only і high-cardinality terms, broad bounded refinement, misses, hostile identity text та forced continuation на глибинах 1-12. Цей bounded physical proof не є повним виконанням 1000 на 1000 і не повинен ним замінюватися.

Структурний бюджет	Значення	Власник	Валідатор
authority_bearing_init_records	5	core/conformance.json	authority_bearing_init_records
base_user_commands_max	10	core/conformance.json	base_user_commands_max
core_normative_artifacts	6	core/conformance.json	core_normative_artifacts
persistent_entity_kinds_max	9	core/conformance.json	persistent_entity_kinds_max
relation_kinds_max	6	core/conformance.json	relation_kinds_max

Структурний бюджет	Значення	Власник	Валідатор
selected_presets	1	core/conformance.json	selected_presets
Граничні значення WorkCard	Значення	Власник	Валідатор
max_bytes	16384	core/conformance.json	bounded-workcard-through-depth-twelve
max_entities	32	core/conformance.json	bounded-workcard-through-depth-twelve
max_fanout_per_entity	8	core/conformance.json	bounded-workcard-through-depth-twelve
max_relations	48	core/conformance.json	bounded-workcard-through-depth-twelve
top_k	12	core/conformance.json	bounded-workcard-through-depth-twelve
Контракт масштабування	Визначення	Власник	Валідатор
event_commit	{"complexity": "O(batch)", "head_lookup": "O(1)"}	core/conformance.json	event_commit
init	{"complexity": "O(core+plan)", "project_tree_passes": 0}	core/conformance.json	init
inventory	{"complexity": "O(files+bytes)", "creditable_consistency_modes": ["immutable-vcs-tree"], "derived_artifact_proxies_per_raw_file": 1, "manifest_verified_jsonl": true, "memory_amplification_max": 32, "observational_mode_pass_credit": false, "project_tree_passes_max": 1, "search_text_bytes_max": 4096, "synthetic_relations_per_raw_file": 0, "synthetic_tasks_per_raw_file": 0}	core/conformance.json	inventory
normal_one_shot_replay	{"checkpoint_authoritative": false, "complexity": "O(delta-since-verified-checkpoint)", "full_replay_allowed": "doctor, validate, recovery, or verified fallback only"}	core/conformance.json	normal_one_shot_replay
projection	{"public_rebuild_ingress": "verified InventoryResult or persisted verified manifest only", "rebuild_product_tree_passes": 0, "reserved_provenance_rejected": true}	core/conformance.json	projection
replay	{"complexity": "O(events)"}	core/conformance.json	replay
saturation	{"fresh_control_state_per_iteration": true, "init_product_tree_scans": 0, "semantic_state_reused": false}	core/conformance.json	saturation
workcard	{"broad_query_behavior": "deterministic bounded top-k seeds, refinement_required=true, no corpus pagination", "complexity": "O(top_k+capped_typed_closure)", "continuation_state_bytes_max": 16384, "continuation_token_bytes_max": 256, "corpus_independent_output_bound": true, "seed_budget": "effective_top_k=min(profile.top_k,max(2,floor(max_entities/(1+observed_branching*requested_depth))))", "selected_closure_union_completeness": 1}	core/conformance.json	workcard
Еталонне вимірювання	Значення	Власник	Валідатор
core_valid_relation_count	198999	core/conformance.json	reference-benchmark
depths_tested	[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12]	core/conformance.json	reference-benchmark
file_count	100000	core/conformance.json	reference-benchmark
forced_continuation	True	core/conformance.json	reference-benchmark
inventory_passes	1	core/conformance.json	reference-benchmark

Еталонне вимірювання	Значення	Власник	Валідатор
performance_profiles	<pre>{ "portable-local-v1": { "commit_bytes_per_changed_record_max": 32768, "commit_p95_ms_max": 500, "commit_p99_ms_max": 1000, "compatible_platforms": ["linux", "windows"], "database_bytes_max": 402653184, "p50_ms_max": 100, "p95_ms_max": 150, "p99_ms_max": 500, "peak_rss_bytes_max": 805306368, "projection_amplification_max": 20, "requires_same_runner_no_degradation": true, "runtime_checkpoint_count_max": 24, "semantic_inflation_max": 3.25, "semantic_ingestion_seconds_max": 600 } }</pre>	core/conformance.json	reference-benchmark
physical_relation_corpus	<pre>{ "artifact_target_coverage": 1, "expected_counts_source": "fixture-manifest", "generation": "explicit-authorized-command-events", "relation_count": 198971, "relation_kind": "READS", "separate_from_raw_inventory": true, "task_count": 1567, "total_core_valid_relations": 198999 }</pre>	core/conformance.json	reference-benchmark
query_count	600	core/conformance.json	reference-benchmark
raw_file_proxy_ratio	1	core/conformance.json	reference-benchmark
rebuild_product_passes	0	core/conformance.json	reference-benchmark
reference_targets_are_environment_specific	True	core/conformance.json	reference-benchmark
semantic_corpus	<pre>{ "depths": [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12], "expected_counts_source": "fixture-manifest", "relation_count": 28, "relation_kind": "DEPENDS_ON", "separate_from_raw_inventory": true, "task_count": 32 }</pre>	core/conformance.json	reference-benchmark
silent_truncations_max	0	core/conformance.json	reference-benchmark
synthetic_task_ratio	0	core/conformance.json	reference-benchmark
Власник: core/conformance.json Валідатор: structural_budgets + workcard_hard_ceiling + scale_contracts + reference_benchmarks			

Набір налаштувань і профілі

Вибирати Semantic Programming і бюджети декомпозиції, retrieval та parallelism Morok/Tower без надання повноважень на дії.

Профіль	Бюджети	Власник	Валідатор
morok-local	<pre>{ "default_dependency_depth": 2, "max_context_bytes": 8192, "max_entities": 16, "max_fanout_per_entity": 4, "max_parallel_tasks": 1, "max_relations": 24, "model_tier": "weak-local", "top_k": 8 }</pre>	presets/semantic-morok-tower.json	profiles

Профіль	Бюджети	Власник	Валідатор
tower-capable	{"default_dependency_depth": 4, "max_context_bytes": 12288, "max_entities": 24, "max_fanout_per_entity": 6, "max_parallel_tasks": 3, "max_relations": 36, "model_tier": "capable", "top_k": 10}	presets/semantic-morok-tower.json	profiles
tower-strong	{"default_dependency_depth": 6, "max_context_bytes": 16384, "max_entities": 32, "max_fanout_per_entity": 8, "max_parallel_tasks": 6, "max_relations": 48, "model_tier": "strong", "top_k": 12}	presets/semantic-morok-tower.json	profiles
Поле preset	Значення	Власник	Валідатор
base_user_commands	["init", "doctor", "status", "next", "validate", "continue", "audit", "refresh", "context", "skills"]	presets/semantic-morok-tower.json	base_user_commands
required_provider_capabilities	["control-runtime", "shape-validation", "content-identity", "local-serialization", "query-projection"]	presets/semantic-morok-tower.json	required_provider_capabilities
forbidden_implicit_behavior	["full-scan-on-init", "provider-discovery-as-authority", "implicit-action-grant", "unbounded-context-expansion", "search-ranking-as-acceptance-proof"]	presets/semantic-morok-tower.json	forbidden_implicit_behavior
semantic_programming	{"default_visibility": "private", "object_promotion_criteria": ["independent-invariant", "independent-policy", "replaceability", "measurable-algorithm", "failure-boundary", "reuse", "domain-significance"], "pipeline": ["validate-input", "prepare-change", "execute-operation", "record-result", "publish-evidence"]}	presets/semantic-morok-tower.json	semantic_programming
additional_provider_capabilities	["filesystem-inventory", "export-scan", "signature", "build-dependency"]	presets/semantic-morok-tower.json	additional_provider_capabilities
Власник: presets/semantic-morok-tower.json Валідатор: Preset schema + preset field identifiers			

Кореневий вхід схеми

#	Прийняте кореневе визначення	Власник	Валідатор
1	#\$defs/StandardManifest	core/contracts.schema.json	#\$defs/StandardManifest
2	#\$defs/SemanticModel	core/contracts.schema.json	#\$defs/SemanticModel
3	#\$defs/AuthorityModel	core/contracts.schema.json	#\$defs/AuthorityModel
4	#\$defs/PolicySet	core/contracts.schema.json	#\$defs/PolicySet
5	#\$defs/Conformance	core/contracts.schema.json	#\$defs/Conformance
6	#\$defs/Preset	core/contracts.schema.json	#\$defs/Preset
7	#\$defs/ProjectInit	core/contracts.schema.json	#\$defs/ProjectInit
8	#\$defs/StandardsInit	core/contracts.schema.json	#\$defs/StandardsInit
9	#\$defs/TechnologiesInit	core/contracts.schema.json	#\$defs/TechnologiesInit
10	#\$defs/AuthorityInit	core/contracts.schema.json	#\$defs/AuthorityInit
11	#\$defs/Activation	core/contracts.schema.json	#\$defs/Activation
12	#\$defs/Grant	core/contracts.schema.json	#\$defs/Grant
13	#\$defs/Lease	core/contracts.schema.json	#\$defs/Lease
14	#\$defs/Task	core/contracts.schema.json	#\$defs/Task
15	#\$defs/Candidate	core/contracts.schema.json	#\$defs/Candidate
16	#\$defs/Artifact	core/contracts.schema.json	#\$defs/Artifact
17	#\$defs/Run	core/contracts.schema.json	#\$defs/Run
18	#\$defs/Finding	core/contracts.schema.json	#\$defs/Finding
19	#\$defs/Decision	core/contracts.schema.json	#\$defs/Decision
20	#\$defs/GateResult	core/contracts.schema.json	#\$defs/GateResult
21	#\$defs/Relation	core/contracts.schema.json	#\$defs/Relation
22	#\$defs/WorkCard	core/contracts.schema.json	#\$defs/WorkCard
23	#\$defs/CommandRequest	core/contracts.schema.json	#\$defs/CommandRequest
24	#\$defs/CommandResult	core/contracts.schema.json	#\$defs/CommandResult
25	#\$defs/EventBatch	core/contracts.schema.json	#\$defs/EventBatch

Власник: core/contracts.schema.json | Валідатор: oneOf + Draft 2020-12